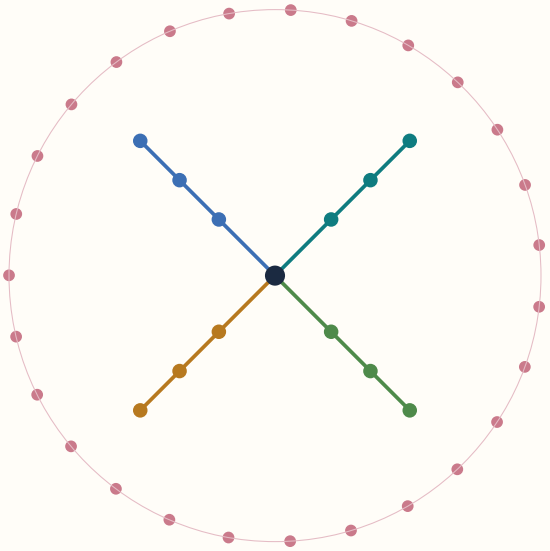


one point 4 lines 12 neighbours 27 far points

Forty Points

One qutrit, the exceptional algebras, and a light cone

An exact and honest account of the $W(3,3)$ programme,
written so that anyone can follow it

The W(3,3) Theory Programme

repository `wilcompute/W33-Theory` · 26 September 2026

Every mathematical statement in this paper carries a status badge and, where it is a computation, the path of the committed script and certificate that proves it. Interpretations are labelled as interpretations. Things that were believed and later shown false are kept on the page, with the measurement that overturned them.

Abstract

For everyone. Take the smallest quantum system that has three settings instead of two — a *qutrit* — and let it act on itself. The ways it can be transformed, and the ways those transformations do or do not interfere with one another, form a finite geometric object with exactly forty points. This paper follows that single object as far as exact mathematics can take it. It turns out to contain a tiny version of space-time with a light cone; four interlocking "clocks"; the largest exceptional symmetry in mathematics, E_8; the octonionic algebra of 3×3 matrices that physicists have long suspected sits behind particle physics; and, inside that algebra, a genuine ten-dimensional Lorentz group whose spinors are the matter particles of one generation. We also say, just as clearly, what the object does *not* do: it does not by itself produce the masses, the forces, or the dynamics of our universe, and it provably cannot decide whether nature is left-handed or right-handed. Finally we settle, by an exhaustive exact computation, the most important open question about the string-theory models built on it: in their supersymmetric vacua the proton is not protected, because a field with the quantum numbers of a right-handed sneutrino is forced to condense — and that field carries exactly the charge that switches the forbidden couplings back on. We follow every escape route we could find to its end: other discrete symmetries, parities built from rotations of the hidden dimensions, and very heavy superpartners. Each fails — one only after we found that the selection rule it relied on is incomplete — and the last would need a top quark lighter than the one we measure.

For specialists. The symplectic generalized quadrangle $W(3,3)$ is the projective commutation geometry of the two-sided Weyl action on one qutrit's Hilbert–Schmidt space; its Bell line is the unitary-channel diagonal. Fixing that line, the 27 transverse Lagrangians form $\mathrm{Sym}_2(\mathbb{F}_3)$ with the finite Minkowski form $t^2-x^2-y^2$, whose null relation is context intersection; the four null directions are $\mathbb{P}^1(\mathbb{F}_3)$, the tetracode is their evaluation code, and the same four labels index the Hesse striations and an $A_2^4\subset E_8$. On the committed executable E_8 Chevalley basis, a Hesse tick line induces the contact grading $1+56+134+56+1$; its 56 is exactly the Freudenthal triple system of the split Albert algebra whose norm is the repository's signed 45-triad E_6 cubic, $Q=(\alpha\beta+T)^2-4\beta N(X)-4\alpha N(Y)+4T(X^{\#},Y^{\#})$, and the 57-dimensional Heisenberg extension carries the quasiconformal cone $N=Q/4-\tau^2$ with the Günaydin–Koepsell–Nicolai inversion law. The Killing form obeys $K(e_\alpha,e_{-\alpha})=-60(-1)^{\mathrm{ht}\,\alpha}$; the three tick lines give commuting $E_{8(-24)}$ involutions with $E_8=80+56+56+56$, restricting to $E_{7(-25)}$, whose Hermitian Jordan triple yields a Euclidean Albert algebra with Peirce split $1+16+10$, a Lorentzian $(1,9)$ Peirce slice, $\mathrm{Der}=F_{4(-52)}$, $\mathrm{str}_0=E_{6(-26)}$, and an $\mathfrak{so}(1,9)$ acting on a Majorana–Weyl 16. The repository's matter-parity charge is $6L_c-2$, and matter parity is the 2π rotation of $\mathrm{Spin}(9)$. The committed split real form, by contrast, is the height twist $\exp(i\pi\rho^\vee)$ and gives a $(5,5)$ slice: signature is a real-form datum, invisible over $\mathbb{F}_3$. For the 215 heterotic $\mathbb{Z}_6$ Standard Models of the $W(3,3)$ twist we recover and freeze the exact spectra and prove, over the full rational $B-L$ freedom and over all 1024 $\mathbb{Z}_2$ characters odd on quarks and leptons, that no FI-cancelling D-flat singlet vacuum preserves matter parity: the never-even singlets have $B-L=\pm1$ and every D-flat vacuum condenses one. The obstruction is one E_6-shaped $U(1)$, $4Q_\psi-\frac{4}{5}Q_\chi$; it extends to every element of the gauge torus times the space group (Smith normal forms), the forced field obeys $Q(n)=-Q(u^cd^cd^c)=-Q(q\ell d^c)=-Q(\ell\ell e^c)$ and regenerates all three couplings at quartic order, and baryon triality is excluded by $Q(u^c)+Q(e^c)=2Q(q)$. Plane-rotation

R-symmetries appear to reopen parity in one model, but only through an R-rule that is incomplete on non-prime planes; with the rules derived from the worldsheet instantons (Bizet et al.) the $\mathbb{Z}_6$-II class closes $(0/128)$, as do the $\mathbb{Z}_2 \times \mathbb{Z}_6$ and $\mathbb{Z}_3 \times \mathbb{Z}_6$ families of the same twist, and $\mathbb{Z}_{12}$-I once the E_6 lattice fixes its admissible rule; in the prime orbifold $\mathbb{Z}_3 \times \mathbb{Z}_3$, where every rule is established, the parity survives and the vacua are supersymmetric — the superpotential is a $\mathbb{Z}_3$-graded function of one or two composites — but a hidden-representation index leaves six massless charge-$\frac{1}{3}$ states in every one, and breaking the hidden group to cure it leaves fractional charges protected by an unbroken discrete 3-group; and two-loop running of the Higgs quartic caps any superpartner scale that could tame the regenerated couplings unless $m_t \leq 171\,\mathrm{GeV}$.

The forty points are forty views of one thing. Looked at through the right lens, the same finite object is a quantum measurement geometry, a finite space-time, a clock, a lattice, an exceptional Lie algebra, and a Lorentzian light cone — and each identification in this paper is an exact computation, not an analogy.

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How to read this paper

This paper is written for two readers at once. One has never studied algebra beyond school and wants to understand what has actually been found. The other wants every statement to be precise enough to check. We have tried not to cheat either of them. The mathematics is stated in full; next to it, in gold-edged boxes, the same idea is explained in ordinary words, and in teal-edged boxes we say what the idea means physically. Rose-edged boxes mark the edge of what is known.

The five badges

Every claim of substance carries one of five labels. They are the most important typographical device in the paper, because the whole value of a speculative subject lies in being able to tell the proved from the hoped-for.

THEOREM	A mathematical theorem: proved by an argument in the text, or a standard result with a reference.
COMPUTATION	An exact finite computation, carried out by a committed script whose output certificate is named. "Exact" means integer or rational arithmetic, or an exhaustive enumeration — never a numerical approximation, unless we say so.
BRIDGE	A bridge: a mathematically true statement *plus* an interpretation of it (for example, calling a mathematical light cone a light cone of time). The mathematics is certain; the interpretation is a proposal.
OPEN	An open question, stated as sharply as we can.
REFUTED	Something that was once believed, written down, and then shown false. We keep these, with the reason.

A map of the journey

The paper follows one chain of exact identifications. Figure 1 is the whole argument in one picture; each box is a section.

What you will need

Nothing beyond patience. We introduce every structure before we use it, and the glossary in Appendix B collects the vocabulary. Readers who want only the story can read the plain-words boxes and the key-idea panels and skip the displayed mathematics; nothing essential will be lost. Readers who want only the mathematics can do the reverse.

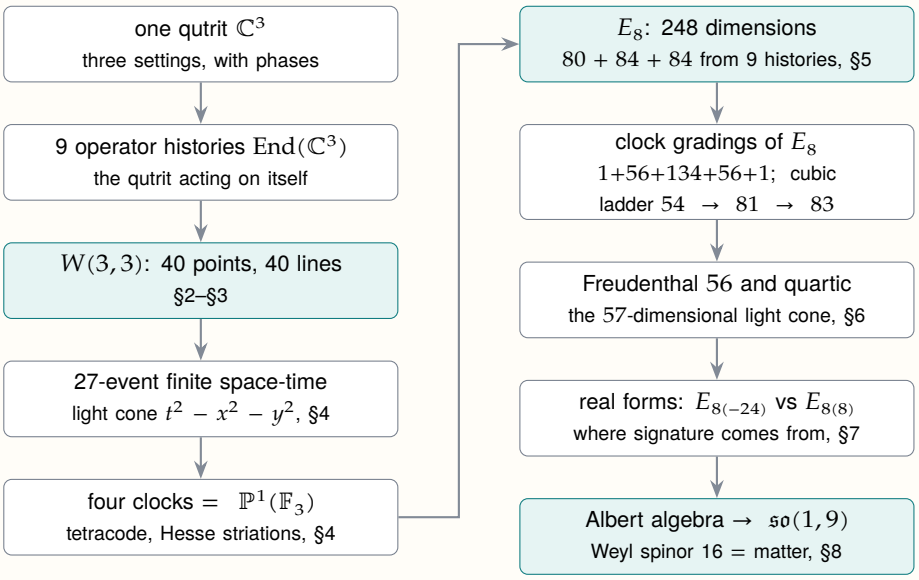


Figure 1. The chain of exact identifications followed in this paper. Shaded boxes are the three places where the story changes scale: from a qubit-sized object to a finite geometry, from the geometry to E_8, and from E_8 to a Lorentzian algebra.

Where the material comes from

Everything here is drawn from the public repository `wilcompute/W33-Theory`, a record of several thousand numbered "passes" by two independent research tracks, and from its three long manuscripts — the $W(3,3)$ paper, the photonic Holonet paper and the Holonet machine blueprint — some of whose pages are adapted here. Passes 10949 and 10950, which supply Sections 6–8, were carried out for this synthesis. A reader who wants to check a statement can find its script and certificate in the ledger of Appendix A.

1 Prologue: a shape made of measurements

1.1 Why look at something finite?

Physics describes the world with continuous quantities: positions, times, field strengths, each a real number that can be divided forever. Yet the deepest facts we know about nature are *discrete*. An electron's spin, measured along any axis, gives one of exactly two answers. Quarks come in exactly three colours. There are exactly three generations of matter. The symmetry groups that organise the forces — $SU(3)$, $SU(2)$, $U(1)$ — are continuous, but their representations, which decide what particles can exist, are labelled by whole numbers.

This programme starts from the discrete end. It asks: is there a single finite object, small enough to write down completely, whose structure already contains the patterns physics keeps finding? The candidate studied here is a configuration of forty points called $W(3,3)$. It is not new — geometers have known it for a century — but it has an unusual habit: every time it is examined through a different lens, it turns out to be a well-known structure from a different part of mathematics or physics. This paper is an account of those lenses, and of exactly how far each one reaches.

In plain words

A useful picture: $W(3,3)$ is like a crystal. A crystal is finite in its unit cell, but the unit cell determines the whole crystal's behaviour — how it breaks, how light passes through it, what sounds it makes. The hope of this programme is that $W(3,3)$ is a unit cell for something larger. The honest answer, as you will see, is that it is a remarkable unit cell for the *symmetries* of physics, and a much weaker one for its *dynamics*.

1.2 Bits, trits and qutrits

Every computer you have used stores information as bits — switches that are either off or on. A *trit* is a three-way switch: 0, 1, or 2. On its own that is not interesting; two bits can imitate a trit. A *qutrit* is different. It is a physical system with three distinguishable states, obeying quantum mechanics, which means it can be in a blend of all three settings at once, with relative phases. In symbols, a qutrit state is a unit vector

$$|\psi\rangle = a_0|0\rangle + a_1|1\rangle + a_2|2\rangle \in \mathbb{C}^3,$$

and the phases of the complex numbers a_j matter.

Two operations generate everything one can do to the "shape" of a qutrit. The *shift* X moves each setting to the next, and the *clock* Z stamps each setting with a phase:

$$X|j\rangle = |j+1 \bmod 3\rangle, \qquad Z|j\rangle = \omega^j|j\rangle, \qquad \omega = e^{2\pi i/3}.$$

They do not commute: $ZX = \omega\, XZ$. That single twist — a phase of one third of a turn picked up when the order of two operations is swapped — is the seed of everything in this paper.

What it means physically

X and Z are the finite versions of the two most important operations in quantum mechanics: translation in position and translation in momentum. The relation $ZX = \omega XZ$ is the three-state version of Heisenberg's $[x,p] = i\hbar$. Physicists call the group generated by X, Z and the phases the *Heisenberg–Weyl group*; its elements are the qutrit *Pauli operators*. They label the possible measurements one can make on a qutrit that are "as simple as possible" — the ones every other measurement is built from.

1.3 Arithmetic with three numbers

Because $X^3 = Z^3 = 1$, exponents only matter modulo 3. We therefore work in the *field with three elements*, $\mathbb{F}_3 = \{0,1,2\}$, where addition and multiplication are done modulo 3 — clock arithmetic on a three-hour clock. It is a genuine number system: every nonzero element has an inverse ($1\cdot 1 = 1$, $2\cdot 2 = 4 = 1$). In particular $2 = -1$ and $\frac{1}{2} = 2$, facts we will use repeatedly.

A qutrit Pauli operator is named by two numbers, X^aZ^b with $(a,b) \in \mathbb{F}_3^2$. Two qutrits need four: $(a_1,b_1,a_2,b_2) \in \mathbb{F}_3^4$, a space with $3^4 = 81$ points.

1.4 The forty in one breath

Here is the whole construction, which Sections 2–3 unpack slowly. Label the 81 two-qutrit Pauli operators by vectors $x \in \mathbb{F}_3^4$. Remove the identity, leaving 80. An operator and its inverse describe the same measurement (one is the other run backwards), and x and $-x = 2x$ label that pair, so the distinct measurements number

$$\frac{3^4 - 1}{3 - 1} = \frac{80}{2} = 40.$$

Two of these forty measurements can be performed together exactly when their operators commute. Recording which pairs commute gives the forty-point geometry $W(3,3)$.

The forty points of $W(3,3)$ are the forty independent directions along which a pair of qutrits — or, as Section 3 shows, one qutrit acting on itself — can be measured. The lines of the geometry are the complete sets of measurements that can be made at the same time.

1.5 What this paper claims, and what it does not

A "theory of everything" should, at a minimum, reproduce the particles and forces of the Standard Model, the geometry of space-time and gravity, and the numbers — masses, couplings, mixing angles — that we measure. It is important to say at the start that nothing in this paper achieves that. What it does achieve is a tight web of *exact* connections between a single finite object and the structures that any such theory seems to need:

- the qutrit Heisenberg group and its Clifford symmetries (Sections 2–3);
- a finite space-time with a light cone, finite clocks, and a rigorous distinction between reversible time and an arrow of time (Section 4);
- the exceptional Lie algebra E_8 and its subalgebras E_7 and E_6 (Section 5);
- the Freudenthal triple system behind four-dimensional black-hole charges, and a 57-dimensional light cone on which E_8 acts by conformal-like maps (Section 6);
- the octonionic Albert algebra, the Lorentz algebra $\mathfrak{so}(1,9)$ and its chiral spinor, which carries exactly the quantum numbers of one generation of matter (Sections 7–8).

Section 9 is devoted to the physics that has been attempted on top of this — string compactifications that realise the Standard Model on the same structure — including the negative results, and Section 10 to a quantum computer architecture read off the same geometry. The Epilogue lists what remains open.

Where this stops

The finite geometry supplies *symmetry and structure*. It does not, by itself, supply a Hamiltonian, an energy scale, a value of Newton's constant, or a reason for any particular mass. Where this paper uses words like "time", "light cone" or "matter", it is naming a mathematical structure that behaves like the physical one; whether nature uses it is a separate, empirical question.

2 The object: the symplectic quadrangle $W(3,3)$

2.1 Definition

Let $V = \mathbb{F}_3^4$ with coordinates $x = (x_1, x_2, x_3, x_4)$ and the *symplectic form*

$$\langle x, y\rangle \;=\; x_1y_3 + x_2y_4 - x_3y_1 - x_4y_2 \pmod 3.$$

It is bilinear, alternating ($\langle x, x\rangle = 0$) and nondegenerate (only $x = 0$ pairs to zero with everything).

Definition 2.1. The *points* of $W(3,3)$ are the one-dimensional subspaces of V (40 of them). Two distinct points $\langle x\rangle, \langle y\rangle$ are *collinear* if $\langle x, y\rangle = 0$. The *lines* of $W(3,3)$ are the two-dimensional subspaces L on which the form vanishes identically (*totally isotropic* or *Lagrangian* planes); each contains $(9-1)/2 = 4$ points.

Proposition 2.1 THEOREM

$W(3,3)$ has 40 points and 40 lines. Every line has 4 points and every point lies on 4 lines. For a point p and a line L not containing it, exactly one point of L is collinear with p.

Proof. Points: $(3^4 - 1)/(3 - 1) = 40$. For a point $p = \langle x\rangle$, the set $x^\perp = \{y : \langle x, y\rangle = 0\}$ is a 3-dimensional subspace containing x; the lines through p are the 2-dimensional subspaces of $x^\perp$ that contain x, and these correspond to the points of $x^\perp/\langle x\rangle \cong \mathbb{F}_3^2$, of which there are $(9-1)/2 = 4$. Double counting point–line incidences, $40 \cdot 4 = (\#\text{lines}) \cdot 4$, so there are 40 lines. Finally, if $p \notin L$ then $x^\perp \cap L$ is a subspace of L of dimension 1 (it cannot be all of L, since then $\langle x\rangle + L$ would be a totally isotropic 3-space, impossible in a nondegenerate 4-space), so exactly one point of L is collinear with p. □

The last property is the defining axiom of a *generalized quadrangle*: there are no triangles made of three lines, but from any point you can always reach any line in one step, in exactly one way. With four points per line and four lines per point, $W(3,3)$ is the generalized quadrangle of order $(3,3)$, usually written GQ$(3,3)$ [41].

2.2 The collinearity graph and its spectrum

Join two points by an edge when they are collinear. Each point has $4 \times 3 = 12$ neighbours, so the graph has $40 \cdot 12/2 = 240$ edges. Two collinear points share exactly 2 further neighbours (the other two points of their line), and two non-collinear points share exactly 4 (the points of $x^\perp \cap y^\perp$, a nondegenerate plane). A graph in which these three numbers are constant is called *strongly regular*; this one is SRG$(40, 12, 2, 4)$ [7].

Proposition 2.2 THEOREM

The adjacency matrix A of $W(3,3)$ has eigenvalues 12 (once), 2 (24 times) and -4 (15 times).

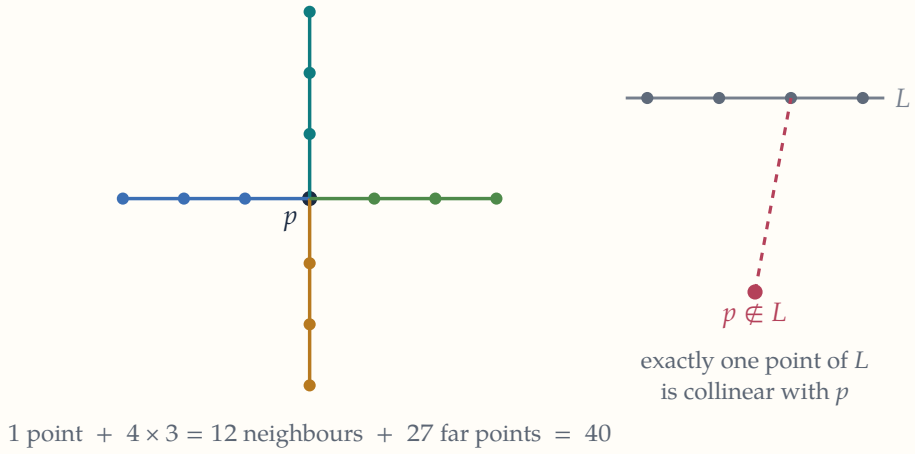


Figure 2. Left: the view of $W(3,3)$ from one point. Right: the quadrangle axiom.

Proof. For SRG(v,k,λ,μ) one has $A^2 = kI+\lambda A+\mu(J-I-A)$, where J is the all-ones matrix. On vectors orthogonal to the all-ones vector, J acts as zero, so every other eigenvalue θ satisfies $\theta^2 = (\lambda-\mu)\theta + (k-\mu)$, i.e. $\theta^2+2\theta-8=0$, giving $\theta \in \{2,-4\}$. Multiplicities f,g satisfy $f+g=39$ and $\operatorname{tr} A = 12+2f-4g=0$, so $f=24, g=15$. □

Since $|-4| < 2\sqrt{11} \approx 6.63$, the graph is *Ramanujan*: it is as good an expander as a 12-regular graph can be. This is why the machine of Section 10 can route between any two of its forty nodes in two hops, and why the graph's minimum bisection is large (100 of the 240 edges, a computed value).

> **In plain words**
>
> "Strongly regular" means that the geometry looks exactly the same from every point, and from every pair of points. There is no privileged measurement. The three eigenvalues are the geometry's "vibration frequencies"; that there are only three is the algebraic signature of this perfect evenness.

2.3 What the points and lines are physically

Let $W(x) = X^{x_1}Z^{x_3} \otimes X^{x_2}Z^{x_4}$ be the two-qutrit Pauli operator with label $x \in \mathbb{F}_3^4$. A direct calculation from $ZX = \omega XZ$ gives

$$W(x)\,W(y) \;=\; \omega^{\langle x,y\rangle}\,W(y)\,W(x). \tag{1}$$

So the symplectic form is not an arbitrary choice: it *is* the commutation phase of two-qutrit measurements.

> **What it means physically: contexts**
>
> - A **point** is a measurement direction: the cyclic group generated by one Pauli operator, which is the same whether it is run forwards or backwards.
> - Two points are **collinear** when their operators commute and can be measured at the same time without disturbing each other.

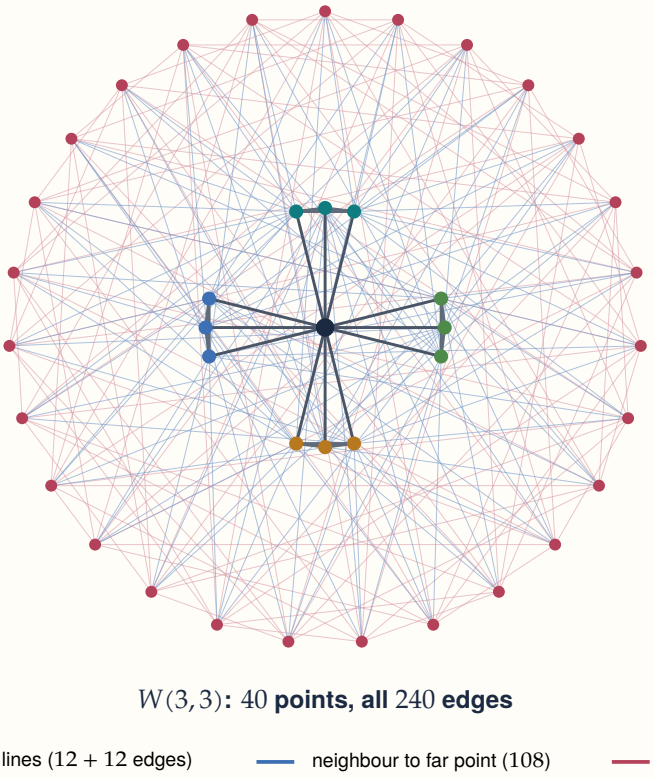


Figure 3. A portrait of $W(3,3)$, computed from the symplectic form and drawn from the point of view of one point p (centre). Its 12 neighbours lie on four lines (four colours, three points each); the 27 far points sit on the outer ring, ordered by which neighbour they see on each of the four lines — by the quadrangle axiom every far point sees exactly one. The edge counts $12 + 12 + 108 + 108 = 240$ are forced by $\mathrm{SRG}(40, 12, 2, 4)$.

- A **line** is a maximal set of mutually compatible measurements: an abelian group of 9 Pauli operators whose common eigenvectors form a basis of the two-qutrit space $\mathbb{C}^9$. Physicists call this a *stabilizer group*, and each of its 9 joint eigenstates a *stabilizer state*. There are 40 lines and so $40 \times 9 = 360$ stabilizer states — the standard count for two qutrits.
- The **quadrangle axiom** says: given any measurement and any complete measurement context not containing it, exactly one member of the context is compatible with it.

The quantum operations that map Pauli operators to Pauli operators (up to phase) form the two-qutrit *Clifford group* [17]. Modulo the Paulis themselves and phases, it acts on the labels x by exactly the linear maps preserving $\langle , \rangle$: the symplectic group $\mathrm{Sp}(4,3)$. So the symmetries of the geometry are, literally, the "cheap" quantum gates.

2.4 Two groups of order 51,840, which are not the same group

The symmetry groups that appear around $W(3,3)$ are easy to confuse, and confusing them has cost this programme real work (Section 5.4). Adapting the blueprint's account:

$$\mathrm{PSp}(4,3) \cong U_4(2) \text{ (order 25 920)}, \qquad \mathrm{Sp}(4,3) \cong 2.U_4(2), \qquad \mathrm{PGSp}(4,3) \cong U_4(2){:}2 \cong W(E_6).$$

The last two both have order 51 840 but are *not isomorphic*: $\mathrm{Sp}(4,3)$ has a centre of order two (the map $x \mapsto -x$), while $\mathrm{PGSp}(4,3)$ — the full automorphism group of $W(3,3)$ — has trivial centre and is the Weyl group of the exceptional root system E_6. THEOREM

In plain words

The two groups have the same size but different "shapes". One of them contains a special element that commutes with everything — a global sign flip. Whether that sign is visible or invisible turns out to decide whether the structure can tell left from right. This is why the same forty points can *host* a handedness without being able to *choose* one (Section 9.6).

2.5 The dual picture and a firewall

Swap the roles of points and lines: the 40 lines of $W(3,3)$, with two lines "adjacent" when they meet, form the dual quadrangle, which is the parabolic quadric $Q(4,3)$. For odd q the quadrangle $W(3,q)$ is *not* isomorphic to its dual, although both have collinearity graph parameters $\mathrm{SRG}(40,12,2,4)$. This distinction matters several times below: the finite space-time of Section 4 lives on the *line* side, while the qutrit operators live on the *point* side. Repeated coincidences of counts between the two sides (for instance 27 vertices, 108 edges, 36 triangles) do not make them the same object — the repository calls this the *point/line firewall*, and it has caught several tempting mistakes. COMPUTATION (`w33_20260924_history_pointline_cospectral_firewall.py`)

2.6 Why three?

A sceptical reader should ask whether "three" was chosen because it works. The repository records five equations in which q appears as an unknown integer and each has exactly one positive solution, $q = 3$. Two can be checked by hand:

- The number of edges of the collinearity graph of $W(3,q)$ is $q(q+1)^2(q^2+1)/2$; the number of elements of $\mathbb{F}_{q^5}$ outside $\mathbb{F}_q$ is $q^5 - q$. Setting them equal gives $2(q-1) = q+1$, so $q = 3$: both equal 240.
- The number of points not collinear with a given point is q^3; it equals 27, the dimension of the fundamental representation of E_6, only for $q = 3$.

Where this stops

These are coincidences of integers, not derivations. They do not prove that three is *necessary* for anything physical. They do show that the programme's most unusual choice — three-level systems rather than two-level ones — is the one value at which several independent structures meet. BRIDGE

3 One qutrit, acting on itself

Section 2 built $W(3,3)$ from *two* qutrits. This section shows that one is enough — provided we look not at the qutrit's states but at the things that can be done to it. That shift of viewpoint is the precise content of the phrase the Holonet papers use, *self-entanglement*, and it is worth getting exactly right, because the phrase can also be read in ways that are not true.

3.1 Operators are states

An operator on one qutrit is a 3×3 matrix: nine complex numbers. So the space of operators $\mathrm{End}(\mathbb{C}^3)$ is itself a nine-dimensional complex vector space, and with the Hilbert–Schmidt inner product $\langle A, B\rangle = \mathrm{tr}(A^\dagger B)$ it is a Hilbert space. Writing $|A\rangle\!\rangle$ for a matrix regarded as a vector,

$$\mathrm{End}(\mathbb{C}^3) \;\cong\; \mathbb{C}^3 \otimes \mathbb{C}^{3*} \;\cong\; \mathbb{C}^9, \qquad |i\rangle\langle j| \;\longleftrightarrow\; |i\rangle \otimes |j\rangle.$$

Under this identification the identity matrix becomes the maximally entangled "Bell" state of two qutrits,

$$|\Omega\rangle = \tfrac{1}{\sqrt{3}}|I\rangle\!\rangle = \tfrac{1}{\sqrt{3}}(|00\rangle + |11\rangle + |22\rangle).$$

This is the Choi–Jamiołkowski correspondence, a standard tool of quantum information: every operation on a system is encoded by a state of two copies of it.

In plain words

Imagine describing a machine not by its current setting but by its complete instruction table: "if the input is j, produce i with amplitude A_{ij}". That table has nine entries. The nine entries behave exactly like the state of two three-way systems — an "input" and an "output". Nothing mystical is happening; it is the same information written in a

different shape. The "two qutrits" of Section 2 can be read as the before-and-after of a single qutrit.

3.2 The two-sided action and its geometry

Let $P_u = X^{u_1} Z^{u_2}$ for $u \in \mathbb{F}_3^2$. A qutrit operator A can be multiplied by Pauli operators from the left and from the right:

$$\Phi_{u,v}(A) = P_u A P_v^\dagger, \qquad (u,v) \in \mathbb{F}_3^2 \oplus \mathbb{F}_3^2.$$

Theorem 3.1 THEOREM COMPUTATION

The 81 maps $\Phi_{u,v}$ form an orthogonal basis of $\mathrm{End}(\mathrm{End}(\mathbb{C}^3)) \cong M_9(\mathbb{C})$; the 80 non-identity maps span $\mathfrak{sl}_9$. They commute up to the phase

$$\Phi_{u,v}\Phi_{u',v'} = \omega^{[u,u']-[v,v']}\,\Phi_{u',v'}\Phi_{u,v}, \qquad [u,u'] = u_1u_2' - u_2u_1'.$$

The form $[u,u'] - [v,v']$ is symplectic on $\mathbb{F}_3^4$, and the resulting commutation geometry on the 40 projective classes is $W(3,3)$, with collinearity graph $\mathrm{SRG}(40,12,2,4)$ and spectrum $12^1\,2^{24}\,(-4)^{15}$. `w33_20260924_temporal_su9_e8_compiler.py`

Proof of the phase. $\Phi_{u,v}\Phi_{u',v'}(A) = P_uP_{u'}AP_{v'}^\dagger P_v^\dagger$ and $\Phi_{u',v'}\Phi_{u,v}(A) = P_{u'}P_uAP_v^\dagger P_{v'}^\dagger$. Using $P_uP_{u'} = \omega^{[u,u']}P_{u'}P_u$ and taking adjoints of $P_vP_{v'} = \omega^{[v,v']}P_{v'}P_v$ gives the ratio $\omega^{[u,u']-[v,v']}$. □

The minus sign is the fingerprint of right multiplication: acting from the right reverses the order of products, so the right-hand copy of the qutrit carries the *opposite* algebra. The two "qutrits" are not two particles, and not even two subsystems in the usual sense: they are the algebra of left actions and its commutant, the algebra of right actions, on one qutrit's operator space.

3.3 The Bell line is the set of physical channels

Among all $\Phi_{u,v}$, which ones are genuine quantum operations? A linear map on operators is physically realisable as a quantum channel only if it is *completely positive*, which for $\Phi_{u,v}$ means its Choi matrix $|P_u\rangle\!\rangle\langle\!\langle P_v|$ is positive semidefinite. A rank-one matrix $|a\rangle\langle b|$ is positive only when b is a positive multiple of a; so up to phase, $\Phi_{u,v}$ is completely positive exactly when $u = v$. The diagonal $\Delta = \{(u,u)\}$ is totally isotropic ($[u,u'] - [u,u'] = 0$) and its four projective points form one line of $W(3,3)$.

Proposition 3.2 THEOREM COMPUTATION

The unitary channels $A \mapsto P_uAP_u^\dagger$ form a distinguished line of $W(3,3)$, the *Bell line*; each fixes $|\Omega\rangle$. The left–right exchange $(u,v) \mapsto (v,u)$ reverses the symplectic form; its eight fixed rays split into two Lagrangian lines, and the + line is exactly the unitary-

channel diagonal. `w33_20260924_liouville_swap_cp_outer_bridge.py`

What it means physically: a built-in time reversal

Exchanging left and right multiplication is the operator version of running a process backwards: $A \mapsto P_u A P_v^\dagger$ becomes $A \mapsto P_v A P_u^\dagger$. Because it reverses the sign of the commutation form, it is an *anti*-symmetry of $W(3,3)$ — the same kind of map as complex conjugation, which is how quantum mechanics implements time reversal. The repository shows that ordinary matrix transposition of the Pauli algebra induces exactly this class of outer symmetry, the one earlier work had identified abstractly (BT172). COMPUTATION (`w33_20260924_intrinsic_transpose_outer.py`, up to a labelling gauge.)

3.4 Three different things called "self-entanglement"

The phrase can mean three distinct things, and the Holonet programme has, at times, slid between them. It is worth separating them once and for all.

Notion	What is entangled	Status here
Intraparticle	Two independent degrees of freedom of one photon (for example its arrival-time bin and its frequency bin).	Standard and experimentally demonstrated.
Operator (Choi)	The input and output legs of one operation, as in §3.1.	The exact algebraic home of $W(3,3)$.
Temporal process	Quantum correlations between interventions at different times, described by a multi-time *process tensor*.	Needs its own witness (§3.5).

A correction the papers needed

An earlier draft of the photonic paper described three optical time bins as containing "a past qutrit and a future qutrit". Three orthogonal bins span only $\mathbb{C}^3$, whereas two qutrits need $\mathbb{C}^9$; the two-bin subsets it named overlap and cannot be independent tensor factors. The architecture now uses two genuinely independent qutrit degrees of freedom of one photon — time bin ⊗ frequency bin — and reads "past" and "future" as the input and output legs of a process, not as simultaneously existing copies. REFUTED (the three-bin reading) THEOREM (the corrected encoding).

3.5 Memory across a causal break

What does it take for a system to carry quantum information from one time to a later one? The repository's cleanest test is a three-step process with a qutrit memory M: swap the system S into M; discard S and replace it with anything at all (a *causal break*); swap back. If the original state reappears, the process had memory. But a classical memory that merely

records "0, 1 or 2" would pass that test too. The decisive witness entangles S with a reference R first.

Theorem 3.3 COMPUTATION

With S initially maximally entangled with a reference qutrit R:

- a coherent memory returns the Bell state exactly (negativity 1, Bell fidelity 1);
- a classical label memory returns $\frac{1}{3}\sum_j |jj\rangle\langle jj|$ (negativity 0, Bell fidelity $\frac{1}{3}$), although it passes every basis-state retrieval test;
- a depolarizing memory $\rho \mapsto (1-p)\rho + pI/3$ returns an isotropic state with Bell fidelity $1 - \frac{8p}{9}$ and negativity $\max(0, \frac{3-4p}{3})$; quantum memory survives exactly while $p < \frac{3}{4}$.

`w33_20260924_temporal_quantum_memory_witness.json`

What it means physically

The threshold $p = \frac{3}{4}$ is the standard point at which a qutrit depolarizing channel becomes entanglement-breaking: beyond it, the link between the earlier and the later system can still carry classical information but no longer quantum information. Retrieval after a break proves that the past is *remembered*; preserved entanglement with a reference proves that it is remembered *quantumly*. Neither involves any influence travelling backwards in time.

3.6 A state picks a clock

A faithful qutrit state $\rho = \mathrm{diag}(p_0, p_1, p_2)$ defines a natural flow on operators, the *modular flow* of Tomita–Takesaki theory, with generator $K = -\log\rho \otimes 1 + 1 \otimes \log\rho$ acting on $\mathrm{End}(\mathbb{C}^3)$. The matrix units are its eigenvectors:

$$K\,E_{ij} = \log\frac{p_j}{p_i}\,E_{ij}.$$

The three diagonal units have frequency zero; the six off-diagonal units E_{ij} $(i \neq j)$ — the six directed transitions between the three levels — come in three opposite pairs. Their pattern of differences is exactly the root system A_2. COMPUTATION (`w33_20260924_modular_a2_event_clock.py`)

In plain words

A perfectly balanced qutrit ($p_0 = p_1 = p_2$) has no preferred direction: every transition has zero frequency, and nothing distinguishes "forward" from "backward". Any imbalance gives each of the six transitions a frequency, and pairs each with its reverse at the opposite frequency. This is the finite form of an old idea of Connes and Rovelli [8]: that a state of a system can itself define a flow of time. It orients change; it does not by itself make change irreversible. That distinction — between an orientation of time and an arrow of time — returns in Section 4.

3.7 Space and time on the same forty points

Section 2 built $W(3,3)$ from two qutrits, with commutation form $[a,a'] + [b,b']$; Theorem 3.1 built it from one qutrit acting on itself, with form $[u,u'] - [v,v']$. The two constructions live on the same forty projective points of $\mathbb{F}_3^4$, and they are not the same structure.

Theorem 3.4 (two quadrangles) THEOREM COMPUTATION

Both forms make the 40 points a generalized quadrangle $GQ(3,3)$ with collinearity $\mathrm{SRG}(40,12,2,4)$. The two quadrangles share exactly 16 lines — the product planes $\ell_1 \oplus \ell_2$ — and each has 24 lines of its own: the spatial ones are the graphs of the 24 maps with determinant -1, carrying the 216 maximally entangled two-qutrit stabilizer states; the temporal ones are the graphs of $SL(2,3)$, the qutrit Clifford group modulo Paulis. Partial transposition exchanges the two sets of own lines, and on states it is a bijection: the partial transpose of each of the 216 maximally entangled stabilizer states *is* the two-time pseudo-density operator of one qutrit passing through one of the 216 Clifford gates (spectrum $(+\frac{1}{3})^6(-\frac{1}{3})^3$, temporal negativity 1). Exactly 2304 of the 103680 similitudes of the spatial form also preserve the temporal one: the local operations and the exchange of the parties. `w33_pass11143_space_time_two_quadrangles.py`

What it means physically: one qutrit across time is a Bell pair across space

Space and time agree on everything classical — the product lines — and disagree on exactly the entangled lines. A maximally entangled pair of qutrits and a single qutrit persisting through a gate are the same line of points, read with the two forms. This is the $W(3,3)$ form of a general fact ("partial transpose as a space–time swap", Fullwood and Li [45]; Marcovitch and Reznik [46]); the two-quadrangle structure, the 16/24 split and the state–gate bijection are specific to $W(3,3)$.

Proposition 3.5 (the spookiness budget) COMPUTATION

For the qutrit Bell inequality of Collins, Gisin, Linden, Massar and Popescu [47] (local bound 2): using only $W(3,3)$'s own resources — stabilizer states, Pauli measurements, Clifford gates — the maximum is exactly 2, both for two qutrits in space and for one qutrit measured at two times. Along the path from the Pauli bases to the optimal bases the value is $2 + \frac{\pi^2}{6}t^2 + O(t^3)$: $W(3,3)$ sits exactly on the edge of the local polytope. One qutrit starting maximally mixed and measured twice gives exactly the Bell-pair value 2.8729; optimised over starts and bases it reaches 3.1628, above the best spatial value 2.9149 [48] — for qubits the two coincide [49]. A classical system with any memory reaches the algebraic 4 in time. `w33_pass11144_spookiness_budget.py`

Spooky action at a distance, in these terms

The correlation geometry of a Bell pair is exactly that of one qutrit correlated with itself across time. The temporal version needs only persistence; its larger values measure how much a measurement disturbs, not nonlocality. The spatial version is "spooky" precisely because it cannot be used to signal, and it needs a resource outside $W(3,3)$ — magic — to exceed the local bound at all (the odd-dimensional discrete Wigner

function of stabilizer states is non-negative, Gross [50], and its phase space is the point space of $W(3,3)$). Nothing here travels backwards in time: what the two readings share is the geometry of the correlations, not a mechanism.

The same swap works in imaginary time. For a Gibbs state $\rho = e^{-\beta K}/Z$ of one qutrit, the partial transpose of the thermofield double $\sum_i \sqrt{p_i}|ii\rangle$ equals the Jamiołkowski matrix of the Euclidean half-evolution $A \mapsto \rho^{1/2} A \rho^{1/2}$, with negativity $((\sum_i \sqrt{p_i})^2 - 1)/2$: the temporal Bell line at $\beta = 0$, a product line as $\beta \to \infty$. The modular clock above is the flow whose Euclidean continuation prepares this entanglement. COMPUTATION (`w33_pass11146_thermal_reading.py`)

Proposition 3.6 (polygamy in time, and where the magic sits) COMPUTATION

A qutrit evolving by the identity is maximally entangled with its past and with its future at once: every pair of times carries the temporal Bell line, so the pairwise temporal negativities of three times sum to 3. Three qutrits in space reach at most 1.1394 for the same sum, and $N_{AB} + N_{AC} \le 1.0328 = 4/\sqrt{15}$ (numerical optimum, identified to 10^{-15}). The temporal advantage of Proposition 3.5 is not special to qutrits: one qudit measured twice beats two qudits for every $d \ge 3$ ($d = 3, 4, 5$: 3.163, 3.283, 3.377 against $2.915 = 1 + \sqrt{11/3}$, 2.973, 3.016), and only qubits coincide. Magic can sit in either place: with Pauli measurements only, the best state reaches $2\sqrt{2}\cos(\frac{1}{3}\arccos\frac{1}{2\sqrt{2}}) = 2.6017$, the largest root of $x^3 - 6x - 2$ (mana 0.39), while the stabilizer state $|\Omega\rangle$ with magic measurements reaches 2.8729. There is no threshold: the least mana that reaches $+\square with Pauli measurements grows linearly from zero, about /usr/bin/bash .65$ per unit of violation. COMPUTATION (`w33_pass11147_temporal_cglmp_by_dimension.py`, `w33_pass11148_polygamy_in_time.py`, `w33_pass11149_magic_for_spookiness.py`)

In plain words

In space, a system can give its full entanglement to only one partner. Across time it gives it to its whole history at once — because it is not sharing anything, it is simply persisting. That is why the spatial kind is scarce and "spooky", and the temporal kind is free.

Theorem 3.7 (perfect space–time gates) THEOREM COMPUTATION

Write a two-qutrit Clifford gate as $S = \left(\begin{smallmatrix} S_{AA} & S_{AB} \\ S_{BA} & S_{BB} \end{smallmatrix}\right) \in \mathrm{Sp}(4,3)$. Its Choi state on $A_{\mathrm{in}}, B_{\mathrm{in}}, A_{\mathrm{out}}, B_{\mathrm{out}}$ is maximally entangled across time (always), carries entanglement rank S_{BA} across space and rank S_{AA} across the space–time diagonal $\{A_{\mathrm{in}} B_{\mathrm{out}}\}$, and

$$\det S_{AA} + \det S_{BA} = 1 \pmod 3,$$

so a gate can never be weak across both space and the diagonal. The gates that are maximal across all three cuts — whose Choi state is the absolutely maximally entangled state of four qutrits — are exactly those with $\det S_{AA} = \det S_{BA} = -1$: there are $13824 = 24^3$ of them (4/15 of $\mathrm{Sp}(4,3)$), forming a single orbit under local Clifford operations. For time-reversing Clifford operations the law reads $\det S_{AA} + \det S_{BA} = -1$,

and the tetracode gate $K \otimes I$, $K = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, is perfect; composed with a local time reversal it is the perfect unitary gate. `w33_pass11156_perfect_spacetime_gates.py`

Entanglement that is spatial and temporal at once

A perfect gate hides everything in pairs: every two-qutrit marginal of its Choi state is maximally mixed, so A's past is uncorrelated with A's future and with B's future separately, and entirely correlated with them jointly — maximal scrambling. No such gate exists for qubits; for qutrits it exists, and in $W(3,3)$ it is the tetracode that sits at every point. That the four-qutrit absolutely maximally entangled state comes from the tetracode and is unique up to local unitaries is known [52]; the complementarity law and the space/time/diagonal reading are specific to $W(3,3)$.

Four refinements of Proposition 3.6. The Pauli-only Bell operator has characteristic polynomial $(x^3 - 6x - 2)(x^3 - 3x + 1)^2$ exactly, its remaining eigenvalues being $2\cos(2\pi k/9)$. In time the magic cannot sit in the start: with Pauli measurements a single qutrit gives exactly 2 whatever its initial state, whereas in space a magic state already violates. The temporal optimum keeps climbing with dimension (3.455, 3.521 at $d = 6, 7$ after the re-optimisation of Section 4, deficit from 4 falling roughly like $1/d$) while the spatial one barely moves (3.050, 3.078). And the $4/\sqrt{15}$ optimum is carried by a state whose invariants are all fifteenths: spectra $(7,4,4)/15$ and $(11,2,2)/15$, $N_{AB} = N_{AC} = 2/\sqrt{15}$, $N_{BC} = 1/15$. COMPUTATION (`w33_pass11152_pauli_bell_charpoly.py`, `w33_pass11153_temporal_magic_budget.py`, `w33_pass11154_polygamy_optimum_structure.py`, `w33_pass11155_temporal_cglmp_trend.py`)

The loose ends close. The ninth roots come from one identity: with $U = (XZ^2) \otimes (XZ^2)$, the joint relabelling of both outcomes, $B^3 - 3B + 1 = (B + 1)(I + U + U^2)$, so on the charged sectors of U the Bell operator trisects the angle $2\pi/3$ and on the neutral sector it gives $x^3 - 6x - 2$. The determinant law is general: for n qutrits every column and every row of blocks satisfies $\sum_i \det S_{ij} = 1$, and perfect three-qutrit gates exist (about 3.1% of $\mathrm{Sp}(6,3)$), each column carrying determinants $(1, 1, -1)$. The value $4/\sqrt{15}$ is derived: on the family $\sqrt{x}\,|000\rangle + \sqrt{(1-x)/4}\sum_{a=1,2}|a\rangle(|0a\rangle + |a0\rangle)$ the negativity is $\sqrt{(1+15x)(1-x)}/4$, maximal at $x = 7/15$. The temporal optimum keeps climbing: 3.576, 3.621, 3.658 at $d = 8, 9, 10$ against 3.101, 3.122, 3.140 in space, its deficit from 4 falling roughly like $3.4/d$ (Section 4; the earlier $d^{-0.75}$ fit to $d \le 10$ is superseded). COMPUTATION (`w33_pass11157_ninth_roots_explained.py`, `w33_pass11158_perfect_three_qutrit_gates.py`, `w33_pass11159_explicit_polygamy_state.py`, `w33_pass11160_temporal_cglmp_high_d.py`)

Three qutrits can be counted exactly. A symplectic matrix is an ordered symplectic basis, the determinant of a block of a column pair is the local symplectic form, and Jacobi's complementary-minor identity shows that invertibility of the nine blocks already makes the Choi state absolutely maximally entangled. The enumeration gives $286\,654\,464 = 2^{17}3^7$ perfect three-qutrit gates, a fraction $128/4095$ of $\mathrm{Sp}(6,3)$ (the same method returns 13824 for two qutrits; the single two-qutrit orbit agrees with Pahari's classification of Clifford dual-unitary gates over finite fields [53]). The tetracode gate has no scalar analogue — no 3×3 matrix over $\mathbb{F}_3$ has all entries and all 2×2 minors nonzero — but it has one over $\mathbb{F}_9 = \mathbb{F}_3[i]$. There $\mathrm{Im}\langle u, v\rangle$ is the symplectic form, so a unitary matrix over $\mathbb{F}_9$ is a symplectic matrix, perfect exactly when all its entries are nonzero: 12288 of the 24192 elements of $U(3, \mathbb{F}_9)$.

The simplest is $K = P^2 + i\mathbf{1}$, a cyclic shift plus i times the all-ones matrix, unitary because $\mathbf{1}^2 = 3\cdot\mathbf{1} = 0$ — the number of parties equals the characteristic (up to the scalar $1+i$ it generates Grassl and Gulliver's double-circulant Hermitian self-dual $[6,3,4]_9$ code [58, 59], so the construction itself is known). Finally, $4/\sqrt{15}$ is a certified local maximum: the partial transposes at the optimal state have no eigenvalue within $2/15$ of zero, and the Hessian has 32 strictly negative directions and exactly 20 flat ones, the local-unitary orbit; 96 further random restarts reach $4/\sqrt{15}$ and nothing higher. Global optimality remains open. COMPUTATION (`w33_pass11162_polygamy_local_maximum.py`, `w33_pass11163_perfect_three_qutrit_count.py`, `w33_pass11164_three_qutrit_tetracode.py`)

Both can be pinned further. On the symmetry class of the optimal state — the states $\alpha|000\rangle + \beta\sum_a|aa0\rangle + \gamma\sum_a|a0a\rangle$ fixed by $U\otimes\bar{U}\otimes\bar{U}$, $U\in U(2)$ on $\mathrm{span}\{|1\rangle,|2\rangle\}$ — the partial transposes split into 2×2 blocks and $N_{AB}+N_{AC} = \sqrt{\gamma^4+4\alpha^2\beta^2}+\sqrt{\beta^4+4\alpha^2\gamma^2}$, whose unique maximum is $4/\sqrt{15}$. Globally, Rana's bound on the number of negative eigenvalues of a partial transpose [55], purity and monotonicity give the rigorous bound $N_{AB}+N_{AC}\le 25/18$, and a dual see-saw from 3000 random starts never exceeds $4/\sqrt{15}$. The perfect three-qutrit gates fall into exactly six local orbits, labelled by a permutation $\pi\in S_3$ that says in which future each past reappears orientation-reversed ($\det S_{\pi(j)j} = -1$); relabelling the parties joins them, so the perfect three-qutrit gate is unique up to local operations and relabelling, as the two-qutrit one is. A perfect four-qutrit gate cannot exist at all — its Choi state would be an AME$(8,3)$ state, which the shadow inequalities exclude [56] — while five qutrits admit one again (an explicit circulant over $\mathbb{F}_9$, cf. [59]): perfect ticks exist for registers of $1,2,3$ and 5 qutrits, never 4. COMPUTATION (`w33_pass11167_polygamy_global.py`, `w33_pass11169_perfect_gate_local_orbits.py`, `w33_pass11170_perfect_gate_ladder.py`)

The polygamy region is now mapped. For every weight λ, the maximum of $N_{AB}+\lambda N_{AC}$ over all states equals its maximum on the same closed-form family (a dual see-saw from 150 starts per weight agrees to 10^{-12}), and for $\lambda\le\lambda_c = 0.9214$ it is simply 1, attained by a maximally entangled pair with the third qutrit idle. Spatial polygamy is a thin bulge beyond the monogamous line $N_{AB}+N_{AC}=1$ — at most 0.0328, and only for nearly balanced sharing — against 2 in time. The bulge is not a qutrit feature: normalised so that a maximally entangled pair has $N=1$, three qubits exceed the line by more ($(8\sqrt{2}-4)/7 = 1.0448$) and the excess shrinks with the local dimension (1.0259 for $D=4$, 1.0159 for $D=7$). The optimising family and the conjecture that it traces the whole negativity region are due to Allen and Meyer [60]; linear monogamy of negativity fails numerically already in [61]. What is new here is the value $4/\sqrt{15}$, its certification and the partial-class proofs (`w33_pass11191_polygamy_across_dimension.py`). The perfect gates also have a geometric home. A tensor factorisation of two qutrits is an octet of W(3,3); the 45 octets are the tritangent planes of the cubic surface and the 27 complete factorisation frames are its lines. A two-qutrit Clifford gate is perfect exactly when it moves the factorisation to a *collinear* one — a tritangent plane sharing a line with the original — while the gates with $I_3 = -1$ move it to a non-collinear one and the local gates fix it: the three classes of the determinant law are the three relations of GQ$(4,2)$, and the perfect gates form a single double coset of the factorisation stabiliser. For three qutrits the 110565 factorisations carry a rank-20 geometry (computed in GAP); the Choi-entanglement signature of a gate distinguishes 15 of its 20 relations, and the perfect relation is a single one, of size 3456. Five qutrits admit perfect ticks but they are sparse: none turns up in 7×10^6 samples of Sp$(10,3)$, one local orbit fills only 10^{-13} of the group, and

the exact count of $\mathbb{F}_9$-linear ones is 2 642 411 520, which realise only two of the five orientation patterns the determinant law allows. Four qutrits, which can never scramble perfectly, come close in a rigid way: the best eight-qutrit stabilizer states found miss exactly two of the 35 balanced cuts, each by two trits, and are otherwise 3-uniform. The register pairs itself up, and read as a tick the only defect is that aligned input pairs reach aligned output pairs with rank 2 instead of 4. COMPUTATION (`w33_pass11172_polygamy_frontier.py`, `w33_pass11177_perfect_gates_tritangent.py`, `w33_pass11178_three_qutrit_orbitals.py`, `w33_pass11175_five_qutrit_patterns.py`, `w33_pass11179_four_qutrit_near_miss.py`)

Who decides what the subsystems are? A theory of everything is not handed a tensor-product structure; it has to produce one, and a tick is non-entangling only relative to the splits it preserves (quantum mereology [57]). On W(3,3) the candidate splits are finite — the 45 tritangent planes for two qutrits, 110565 factorisations for three — so the question has an exact answer. A tick is local in a split exactly when it lies in that split's stabiliser, and the stabiliser is a $\{2,3\}$-group; hence every tick whose period involves a prime ≥ 5 entangles in *every* split, and for two qutrits so does every tick of period 9. Such intrinsically entangling ticks are 19/45 of all two-qutrit Clifford ticks and $7922/12285 \approx 64.5\%$ of three-qutrit ones, while ticks that single out a unique split fall from 19/48 to 7/48. Perfection is relative: some two-qutrit ticks are product gates for 6 splits and maximal scramblers for 27. The paper's clock is a product of three independent qutrit clocks in exactly 108 splits — none of them the light-cone coordinates in which it is written; 4 are the orthogonal frames of the Lorentzian form and 104 mix positions with momenta — while the perfect interacting tick is local in none: the interaction that makes it scramble maximally removes every split in which it could look local. COMPUTATION (`w33_pass11180_mereology.py`, `w33_pass11181_intrinsic_entanglement.py`, `w33_pass11182_paper_ticks_mereology.py`)

The polygamy maximum is now proved on the four-parameter class of states fixed by the maximal torus of its symmetry: a certified branch and bound — AM–GM, a concavity core checked with interval Hessians, and upper bounds everywhere else — closes every box, so $4/\sqrt{15}$ is the maximum there; globally it remains the value every search finds. In the E_6 reading, relative to one complete frame the 45 splits are the 5 couplings $1 \cdot 10 \cdot 10$ and the 40 couplings $16 \cdot 16 \cdot 10$ of the E_6 cubic; a partially entangling tick never keeps a singlet-sector split in the singlet sector, and the frame stabiliser $W(D_5)$ — the $SO(10)$ Weyl group — acts on such a split only locally or perfectly, never partially (a counting dictionary; no field content is claimed). The rank-20 geometry of three-qutrit splits is recovered independently, and a finer invariant — where the rank-one blocks send their images and kernels — separates almost everything the signature merged: what remains are exactly two pairs of relations that are each other's time reverse, a gate and its inverse. Block ranks, orientations and incidences thus classify how two subsystem splits relate completely, except for the direction of time. Beyond $\mathbb{F}_9$, 71 absolutely maximally entangled ten-qutrit graph states found by tabu search give 17 892 further perfect five-qutrit gates, and they realise the same two orientation patterns as the $\mathbb{F}_9$ census; behind this sits the determinant law of every order, $\sum_{|I|=k} \det S[I,J] = 1$ for every set J of k parties, with $\det S[I,J] = \det S[I^c,J^c]$. COMPUTATION (`w33_pass11187_torus_class_bnb.py`, `w33_pass11185_e6_yukawa_sectors.py`, `w33_pass11184_merged_orbitals.py`, `w33_pass11186_five_qutrit_beyond_f9.py`)

The five-qutrit pattern question is now settled, by a structure every ten-qutrit absolutely maximally entangled stabilizer state carries. For any four parties, the stabilisers trivial on them restrict isomorphically to each of the other six, and the six symplectic signs sum to zero

mod 3: every six parties split 3|3 or not at all. For any three parties these splits assemble into a graph on the other seven whose degrees lie in $\{0, 3, 6\}$; the Klein quadric forbids two of the four such graphs (their Gram matrices are elliptic), and an exact constraint solve with the isotropy duality between complementary parties forbids a third. What remains is always a prism plus an isolated vertex, so the 30 uniform four-sets form a Steiner system $S(3, 4, 10)$ — the inversive plane of order 3 — and the sign structure is unique, with automorphism group $\mathrm{PGL}(2, 9)$: the ten parties behave as the projective line over $\mathbb{F}_9$. Consequently a perfect five-qutrit gate is a cross-plus-permutation exactly when its inputs contain a block (180 of 252 choices) and a ten-cycle otherwise; the other three patterns never occur. The ties of the three-qutrit relations fall to oriented data: the $\mathrm{SL}(2, 3)$ class of a loop holonomy distinguishes N from $-N$, and for the one pair whose holonomies all vanish a symplectic twist $Q = \pm 1$ does. Holonomy and twist classify all 20 relations including their direction in time, and the anti-unitary time reversal $(x, z) \mapsto (x, -z)$ exchanges exactly the two pairs no unoriented invariant separates. THEOREM COMPUTATION (`w33_pass11190_ame10_sign_structure.py`, `w33_pass11189_oriented_holonomy.py`)

The same signs organise the whole family of maximally entangled qutrit registers. For six qutrits every local graph is a pentagon and the sign structure is unique with symmetry $\mathrm{PGL}(2, 5)$ — the projective line over $\mathbb{F}_5$, as ten qutrits were the line over $\mathbb{F}_9$. For eight qutrits, which cannot be absolutely maximally entangled [56], the signs alone are consistent: they would force the affine geometry $\mathrm{AG}(3, 2)$ (a Steiner system $S(3, 4, 8)$, symmetry $\mathrm{AGL}(3, 2)$), so that obstruction lies beyond the signs. Locally, the stabilisers trivial on any three of the ten parties always carry a unique regular spread of $\mathrm{PG}(3, 3)$ [66], but its reguli are not the state's Steiner blocks (they share two of five local circles). Finally the torus-class theorem for $4/\sqrt{15}$ is now certified with directed rounding (9 219 253 boxes, every closing inequality evaluated with outward-rounded arithmetic). THEOREM COMPUTATION (`w33_pass11200_sign_method_ame6_ame8.py`, `w33_pass11199_ame10_local_spreads.py`, `w33_pass11203_certified_bnb.py`)

Both structures are explicit cross-ratio geometry. Labelled by the projective line over $\mathbb{F}_5$, every four of six maximally entangled qutrits split into their harmonic pair (cross-ratio -1). Labelled by the line over $\mathbb{F}_9$, the uniform four-sets of ten are the Baer sublines, and two points outside a non-subline four-set share a sign class exactly when they are harmonic to one pair of it whose pairing avoids one Galois class of cross-ratios ($\{1 \pm i\}$ versus its conjugate) — the structure is chiral, which is why its symmetry is $\mathrm{PGL}(2, 9)$ and not $\mathrm{P\square L}(2, 9)$. COMPUTATION (`w33_pass11214_projective_line_rule.py`)

The ten-qutrit state itself is a classical object, though not the expected one. Every ten-qutrit stabilizer state in the census becomes $\mathbb{F}_9$-linear after local Cliffords, with an essentially unique $\mathbb{F}_9$ structure. As a $[10, 5, 6]_9$ code it is not Reed–Solomon: its Schur square has dimension 10, not 9. By Glynn's classification of ten-arcs in $\mathrm{PG}(4, 9)$ [71] it is therefore Glynn's non-classical arc. No Reed–Solomon code of this length is Hermitian self-dual, so no other $\mathbb{F}_9$-linear candidate exists. In particular, the Hermitian self-dual code behind the known AME(10,3) state [58, 59] is necessarily Glynn's. The state's local-Clifford automorphism group has order 2880 and permutes the qutrits as $\mathrm{PGL}(2, 9)$, the sign structure's group. Together with the exhaustive $\mathbb{F}_9$ count this makes the state unique up to local Cliffords and relabelling: there are $24^{10} \cdot 10!/2880 = 79\,888\,260\,016\,373\,760$ perfect five-qutrit Clifford gates of this type, all in one orbit. Whether a stabilizer state that is not $\mathbb{F}_9$-linear exists is open; none appeared in 300 searches. Algebraically, every AME(10,3) state with a ten-cycle sym-

metry, with the cycle's local twist of order 1, 2 or 4, is $\mathbb{F}_9$-linear. Of the 1600 codes invariant under the plain shift, 48 are AME and all 48 are $\mathbb{F}_9$-linear (`w33_pass11249_ame10_cyclic_search.py`). THEOREM COMPUTATION (`w33_pass11224_ame10_f9_structure.py`)

The orientation of a time-directed relation also has a uniform expression. The Kashiwara–Maslov index of three product Lagrangians, two from one split and one from the other, lives in the Witt group $W(\mathbb{F}_3) = \mathbb{Z}/4$ and is odd under time reversal. Oriented by a gauge-invariant profile, its net chirality is ∓ 54 on the 256 pair, ± 18 on the 6912 pair and 0 on the other 16 relations. The index is the phase of the stabilizer Bargmann invariant, $\operatorname{tr}(P_{L_1} P_{L_2} P_{L_3}) \propto (-i)^{\tau(L_1,L_2,L_3)}$: the finite-field form of the link between the Maslov index and the Weil representation [74], checked here on random triples. Together with the third-moment Gauss sum $\pm i\sqrt{3}/243$ above, the direction of time between two subsystem descriptions is carried by an imaginary Gauss-sum — Pancharatnam–Berry — phase, and by nothing time-symmetric. The two are one chirality: on all 512 members of the 256 pair the Maslov count is exactly $54Q$. Twenty-seven divides it because the relation's symmetry group moves every odd-index triple in an orbit whose effective 3-part has size divisible by 27. The remaining factor 2 is now structural. After quotienting the projectively invisible scalar $\{\pm I\}$, the effective relation stabiliser is the flag Borel $U_{81} \rtimes C_2$ of order 162. Restricting to its normal Sylow-3 subgroup leaves exactly one unmatched 27-orbit and one oppositely signed 81-orbit, so $\chi = \pm(81 - 27) = \pm 54$ and $|\chi|/27 = 3 - 1 = 2$. The residual C_2 only fuses the U_{81} orbits. The raw order-324 matrix stabiliser is $C_2 \times (U_{81} \rtimes C_2)$, not the distinct order-324 similitude normaliser. COMPUTATION (`w33_20261001_maslov_borel_factor54.py`) The Choi-state phase needs labelled registers. On the 6912 pair it vanishes on most labellings and has the relation's sign on the rest. Sorting the 36 relabellings by block type removes the labels: the resulting multiset is chiral on exactly the four time-directed relations and separates all 20. COMPUTATION (`w33_pass11212_one_chirality.py`, `w33_pass11222_why_54.py`) An independent CNF encoding solved with two further SAT solvers (CaDiCaL, Glucose) reproduces the two infeasibility results behind the $S(3,4,10)$ theorem. COMPUTATION (`w33_pass11209_maslov_chirality.py`, `w33_pass11211_ame10_sat_replication.py`)

4 Time: a finite light cone and four clocks

Section 3 ended with a qutrit whose operators behave like the before-and-after of a process. This section asks what *time* looks like inside $W(3,3)$. The answer comes in three layers, and keeping them apart is the main lesson of the section:

1. a finite **present**: 27 "events" carrying an exact Minkowski-type form $t^2 - x^2 - y^2$, with a light cone and four null directions;
2. an unbounded **history**: the paths through those events, which never close up and which remember a winding number that the present forgets;
3. an **arrow**: the direction of time in the thermodynamic sense, which appears only when information is thrown away.

Everything in layers 1 and 2 is reversible. Layer 3 is where irreversibility enters, and we shall see exactly what it costs.

4.1 Choosing a "now"

Fix one line B of $W(3,3)$ — the Bell line of Section 3, the unitary channels. The other 39 lines fall into two kinds: 12 lines meet B (three through each of its four points), and 27 lines miss it entirely. Every line that misses B can be written uniquely as

$$L_S = \{(Sy, y) \,:\, y \in \mathbb{F}_3^2\}, \qquad S = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad a,b,c \in \mathbb{F}_3,$$

for a *symmetric* 2×2 matrix S. There are $3^3 = 27$ such matrices. So the lines of $W(3,3)$ decompose as

$$40 \;=\; \underbrace{1}_{B} \;+\; \underbrace{12}_{\text{meet } B} \;+\; \underbrace{27}_{\text{histories } S}\,.$$

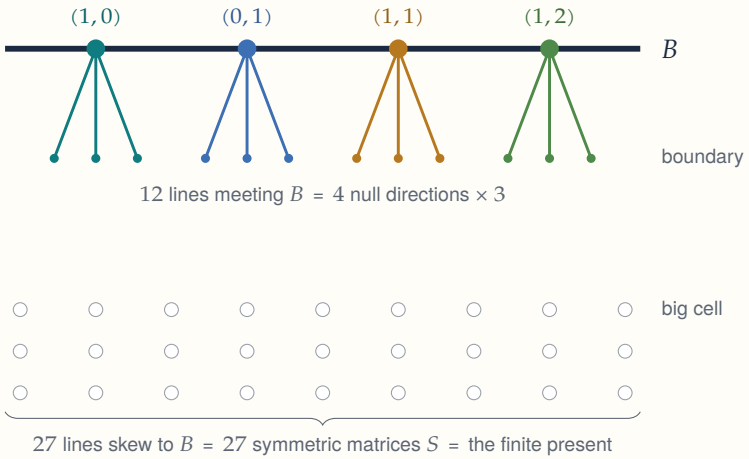


Figure 4. The forty lines of $W(3,3)$ seen from the Bell line B. The four points of B are labelled by the four projective directions of $\mathbb{F}_3^2$; they will turn out to be the four null directions of a finite light cone. The $1 + 12 = 13$ lines touching B form the "boundary"; the 27 others are the events of the finite present.

The 13 boundary lines are not an arbitrary leftover. In Plücker coordinates the 40 lines of $W(3,3)$ lie on the quadric $x_0x_4 - x_1x_3 + x_2^2 = 0$ in $\mathbb{P}^4(\mathbb{F}_3)$ — this is the dual quadrangle $Q(4,3)$ of Section 2 — and the 27 histories are its "big cell" $x_0 \neq 0$, reached by $S \mapsto [1 : a : b : c : ac - b^2]$. The 13 points with $x_0 = 0$ form the tangent cone of the quadric at B: B itself plus four three-point generators, one for each point of B. COMPUTATION (`w33_20260924_history_bigcell_q43_compactification.py`)

4.2 The finite light cone

Two histories L_S and L_T are either disjoint or they meet in a point. A short calculation shows they meet exactly when

$$\det(S - T) = 0.$$

Write $S = \begin{pmatrix} t+x & y \\ y & t-x \end{pmatrix}$ (possible because 2 is invertible mod 3). Then

$$\boxed{\det S \;=\; t^2 - x^2 - y^2}$$

— the Minkowski form of a space-time with one time and two space dimensions. Two events are *null-separated* ("light can travel between them") exactly when their lines meet, i.e. when the two contexts share a measurement.

Theorem 4.1 THEOREM COMPUTATION

The 27 histories form the finite Minkowski space $(\mathbb{F}_3^3,\ t^2 - x^2 - y^2)$. Relative to any event, the other 26 split as 8 null, 6 timelike ($\det = 1$) and 12 spacelike ($\det = -1$). The null directions through an event are the four rank-one matrices vv^{T}, $v \in \mathbb{P}^1(\mathbb{F}_3)$, and they correspond bijectively to the four points of B. `w33_20260924_history_bigcell_q43_compactification.py`

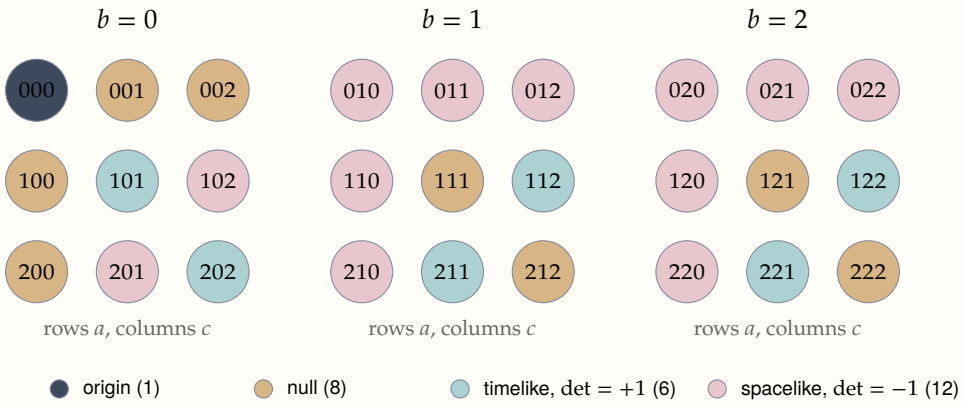


Figure 5. The 27 events $S = \left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$, labelled *abc* and coloured by their separation from the origin 000. The counts $1 + 8 + 6 + 12$ are those of a light cone over the field with three elements.

In plain words

In ordinary relativity, two events are "null-separated" if a flash of light could connect them. Here there is no light and no distance — only three possible values of each coordinate — yet the same algebraic form $t^2 - x^2 - y^2$ organises the 27 events, and "null" has a sharp operational meaning: two complete measurement contexts share exactly one measurement. The *celestial circle* — the set of directions light can travel — is in ordinary $2 + 1$ dimensions a circle of directions; here it is the four points of the projective line over $\mathbb{F}_3$.

4.3 Four null pulses make one tick

The four null directions have integer representatives

$$D_0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad D_\infty = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad D_+ = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad D_- = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

each positive semidefinite of rank one. Over $\mathbb{F}_3$ they satisfy a single relation, and it is worth staring at:

Proposition 4.2 (the common mode) THEOREM COMPUTATION

The counter map $\mathbb{F}_3^4 \to \mathrm{Sym}_2(\mathbb{F}_3)$, $(n_0, n_\infty, n_+, n_-) \mapsto \sum n_i D_i$, is onto with kernel exactly $\langle(1,1,1,1)\rangle$, so

$$\mathrm{Sym}_2(\mathbb{F}_3) \;\cong\; \mathbb{F}_3^4/\langle 1111\rangle.$$

Over the integers, however,

$$D_0 + D_\infty + D_+ + D_- \;=\; 3I,$$

a strictly timelike matrix of trace 6. `w33_20260924_positive_null_counter_cover.py`

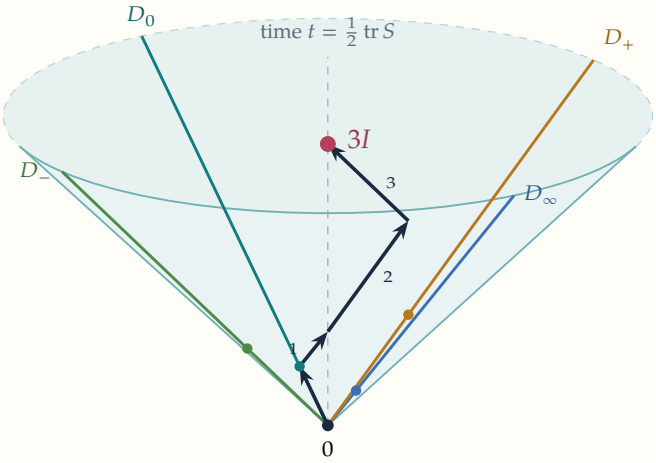


Figure 6. The real light cone of 2×2 symmetric matrices, drawn with time $t = \frac{1}{2}\,\mathrm{tr}\,S$ upward. The four null rays $D_0, D_\infty, D_\pm$ are the lifts of the four points of the Bell line. The black zig-zag adds one pulse in each direction: it ends on the time axis at $3I$. Modulo three this is the zero matrix — the finite present cannot see it — but the integer lift records it as elapsed proper time.

What it means physically: two kinds of time

Proposition 4.2 separates two notions that ordinary physics merges. The *state* of the finite present is a counter of null pulses modulo their common mode: 27 values. But a history that fires one pulse in every direction returns to the same present while advancing by a pure time step upstairs. The finite chart measures *where* you are; the lifted history measures *how long* it took. This is the structural shape of the Page–Wootters idea that time is a relation between a clock and a system, and of the distinction, in relativity, between coordinate position and proper time along a world line. BRIDGE

4.4 Light-like steps: the null graph and its clock

Join two of the 27 events when they are null-separated. The resulting graph is 8-regular with 108 edges, and it can be diagonalised exactly.

Theorem 4.3 (spectral clock) COMPUTATION

The null graph has adjacency spectrum $8^1\,2^{12}\,(-1)^8\,(-4)^6$ and Laplacian spectrum $0^1\,6^{12}\,9^8\,12^6$. At $t^\star = 2\pi/9$ the continuous-time walk $U = e^{-it^\star L}$ satisfies $U^3 = I$, with eigenvalue multiplicities 1^9, ω^{12}, $(\omega^2)^6$. The nine fixed modes are the constant mode and the eight nonzero dual-null vectors of $\mathbb{F}_3^3$. `w33_20260924_null_history_spectral_clock.py`

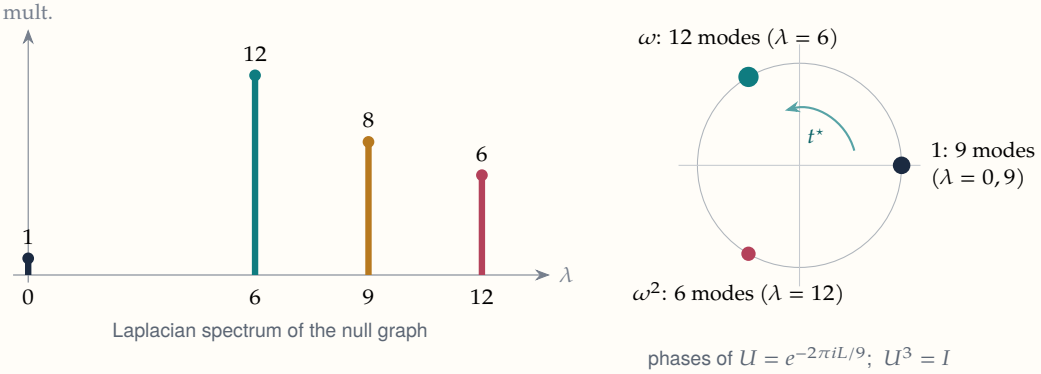


Figure 7. Left: the four Laplacian frequencies of the null graph. Right: running the graph as a quantum walk for time $2\pi/9$ sends every mode to a cube root of unity — the 27-event present carries an exact qutrit-valued clock.

A tempting identification that fails

The nine fixed modes invite the reading "9 = the nine operator histories of one qutrit" from Section 3. It is false. The symmetry group acting on the fixed nine is faithful of order 648, whereas the projective qutrit Clifford group has order 216, and the natural Bell-line S_4 splits the eight null modes $4 + 4$ while $SL(2,3)$ is transitive on the eight nonidentity qutrit Pauli labels. The count matches; the module does not. REFUTED (as an identification)

4.5 History is a path, and paths do not close

The null graph is connected with 27 vertices and 108 edges, so its cycle space has rank $108 - 27 + 1 = 82$. Its fundamental group is the free group F_{82} and its universal cover is the infinite 8-regular tree.

Theorem 4.4 (orientation and homology) COMPUTATION

(i) The 108 null edges are partitioned into 36 edge-disjoint triangles, one for each point of $W(3,3)$ off the Bell line.

(ii) A signed sum of the 36 triangle boundaries is a primitive cycle through all 108 edges with coefficients ± 1. The index-2 subgroup of the Bell stabiliser (order 648) fixes it, and the other coset negates it: the cycle space splits objectwise as $82 = 1_{\text{orientation}} + 81$.

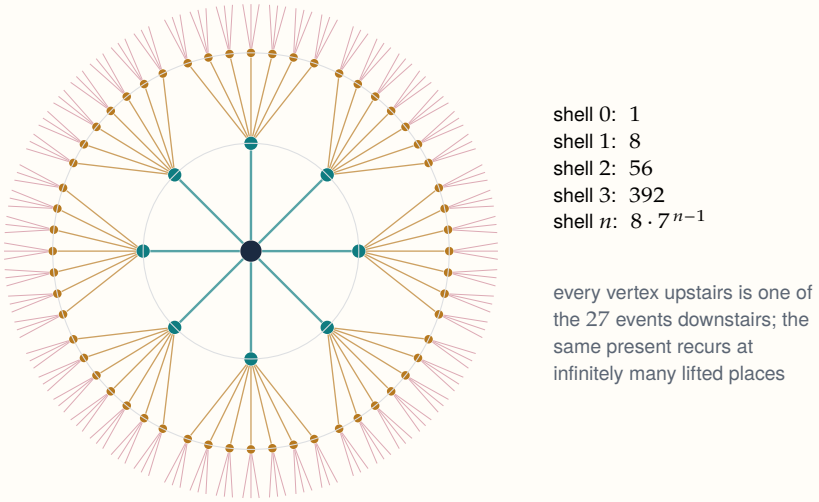


Figure 8. The universal cover of the null graph (only three grandchildren of each vertex are drawn in the outer shell). The present is finite; the history is an unbounded tree.

(iii) Filling the 36 triangles leaves first Betti number 46, and the reduced 81-dimensional module splits as $35 + 46$. The 46 injects, by an explicit chain map, into $H_1(W(3,3))$, which is itself 81-dimensional and irreducible — the Steinberg module of $\mathrm{PSp}(4,3)$.

`w33_20260924_history_invariant_cycle_orientation.py`, `w33_20260924_history_to_global_h1_intertwiner.py`

So there is exactly one global orientation of the history graph, and the symmetry group of the "now" either preserves it or reverses it. This is a *direction* for time, but not yet an *arrow*: reversing it is a symmetry of the geometry.

Temporal observable	**Lives on**	**Reversible?**
position in the present	one of 27 events	yes
step count	length of a path in the null graph	yes
reduced word	element of the free group F_{82}	yes
winding	homology class in $\mathbb{Z}^{82}$; the common mode of Prop. 4.2	yes
orientation	the sign of the unique invariant cycle	yes (up to a symmetry)
coherent phase	the Weil phase μ_{12} below	yes (complex conjugation)
records	classical outcomes kept by an observer	**no**: discarding them is the arrow

4.6 A coherent twist: the twelfth roots of unity

For the qutrit Fourier transform F and the chirp $G = \mathrm{diag}(1, \omega, \omega)$ one has $(FG)^3 = iF^2$. The metaplectic (Weil) representation that implements the symplectic symmetries on quantum states therefore carries phases in $\langle i, \omega \rangle = \mu_{12}$. Across the 27 quadratic forms of the

present, the Weil phases occur with census 1^{13}, i^4, $(-1)^6$, $(-i)^4$. Complex conjugation — the quantum-mechanical time reversal — sends $i \mapsto -i$ and $\omega \mapsto \omega^{-1}$. This is a reversible, coherent orientation mechanism, distinct from entropy production. COMPUTATION (`w33_20260924_temporal_maslov_mu12.py`)

4.7 Four clocks: the Hesse striations and the tetracode

The four points of the Bell line — the four null directions — reappear in a second, seemingly unrelated place. Identify the nine operator histories $|i\rangle\langle j|$ with the nine points of the affine plane $AG(2,3) = \mathbb{F}_3^2$. Its 12 lines fall into four *parallel classes* of three lines each, one class for each direction $v \in \mathbb{P}^1(\mathbb{F}_3)$. Each class is a way of reading the nine cells as "three ticks of a clock"; we call the four classes the four *Hesse clocks*.

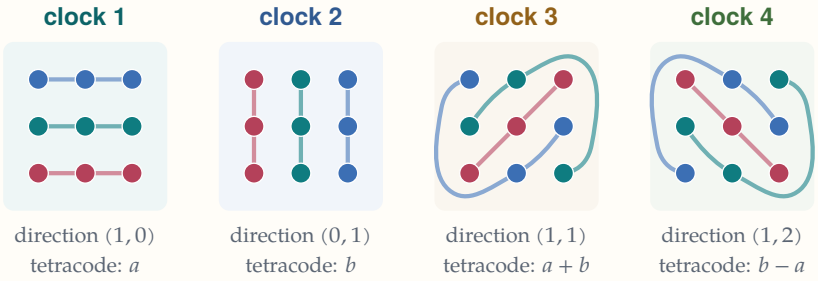


Figure 9. The four Hesse clocks. Each panel shows the nine operator histories of one qutrit as the affine plane $\mathbb{F}_3^2$, split into three parallel lines (the three "ticks"). Lines of the two diagonal classes wrap around, as on a torus. The four panels are labelled by the four points of $\mathbb{P}^1(\mathbb{F}_3)$ — the same four labels as the null directions of Figure 4 — and by the coordinate of the tetracode they read.

Theorem 4.5 (one S_4 for three structures) COMPUTATION

The four projective null directions of the present, the four Hesse clocks, and (Section 5) four orthogonal A_2 root systems inside E_8 are all indexed by $\mathbb{P}^1(\mathbb{F}_3) = \{(1,0),(0,1),(1,1),(1,2)\}$. For every one of the 24 elements of $PGL(2,3) \cong S_4$, the induced permutations of the three four-element sets are identical. The larger groups differ (the Bell stabiliser acts with kernel 27, $AGL(2,3)$ with kernel 18); the common object is the quotient S_4 and its action. `w33_20260924_null_hesse_4a2_s4_intertwiner.py`

The four clocks also carry a code. Evaluate the linear "clock reading" $\ell_{(a,b)}(x,y) = ax + by$ at oriented representatives $(1,0),(0,1),(1,1),(2,1)$ of the four directions. The result is

$$(a,\ b,\ a+b,\ b-a) \in \mathbb{F}_3^4,$$

and the nine such words are *exactly* the repository's frozen *tetracode*, the $[4,2,3]_3$ code at the heart of the ternary Golay code and of the E_8 lattice's tetracode construction — no coordinate permutation needed. COMPUTATION (`w33_pass10946_clock_code_cone_objectwise.py`)

What it means physically: a frame and an orientation

A projective direction picks a clock *frame*: which three-way split of the nine histories

(a,b)	$(1,0)$	$(0,1)$	$(1,1)$	$(2,1)$
$(0,0)$	0	0	0	0
$(1,0)$	1	0	1	2
$(2,0)$	2	0	2	1
$(0,1)$	0	1	1	1
$(1,1)$	1	1	2	0
$(2,1)$	2	1	0	2
$(0,2)$	0	2	2	2
$(1,2)$	1	2	0	1
$(2,2)$	2	2	1	0

nine clock readings $\ell_{(a,b)}$ evaluated at the four oriented null directions:
every nonzero word vanishes on exactly one direction (one clock is "stopped"),
minimum distance 3: the tetracode $[4,2,3]_3$.

Figure 10. The tetracode as a table of clock readings (colour: 1 teal, 2 rose).

counts as the ticks. Choosing a representative of the direction — $(1,2)$ or its negative $(2,1)$ — picks an *orientation* of that clock. The tetracode's peculiar sign $b-a$ (rather than $a-b$) in its last coordinate is exactly this orientation choice. Each nonzero codeword stops exactly one of the four clocks; 4 stopped clocks × 2 orientations × 3 origins give 24 complete clock calibrations.

The other research track found a further layer (Pass 10948, which we cite rather than re-derive). Over $\mathbb{F}_3$, $x^4+1=(x^2+x-1)(x^2-x-1)$, and after flipping the orientation of one clock the tetracode becomes the ideal $(x^2\pm x-1)$ in $\mathbb{F}_3[x]/(x^4+1)$: the cyclic shift of clock readings with a sign, $N(c_0,c_1,c_2,c_3)=(-c_3,c_0,c_1,c_2)$, preserves it and satisfies $N^4=-I$, $N^8=I$. The two orientations give two transverse self-dual tetracodes forming a hyperbolic pair; the negacyclic presentation of the tetracode was used in a cyclotomic E_8 construction by Watanabe *et al.* [44]. COMPUTATION (`w33_pass10948_clock_tetracode_negacyclic_hyperbolic_bridge.py`) A clock whose fourth tick is a sign flip and whose eighth tick is the identity behaves exactly like a spinor under rotation, and this is literal: the tick lifts to an element of order eight in $GL(2,3)$ whose fourth power is the central 2π spin sign, and in Section 8 the same sign is matter parity (Theorem 8.3).

4.8 Clocks as relations: the Page–Wootters weld

Label a clock qutrit's phase by $c\in\mathbb{F}_3$ and a system's phase by $p\in\mathbb{F}_3$. The four directions of $\mathbb{F}_3^2$ are c, p, $c+p$ and $c-p$. For an order-three evolution U of the system, the two stationary "history states"

$$(X_{\text{clock}}\otimes U)|\Psi_+\rangle=|\Psi_+\rangle,\qquad (X_{\text{clock}}\otimes U^{-1})|\Psi_-\rangle=|\Psi_-\rangle$$

are supported, in the clock's phase basis, exactly on $c+p=0$ and $c-p=0$: the two correlated directions. The two remaining directions ($c=0$, $p=0$) are the uncorrelated axes. On the committed E_8 basis, pairs of axis-only backgrounds generate a subalgebra of dimension 24, whereas any graph-correlated background generates all 248 dimensions. COMPUTATION (`w33_20260924_page_wootters_diagonal_weld.py`)

In plain words

In 1983 Don Page and William Wootters proposed [40] that the universe as a whole might be static, with time emerging as a correlation between a "clock" part and the rest. The finite version above is exact: a state that is globally unchanging, yet in which the clock reading and the system's phase are locked together. Only the two *correlated* clock directions produce such states — and only they, in the E_8 computation, generate the whole algebra.

4.9 A second light cone: Hermitian $3+1$, and a weld that fails

Replace real symmetric matrices by Hermitian ones over $\mathbb{F}_9 = \mathbb{F}_3(u)$, $u^2 = -1$: $X = \left(\begin{smallmatrix} a & z \\ \bar{z} & c \end{smallmatrix}\right)$ with $z = x + uy$. Then $\det X = ac - x^2 - y^2$ is a form in four variables — a $3+1$-dimensional finite Minkowski space.

Theorem 4.6 (the Hermitian present is a spread) COMPUTATION

The 81 Hermitian events split $21 + 30 + 30$ by determinant, and the 40 projective directions split $10 + 15 + 15$. An explicit map sends the Hermitian quadric to the elliptic hyperplane section of $Q(4,3)$, and under the line dictionary the ten null directions become ten pairwise disjoint lines of $W(3,3)$ covering all 40 points — a *spread*. The 36 elliptic sections are exactly the 36 spreads of $W(3,3)$. `w33_20260924_hermitian_3p1_spread_bridge.py`

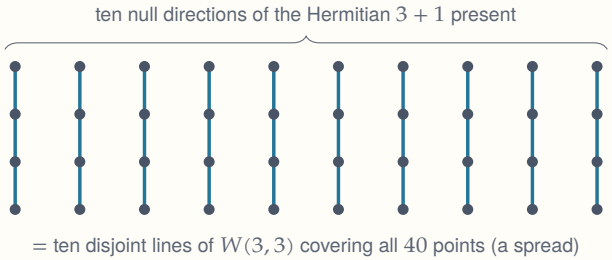


Figure 11. The Hermitian $3+1$ light cone is a partition of the forty points into ten complete measurement contexts — a complete "schedule" for measuring every two-qutrit direction exactly once.

The complex-structure weld does not exist

One might hope to weld the Hermitian light cone to the symplectic geometry through a single complex structure J ($J^2 = -I$) with $\Omega(x,y) = B(Jx,y)$. Exhausting all 729 candidates finds none. The light cone and the qutrit geometry meet through the point/-line duality of Section 2, not through one compatible J. REFUTED (for the direct weld)

4.10 The arrow: forgetting two trits

Every structure so far is reversible. Where does irreversibility come from? There is a sharp answer.

Theorem 4.7 (the price of an arrow) THEOREM

For every qutrit state ρ,

$$\frac{1}{9}\sum_{u\in\mathbb{F}_3^2} P_u\,\rho\,P_u^\dagger \;=\; \mathrm{tr}(\rho)\,\frac{I}{3}.$$

Each map $\rho \mapsto P_u \rho P_u^\dagger$ is unitary and reversible; their average, obtained by forgetting which u occurred, is the completely depolarising channel, which erases every state to the same fixed point. The forgotten record is two trits, so by Landauer's principle [32] making a qutrit clock irreversible in this way dissipates at least $2\,k_B T \ln 3$ per step.

Proof. The left side, as a function of ρ, commutes with conjugation by every P_v (which merely permutes the u's up to phases that cancel). Its image Σ therefore commutes with all nine Paulis, which span $\mathrm{End}(\mathbb{C}^3)$, so Σ is a scalar; taking traces gives the scalar $\mathrm{tr}(\rho)/3$. □

What it means physically

The finite present has a light cone; the history has an orientation; the Weil phase has a handedness. None of these is an arrow of time, because each can be reversed by a symmetry. The arrow appears exactly when a record is discarded. For a qutrit clock the minimal record is the two-trit label of the Pauli step — the photonic papers' "two trits per cycle" (`bt823_the_closure.py`; see also `w33_20260924_adqc_record_arrow.py`) — and discarding it turns a reversible tick into the maximally irreversible channel. This is the finite form of the modern view that the thermodynamic arrow is information loss, and it is *consistent* with the Connes–Rovelli idea of Section 3: a state can orient time, but only forgetting makes it flow one way. BRIDGE

The two-trit record need not be put in by hand. For a two-qutrit Clifford gate the tripartite information of its Choi state is $I_3 = 2 - \mathrm{rank}\,S_{AA} - \mathrm{rank}\,S_{BA}$, which the determinant law of Theorem 3.7 makes never positive: no gate concentrates information, and the perfect gates are the maximal scramblers ($I_3 = -2$). Whenever S_{BA} is invertible, a tick with a partner in a maximally mixed state, followed by forgetting the partner and its purification — two trits — is exactly the completely depolarising channel of Theorem 4.7. The arrow appears as soon as a maximally entangling tick is followed by discarding what it entangled with. COMPUTATION (`w33_pass11161_scrambling_and_arrow.py`)

With three qutrits the record is shared rather than handed over. In a perfect three-qutrit tick each qutrit's past is invisible to every proper subset of the three futures and fully present — two trits — in all three together. This is the secret-sharing property of absolutely maximally entangled states [54] ("exact local masking" [53]). So forgetting *any one* future erases the past completely, and the erased record is still two trits: the price of the arrow belongs to one qutrit, not to the number of partners. The spectral clock of Theorem 4.3 is not such a tick. Read on the three coordinates of $\mathbb{F}_3^3$, it is a Clifford gate, $X_a \mapsto X_a$ and $Z_b \mapsto Z_b X_{Mb}$ with M the Gram matrix of q — the free propagator $e^{-iq(p)}$ of the Lorentzian form, coupling the two light-cone coordinates by one trit and acting locally on the transverse one. This is forced: a Clifford gate that commutes with all translations has off-diagonal blocks of rank at most one, so no translation-invariant clock is perfect. Conserved momentum caps scrambling at one trit per pair per tick, and forgetting a partner after a clock tick erases at most one trit; the

two-trit arrow needs an interaction that breaks translation invariance. COMPUTATION (w33_pass11165_three_qutrit_arrow.py, w33_pass11166_clock_tick_clifford.py)

What does make a tick perfect? Nothing built from the clock's own symmetry. With its Fourier-dual potential kick $V = e^{-iq(x)}$ and the 48 Lorentz transformations of the finite light cone, the clock generates a group of order 576 with no perfect element: however ticks and kicks alternate, the transverse qutrit never exchanges information with the light-cone pair. A perfect step needs a potential N that breaks Lorentz invariance. Tick–kick–tick is never perfect (one block determinant cancels identically), but kick–tick–kick $V(N)\,K\,V(N)$ is perfect for exactly 108 of the 729 quadratic potentials, the simplest being $x_1(x_0 + x_2)$. All 108 share one orientation pattern: the past of each light-cone coordinate reappears time-reversed in the future of the *other*, and the transverse coordinate's in its own. The pattern is forced — the light-cone diagonal blocks have square determinants — and it is the light-cone reflection $v_0 \leftrightarrow v_2$, the only nontrivial coordinate symmetry of q. In time, one qudit's Bell value never reaches the algebraic maximum: $I_d = 4$ would force two vectors of one basis to coincide, although a classical system with memory reaches 4 — the qudit's only memory between the two times is the post-measurement vector, too small to carry both the outcome and the setting. It comes close, though: with the optimal state fixed as the top eigenvector of one operator and only the four bases varied, the temporal CGLMP value is 3.715, 3.787, 3.831, 3.853 at $d = 12, 16, 20, 24$, the earlier $d = 6, 7$ values improve to 3.455, 3.521, and the deficit from 4 falls like $3.4/d$ — consistent with 4 as a supremum that is never attained. COMPUTATION (w33_pass11168_clock_dual_pair.py, w33_pass11171_temporal_cglmp_large_d.py)

The approach to 4 can be written down. With d outcomes every CGLMP term is one linear ramp in the cyclic lag δ of the second outcome from its target, so $I_d = 4 - \frac{2}{d-1}L$ with L the total expected lag (a local model has $L \geq d - 1$). A strategy read off the optimisers — a rest vector shared by the first basis and both later bases, and a shift register on its complement, spread by one Householder reflection — has exactly $L = 3 + 2(\sqrt{d} - 2)/(\sqrt{d}(\sqrt{d} - 1))$, so $I_d \geq 4 - 6/(d - 1) - O(d^{-3/2})$: with the no-go above, $\sup_d I_d = 4$, never attained. The post-measurement vector carries the setting (rest vector or register) as well as the outcome, at a cost $O(1/d)$. Three further facts tie this to the clock. The orientation pattern forced on every perfect interacting tick is the symmetric-square image of the swap in $GL(2, 3)$, an element of determinant -1 — the coset that the tetracode clock group marks as time-reversing — and a single oblique Hesse clock, squared, is already a perfect potential. Entanglement between a qutrit and its own past through a Clifford tick is all-or-nothing: it survives in full exactly when the tick acts locally on that qutrit, and a perfect tick erases it until an echo (for the tick above, at the ninth tick). And a temporal Bell violation does not need it: through one clock tick a light-cone qutrit has no temporal negativity yet still reaches $3.061 > 2$. In time, a violation witnesses the setting-dependent memory of the first measurement, not entanglement. COMPUTATION (w33_pass11173_temporal_cglmp_to_four.py, w33_pass11174_lightcone_reflection_time_reversal.py, w33_pass11176_temporal_echo.py)

The arrow of Theorem 4.7 needs a partner to forget, and which qutrit is the partner depends on the subsystem split. For every tick and split, forgetting everything but qutrit q destroys rank $S_F[\text{others}, q]$ trits; the *intrinsic arrow* of a tick is the minimum of the total over all splits — the loss no choice of subsystems avoids. It is quantised: for two qutrits it is 0 or exactly 2 trits (never 1, since symplecticity makes the two exports equal), and 2 for 227/360 of all ticks; for three qutrits it takes only the values 0, 2, 3, because information never leaks out of a single subsystem alone. The paper's clock has no intrinsic arrow (it is arrow-free in its 108

splits); the perfect interacting tick costs at least 2 trits in every split, and the $\mathbb{F}_9$ gate 3. An arrow of time that survives every change of subsystems exists exactly for these ticks, and comes in units of at least one full qutrit record. COMPUTATION (w33_pass11183_intrinsic_arrow.py)

The three-qutrit values are now exact on every conjugacy class of $\mathrm{PSp}(6,3)$ (GAP representatives, each cross-checked against its fixed splits): the arrow is 0, 2 or 3 trits for fractions 55241/884520, 581/1080 and 8836/22113, never 1 and never more than one trit per qutrit. Every tick whose period involves 7 or 13 — the primes of $3^3 \pm 1$ — carries the maximal arrow, as does the cyclic relabelling of the three qutrits, which in characteristic 3 is unipotent; the two-qutrit swap has none. Time reversal itself is anti-unitary: it is the anti-symplectic half of $W(E_6) = \mathrm{PSp}(4,3).2$ [63, 64]. Measured the same way, its elements have arrow 0, 2 or 4, and 4 — everything exported in every one of the 45 splits — occurs for exactly one class, of size 36: the reflections of E_6. The time reversals no choice of subsystems can tame are the root reflections. Choosing subsystems by minimising what the dynamics does across them is the idea of operational mereology [62]; here the minimisation is exact and finite. THEOREM COMPUTATION (w33_pass11188_exact_arrow_by_class.py, w33_pass11192_time_reversal_mereology.py)

Each such reflection moves exactly the twelve lines of one double-six, swaps the two qutrits in the 15 splits built from the remaining lines — a copy of the quadrangle $\mathrm{GQ}(2,2)$ — and exports everything elsewhere. The Clifford dynamics has no T-violation: every tick is inverted by some time reversal, as Wonenburger's theorem [65] implies; but for 77/162 of two-qutrit and 84013/157464 of three-qutrit ticks the reversal cannot be unitary — exactly the ticks whose period is divisible by 3 — so Wigner's anti-unitarity is forced by the dynamics. Any CP violation must therefore come from the non-Clifford layer. Two further readings of the arrow. Arrow-free dynamics becomes rare as the register grows: 1, 133/360 and 55241/884520 of all ticks for one, two and three qutrits, about 0.0058 ± 0.0005 for four and 0 of 3000 samples for five — a second law by counting. And a walk that re-chooses the subsystem split along a time-oriented relation is a clock that runs one way: every directed relation between three-qutrit splits has non-real eigenvalues, its reverse the conjugates, with rotation frequencies in the Eisenstein field (for the smallest, $-14 \pm 18\sqrt{3}\,i$). The twist is a measurable phase: a third-moment local-unitary invariant of the gate's Choi state is purely imaginary on the smallest time-oriented relation, $\pm i\sqrt{3}/243$ — a quadratic Gauss sum — with sign Q on all 256 members, and it changes sign under time reversal; invariants whose three-cycles all turn the same way are real, and averaging over all relabellings of the registers erases the sign — the arrow is locally measurable only on labelled registers. COMPUTATION (w33_pass11202_reflection_double_six.py, w33_pass11206_t_violation.py, w33_pass11204_second_law_counting.py, w33_pass11205_one_way_clock.py, w33_pass11201_twist_chiral_invariant.py)

Time-reversal symmetry breaks exactly where the substrate stops being Clifford. With the cubic phase $\zeta_9^{x^3}$ switched on, a growing fraction of single-qutrit dynamics has no substrate time reversal (1/12 with one cubic gate, 0.92 with eight), and the minimal violating ticks are precisely the cubic phase composed with a cyclic shift of the three levels — a phase together with mixing of three states, the structural shape of CP violation. The mean number of protected qutrits (Theorem $A = n - c$) is now exact for every n from the cycle index of $\mathrm{Sp}(2n,3)$ and converges super-exponentially to $0.725535350799058349449\ldots$, so a large Clifford dynamics carries an unavoidable arrow of $n - 0.7255\ldots$ trits per tick; the one-way re-

lations between splits circulate with current spectra $\pm 24\sqrt{3}\,i$, $\pm 36\sqrt{3}\,i$ and $\pm 72\sqrt{3}\,i$, $\pm 324\sqrt{3}\,i$. COMPUTATION (w33_pass11213_cubic_t_violation.py, w33_pass11215_protected_qutrit_limit.py, w33_pass11216_clock_circulation.py)

With two qutrits the entangler can do the mixing: $(T \otimes T)$ SUM has no substrate time reversal, while a single cubic phase with an entangler always has one (exact search over the anti-unitary two-qutrit Clifford group). Exhaustively over all short words, phases on the control qutrit alone never break T, phases on the target alone need at least three cubic gates, and phases on both break it with two — the cheapest T-violation uses both sectors, as CP violation uses both quark sectors (an analogy, not a derivation). Every minimal violating tick, on one qutrit or two, has the same best time-reversal fidelity $(1 + 2\cos\frac{2\pi}{9})/3$ — the phase $2\pi/9$ of $\zeta_9^{x^3}$ — and with at most two cubic gates no violation is milder; with three or more the fidelity spectrum widens in both directions (up to 0.939 and down to 0.567 on two qutrits), while violation becomes generic (8%, 65%, 93%, 98% of random two-qutrit words with 1–4 cubic gates). The natural cubic Pauli invariant cannot detect it (it is identically real); time-reversal interferometry does. COMPUTATION (w33_pass11227_two_qutrit_cp_mixing.py, w33_pass11228_t_witness.py, w33_pass11235_two_qutrit_t_spectrum.py)

The fidelity levels are algebraic: the minimal one has F^2 a root of $81x^3 - 81x^2 + 18x - 1$, a second level lies in the same cubic field $\mathbb{Q}(\cos\frac{2\pi}{9})$, and the stronger level 0.712386 reached by single-sector violators is exactly the square of the minimal one: the optimal reversal leaves nine residual phases forming the sumset $\{-1,0,1\} + \{-1,0,1\}$ (units of $2\pi/9$), so its trace is $(1 + \zeta + \zeta^{-1})^2$, two minimal quanta multiplied (exact arithmetic in $\mathbb{Z}[\zeta_9]$, with a Galois-conjugate gap closing the float search). As a residual flavour symmetry the cubic gate adds no complete mixing pattern: its eigenbasis is a stabilizer basis, and even the non-stabilizer eigenbases of one-cubic-gate words give no PMNS or Cabibbo pattern, so the finite-symmetry no-go (w33_pass11225_flavour_from_symmetry.py) extends to the magic layer at that depth; the cubic resource supplies CP-odd phases, not angles. Generic search over the three-qutrit Clifford group is blind (it reaches 0.20 on a tick that is reversible), but reversibility is decidable exactly: in the symmetric Weyl basis $U = 3^{-n}\sum_p c(p)W(p)$ the Weil representation acts without phases, so U has a substrate time reversal if and only if $c(Lp) = \mu\,\omega^{\langle b,Lp\rangle}c(p)$ for some anti-symplectic L, $b \in \mathbb{F}_3^{2n}$ and $|\mu| = 1$. A backtracking search pruned by this condition agrees with the exhaustive two-qutrit search on every word tested and decides three qutrits in seconds: $(T\otimes T\otimes T)$ $\mathrm{SUM}_{23}\mathrm{SUM}_{12}$ violates, $(T\otimes 1\otimes T)$ $\mathrm{SUM}_{23}\mathrm{SUM}_{12}$ is reversible, and random three-qutrit ticks with $1,2,3,4$ cubic gates violate in $10/100$, $94/100$, $100/100$, $100/100$ cases. Finally, for any number of qutrits every best reversal fidelity has F^2 in the cubic field $\mathbb{Q}(\cos\frac{2\pi}{9})$ (traces lie in $\mathbb{Q}(\zeta_9)$, and $|t|^2$ is real); 148 levels are identified exactly, e.g. $F = (5+c)/9$ and $F = (-4+2c+4c^2)/9$ with $c = 2\cos\frac{2\pi}{9}$, and F itself lies outside the field in five of them. COMPUTATION (w33_pass11236_cubic_flavour.py, w33_pass11237_fidelity_levels.py, w33_pass11238_one_sector_law.py, w33_pass11239_three_qutrit_t.py, w33_pass11251_exact_reversal.py, w33_pass11252_exact_reversibility.py, w33_pass11253_cubic_field_levels.py)

The level shared by the magic vacuum and the dynamics is one number. T has spectrum $\zeta^{0,\pm1}$, so $\langle +|T|+\rangle = \mathrm{tr}\,T/3 = F_{\min}$: the largest stabilizer probability of the T-magic ray $T|+\rangle$ is $|\,\mathrm{tr}\,T/3|^2$, and the optimal residual of $(1\otimes T)\mathrm{SUM}(1\otimes T^2)$ has the spectrum of $T\otimes T$, so its reversal fidelity is $|\,\mathrm{tr}(T\otimes T)|/9$ — the same $|\,\mathrm{tr}\,T/3|^2$. Closeness to the Clifford group bounds the violation: with $F_{\mathrm{Cl}}(U) = \max_C |\,\mathrm{tr}(C^\dagger U)|/d$, $F_T \geq 2F_{\mathrm{Cl}}^2 - 1$ for every n, and $F_T \geq F_{\mathrm{Cl}}^2$ on all 1503 violating words tested (while $F_T \geq F_{\mathrm{Cl}}$ fails). The one-magic-gate census obeys

an affine law: on two qutrits $C(T \otimes 1)$ breaks the arrow exactly when the Pauli frame of C avoids an affine subspace attached to its symplectic class, so it depends on the frame only through that subspace; exactly 6480 of the 51 840 classes — one in eight, as for one qutrit (3/24) — carry a proper subspace (codimension 1, 2 or empty in 5160, 24, 1296 classes), and the probability is exactly $\frac{223}{2430}$ — now unconditionally: conjugation by the 1944 Cliffords commuting with $T \otimes 1$ splits all 4 199 040 cases into 4110 orbits, one exact decision per orbit verifies the affine law on every class, and an independent derivation agrees class by class. That derivation rewrites reversibility of $C(T \otimes 1)$ as a twisted fixed point, $T_1 E T_1^{-1} \propto C^{-1} E C^T$ for a Clifford E, which forces the symplectic part of E into the stabilizer of the magic axis; the 1296 classes in which every frame violates are exactly those with no solution even at the symplectic level. The bad set is geometry of the magic axis z_1 as a point of $W(3,3)$: every class fixing z_1 is bad, none reversing it, none sending z_1 and its preimage to different lines through it, exactly those with $M^2 z_1 = z_1$ (bad) or the symplectically unsolvable ones along a single line, and one in nine of those moving z_1 to a non-collinear point — $\frac{1}{80} + \frac{3}{80} + \frac{6}{80} = \frac{1}{8}$. On one qutrit the probability for $1, \ldots, 4$ cubic gates is exactly $\frac{1}{12}, \frac{13}{32}, \frac{421}{768}, \frac{1425}{2048}$. Over the whole two-qutrit Clifford group the perfect-tick edge pattern does not survive: one magic leg breaks a perfect tick in 7% of cases, but magic on three of the four legs breaks every $\mathbb{F}_9$-linear perfect tick (exhaustive: 165 888 reduced cases covering all 13 436 928 dressings, since $T^3 = Z$ lets the Pauli parts be absorbed into the frame; one magic leg breaks exactly $\frac{1}{12}$ of them), against 91% for other perfect ticks. Finally, the chamber group's cocycle orientation is not the orientation of the magic: the one-qutrit group $\langle X, S, T \rangle$ modulo scalars is SmallGroup(81, 9), with 18 elements of order 9, while $U_{81} = C_3 \wr C_3 = \mathrm{Syl}_3 \mathrm{Sp}(4,3)$ has 36. On three qutrits 11.1% ± 0.9% of sampled symplectic classes are flagged bad by seven frames (one in eight is 12.5%; the seven-frame screen is exact if the affine law holds at $n = 3$, checked on two classes). The uniform Clifford+T random word is reversible with probability tending to 0 (equidistribution in $PU(3)$ and the portmanteau theorem on the closed, Haar-null reversible set), although reversible circuits exist at every depth $9m$ because $T^9 = 1$; no rate is proved. **COMPUTATION** (`w33_pass11308_shared_level_and_clifford_bound.py`, `w33_pass11309_one_gate_law.py`, `w33_pass11310_magic_orientation_cocycle.py`, `w33_pass11311_perfect_edge_law.py`, `w33_pass11312_depth_law.py`, `w33_pass11330_orbit_census.py`, `w33_pass11331_twisted_fixed_point.py`, `w33_pass11332_reversibility_limit.py`, `w33_pass11333_three_qutrit_one_gate.py`, `w33_pass11334_f9_three_leg_law.py`)

The one-gate question is linear algebra over $\mathbb{F}_3$. Writing $E = W(e) V_Q$, the twisted fixed point splits into a symplectic condition $M s^{-k} Q (JMJ) = Q$, $Qz_1 = z_1$ (linear in Q, plus symplecticity) and an affine frame condition, so the reversible frames of a class are a union of affine subspaces, one per solution. This exact linear decider agrees with the exhaustive census on all 51 840 two-qutrit classes and with the Weyl criterion on every three-qutrit check. It explains the one-in-nine of the non-collinear cell: only classes whose symplectic condition has exactly three solutions (a transvection coset, two ninths of the cell) can violate, and exactly half do, the halves being exchanged by the central parity -1 (M violating $\Leftrightarrow$ $-M$ reversible), which likewise exchanges the cells where z_1 is fixed (always violating) and reversed (never). On three qutrits the decider gives the violating fraction 0.1333 ± 0.0019 over 31 994 classes — one in eight is excluded at 4.4σ — while the fixed/reversed and $M^2 z_1 = \pm z_1$ rules persist; and the union of affine pieces is genuinely a union there (2 classes whose reversible frames are not an affine subspace, 2 confirmed frame by frame by the Weyl criterion). The arrow is

also spectral: reversible magic Floquet ticks show orthogonal-class level statistics and violating ones unitary-class ($\langle r\rangle = 0.536$ and 0.600 at five qutrits, against 0.531 and 0.600), while the symplectic class is impossible because 3^n is odd ($\Theta^2 = +1$). On the representations of $PU(3)$ the Clifford-averaged cubic gate contracts the first Clifford-anisotropic sector by exactly 3/8 per gate; reversibility, a singular observable, decays more slowly (about 0.72 per gate). COMPUTATION (w33_pass11350_linear_decider.py, w33_pass11351_one_ninth.py, w33_pass11352_three_qutrit_geometry.py, w33_pass11353_level_statistics.py, w33_pass11354_decay_rate.py)

The arrow has a Jarlskog invariant. A tick is substrate-reversible exactly when its transpose is Clifford-conjugate to it ($V\bar{U}V^\dagger = \lambda U^{-1} \Leftrightarrow U^T = \bar{\lambda}V^\dagger UV$), so any Clifford-invariant function that separates U from U^T witnesses the violation. All such polynomials of degree (t,t) are packaged in one closed form,

$$J_{2t}(U) = \operatorname*{avg}_C |\mathrm{tr}(U^\dagger C U C^\dagger)|^{2t} - \operatorname*{avg}_C |\mathrm{tr}(U^\dagger C U^T C^\dagger)|^{2t} \geq 0,$$

half the squared distance between the Clifford twirls of $U^{\otimes t} \otimes \bar{U}^{\otimes t}$ and of its transpose. For a qutrit $J_2 \equiv J_4 \equiv 0$ and the first witness is cubic, J_6 — the degree of the quark-sector invariant $\mathrm{Im}\,\mathrm{tr}[H_u, H_d]^3$. On every one-qutrit Clifford+T word with up to three cubic gates (checked word by word against the exact criterion) and four (by exact count, 2 077 650 of 2 985 984), $J_6 > 0$ exactly for the violators; the same holds on all two-qutrit ticks checked. A single magic gate breaks the arrow with strength $J_6 = 9/8$, and every value is an element of $\mathbb{Q}(\cos\frac{2\pi}{9})$ with denominator dividing 216. The degree is a property of the dimension: the first non-vanishing witness has degree 10 for qubits, 6 for qutrits and 4 for $d = 5$. On the Clifford+T walk the higher tensor degrees contract by exact rationals, 3/8, 11/24, 1/2, growing with degree. On three qutrits the one-gate law keeps its two universal rules (no symplectic solution: always violating; exactly one: never) while the half-rule and the parity exchange loosen. COMPUTATION (w33_pass11355_substrate_jarlskog.py, w33_pass11356_jarlskog_two_qutrits.py, w33_pass11357_jarlskog_degree_by_dimension.py, w33_pass11358_three_qutrit_mechanism.py, w33_pass11359_higher_degree_rates.py, w33_pass11360_jarlskog_arithmetic.py)

Three corrections and sharpenings follow. First, J_6 is complete on every Clifford+T word checked (now 1.4 million random words of depth up to 20, each decided exactly) but *not* on $PU(3)$: gradient descent finds a smooth four-dimensional family of non-reversible ticks with $J_6 = 0$ (certified by $J_8 > 0$; one point refined until $\|A_3(U) - A_3(U^T)\| \approx 10^{-14}$), while the same search on J_8 finds none, and a control at $d = 5$ finds spurious zeros of J_4 every time. Invariants of the finite group Cl/phase separate orbits in degree at most 216, so the hierarchy is complete at some degree $t^* \in [4, 216]$; we conjecture $t^* = 4$ — the arrow of a generic qutrit tick is a four-copy observable. Second, the witness degree 10, 6, 4 for $d = 2, 3, 5$ is a twisted Frobenius–Schur count: $\tau : f(U) \mapsto f(U^T)$ acts on the Clifford invariants inside each irrep λ with trace $|\mathrm{Cl}|^{-1}\sum_C \chi_\lambda(\bar{C}C)$, so $J_{2t} \not\equiv 0$ exactly when some λ of degree $\leq t$ has a τ-odd invariant; for qubits this happens first at spin $5 = E + 2T_1 + T_2$ of the octahedral group, and for $d = 7$ the count predicted degree 4 before it was checked. Third, the substrate's discrete symmetries $\{1, {}^T, {}^-, {}^\dagger\}$ are the Altland–Zirnbauer symmetries T, C, $S = TC$ of the Floquet operator; in odd dimension $T^2 = C^2 = +1$, so a tick lies in one of the classes A, AI, AIII, D, BDI, and the arrow is its class: every one-gate violator of a qutrit is chiral (AIII), class D never occurs on words, and at two qutrits 151 of 274 sampled one-gate violators are not chiral at all. Read as a mixing matrix between the clock basis and its image, a tick

has flavour Jarlskog J_{CKM}, which is even under $U \mapsto U^T$ and therefore blind to the arrow; at one gate the two are exclusive (all 18 violators have $J_{\mathrm{CKM}} = 0$, all 162 maximal-CP ticks are reversible). On the Clifford+T walk, $\mathbb{E}|\mathrm{tr}\, U_k|^6 = 6 + (3/8)^k$ and $\mathbb{E}|\mathrm{tr}\, U_k|^8 = 23 + 16(3/8)^k$ exactly, and the walk approaches a unitary t-design at the exact rates $0, 0, \frac{3}{8}, \frac{3}{8}, \frac{11}{24}, \frac{7}{12}, \frac{1}{8} + \sqrt{7/32}, \rho_{(8,8)}$ for $t = 1, \ldots, 8$, with $\rho_{(8,8)}$ the largest root of $2592x^3 - 1332x^2 - 585x + 140$; the largest irrep rate found, $\rho_{(18,0)}$ (root of $864x^3 - 984x^2 + 230x + 9$, 0.781), exceeds the empirical 0.72 decay of the reversible fraction, which is therefore not a representation-theoretic rate. Finally, the three-qutrit one-gate fraction is exact: 2308 orbits of the magic-axis stabiliser on $\mathrm{Sp}(6,3)$ (GAP), decided by the linear decider and, for 109 high-symmetry classes, by the Weyl criterion on centraliser orbits of frames, give a violating fraction of exactly $\frac{240857}{2388204} \approx 0.10085$ and a bad-class fraction of exactly $\frac{437}{3276} \approx 0.13339$ (vs. 1/8 at two qutrits). THEOREM COMPUTATION (`w33_pass11369_j6_completeness.py`, `w33_pass11370_twisted_indicator_degree_law.py`, `w33_pass11371_flavour_vs_arrow.py`, `w33_pass11372_irrep_contraction_rates.py`, `w33_pass11373_three_qutrit_exact_fraction.py`)

The blind spot of J_6 has a precise anatomy. Its non-reversible zeros form two four-dimensional pieces. One is explained exactly: on a pseudo-reflection $U = 1 + \lambda|\psi\rangle\langle\psi|$ all traces entering J_6 are real, the Clifford orbit moments of ψ and $\bar{\psi}$ agree up to order five because the first T-odd Clifford-invariant polynomial of a qutrit ray has bidegree $(6,6)$ (and is unique there), and therefore $J_6(U) = |\lambda|^{12}\, \Delta_6(\psi)$ with $\Delta_6 = \mathrm{avg}_C |\langle\psi|C\psi\rangle|^{12} - \mathrm{avg}_C |\langle\psi|C\bar{\psi}\rangle|^{12} = 2|h_6(\psi)|^2$; its zero set is a real hypersurface of non-real-type rays. The degree-eight witness survives every test that breaks J_6 (no spurious zero on the pseudo-reflection locus, none reached from 307 spurious starts, positive minimum away from the reversible set), so $t^* = 4$ remains the conjecture; near the identity every witness is a twelfth-order effect. The four magic-axis rules reduce to one law, verified exhaustively at $n = 2$, on every orbit at $n = 3$ and on samples at $n = 4$: if $M^2 z_1 = z_1$ then $v = z_1 + M z_1$ is fixed by M and every frame with $\omega(a, v) \neq 0$ breaks the arrow. At four qutrits the rules hold on every decided targeted class and the bad-class fraction is 0.118 ± 0.006, below the exact three-qutrit $437/3276$. Finally the exact one-qutrit reversible fraction at depth five is $11003/49152$; the denominators $2^{3k-1}3^{[k \text{ odd}]}$ hold for $k \leq 5$, and the local decay rate appeared to drift upward with depth (withdrawn: a decider-threshold artefact, corrected in Pass 11459). THEOREM COMPUTATION (`w33_pass11418_j8_completeness.py`, `w33_pass11419_spurious_family.py`, `w33_pass11420_magic_axis_rules.py`, `w33_pass11421_four_qutrits.py`, `w33_pass11422_depth5_exact.py`)

The T-odd invariant of a qutrit state is a chirality of its mutually unbiased bases. With $p_s = |\langle s|\psi\rangle|^2$ over the twelve stabiliser states, the unique odd invariant of bidegree $(6,6)$ is the signed average $h_6(\psi) = \frac{1}{432} \sum_g \mathrm{sgn}(g)\, p^3_{ga} p^2_{gb} p_{gc}$ over the extended Clifford group, with a, b, c in three different bases, and $\Delta_6 = 349920\, h_6^2$; an ordered triple of bases and its reverse are inequivalent under the Clifford group, and complex conjugation reverses it. On the pseudo-reflection component of J_6's blind spot the degree-eight witness obeys the exact law $J_8 = |\lambda|^{14}[24\Delta_7 + |\lambda|^2(\Delta_8 - 8\Delta_7)]$ (correcting the expansion displayed earlier), positive for every phase because $\Delta_7 > 0$ and $\Delta_8/\Delta_7 \geq 3.76$ on every sampled point; J_8 is positive on the generic component too. The magic-axis law has a proved grading behind it — when $M^2 z_1 = z_1$ the Pauli $W(z_1 + M z_1)$ commutes with $V_M T_1$ and the frames it grades nontrivially are exactly the ones the law calls violating (proved); that they do violate is verified, not proved, now on every conjugacy class of $\mathrm{Stab}(z_1) \subset \mathrm{Sp}(8,3)$ that the decider reaches (97.6% of the group by class size), without exception. Each magic gate offers only $216/27 = 8$ effective Clifford choices, because the 27 Cliffords κ with $T\kappa T^{-1}$ Clifford can

be pushed through; this explains the powers 2^{3k} in the denominators and yields the exact reversible fractions 20327/131072 and 353927/3145728 at depths six and seven, with the denominator law $2^{3k-1}3^{[k\text{ odd}]}$ intact through $k = 7$. At four qutrits the excess of 437/3276 over 1/8 disappears cell by cell: it lived in the same-line cell, which falls from 77/342 to 0.14. THEOREM COMPUTATION (`w33_pass11433_magic_axis_law.py`, `w33_pass11434_h6_explicit.py`, `w33_pass11435_j8_positivity.py`, `w33_pass11436_depth_reduction.py`, `w33_pass11437_four_qutrit_cells.py`)

The magic-axis law is now a theorem on a quarter of each bad cell. The moduli of the Weyl coefficients of a tick are invariant under any reversal up to an anti-symplectic relabelling; a Clifford V_M has coefficients of constant modulus on $\mathrm{Im}(M-1)$, and the cubic gate splits them over the three cosets $\mathrm{Im}(M-1)+jz_1$ with the three distinct moduli $|\tau_j| = 0.844, 0.449, 0.293$. When $\mathrm{Im}(M-1)$ is nondegenerate the relabelling must fix $z_1 + Mz_1$, and $\omega(a, z_1 + Mz_1) = -\omega(a, z_1 + Mz_1)$ forces the stated frames to violate. The proof is sound on every two-qutrit class it covers; magnitudes alone certify all violating frames of 82% of sampled two-qutrit classes, and about 12% are magnitude-blind and need phases. At four qutrits the fixed-axis case is settled on 97.6% (decider) plus 0.6% (theorem) of $\mathrm{Stab}(z_1)$, and on every decided same-line class. The generic component of J_6's blind spot is not a hidden symmetry (no continuous supergroup with the same cubic invariants exists, and the Hessian group is maximal) and not uniformly an achiral-eigenvector phenomenon. The one-gate bad-class fraction at five qutrits is 0.138 ± 0.004, 3.3σ above 1/8, so the fraction does not settle at 1/8. Correction: the earlier report of a decay rate drifting upward with depth was an artefact of a loose decider cut (2.999), which counts dense near-miss violators as reversible (32% false at depth 24, 79% at depth 30); with the cut $3 - 10^{-7}$ the reversible fraction decays at a constant rate to depth 30 (local rates 0.7209, 0.7231, 0.7278), within one standard error of $\rho_{(8,8)} = 0.7228$ — a sharpened coincidence without a mechanism. THEOREM COMPUTATION (`w33_pass11456_n4_closure.py`, `w33_pass11457_magic_axis_theorem.py`, `w33_pass11458_generic_component.py`, `w33_pass11459_deep_decay.py`, `w33_pass11460_five_qutrits.py`)

The magic-axis law now has a uniform proof on every class of two qutrits. Where the magic axis lies inside $\mathrm{Im}(M-1)$ and the twisted moduli $|\sum_j \tau_j \omega^{js+qj^2}|$ separate ($q \neq 0$), their level sets are the three cosets of a hyperplane K, and a reversing relabelling must move $v = z_1 + Mz_1$ inside $\mathrm{Rad} \cap K$; this certifies 704 classes outright. Where the moduli are blind ($q = 0$), the coefficients are pure phases $\omega^{Q(r)}\zeta^{s(r)^3}$ (the cubic phase of T itself), and the exact reversal criterion becomes two $\mathbb{F}_3$ conditions, $\ell \circ L = \ell$ and $B(Lr, Lr') = B(r, r') - 2\ell(t)\ell(r)\ell(r')$; with $\ell = \mu B(v, \cdot)$ and $\ell(v) = 0$ they force $Lv = v$, hence $2\omega(a, v) = 0$. The same phase argument closes the 48 partial hyperplane classes, and a Wall-form argument closes the remaining 24 (transvections). In all, $352 + 704 + 48 + 168 + 24 = 1296$ classes, each under a theorem whose hypotheses are $\mathbb{F}_3$ facts checked per class; the hypotheses can fail at three qutrits, so this is not yet an n-independent proof. At four qutrits the same theorems leave 0.24% of $\mathrm{Stab}(z_1)$ open (was 1.79%). Exact data: the reversible fraction at eight cubic gates is $\frac{665275}{8388608}$ (weighted deduplication of 959 040 distinct depth-six operators, validated on $P_1, \ldots, P_7$); the one-gate bad-class fraction at six qutrits is 0.123 ± 0.011 (1000 classes). A re-audit of every sampled cut found no further artefact, though the J_6 cut of the deep-word scan has less than a decade of margin. Time reversal acts on the Clifford invariants of a qutrit *state* with the exact Hilbert–Molien series

$$\sum_k O_k t^k = \frac{t^6(1+t)}{(1-t)(1-t^2)(1-t^3)(1-t^4)(1-t^6)}, \qquad \sum_k (E_k - O_k) t^k = \frac{1+t^5}{(1-t)(1-t^3)(1-t^4)},$$

proved by a quasi-polynomial bound; asymptotically half of all state invariants are time-odd. The two lowest arrows are the $(3,2,1)$ and $(4,2,1)$ chiralities $\sum_g \operatorname{sgn}(g)\, p_{ga}^k p_{gb}^2 p_{gc}$ of an oriented triple of mutually unbiased stabiliser bases ($k = 3,4$), and explicit invariants reproduce $O_6, O_7, O_8 = 1,2,3$. By dimension, the first time-odd state invariant has degree $9,6,6,5,5$ for $d = 2,3,5,7,11$, unique only for $d \le 3$; the two-qutrit register (all 4 199 040 Cliffords) again has a unique one in degree 6, and on $\psi \otimes s$ (s a stabiliser state) its frame-potential difference is exactly $\Delta_6(\psi)/108$: compressed to that slot, a two-qutrit Clifford is a one-qutrit Clifford on $1/120$ of the group and $3^{-1/2}$ times one on $27/40$, uniformly, so $1/108 = 1/120 + 1/1080$ — the register's arrow, seen from one qutrit, is the qutrit chirality h_6. THEOREM COMPUTATION (`w33_pass11486_blind_classes_phases.py`, `w33_pass11487_magic_axis_extension.py`, `w33_pass11488_exact_depth8.py`, `w33_pass11489_six_qutrits.py`, `w33_pass11490_cut_audit.py`, `w33_pass11491_time_odd_molien.py`, `w33_pass11492_time_odd_by_dimension.py`)

The magic-axis law is a theorem for every number of qutrits. In the linear decider's variables a reversal is a symplectic Q with $Ms^{-k}Q(JMJ) = Q$ and $Qz_1 = z_1$, plus a frame condition. If $M^2 z_1 = z_1$ then $\omega(Mz_1, z_1) = 0$, so the shear fixes $v = z_1 + Mz_1$; the anti-symplectic $A = QJ$ satisfies $A = Ms^{-k}AM$, whence $Av = -v$; Q commutes with the cubic gate because both preserve $W(z_1)$; and pairing the frame condition with $\omega(v,\cdot)$ leaves $2\,\omega(v,a) = 0$. So every frame with $\omega(v,a) \neq 0$ violates — for all n, superseding the class-by-class proofs above. The identity $QJv = -v$ was checked on every affine solution space at $n = 2,3,4$ (4 684 spaces, including every four-qutrit class beyond the decider's reach). The same variables give the good rules: when $Mz_1 = -z_1$ or $M^2 z_1 = -z_1$, a frame is reversible whenever some solution has $(A^{-1} - 1)\ker(M-1) \perp a$; at $n = 2$ these criteria cover every frame of all 2592 good classes, at $n = 3$ about 96% of the good mass. The four rule cells have exact shares $2,1,3/(3^{2n}-1)$ of the one-gate classes, so they are a vanishing corner of the landscape and do not explain the bulk of the bad-class fraction. Exact data continue: $P_9 = \frac{11519231}{201326592}$ (the denominator law holds through $k = 9$; two-step rates 0.713 and 0.715 still climbing toward the sampled 0.72), and a cubed-phase test, $|c(Lp)| = |c(p)|$ with $(c(Lp)/c(p))^3$ constant, decides every word to depth nine exactly; on $PU(3)$, a search of 26958 non-affine phase relations found 4 unitaries, all reversible, so no counterexample. The one-gate bad-class fraction at six qutrits is 0.128 ± 0.007 from 2 500 classes. Finally, $\Delta^{(2)} = \Delta_6/108$ is now proved outright (the rank-one compressions are uniform over the 144 stabiliser pairs), and the time arrows of a qutrit state form a module with one new generator in each degree $6,7,8,9,10$ and none in 11–13, over an even ring generated in degrees $1,3,\ldots,9$. THEOREM COMPUTATION (`w33_pass11498_magic_axis_law_all_n.py`, `w33_pass11499_good_rules.py`, `w33_pass11500_exact_depth9.py`, `w33_pass11501_six_qutrits_extended.py`, `w33_pass11502_time_odd_module.py`, `w33_pass11503_rule_cell_shares.py`, `w33_pass11504_cubed_phase_test.py`)

The good rules reduce to blocks. Splitting $V = V_1 \oplus V'$ into the generalised 1-eigenspace of M and the rest (they are M-invariant and ω-orthogonal, and every reverser respects the split), a frame is reversible as soon as one reverser exists on the block of z_1 that negates z_1, and, on V_1, the reversers of the unipotent part cover every frame through the criterion of Pass 11499. The second condition holds in any dimension when M is semisimple at 1 (an anti-symplectic involution whose (-1)-space is a Lagrangian through the frame) and was checked for all 3^2 and 3^8 unipotent elements of $\mathrm{Sp}(2,3)$ and $\mathrm{Sp}(4,3)$; the first was checked for every block of dimension ≤ 4. Hence the good rules hold for every n whenever the unipotent part at 1 and the block of z_1 have dimension at most four, which covers

every class at two and three qutrits. The cubed-phase test is *not* exact on $PU(3)$: searching the 816 symmetry classes of non-affine phase relations exhaustively (one-dimensional eigenspaces exactly, the rest by Levenberg–Marquardt) finds 1212 eigenspaces containing unitaries, and two of them hold unitaries that pass the test yet are not reversible (Clifford overlap $2.985 < 3$, confirmed by the Weyl criterion); the earlier report of none was an optimiser that missed nearly all of them. On Clifford+T words the test remains exact through depth nine. The time arrows of a qutrit state are one family: the five generators are the chiralities $\sum_g \operatorname{sgn}(g)\, p_{ga}^k p_{gb}^2 p_{gc}$, $k = 3, \dots, 7$, over three mutually unbiased bases, and they generate freely through degree 12 with a single relation in degree 13 (correcting the earlier statement that the degree-8 generator needs two states of one basis). Exact data: $P_{10} = \frac{21873707}{536870912}$ (denominator law through $k = 10$); six qutrits give 0.131 ± 0.006 from 4 000 classes. **THEOREM** **COMPUTATION** (`w33_pass11511_good_rules_all_n.py`, `w33_pass11512_cubed_phase_exhaustive.py`, `w33_pass11513_exact_depth10.py`, `w33_pass11514_six_qutrits_4000.py`, `w33_pass11515_odd_generators_and_syzygy.py`)

One-gate reversibility is linear algebra. Let K_0 be the vectors fixed by M and orthogonal to the magic axis. Pairing the frame condition of any reversal Q with $\omega(w, \cdot)$, $w \in K_0$, shows that a reversible frame satisfies $\omega((1 - (QJ)^{-1})w, a) = 0$ for all $w \in K_0$ — for every n. Conversely, every frame satisfying these conditions for some solution Q is reversible on all 51 838 two-qutrit classes and all 2 117 checked three-qutrit orbits (99.97% of $\mathrm{Sp}(6,3)$): the reversible frames are exactly the union over Q of the subspaces $((1 - (QJ)^{-1})K_0)^{\perp}$. The magic-axis law, the good rules and their exceptions are special cases. Its uniform part — a subspace W of vectors anti-fixed by every reversal — is a fast, rigorous sufficient condition for violation: it gives $P_2 = 1/8$ exactly and $P_3 \geq 97/728$ (the exact value $437/3276$ exceeds it by $1/6552$, nine orbits with affine violating sets), and on 1.2×10^6 samples per size $P_4 \geq 0.1337 \pm 0.0003$, $P_5 \geq 0.1342 \pm 0.0003$, $P_6 \geq 0.1347 \pm 0.0005$. So the one-gate bad-class fraction does not settle at $1/8$, and the conjectured non-collinear law $(1 - 9^{1-n})/8$ is refuted (0.1327 at $n = 4$, $z = +21$). The cubed-phase test is exact on words because only Cliffords and single-T operators ever carry a non-affine relation (depth ≤ 7); every word with two or more T gates is decided by affine relations alone. The invariants of a qutrit state are now proved through degree 20: the time-symmetric ring is generated by the stabiliser moments $\sum_s |\langle s|\psi\rangle|^{2k}$, $k = 1, 3, \dots, 9$, and the time arrows by the five chiralities, with exact relation counts. The good rules extend to blocks of dimension six (and the regular unipotent blocks of dimension eight) by explicit witnesses. **THEOREM** **COMPUTATION** (`w33_pass11531_cubed_test_words_vs_pu3.py`, `w33_pass11532_good_rules_dim6.py`, `w33_pass11533_one_gate_law.py`, `w33_pass11534_time_arrow_module_exact.py`, `w33_pass11535_decay_exact_data.py`, `w33_pass11536_good_rules_dim8.py`, `w33_pass11537_union_law.py`)

The Hesse CP sign is an arrow of time of a qutrit. The Hesse doublet that carries the flavour CP order parameter of the companion track is exactly the projection of $\psi^{\otimes 3}$ onto the Hesse pencil: the Clifford generators act on it by the stored doublet matrices, complex conjugation of ψ is its coefficient CP, and the Heisenberg group acts trivially. Its Bloch data are the four MUB triple products $\Pi_b = \prod_{s \in b} |\langle s|\psi\rangle|^2$, linearly: $\rho = 3\sum_b \Pi_b$ and $r = -9\sum_b \Pi_b m_b$ with m_b the MUB vertices. Hence $(\sum_b \Pi_b)^2 = 3\sum_b \Pi_b^2$ for every pure qutrit; $\rho = M_3 - 1$ lies in $[0, 1/3]$, with $1/3$ exactly on stabiliser states and 0 exactly on the nine Hesse SIC states; and the CP-odd discriminant is the Vandermonde $W = -108^3 \prod_{a<b}(\Pi_a - \Pi_b)$. The 24 CP chambers are the 24 orderings of the four MUBs by their triple products, and time reversal swaps two MUBs. Equivalently, every qutrit state lies on one Hesse cubic E_ψ; time reversal conjugates

its j-invariant, and $\operatorname{sign} W = \operatorname{sign} \operatorname{Im} j(E_\psi)$ (a ramification argument), with $|j|$ and $|j - 1728|$ explicit in the Π_b. The time-symmetric invariants of a qutrit are all functions of the four outcome spectra (through degree 20), but its lowest arrows (degrees 6–8) are not: label-blind arrows start at degree 9. The eight null vectors of the Hamming arrow are the eight Clifford transvection ticks: the stored Weil phase is the trace phase of the tick, the outer reversal runs the tick backwards, and the arrow is the tick's handedness about its stabiliser vertex on the Hesse sphere. Finally, the open half of the union law reduces to a splitting: off the isotropic locus every reversal has $k = 0$, the frames of one reversal are $N_Q^\perp$ cut by one explicit affine condition, and they equal $N_Q^\perp$ whenever the reverser's order on $K_0 + \langle x_Q \rangle$ is prime to 3. This proves the union law for every two-qutrit class outside the two eigenvector cells and for 99.67% of $\mathrm{Sp}(6,3)$. THEOREM COMPUTATION (`w33_pass11641_hesse_cp_is_mub_vandermonde.py`, `w33_pass11642_label_blind_time.py`, `w33_pass11643_hamming_cube_is_transvection_clock.py`, `w33_pass11644_union_law_splitting.py`, `w33_pass11645_time_reversal_is_conjugation_of_j.py`)

The union law is a theorem. Every reverser A of a one-gate tick has a semisimple part A^N (N odd) that is again a reverser, splits, and reverses at least the frames A does; reversals with a nonzero shear exponent are dominated the same way (their cube is shear-free in the eigenvector cells). So the reversible frames are exactly the union of $((1 - A^{-1})K_0)^\perp$ over the *semisimple* anti-symplectic reversers negating the magic axis — proved for every two-qutrit class and 99.97% of $\mathrm{Sp}(6,3)$, with 108 same-line reversals at $n = 3$ checked individually. A tick has no reversal at all exactly when a linear system is unsolvable: 1/40 of $\mathrm{Sp}(4,3)$, 1/28 of $\mathrm{Sp}(6,3)$, and about 3.7% at $n = 4, 5$. The label-sensitive arrows of a qutrit are exactly the arrows that survive where two MUBs have equal outcome spectra, with Hilbert series $t^6(1 + t - t^3)/((1 - t)(1 - t^2)(1 - t^3))$. The strongest arrow is attained at $\psi^* = (0, \cos\frac{\pi}{8}, \sin\frac{\pi}{8} e^{5\pi i/6})$: $\max|h_6| = \sqrt{3}/62208$ on a 144-ray orbit (72 per time orientation), where three MUBs share a spectrum, so the Hesse CP order parameter and every label-blind arrow vanish; $-h_6^2$ is a CP potential with no free target ratios and $h_6\chi$ a diagonal Weyl selector. Finally, the Hesse doublet generalises: the Heisenberg-invariant cubics of n qutrits carry the dual even Weil representation, whose image is the reflection group G_6 for one qutrit and G_{33} (order 51840, the Burkhardt group) for two, and the 40 two-qutrit stabiliser bases map to 40 rays whose overlap-$\frac{1}{3}$ graph is $\mathrm{SRG}(40, 12, 2, 4)$ in its antiregular $Q(4,3)$ form — the graph of the lines of $W(3,3)$, dual to (and not isomorphic with) the $W(3,3)$ point graph; the norm is the Gross–Nezami–Walter third moment. THEOREM COMPUTATION (`w33_pass11647_union_law_closed.py`, `w33_pass11648_label_blind_sigma.py`, `w33_pass11649_maximal_arrow_state.py`, `w33_pass11650_no_reversal_fraction.py`, `w33_pass11651_n_qutrit_hesse_space.py`)

The Burkhardt geometry of two qutrits. In the two-qutrit Hesse space ($\mathbb{P}^4$, symmetry G_{33}) the whole of $W(3,3)$ becomes projective geometry: the 40 stabiliser bases (lines of $W(3,3)$) are points, the 40 transvection ticks (points of $W(3,3)$) are the eigen-lines and eigen-planes of their gates — two tick-lines meet iff the points are collinear, a stabiliser point lies on a tick-line iff the point lies on the Lagrangian, and the tick-planes are exactly the 40 planes of the Burkhardt quartic. Each of the 45 tensor factorisations acts by a local parity that is a reflection (SWAP is one); the 45 mirrors are the Burkhardt nodes, mutually orthogonal exactly when the factorisations are collinear in $GQ(4,2)$ (the perfect-gate relation). Each node is a determinant: $\langle n_F, u(\psi) \rangle = \det \psi_F$, the qutrit concurrence of the factorisation; the 45 determinants span only five dimensions, and for each of the 27 factorisation frames

$\sum_F |\det_F \psi|^2 = (M_3 - 2)/6 \le 1/9$ — a monogamy rule allowing at most three of five factorisations to be maximally entangled. Two qutrits are the only size at which factorisations act as reflections. The Hesse CP sign of two qutrits starts, as for one, at degree 18, so it misses the lowest arrows; on the equal-spectra locus the label-sensitive arrows are generated by the two lowest arrows with one relation, $g_7(4e_s - N^2) = g_6(8Ne_s - 2N^3 - \Pi_s)$. Shear-twisted reversers reduce to one conjugacy $YMY^{-1} = s^k M$ per class, which closes the same-line case of the union law. THEOREM COMPUTATION (`w33_pass11655_shear_twisted_reversers.py`, `w33_pass11656_sigma_module_two_arrows.py`, `w33_pass11657_hesse_space_geometrises_w33.py`, `w33_pass11658_hesse_cp_tower.py`, `w33_pass11659_burkhardt_nodes_are_concurrences.py`)

E_8 from two qutrits. Three copies of an n-qutrit system, averaged over the Pauli group, return one conjugated copy of the system: the invariant cubic tensors are $T_u = \sum_a |a\rangle|a+u\rangle|a-u\rangle$, three-cycles fix them and transpositions send $u \mapsto -u$, so the Weil parity *is* the permutation symmetry of the triple — symmetric triples carry the even half, antisymmetric triples the odd half. For two qutrits the even half is the Burkhardt space above and the odd half is the Cartan subalgebra of $\mathfrak{e}_8 = \mathfrak{sl}(9) \oplus \Lambda^3\mathbb{C}^9 \oplus \Lambda^3\mathbb{C}^{9*}$, the Lie algebra of two-qutrit operators together with the creation and annihilation of three fermions in the nine two-qutrit modes (Jacobi fixes the one constant). The Pauli-singlet trivectors — the four parallel classes of lines of $AG(2,3)$ — commute and form a Cartan subalgebra; every nonzero Pauli charge carries exactly three roots; and read on that Cartan the 240 roots are the Witting polytope, 40 rays times the six Eisenstein units, each ray a point of $W(3,3)$ and two rays orthogonal exactly when the Paulis commute. Complex conjugation of the qutrit amplitudes is an antilinear automorphism selecting the split form $E_{8(8)}$. The forty points of this paper are thus the root directions of the Lie algebra generated by two qutrits and three fermions. THEOREM COMPUTATION (`w33_pass11680_heisenberg_cubic_tensors.py`, `w33_pass11681_e8_from_two_qutrits.py`)

The two-qutrit E_8 as a dictionary. The centraliser of a Pauli subgroup is the Cartan plus the root spaces on its symplectic complement, so the geometry of $W(3,3)$ becomes a lattice of E_8 subalgebras: a point gives $E_6 \times SU(3)$, a line gives $SU(3)^4$, and a tensor factorisation gives a dual pair $D_4 \times D_4$ whose three mixed blocks $(8_v, 8_v) + (8_s, 8_s) + (8_c, 8_c)$ are cycled by the $\mathbb{Z}_3$ grading — triality is the passage between operator, three-fermion and three-hole. The qutrit T gate is the E_8 $|3|$-grading element: it keeps E_6 and breaks the point's $SU(3)$ to $SU(2) \times U(1)$. Matter chirality (27 versus $\overline{27}$) is the character $\varepsilon\tau$ of Pauli inversion times antiunitarity, the finite form of $P = CT$, so a Clifford-invariant arrow of time cannot select it, while $\mathrm{Im}\langle P\rangle$ can. Codex's Hilbert-space Witting rays are the E_8 root rays under the unique antilinear intertwiner, and the construction closes only for nine modes. As E_8 group elements, elementary ticks leave the shape $\mathfrak{su}(3) \oplus \mathfrak{su}(2)$ unbroken in about 0.2% of magic ticks and in none of the sampled Clifford ticks. None of this is a dynamics: the finite geometry is the symmetric skeleton, and every physical selection remains a symmetry breaking still to be derived. THEOREM COMPUTATION (`w33_pass11687_11691_two_qutrit_e8_dictionary.py`, `w33_pass11692_tick_symmetry_in_e8.py`)

The vacuum and the clock. The quadratic intensity of the chirality order parameter is fixed by the Weil parity: for every n-qutrit state $\sum_{v\neq 0}(\mathrm{Im}\langle D_v\rangle)^2 = \frac{d}{2}(1 - \langle\Pi\rangle^2)$, so it cannot select anything. Its quartic can: for two qutrits $\sum_{v\neq 0}(\mathrm{Im}\langle D_v\rangle)^4 \le 27/8$, with equality exactly on the 320 stabiliser states of a line with a nontrivial character (the proof uses that $W(3,3)$ has no triangles). The bounded potential built from it therefore has vacua that choose a

line, a point on it and a sign of the $C \times T$-odd order parameter, spontaneously. Symmetry breaking to the Standard-Model shape needs the opposite resource. By Kac's theorem an E_8 element with unbroken $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ has order at least 16, and every third-level diagonal two-qutrit clock has order at most 9. A single such clock thus stops at the grand-unified algebras E_7, $E_6 \times SU(3)$, $Spin(10)$ and $SU(5) \times SU(2)$; the Standard-Model shape needs a fourth-level clock or two commuting third-level clocks, the simplest being $Z \otimes I$ with $(I \otimes T)\,CZ$, and $E_8 \to E_6 \times SU(3) \to Spin(10) \times SU(2) \to$ SM shape is realised by commuting clocks. Every such breaking from $Z \otimes I$ is trinification breaking on a single line: colour is the $SU(3)$ of one point, the weak $SU(2)$ sits in the $SU(3)$ of another, and the clocks fix that line's chirality vacua. On the Burkhardt side, the Gruson–Sam–Weyman Coble covariant of the E_8 Cartan is -2 times the Maschke quartic, the unique equivariant one. No two-qutrit gate exchanges 8_s and 8_c; the only triality realised is the central $\mathbb{Z}_3$. These are compatible selections, not a dynamics that forces them. THEOREM COMPUTATION (`w33_pass11697_11698_chirality_vacuum.py`, `w33_pass11699_coble_covariant_is_maschke_map.py`, `w33_pass11700_triality_is_central.py`, `w33_pass11701_11702_clock_symmetry_breaking.py`, `w33_pass11703_vacuum_and_clock_share_a_line.py`)

A consistent Standard-Model configuration. The line-trinification breaking above makes colour out of three-fermion generators: the colour $SU(3)$ of a point is spanned by the states that fill the three eigenspaces of its Pauli. A pure-state vacuum, whose E_8 centraliser is $\mathfrak{su}(8) \oplus \mathfrak{u}(1)$ inside the operator grade, can never leave that colour unbroken. The consistent configuration is operator-type. In 83 592 pairs of commuting third-level clocks, colour rotates three of the nine two-qutrit levels and the weak $SU(2)$ two more. Exactly one of the six $SU(5)$ completions is then a Georgi–Glashow $SU(5)$ on five levels, and a chirality vacuum on one of the remaining four levels leaves $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ unbroken (checked with the full E_8 bracket). Requiring the vacuum to be annihilated by the hypercharge leaves exactly one hypercharge, under which the five levels carry the charges of a 5 or $\overline{5}$ and the whole $\mathfrak{e}_8$ has the standard spectrum without exotic charges. The handedness is fixed either by preserving the Standard Model alone or by the clocks' direction of time. A single tick also suffices at the fourth level of the Clifford hierarchy: $R^2 \otimes R^{-1}$ with $R = \zeta_{27}^{x^3}$, of E_8 order 27. For every number of qutrits, the quartic chirality potential is bounded by $\frac{9}{8}\,3^{n-1}$ and condenses exactly on Lagrangian stabiliser states with a nontrivial character. None of this chooses the clocks, the sign of the hypercharge or the observed hand. THEOREM COMPUTATION (`w33_pass11706_11709_vacuum_clock_hypercharge.py`, `w33_pass11710_11711_single_tick_and_n_qutrit_vacua.py`)

The heterotic W(3,3) vacua in two-qutrit levels. The W(3,3) heterotic twist is the A8 Kac class, whose gauge shift has centraliser $SU(9)$ and leaves the roots $84+\overline{84}$: the decomposition $\mathfrak{e}_8 = \mathfrak{sl}(9) \oplus \Lambda^3 \oplus \Lambda^{3*}$ of two qutrits. On $T^6/\mathbb{Z}_3$ the string spectrum is $3\,(84,1) + 3\,(1,14) + 3\,(1,64)$ untwisted and $27\,(\overline{9},1)$ twisted, with gauge group $SU(9) \times SO(14) \times U(1)$. The chiral matter is the three-fermion grade once per complex plane, the 27 fixed points $\mathbb{F}_3^3$ carry dual two-qutrit states, and the $SU(9)$ anomaly cancels as $3 \cdot 9 = 27$. Of the 87 $\mathbb{Z}_6$-I W(3,3) Standard Models, 33 have colour on three levels, the weak $SU(2)$ on two and all three quark doublets untwisted three-fermion states. In 31 of them the four remaining levels split as $\{a\}\,|\,\{b\}\,|\,\{c,d\}$: the two degenerate planes carry families on one level, the order-three plane on the other, and the untwisted fives sit on the pair. The flagship's hypercharge vanishes on the four non-GUT levels. The top Yukawa $y_t = g$ is the nine-level determinant: Q, u^c and H fill all nine levels between them. The Higgs and its colour partner are the two halves of one level-pair multiplet,

split by planes. These are readings of actual spectra; they do not supply flatness, masses beyond the top, or a selection of these vacua. COMPUTATION (w33_pass11713_family_e6_grade_dictionary.py, w33_pass11714_11716_heterotic_two_qutrit_levels.py)

Charges, exotics and the top in levels. Hypercharge and T_3 are diagonal on the nine levels, so electric charge is the level occupation weighted by $\frac{1}{3}$ on colour levels, 0 and -1 on the weak levels and 0 on the flavour levels: integral occupation forces standard charges. In the 33 $\mathbb{Z}_6$-I W(3,3) Standard Models with colour and weak inside the $SU(9)$, every fractionally charged state has fractional occupation and is twisted. The twisted half of each family is a single fermion in one level, so a family is a three-fermion ten plus a one-fermion antifive. Because couplings conserve occupation modulo integers, and the charged fractional states are unbalanced class by class in all 33 models, some exotics stay massless at every order unless the vacuum condenses fractional-occupation fields. The tree-level up Yukawa is $g\,\varepsilon_{\alpha\beta}$ on the two degenerate planes (singular values $g, g, 0$): $m_t = m_c$ at string tree level, so a realistic hierarchy needs further breaking. The level traces give $\sin^2\theta_W = \frac{3}{8}$, and the trivector cubic of the two-qutrit E_8 is exactly the nine-level determinant (1680 couplings, constant 1). The Codex $\mathbb{Z}_6$-II benchmark lies in the W(3,3) anomaly class but not in the strict $SU(9)$ class, so this dictionary does not apply to it. THEOREM COMPUTATION (w33_pass11802_11807_levels_charges_textures.py)

Yukawa textures of the class. Exact integer programming over all insertion orders gives the following, using the necessary selection rules: gauge charges, hidden N-ality, point-group sector and H-momentum. In every core model the tree-level up Yukawa is $g\,\varepsilon_{\alpha\beta}$, so $m_t = m_c$. In ten models, the flagship among them, it stays exactly $g\,\varepsilon_{\alpha\beta}$ to all orders whatever the hidden-neutral condensates; in each of them no gauge-neutral singlet monomial carries H-momentum. With hidden-neutral condensates the down and lepton Yukawas vanish to all orders by gauge invariance alone in 22 of 33 models. With hidden composites they first appear at degree 12 or more (degree 4 in two models), using the corrected H-momentum rule of Cabo Bizet et al. Fractional electric charges are labelled by $\mathbb{Z}_2 \times \mathbb{Z}_3$: one bit of twist parity and one qutrit, the fixed point of the order-three plane. These are degeneracy and suppression statements valid for every coefficient, not mass predictions. As perturbative superpotential vacua they disfavour the class. THEOREM COMPUTATION (w33_pass11810_11813_yukawa_textures.py; Lean W33.Pass11815LevelCharges)

Why the even-order class splits. The W(3,3) vacuum $3(84)+27(\bar{9})$ is three copies of the minimal anomaly-free three-generation content $84+9\times\bar{9}$ of $SU(9)$ flavour unification. Two theorems of the two-qutrit E_8 then control its Yukawa sector. First, the cubic $\varepsilon(x \wedge y \wedge z)$ on $\Lambda^3\mathbb{C}^9$ is alternating, so when families are copies the top Yukawa is $\sum_k \varepsilon_{ijk}h_k$, with singular values $(|h|, |h|, 0)$; with a single copy it vanishes. Second, the $\mathbb{Z}_9$ centre forbids $84\cdot\bar{9}\cdot\bar{9}$, so the bottom Yukawa needs the quartic $84\cdot\bar{9}^{-3}$, one field beyond the top. In $\mathbb{Z}_6$-I the two planes with equal twist are such copies, giving $m_t = m_c$. The hidden $SU(5)$ confines at $\Lambda/M_s \approx 4\times10^{-9}$, so composite bottom and tau masses are negligible and the class is excluded. In $\mathbb{Z}_6$-II all twists differ: the W(3,3) $SU(9)$ models and the Codex benchmark have hierarchical up matrices with a single cubic top, and down entries with ordinary singlets. The live branch is therefore the one whose geometry breaks the copy symmetry. THEOREM COMPUTATION (w33_pass11818_11821_su9_flavour_and_corrected_textures.py; Lean W33.Pass11822SU9Flavour)

The live branch does not survive its whole Yukawa sector. Ranked by a Froggatt–Nielsen estimate of all ten mass ratios and mixings, no $\mathbb{Z}_6$-II model fits. The single-cubic-top models get

the hierarchy ($m_c/m_t \approx 0.002$, $m_s/m_b \approx 0.016$, $m_b/m_t \approx 0.02$) but invert V_{us} and V_{cb}. Every one of them also allows a renormalisable QLd^c, so a further matter parity is required. Exact two-qutrit flavour does not help: the Pauli group has no invariant in the 9 or the 80 and four in the 84 (the E_8 Cartan), and the $\mathbb{Z}_3 \times \mathbb{Z}_3$ line it leaves after a level-diagonal GUT breaking allows at most one up entry per family row. The $T^6/\mathbb{Z}_6$ route of the W(3,3) $SU(9)$ vacuum therefore closes at the level of necessary rules. COMPUTATION (`w33_pass11826_11830_z6ii_candidate_and_qutrit_flavour.py`)

All of this is one law.

Theorem 4.8 (the intrinsic arrow for every number of qutrits) THEOREM

Let $c(S)$ be the largest number of mutually orthogonal S-invariant nondegenerate planes: the qutrits a tick can keep to themselves. For every n and every Clifford tick $S \in \mathrm{Sp}(2n, 3)$,

$$A(S) \;=\; n - c(S), \qquad c(S) = \sum_{\lambda=\pm 1} \left(m_1(\lambda)/2 + m_2(\lambda)\right) + m_1^{\mathbb{F}_9}(x^2+1),$$

with m_s the number of Jordan blocks of size s. So $A \le n$, $A = n$ exactly when no qutrit is dynamically protected, and $A \neq 1$.

Sketch. In any split, $a \le c$ planes are invariant and the rest meet their image at most in a line, so $E \ge n - a \ge n - c$. Conversely, split off a maximal invariant family and decompose the rest orthogonally by the characteristic polynomial and the unipotent and unitary Jordan structure. In each indecomposable piece, a Lagrangian for both ω and $\omega(\cdot, (S + S^{-1})\cdot)$ that is transverse to its image, or meets it in one eigenvector, yields a split in which every qutrit meets its own image. Such a Lagrangian comes from a Taussky–Zassenhaus graph [67], a Cayley-transform parity subspace, an eigenvector reduction, or a Hermitian real form (structure theory of [70, 73]). □

In every optimal split each protected qutrit exports nothing and each unprotected one exports *exactly one* of its two trits. Theorem 4.7 prices a forgotten record at two trits. The best choice of subsystems halves that price: one trit, or $k_B T \ln 3$ by Landauer, per unprotected qutrit per tick. BRIDGE

The theorem is checked on every conjugacy class for $n \le 5$ (20, 74, 278 and 940 classes). It turns the second law by counting into exact numbers: the arrow-free fraction is $P(c = n)$, which is $43815911/8703676800 \approx 0.00503$ for four qutrits and ≈ 0.00020 for five. A random tick protects 0.7255 ... qutrits on average (exactly 0.725322, 0.725518, 0.725535 for $n = 3, 4, 5$), so $E[A] = n - 0.7255\ldots$ and the maximal arrow has probability about 0.45. The same law holds class by class for qubits through $n = 5$ and for 5- and 7-dits. It is a law of Clifford dynamics, not a special property of qutrits. The proof above covers every odd local dimension; qubits need a separate argument. Among the time reversals, the 36 reflections are also the only *Kramers* ones: an anti-symplectic θ with $\theta^2 = -1$ exists only for an even number of qutrits and is local only by exchanging them. Those with $\theta^2 = +1$ are complex conjugation in suitable subsystems. THEOREM COMPUTATION (`w33_pass11207_arrow_theorem.py`, `w33_pass11207_arrow_by_planes.py`, `w33_pass11208_arrow_universality.py`, `w33_pass11210_kramers_time_reversal.py`)

For qubits the diagonalisation step fails. In characteristic two, $q(x) = \omega(x, Sx)$ can vanish on every Lagrangian of that kind, as it does on the four-dimensional pieces $V(4)$ and $W(2)$. The law survives. A split in which every qubit meets its image exists exactly when some Lagrangian for ω and $\omega(\cdot, (S+S^{-1})\cdot)$ meets its image only in fixed vectors and either carries $q \neq 0$ or is fixed pointwise. Every indecomposable piece has such a Lagrangian:

- an $\mathbb{F}_2[s]$-lattice, $s = t^2/(1+t)$, in the t-adic model of the unipotent blocks $V(2k)$ and $W(k)$ [72];
- a graph or a real form for the other pieces;
- for three small pieces, one found by search.

So $A(S) = n - c(S)$ for every number of qubits, with $c = m_1(1)/2 + \chi_2\, m_2(1) + m_1(x^2+x+1)$, where χ_2 is Hesselink's index at two (exact on all 320 classes with $n \leq 5$). The unavoidable record is one bit, $k_B T \ln 2$, per unprotected qubit per tick. A random qubit tick protects about 0.665 of a qubit, fewer than a qutrit tick. The characteristic-two cycle index gives this exactly through two identities, checked on every unipotent class of $\mathrm{Sp}(2m, 2)$, $m \leq 6$, and on all of $\mathrm{Sp}(4, 4)$, and both theorems: Lusztig's unipotent pieces of Sp_{2n} in characteristic two are the Jordan strata, with point counts given by characteristic-free polynomials [68, 69], and in Lusztig's parametrisation the form on the size-two blocks ranges uniformly over the nondegenerate symmetric forms:

- the Jordan types are weighted as in odd characteristic;
- a block pattern with an even number m_2 of size-two blocks has Hesselink index zero with probability 2^{-m_2}.

They give $\lim E_n[c] = 0.66516065\ldots$, unconditionally. The unitary part is $0.27261186563783750\ldots$. The means are not monotone, $E_7 < E_6$. COMPUTATION (`w33_pass11233_qubit_limit_cycle_index.py`, `w33_pass11244_hesselink_qubit_limit.py`, `w33_pass11250_qubit_identities_proved.py`) THEOREM COMPUTATION (`w33_pass11217_qubit_arrow_law.py`)

The same law governs the continuum. Every step of the odd proof works over any field of characteristic $\neq 2$, so $A(S) = n - c(S)$ holds for the real symplectic matrices of Gaussian (free bosonic) dynamics. There A is the least number of quadrature channels through which the modes must leak into each other per tick, and c counts the normal modes, elliptic and hyperbolic. Stable free dynamics, and even pure squeezing, has no intrinsic arrow: its normal modes are the subsystems it keeps to itself. The arrow appears only with complex instability, two channels for each loxodromic quartet, and with Jordan defects. It switches on exactly at a Krein collision of modes of opposite signature, which is Krein's stability theorem read as a statement about time. THEOREM COMPUTATION (`w33_pass11219_gaussian_arrow.py`) The Gaussian arrow is also an entanglement rate. On the Choi state of the evolution, made with squeezing r, each mode's entanglement with the rest grows as $2r$ nats per channel. So A is half the least total operator-entanglement rate over all choices of modes. COMPUTATION (`w33_pass11221_gaussian_choi_entropy.py`)

Being maximally entangling is a property of a gate in a given factorisation; the arrow is what remains when the factorisation is chosen freely. Among two-qutrit gates that are perfect in the standard factorisation, 17/48 have no arrow: they are products in another one. The geometry's own perfect tick p is one of them. Exchanges carry arrows instead. The local gates with $A = 2$ all swap the two qutrits, and of the three-qutrit ticks the largest arrow,

$A = 3$, belongs to the $\mathbb{F}_9$ gate, which cycles the qutrits where it acts locally. The perfect interacting tick has $A = 2$: one qutrit stays dynamically protected. For qubits the mean number of protected qubits is now exact through six, $E_6[c] = 0.665161$, tending to about 0.66517. COMPUTATION (`w33_pass11223_named_tick_arrows.py`, `w33_pass11220_qubit_mean_protection.py`)

A finite AdS_4. One level above the phase space the symmetry is no longer transitive. The ω-traceless bivectors b of $\mathbb{F}_3^4$ give operators J_b obeying the Clifford relation $J_aJ_b + J_bJ_a = 2\beta(a,b)$, so the phase space is the spinor of a five-dimensional quadratic space, as $Sp(4) \cong \mathrm{Spin}(5)$, whose real form $\mathrm{Spin}(2,3)$ is the isometry group of anti-de Sitter space. Its 121 points are the 40 measurement contexts (null: the conformal boundary), the 45 tensor factorisations, and the 36 Kramers time reversals. Orthogonality is anticommutation, and it reproduces three earlier results at once. A Kramers reversal is local exactly in the 15 splits orthogonal to it, always as a swap. A gate is perfect exactly when it moves a split to an orthogonal one. The 27 orthonormal frames of splits are the 27 complete factorisation frames. The time reversals are the bulk: the stabiliser of each is $SL(2,9) \cdot 2$, the finite $SL(2,\mathbb{C})$ with $\mathbb{F}_9$ for $\mathbb{C}$. Its sky is the 10 contexts it preserves, a projective line $P^1(\mathbb{F}_9)$ on which it acts as $S_6 \cong P\Sigma L(2,9)$. In the tangent space of a time reversal the Lorentz orbits are $1+20+30+30$ (origin, light cone, two non-null classes). The light cone is the image of the spinor map $\psi \mapsto \psi \wedge J\psi$, which is $4:1$, and the light-cone graph is the Brouwer–Haemers graph $\mathrm{SRG}(81,20,1,6)$. Two time reversals compose to a Clifford gate, a quarter-period Fourier element or a unipotent tick. This is a kinematic skeleton — Lorentz group, light cone, spinors, a boundary made of measurements — and not a dynamics. THEOREM COMPUTATION (`w33_pass11831_11833_finite_ads4.py`)

The skeleton carries fields. The Weil representation of $Sp(4,3)$ on $\mathbb{C}^9$, built here exactly as a linear representation, splits into two irreducible singletons, Rac = 5 and Di = 4, on which a bulk time reversal squares to $+1$ and -1. As in the Flato–Fronsdal theorem, Rac⊗Rac is multiplicity-free, $15 \oplus 10$. Its symmetric part is exactly the one mode that the bulk (functions on the 36 reversals, $1 + 15 + 20$) shares with the boundary (functions on the 40 contexts). The boundary therefore reconstructs the 15 and is blind to the 20, which only the 45 subsystem splits see: bulk reconstruction needs measurements *and* factorisations. On the tangent Minkowski space the invariant operators are functions of the mass shell. A massless momentum has the little group $3^2 \rtimes \mathbb{Z}_4$, with four helicities and eight "continuous-spin" representations, and every massive momentum has $SL(2,3)$, the finite $SU(2)$, which is also the one-qutrit Clifford group. Inside the two-qutrit E_8, the only element commuting with the finite Lorentz group is the Weil parity $P = T^2$: one boson/fermion charge, and no non-abelian internal algebra. THEOREM COMPUTATION (`w33_pass11834_11837_singletons_bulk_poincare_e8.py`; Lean `W33.Pass11838CliffordQuadric`)

Bulk points are crosscaps. A time reversal T_p defines the bilinear form $B_p(\psi,\phi) = \langle T_p\psi, \phi\rangle$, symmetric on Rac and antisymmetric on Di, and $p \mapsto B_p$ is exactly equivariant. So $v \mapsto (p \mapsto B_p(v))$ is an explicit holographic map from $\mathrm{Sym}^2\mathrm{Rac}$ onto the 15-dimensional bulk mode. The resulting bulk two-point function is $+1$ between orthogonal (anticommuting) reversals and -1 between null-separated ones. This is the finite form of the crosscap description of local bulk states (Verlinde; Nakayama–Ooguri). The massless shell propagator takes its one distinct value exactly on the light cone. Massless fields of helicity $n/2$ are the functions on the 80 spinors that pick up i^n under J: four irreducible, inequivalent 20-dimensional representations, a finite spinor-helicity formalism. The hidden bulk 20 has no Rac·Rac vertex and

is reached only through boson–fermion pairs $(\mathrm{Rac}\otimes\overline{\mathrm{Di}})^2$. The massive little group never carries three labelled doublets, so the "three generations from spin" reading fails. In E_8 the answer depends on the embedding. Through the Weil route the finite Lorentz group commutes with a single $\mathfrak{u}(1)$; through $PSp(4,3) \cong W(E_6)'$ and the Tits lift extended by its order-2 torus elements it commutes with exactly $\mathfrak{su}(3)\oplus\mathfrak{su}(2)$: the A_2 centralising E_6, and the $SU(2)$ of the time reversal's own root (the faithful lift is treated below). THEOREM COMPUTATION (`w33_pass11839_11842_crosscap_fields_couplings_spin.py`, `w33_pass11843_e8_tits_lorentz_commutant.py`)

The faithful lift sharpens this. The finite Lorentz group A_6, lifted to E_8 as a group of order exactly 360, commutes with $\mathfrak{su}(5)$, the Georgi–Glashow algebra. The order-2 torus elements of the Tits extension break it to $\mathfrak{su}(3)\oplus\mathfrak{su}(2)$. The one direction they break has ad-spectrum on E_8 exactly $0^{36}, (\pm\frac{1}{6})^{30}, (\pm\frac{1}{3})^{30}, (\pm\frac{1}{2})^{20}, (\pm\frac{2}{3})^{15}, (\pm\frac{5}{6})^{6}, (\pm1)^{5}$, which is the Standard-Model hypercharge; the faithful A_6 together with an order-7 hypercharge rotation commutes with exactly $\mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1)_Y$. Under hypercharge, colour, weak isospin and Lorentz, E_8 arranges (Q, u^c, e^c) in a Lorentz 5 and (d^c, L) in a Lorentz 10. Translations do not split off: the Minkowski module occurs only as the quotient of the A_5 lattice mod 3 by a central $\mathbb{Z}_3$. The resulting finite Poincaré group commutes with exactly $\mathfrak{su}(3)\oplus\mathfrak{su}(2)$. A finite bulk path integral closes the loop. With the crosscap propagator, the boundary two-point function is 216 times the projector onto the boundary 15, and the tree cubic Witten diagram is exactly proportional to the boundary's own cubic, as multiplicity one requires. The hidden bulk 20 never reaches the boundary at tree level but circulates in loops. THEOREM COMPUTATION (`w33_pass11844_11846_e8_poincare_lifts_matter.py`, `w33_pass11847_finite_bulk_path_integral.py`; Lean `W33.Pass11848CrosscapProjector`)

No hand-chosen rotation is needed. The central element of the finite translations, the radical of the A_5 lattice mod 3, commutes with the Lorentz group, and together they commute with exactly $\mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1)_Y$. The spinorial Lorentz group $SL(2,9)$ acts through Di, which is quaternionic, so it lies in $Sp(4) \cong \mathrm{Spin}(5)$. As the parallel track found first with a Clifford lift, its commutant in E_8 is $\mathfrak{so}(11)$; the two-qutrit Weil realisation reproduces this, and $E_8 = (55,1)+(11,5)+(32,\mathrm{Di})+(1,10)$: the Lorentz spinor is carried exactly by the $\mathrm{Spin}(11)$ spinor, which contains one $SO(10)$ family 16 with its conjugate. In these embeddings the threefold index of $E_6\times SU(3)$ is colour, so no embedding examined gives three generations. At four points the boundary sees the whole bulk. There are exactly three invariant quartics on the boundary 15, and contact, 15-exchange and 20-exchange diagrams span them. The hidden 20, absent from two- and three-point functions, appears in the four-point function. THEOREM COMPUTATION (`w33_pass11849_11851_spinor_lorentz_central_z3_generations.py`, `w33_pass11852_bulk_four_point.py`; Lean `W33.Pass11853HyperchargeSpectrum`)

Three sharp limits follow.

- *No exact family symmetry.* With the family $SU(3)$ inside $SU(5)_L$, the centraliser of grand unification and family in E_8 is $\mathfrak{su}(2)\oplus\mathfrak{u}(1)$, which no copy of A_6 or $SL(2,9)$ fits into. Three generations cannot come from an exact family $SU(3)$ that commutes with both the grand-unified group and the Lorentz group.
- *No chirality from Lorentz-commuting projections.* An automorphism commuting with the spinorial Lorentz group acts on Di by ±1. The $\mathfrak{so}(11)$ spinor is quaternionic, so every such projection leaves the spinorial matter vector-like. (In fact no single automorphism of E_8 yields a chiral fixed spectrum, since the adjoint is real; chirality must come from

pairing complex E_8 eigenspaces with complex geometric sectors.)

- *The boundary fixes the bulk.* The s-channel decomposition of the four-point function has channels $1, 15, 20, 24, 60$. The 60 channel is fed only by 15-exchange and the 24 channel by both exchanges but not by contact, so the boundary data determine the cubic coupling, then the hidden 20's coupling, then the quartic, with two crossing relations to spare.

Spinorial Lorentz together with the central translation commutes with $\mathfrak{su}(3) \oplus \mathfrak{so}(5) \oplus \mathfrak{u}(1)$, which contains the left–right-symmetric algebra. THEOREM COMPUTATION (`w33_pass11854_11857_family_nogo_chirality_firewall_combined.py`, `w33_pass11856_boundary_bootstrap_channels.py`; Lean `W33.Pass11858A5LatticeMod3`)

The central translation has a familiar name. On E_8 it acts as $\exp(8\pi i Y)$, which is colour triality: phase $\omega^{\mp 1}$ on quarks and antiquarks and 1 on every colour singlet. It removes exactly the coloured X, Y bosons of $\mathfrak{su}(5)$, which is why Lorentz and the central translation together commute with the Standard-Model algebra. The translations' characters fall into three central-charge sectors. The neutral one is Minkowski momentum space $(1 + 20 + 30 + 30)$; the two charged ones are conjugate, with orbits $6 + 15 + 60$. So the geometry supplies a $1 + 2$ pattern, not three identical copies. In the spinorial structure the extra $\mathfrak{u}(1)$ is $B - L$: lepton charge three times quark charge. Lorentz-invariant vevs cannot remove it, because they preserve rank. Reflection positivity of the boundary four-point function gives exact integer channel data, hence the bounds $g_3^2 \le 1/48$, $g'^2 \le 1/8 + 6g_3^2$ and $g_4 \le 1/12$ on the bulk couplings. THEOREM COMPUTATION (`w33_pass11859_11862_parity_orbifold_sectors_breaking.py`, `w33_pass11861_bootstrap_positivity_bounds.py`; Lean `W33.Pass11863A6NotInSL2`)

The finite Lorentz group is ambivalent: every element is conjugate to its inverse, so every representation is self-conjugate and there is no analogue of a Weyl spinor. Its spinor Di carries no helicity at all on the light cone, since the massless little group acts on it only through continuous-spin-type representations. So no chiral fermion can be built from Di. In the other direction the structure is generous. Every $B - L$-charged singlet of the Standard Model is a Lorentz spinor, and exactly one Lorentz-invariant bilinear of them exists: a Majorana-type $\nu^c \nu^c$ condensate. Its unbroken symmetry is exactly $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)_Y$, the left–right seesaw pattern, here forced as the unique option. Coloured (charged-sector) momenta have only rotation little groups (icosahedral, octahedral, dihedral); only colour singlets reach the light cone. At one loop the bubble and box are positive in every channel, so the tree-level positivity bounds mark loss of perturbative control, not exclusions. THEOREM COMPUTATION (`w33_pass11864_11866_weyl_chirality_bl_quark_sectors.py`, `w33_pass11867_one_loop_positivity.py`; Lean `W33.Pass11868A6NotInGL2RealFixed`)

Three layers of time, exactly separated: a finite present with a light cone (27 events, $t^2 - x^2 - y^2$); an unbounded, oriented history (a free group, an 8-regular tree, a winding the present cannot see); and an arrow that costs exactly two trits of forgotten record per tick, of which one trit per unprotected qutrit no choice of subsystems avoids (Theorem 4.8).

Where this stops

None of this derives a continuum space-time, a speed of light, a metric that curves, or gravity. The light cone here is an algebraic form over a three-element field; "null" means "shares a measurement". Whether the present, history and arrow of the finite model are the finite shadows of physical time is a bridge hypothesis, and the dynamics that would make it testable — a Hamiltonian, an energy scale — is not supplied by the geometry. OPEN

5 E_8 from nine histories

E_8 is the largest of the five exceptional simple Lie algebras: 248 dimensions, 240 roots, a symmetry so rigid that it appears, often unexpectedly, in string theory, in the densest packing of spheres in eight dimensions, and in the physics of certain magnetic materials. This section shows that one qutrit's nine operator histories build E_8 in a way in which every piece has a meaning in the finite geometry — and it records carefully one famous coincidence that turned out to be a trap.

In plain words

A Lie algebra is the "infinitesimal" version of a continuous symmetry: the set of all directions in which you can nudge a system while keeping its rules intact, together with the rule (the *bracket*) for what happens when two nudges are combined in opposite orders. Its *roots* are the special directions that the symmetry's own "clock" (its maximal commuting part) rotates at definite rates. E_8 has 240 of them.

5.1 Eighty plus eighty-four plus eighty-four

Recall from Section 3 that the 80 non-identity two-sided Weyl maps on one qutrit span $\mathfrak{sl}_9$, the Lie algebra of traceless 9×9 matrices acting on the nine operator histories $|i\rangle\langle j|$. Now add *packets*: unordered triples of histories. There are $\binom{9}{3} = 84$ of them, forming the representation $\Lambda^3\mathbb{C}^9$; their duals form $\Lambda^6\mathbb{C}^9$, another 84. A classical construction [1, 3] makes

$$E_8 \;=\; \mathfrak{sl}_9 \;\oplus\; \Lambda^3\mathbb{C}^9 \;\oplus\; \Lambda^6\mathbb{C}^9, \qquad 248 = 80 + 84 + 84,$$

into a Lie algebra. On the nine history cells the brackets are literal packet operations: E_{ij} replaces cell j by cell i inside a packet; two disjoint triples bracket to the complementary six-set; a triple and a six-set differing by one cell contract to an E_{ij}; opposite packets return a Cartan direction. COMPUTATION (`w33_20260924_temporal_su9_e8_compiler.py`, `w33_20260924_temporal_a8_e8_process_bracket.py`)

5.2 Hesse lines are copies of A_2

Place the nine histories on the affine plane $\mathbb{F}_3^2$ as in Figure 9. A triple S of cells labels the root

$$w_S = \sum_{i\in S} e_i - \tfrac{1}{3}\sum_{i=1}^{9} e_i, \qquad w_S \cdot w_T = |S \cap T| - 1.$$

(Indeed $|w_S|^2 = 3-2+1 = 2$.) The 84 triples fall into two orbits of the affine group $AGL(2,3)$ (order 432): the 12 *lines* of the plane and the 72 non-collinear *triangles*.

Theorem 5.1 (four clocks, four A_2's) THEOREM COMPUTATION

The three lines of one parallel class give three roots with pairwise products -1 and sum 0; with their negatives they form a root system A_2. Lines of different classes meet in exactly one cell, so their roots are orthogonal. The 12 Hesse lines therefore span a reflection-closed subsystem $4A_2 \subset E_8$, one A_2 per clock, and the root shell refines as

$$240 \;=\; \underbrace{72}_{A_8} \;+\; \underbrace{24}_{\text{Hesse } 4A_2} \;+\; \underbrace{144}_{\text{triangles}}\,.$$

`w33_20260924_temporal_hesse_4a2_a8_bridge.py`

Proof. Disjoint triples have $|S \cap T| = 0$, hence product -1; the three lines of a class partition the nine cells, so $\sum w_S = \sum_i e_i - 3 \cdot \frac{1}{3}\sum_i e_i = 0$. Two non-parallel lines of an affine plane meet in one point, giving product $1 - 1 = 0$. □

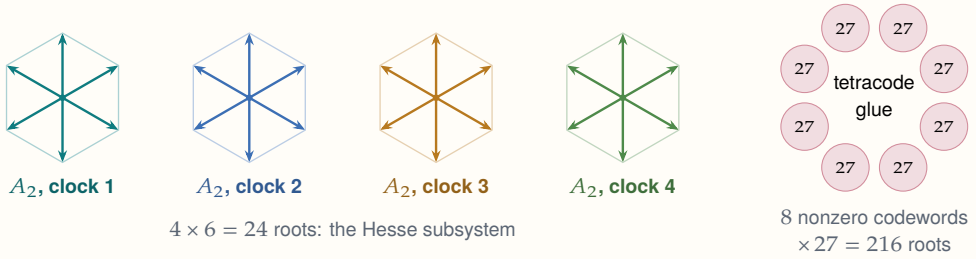


Figure 12. E_8 assembled from four clocks. Each Hesse clock contributes one A_2 root system (left). The remaining 216 roots come in eight sectors of 27, one for each nonzero word of the tetracode (right): $240 = 4 \cdot 6 + 8 \cdot 27$.

5.3 The glue is the tetracode, and the tetracode is the clock

The lattice E_8 contains the sublattice A_2^4 with quotient $\mathbb{F}_3^2$, and the well-known construction of E_8 from the tetracode [9] reads it as

$$E_8 = \bigcup_{c\in\mathcal{T}} (c + A_2^4), \qquad 240 = 4 \cdot 6 + 8 \cdot 27,$$

where $\mathcal{T} \subset \mathbb{F}_3^4$ is the tetracode: each of the eight nonzero codewords has weight three, and contributes $3^3 = 27$ minimal vectors. Combined with Section 4, the four glue coordinates are the four clocks and the eight nonzero sectors are the eight oriented, nonzero linear clock readings. THEOREM (classical) COMPUTATION (the clock reading, `w33_pass10946_clock_code_cone_objectwise.py`)

5.4 A trap: $240 = 240$

The collinearity graph of $W(3,3)$ has 240 edges, and E_8 has 240 roots. For years the repository looked for a natural bijection between them. The question has a precise and negative answer.

Theorem 5.2 (no equivariant bijection) COMPUTATION

The two groups of order 51 840 of Section 2 each do only half the job:

- $\mathrm{PGSp}(4,3) \cong W(E_6)$ acts faithfully and transitively on the 240 edges of $W(3,3)$, but *intransitively* on the 240 roots of E_8;
- $\mathrm{Sp}(4,3)$ acts faithfully and transitively on the 240 roots, but its centre acts trivially on the edges, so its edge action is not faithful.

Hence no bijection between edges and roots is equivariant for a common group. `w33_pass1020_e8_transitive_51840.g`

What *does* exist is better. The Weyl group of E_8 contains a regular element w of order three. By Springer's theory [42] its centraliser is the complex reflection group $G_{32} \cong \mathbb{Z}_3 \times \mathrm{Sp}(4,3)$ of order 155 520 — the symmetry group of the *Witting polytope*, whose 240 vertices in $\mathbb{C}^4$ are the E_8 roots.

Theorem 5.3 (the $6:1$ fibration) COMPUTATION

The 240 roots of E_8 fall into 40 complex lines $\{\pm v, \pm\omega v, \pm\omega^2 v\}$, each a copy of the six Eisenstein units. $\mathrm{Sp}(4,3)$ permutes the 40 lines with subdegrees $1, 12, 27$: the quotient is the collinearity graph of $W(3,3)$ — on its *points*, not its lines. The fibre group is $\langle c^5 \rangle \cong \mathbb{Z}_6$ for a Coxeter element c of $W(E_8)$. `w33_pass1021_e8_fibration_over_forty.g`

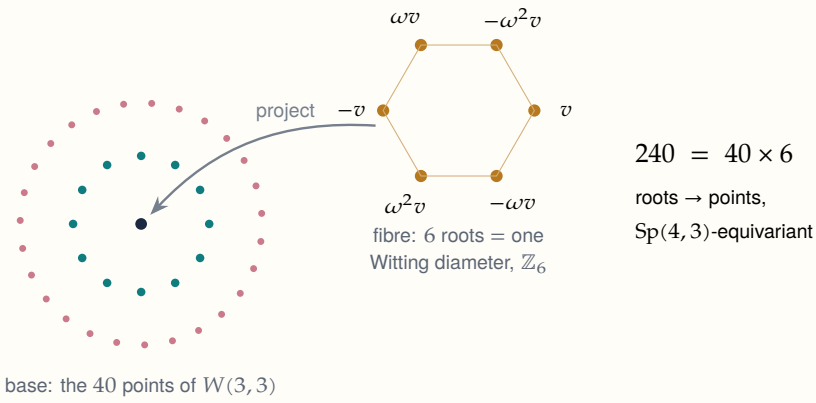


Figure 13. The correct relation between E_8 and $W(3,3)$: not edges to roots, but a six-to-one map from roots to points, whose fibres are the Eisenstein unit hexagons.

What it means physically: a photon's four-dimensional state space

The 40 Witting diameters are 40 rays in $\mathbb{C}^4$ — the state space of a photon's path and polarisation, or of two qubits. Two rays are orthogonal exactly when the corresponding points of $W(3,3)$ are collinear. So the same geometry is drawn twice: by *states* of a four-level system (the Witting rays) and by *operators* on two qutrits (the Pauli classes). The

photonic Holonet papers call this operator–operand duality. COMPUTATION (bt817_self_entangled_photon_atlas.py, bt821_operator_operand_duality.py)

5.5 $E_6 \times A_2$, the twenty-seven, and a cubic

Under its subalgebra $E_6 \oplus A_2$,

$$E_8 = (\mathbf{78},\mathbf{1}) \oplus (\mathbf{1},\mathbf{8}) \oplus (\mathbf{27},\mathbf{3}) \oplus (\overline{\mathbf{27}},\overline{\mathbf{3}}),$$

which is a $\mathbb{Z}_3$-grading $248 = 86 + 81 + 81$. The 27 of E_6 carries a unique invariant cubic form N; in the repository's committed basis it has exactly 45 signed terms (*triads*), each coordinate in 5 triads and each pair in at most one — precisely the incidence of the 27 lines and 45 tritangent planes of a smooth cubic surface, whose symmetry group is $W(E_6)$, the automorphism group of $W(3,3)$ [16]. THEOREM (classical) COMPUTATION (canonical_su3_gauge_and_cubic.json)

What it means physically

In grand unified theories built on E_6, the **27** holds one generation of matter: under $SO(10)$ it splits $\mathbf{27} = \mathbf{16} + \mathbf{10} + \mathbf{1}$ — sixteen fermions (including a right-handed neutrino), a ten of Higgs-type fields, and a singlet. The cubic N is then the unique renormalisable Yukawa coupling $\mathbf{27}^3$. Section 8 shows this same split arising as scalar, vector and spinor of a ten-dimensional Lorentz group.

5.6 The explicit semisimple three-plane and its qutrit mirrors

The $\mathbb{Z}_3$ grading has rank three in Vinberg's sense, but that abstract statement does not identify a Cartan subspace in the repository's signed $3 \otimes 27$ coordinates. An earlier displayed commuting kernel was nilpotent and its Cartan interpretation was withdrawn. The missing object can now be written and checked exactly.

Theorem 5.4 (the signed cubic contains its G_{26} Cartan plane) COMPUTATION

There is an integral signed-cubic background v whose cubic Jacobian has rank 78 and whose grade-one centraliser is a three-dimensional commuting plane $\mathfrak{c}$. For $N = 18\,\mathrm{ad}(v)$,

$$\operatorname{rank} N = \operatorname{rank} N^3 = 234.$$

The three diagonal blocks of N^3 have rank 78 and the same nonzero characteristic polynomial. The only repeated factors have degree nine and each has its full 27-dimensional kernel, so N^3, hence N, is diagonalizable. Thus $\mathfrak{c}$ is a regular semisimple Cartan subspace. w33_pass11260_exact_semisimple_cartan.py

If (x,y,z) are its committed rational coordinates, an explicit invertible map over $\mathbb{Q}(\sqrt[3]{2},\zeta_{18})$ carries them to standard qutrit coordinates (X,Y,Z). On $\mathfrak{c}$ the complementary Pfaffian is

$$2^{17} C_3 N_9 S_9^3,$$

where C_3N_9 is, up to -27, the product of the twelve stabilizer-state mirrors and S_9 is the product of the nine SIC mirrors. Consequently the Pfaffian weights the two mirror orbits by 1 and 3; the G_{26} Jacobian weights them by 2 and 1. `w33_pass11261_g26_cartan_coordinate_map.py`

The two relative invariants separate the two mirror orbits arithmetically. Let A be the product of the nine SIC mirror forms and B the product of the twelve stabilizer/MUB mirror forms. Their divisor valuations are

$$\operatorname{div} P = (3,1), \qquad \operatorname{div} J = (1,2),$$

so the valuation matrix has determinant 5 and

$$2\operatorname{div} P - \operatorname{div} J = (5,0), \qquad 3\operatorname{div} J - \operatorname{div} P = (0,5).$$

Equivalently P^2/J is a scalar multiple of A^5 and J^3/P a scalar multiple of B^5; with the frozen normalisations these become exact cleared polynomial identities of degrees 78 and 99. Thus the signed-E_8 Pfaffian and the G_{26} Jacobian form two mixed probes whose integer combinations isolate the SIC and stabilizer wall arrangements separately. The index five is only the determinant of this divisor lattice; it is not a physical $\mathbb{Z}_5$, particle count or coupling. **COMPUTATION** (`w33_20261001_g26_e8_mirror_separation.py`)

The same exact Cartan plane admits a reconstructed positive grade-pair form from the numerical E_8 bracket coordinates. In the committed basis,

$$G_{ij} = \operatorname{Tr}\big(\operatorname{ad}(q_i \in \mathfrak{g}_1)\operatorname{ad}(q_j \in \mathfrak{g}_2)\big) = \begin{pmatrix} 590 & 20 & 0 \\ 20 & 980 & 0 \\ 0 & 0 & 300 \end{pmatrix},$$

whose principal minors are 590, 577800 and 173340000. Thus, after the natural $\mathfrak{g}_1 \leftrightarrow \mathfrak{g}_2$ grade-exchange identification, the semisimple slice has an positive kinetic candidate for this declared identification. This choice is additional structure and is not identified with the unitary G_{26} metric. Using it without fitted coefficients gives a useful failure mode. If $F = C_3N_9S_9$ is the product of all 21 mirror forms, the projective functional $|F|^2/(q^\dagger G q)^{21}$ has an exact strict local maximum at [0:0:1]: both real and imaginary tangent Hessians of its logarithm are negative definite conditional on the displayed reconstructed Gram matrix. That ray maps to the qutrit $[1, \zeta_9, \zeta_9^{-1}]$ and its cubic postselection interface has zero Jarlskog invariant. Equal mirror repulsion therefore does *not* generate CP; an additional oriented datum is required. **COMPUTATION** (`w33_20261001_cartan_kinetic_wall_vacuum.py`)

The qutrit ray just displayed is exactly the standard qutrit T-magic ray $|T_3\rangle = (1, \zeta_9, \zeta_9^{-1})/\sqrt{3}$. For the coefficient-free equal-wall logarithmic barrier in the standard unitary G_{26} metric it is an exact stationary point whose real projective Hessian has spectrum $30, 30, 54, 54$. Its projective Clifford orbit has 72 rays, stabiliser C_3, and every ray singles out exactly one of the four stabilizer MUBs by the purity pattern $(1/3, 5/9, 5/9, 5/9)$, with 18 rays per MUB. Through the dictionary below this selects one tetrahedral ϕ axis; the residual C_3 cycles the other three axes, so one further ternary selector chooses an incident oriented chord, all of which lie in the TM_1 class of Pass 11267. Complex conjugation stays inside the same 72-ray orbit and leaves $(u_6, u_{12}, u_{18}) = (0, 0, -1/729)$ unchanged, so no scalar G_{26}-invariant Cartan

potential can choose the CP/time orientation. COMPUTATION (w33_20261001_balanced_g26_tmagic_vacuum.py)

There is also a regular polynomial selector with an exact global statement. For $\lambda, \alpha, \beta, r_0 > 0$,

$$V(v) = \lambda(v^\dagger v - r_0^2)^2 + \alpha|u_6(v)|^2 + \beta|u_{12}(v)|^2$$

has precisely the 72 projective T-orbit rays as global minima at radius r_0, with a common phase still free. In the chart $z = 1$, the cube ratios $A = x^3$, $B = y^3$ have eight simple solutions; the exact eliminant factors as $(A^2 + A + 1)(A^3 - 3A^2 - 24A - 1)(A^3 + 24A^2 + 3A - 1)$. All coordinates are nonzero and there are no points at infinity, so the nine cube-root lifts give 72 reduced rays. The invariant quotient identifies these with the T orbit. Their location is independent of positive angular coefficients. In standard real kinetic coordinates the Hessian spectrum is $0, 8\lambda r_0^2, 32\alpha r_0^{10}$ (twice) and $200\beta r_0^{22}$ (twice). This is a constructed scalar effective potential; its higher-degree interactions, radius, couplings and phase require physical input. Neither these curvatures nor its chosen zero energy predict masses or the cosmological constant. COMPUTATION (w33_20261001_g26_global_polynomial_vacuum.py)

An alternative normalization can now be certified without reconstructing the large numerical gauge map. In the canonical flat 27×3 order, let Q have columns supported at $(0, 52, 80)$, $(5, 37, 75)$ and $(26, 40, 54)$, with entries one. The integer E_6 basis gives $Q^\dagger Q = 3I$, vanishing cross moment maps and pairwise zero brackets. Thus $\Phi = Qq/\sqrt{3}$ has canonical norm $\|q\|^2$ and is exactly D-flat. Its mass pencil splits into three cycle blocks of size three, nine bipartite blocks of size six and one of size eighteen. Relative signed permutations have order three (up to a minus sign in the six-blocks); the eighteen-block carries the regular $\mathbb{Z}_3^2$ character. Consequently, for every complex q, its squared singular spectrum is the twelve normalized MUB overlaps twice each, the nine normalized SIC overlaps with factor $2/3$ six times each, and three zeros. In particular the moments are $20\|q\|^2$ and $8\|q\|^4$. This certifies the earlier numerical mirror law on a new exact plane, not the earlier floating gauge transformation. COMPUTATION (w33_20261001_exact_cartan_mirror_completion.py)

There is an explicit mass mechanism on this normal slice. The signed E_6 cubic supplies a global degree-six contraction C_6 of three quadratic cross products; the ternary cubic $d(\Phi a)^3$ supplies an Aronhold scalar S. Exact symbolic restriction gives $C_6|_Q = 12u_6$ and $S|_Q = (152u_6^2 - 32u_{12})/5$. Thus $I_6 = 9C_6/4$ and $I_{12} = (152I_6^2 - 3645S)/32$ restrict to the standard invariants on $Q/\sqrt{3}$. All lower/dual E_6 and volume-form generator checks vanish exactly, and their global analytic differentials are executable. At unit T, the covectors $\overline{T}, \nabla u_6, \nabla u_{12}$ have Gram $\mathrm{diag}(1, 16, 100)$. Their rank-one mass operators therefore lift all three null modes without changing the 78 heavy singular masses. With specified EFT coefficients, the light masses are $|c_5|r^2/\Lambda$, $16|c_6|r^{10}/\Lambda^9$ and $100|c_{12}|r^{22}/\Lambda^{21}$. The global covariant circuits are explicit; measured scales, particle assignments and complete EFT-loop stability remain open. These are nonsupersymmetric EFT interactions. COMPUTATION (w33_20261001_global_e6_cartan_covariants.py) Adding positive moment squares to the global radial/invariant selector names an 81-field classical potential. Its zero set reduces to the earlier Cartan intersection by moment-zero orbit theory. The 72 Cartan rays are little-Weyl-related gauge representatives, not 72 distinct physical vacua or prepared quantum resources; the common phase remains flat.

A separate declared gauge/scalar/fermion model does not automatically select this vacuum. With $g_{SU(3)} = g_{E_6}/\sqrt{6}$, exact rational inverse-Gram identities match the gauge squared

masses to one third of the cubic squared masses. The moment potential supplies matching massive real scalars, and conjugate fermion multiplets cancel perturbative gauge anomalies. The complete Landau-gauge one-loop angular potential is $(4g^4/9 - 8|y|^4)F/(64\pi^2)$, where the earlier interval witnesses show T is a saddle of F. A zero prefactor leaves angular flatness. Higher EFT operators, additional sectors and stable quantum alignment remain open. COMPUTATION (w33_20261001_complete_gauge_scalar_fermion_loop.py)

The invariant compiler now also supplies the global degree-eighteen coordinate. If $\mathcal{T} = dS_F[\mathrm{Hess}(F)]$ is the classical Hessian polarization of the ternary cubic's Aronhold scalar, exact restriction gives

$$I_{18} = \frac{19683\mathcal{T} - (48384/5)I_6^3 + (13824/5)I_6 I_{12}}{1492992}, \qquad I_{18}(Qq/\sqrt{3}) = u_{18}(q).$$

Its analytic differential is executable on all 81 fields. The real term $\gamma|I_{18} + r_0^{18}/729|^2$ lifts the common phase, but all selected representatives share one gauge orbit and do not choose physical CP. COMPUTATION (w33_20261001_degree18_phase_completion.py)

Actual charges impose a stronger physical boundary. Eight exact stabilizer generators annihilate the entire Q plane, so the three completed light modes are singlets of the connected unbroken gauge group. Its dimension eight cannot contain an unbroken Standard Model algebra of dimension twelve. The declared conjugate-paired matter has zero chiral index under every subgroup restriction. Standard 27 branching does not identify these normal modes as charged families. COMPUTATION (w33_20261001_chiral_mass_assignment_audit.py) Including the scalar selector and all three lifts in the full declared one-loop determinant yields 83 positive real scalar modes and 81 massive fermion singular pairs at T. The tested finite-step quantum curvature is negative and infrared-sensitive because of vanishing Goldstone masses; no smooth quantum stability theorem follows. COMPUTATION (w33_20261001_completed_eft_one_loop.py) The same inventory gives standard one-loop $b_{E_6} = 17$ and $b_{SU(3)} = -59/2$, so equal standard couplings drift and family $SU(3)$ has a Landau pole. Neither transmutation nor a finite vacuum counterterm is fixed. COMPUTATION (w33_20261001_uv_running_scale_audit.py)

There is a distinct conditional nonperturbative model. The E_6-only stabilizer has dimension fourteen on the generic SIC wall $(1, 1, z)$ over $\mathbb{Q}(z)$, but remains eight at the MUB-only control $(0, 1, 2)$. A degree-eighteen invariant condensate denominator vanishing on the SIC orbit is therefore proportional to u_{18}, using its order-two mirror multiplicities. This provides a normalization interface to the near-SUSY three-27 condensate studied at https://arxiv.org/abs/2505.07931; it does not import that theory's dynamics into the preceding paired EFT. For the declared canonical Cartan action

$$W = \Lambda^9(-729I_{18})^{-1/3}, \qquad V = \sum_i |\partial_i W|^2 - 18m\,\mathrm{Re}\,W,$$

the chosen branch has a stationary radius $r^8 = 14\Lambda^9/(3m)$. All six real slice gradients vanish exactly, and the canonical Hessian has eigenvalues

$$m^2\left(\frac{162}{49}, \frac{162}{49}, \frac{486}{49}, \frac{486}{49}, \frac{648}{7}, \frac{486}{7}\right).$$

The invariant-gradient angular basis has nonzero two-by-two mixing blocks; these are scalar modes, not CKM or PMNS angles. This result is conditional on the denominator and canoni-

cal slice Kähler metric. Full eleven-modulus stability, quantum Kähler corrections, the AMS-B/strong scales, Standard Model charges and nonzero vacuum energy remain separate questions. COMPUTATION (w33_20261001_condensate_cartan_potential.py)

Full-modulus and Standard Model interfaces (Passes11270–11274). The signed81-field action has an E_6 orbit of complex dimension70, so its physical tangent quotient has eleven complex directions. A numerical full Hessian and fermion calculation reproduces the published canonical spectrum of Goh et al.: eight scalar Goldstones, ten squared masses $162m^2/49$, two $486m^2/49$, and $486m^2/7, 648m^2/7$. Two small radial Kähler deformations are tested with the full compensator potential, rather than an inverse-metric-only substitution; general corrections remain open. In a separate Higgs configuration, the actual canonical vectors e_0, e_1 and an adjoint with Cartan coefficients $(1/3, 2/3, 1, 0, 1/2, 0)$ leave exactly $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ with the standard hypercharges. The signed cubic d_{AB0} is an invariant rank-ten exotic mass block. Symmetric family-sextet coupling gives rank30 across three27s, preserving three chiral16s and three singlets. A positive compact orbit-distance potential selects this imposed Higgs orbit; its extra fields and nonpolynomial EFT are not a dynamically derived renormalizable vacuum. COMPUTATION (w33_pass11270_full_condensate_moduli.py, w33_pass11271_canonical_sm_higgs_bridge.py)

In the canonical eleven-modulus EFT a computed hard one-loop radial tadpole gives $\Delta_{\mathrm{hard}} = V'_{1,\mathrm{hard}}/(2r)$; at the shifted stationary point $G_{\mathrm{tree}} + \Delta_{\mathrm{hard}} = 0$ for the eight global Goldstones. Heavy gauge thresholds and momentum-dependent nonzero poles are excluded. For the supplied torus benchmark, uniform Gaussian-polynomial domination now proves the existing Wilson curvature response converges at fixed heat time. At $t = 1/5$ the directed outside-$K = 64$ tail is below 1.083×10^{-25}; no spacetime emergence or finite-spacing error enclosure is implied. Finally the canonical spectrum obeys $\mathrm{STr}\, M^2 = 0$ but $\mathrm{STr}\, M^4 = (852930/2401)m^4 > 0$, retaining vacuum-energy running. An external global-sequestering action passes the constant-shift test, but its constraints and historic residual are not supplied by W33. COMPUTATION (w33_pass11272_resummed_condensate_radial_control.py, w33_pass11273_wilson_uniform_tail.py, w33_pass11274_scale_and_sequestering.py)

Polynomial local alignment and new physical tests (Passes11275–11279). A separate polynomial Higgs EFT on two complex27 vectors and a real78 has a nonnegative sum-of-squares potential whose exact constraint differential has rank120 in186 real fields. Its kernel is exactly the66-dimensional compact E_6 orbit of the canonical SM point, so the orbit is isolated locally with positive normal Hessian. This uses an imposed degree8 cubic-covariant selector; global uniqueness and selected scales remain open. Two added complex matrix scalars give a degree-four sum-of-squares lift with the same local gauge kernel, but their large inventory changes the combined E_6 beta coefficient to-80. This renormalizable scalar construction is not an established viable UV model. An independent family sextet produces a rank-ten spectator mass through a dimension-six composite operator, improving the family beta coefficient to $-79/3$ without restoring asymptotic freedom. An actual momentum bubble in a *separately declared radial-only scalar EFT* shifts its squared pole by about -0.144% from curvature; other modulus, fermion and gauge loops are excluded. A mixed-characteristic symplectic lift is explicit, but does not select an invariant positive real metric. The known clique topology also makes the top boundary of its circle-history product injective: this candidate has no intrinsic topological four-flux for sequestering. Neither geometry test excludes all pos-

sible W33-derived spacetime constructions. COMPUTATION (w33_pass11275_polynomial_sm_higgs.py, w33_pass11276_sextet_composite_spectator.py, w33_pass11277_radial_momentum_pole.py, w33_pass11278_symplectic_ring_geometry.py, w33_pass11279_history_flux_obstruction.py)

Smaller field and continuum architectures (Passes11280–11284). A degree-four factor lift of the rank-five cubic selector uses three complex 27×5 fields with an added auxiliary $U(5)$ gauge factor. Its996 real fields have91 gauge and905 positive normal directions locally; the combined E_6 coefficient becomes13 and auxiliary $SU(5)$ coefficient29/6, while the auxiliary Abelian and existing family $SU(3)$ UV problems remain. A degree-twelve family-sextet potential selects an imposed orbit with singular values$(1, 2, 3)$; a single sextet tree renormalizable potential cannot isolate three distinct nonzero values. The mixed bilinear $v^A H_A^\dagger$ avoids the earlier neutral-Higgs cubic zero, but one aligned texture gives no CKM mixing. The radial self-energy now includes all22 scalar and11 Weyl internal modes in the declared fixed-slice EFT: its open Goldstone channels give $\Gamma(h \to 8GG) = M_h^3/(8\pi r_0^2)$ and a complex pole with leading $\Gamma/M_h \simeq 0.003606$ for the stated parameters. The full external self-energy matrix and physical UV matching remain open. An added real continuum metric chart needs a relative symplectic-plane scale and an overall scale to span all ten local Lorentz-metric variations. An added identity double of the genus81 Levi-graph handlebody, times a history circle, has one top-flux class and supports a conditional linear-sequestering action. These are explicit architectures with additional real-base, gluing, coupling and flux inputs; they do not derive gravity or the observed vacuum energy. COMPUTATION (w33_pass11280_factor_higgs_mediator.py, w33_pass11281_family_sextet_selection.py, w33_pass11282_all_modulus_radial_self_energy.py, w33_pass11283_symplectic_metric_field.py, w33_pass11284_closed_history_flux.py)

A further semisimple auxiliary SO(10) completion needs explicit frame projection constraints to remove quartic-flat unused columns. Its local normal count is1740; replacing the elementary family Higgs by a composite leaves $b_{E_6} = 4, b_{SO(10)} = 1$, with UV mediators and family running still open. A separately imposed CP-even mixed-moment potential selects Fourier mixing with $J^2 = 1/108$, rather than the observed CKM hierarchy. The canonical Kähler quotient now supplies the full22-field nonanalytic self-energy matrix, checked by an independent nonradial evaluation on the actual D-flat sheet; local matching, vector thresholds and physical poles remain open. The imposed Einstein–Hilbert metric pullback has two graviton polarizations under its fiber and ADM constraints. Nontrivial Levi-cycle gluing preserves one top-flux class but does not select its magnitude. The cycle determinant and critical group are prior results (Passes5031/5441); their separately declared Abelian Chern–Simons model has a gapped-boundary obstruction repaired by doubling. These constructions retain additional potential, geometry and coupling inputs. COMPUTATION (Passes11285–11292; scoped certificates and independent controls)

Explicit tree matching of the composite Yukawa costs six or twelve E_6 beta units, exceeding the old budget of four. Replacing the third large frame by a real auxiliary symmetric54 restores a quartic alignment potential with 1254 positive normal directions; the two mediator choices retain positive $(b_{E_6}, b_{SO(10)}) = (8, 8)$ or $(2, 8)$. Family running and complete-coupling UV stability remain open. A separately declared hierarchical CP potential uses adjugate-trace invariants to remove the four relative sextet flats at four tested vacua; its hierarchy and root coefficients remain inputs. Local two-point Ward matching yields four-

teen leading complex massive poles plus eight Goldstones in a declared spacelike scheme, with invariant tadpole and UV matching still open. Minimal covariant quotient matter preserves the metric fiber and continuum ADM constraints. An added membrane-rate model selects a nearest-lattice flux sector but retains bare vacuum energy. The old uniform linear-volume branch fixes the intensive Q/Vol, so its extensive flux is not that membrane ladder. These are scoped EFT constructions and obstructions, not observed-parameter predictions. COMPUTATION (Passes11293–11297; matching, phase, constraint and flux controls)

The next audit contracts the actual single-channel mediator Yukawas, but finds that one vectorlike pair has a rank-one channel map and cannot implement two independent flavour sectors. A second pair gives $b_{E_6} = -10$; two scalar mediators instead retain $b_{E_6} = 2$, with their complete running open. Mandatory symmetric54 quartics expose missing counterterms; full portal closure remains open. For the canonical two-Higgs mass map, five signed paths prove that its entire spectrum depends only on $S_u + S_d$, so this fermion determinant cannot lift relative Takagi phases or derive the imposed hierarchy. An invariant Hermite counterfunctional matches the vector-sector tadpole and Hessian in a declared higher-degree EFT; scalar/Weyl and UV matching remain open. Native Levi site constraints generate missing edge currents, rather than inheriting continuum ADM closure. A separately added Maxwell membrane form admits a closed two-cap junction solution with volume and curvature-flux constraints and common bare-shift cancellation. Its geometry, flux sectors and scales remain inputs; no observed-parameter prediction is claimed. COMPUTATION (Passes11298–11302; scoped tensor, matrix, bracket and junction controls)

A single scalar mediator with two sources escapes the fermion-pair channel-rank obstruction. An exact Riccati inequality excludes one-loop complete weak-coupling asymptotic freedom of the unchanged54 inventory including scalar portals; eight added SO10 vector Weyls, two coupled to54, supply a bounded AF ray in the isolated scalar/fermion block. Full portal completion and exotic-field decoupling remain open. Two vectorlike E6-singlet family probes detect all four old Takagi-phase flats, without deriving a stationary physical vacuum. Full vector cut matrices extend conditional massive poles, but a rank11 scalar Goldstone infrared logarithm prevents unqualified regulator-free total local matching. Added native Levi clocks admit exactly first-class relational constraints, retaining80 oscillators rather than deriving gravity. In a separately supplied two-cap membrane model, integer flux leaves one constrained radial negative direction with curvature about−204.71; full fluctuation stability, geometry and observed cosmological-constant selection remain open. COMPUTATION (Passes11303–11307; scoped EFT, tensor and constrained-action controls)

A subsequent complete seven-quartic 54 + 45 restriction exposes a further obstruction: the original eight-Weyl texture has a positive 45 Riccati barrier, of minimum 298418/295695, including its scalar portals. An allowed joint 54/45 Yukawa texture instead has a positive 54 barrier. The earlier isolated AF ray remains valid; neither result closes the full inventory. Its isolated 54 model admits a positive leading radiative radial curvature on an input RG crossing, but selects an $\mathrm{SO}(9)$ orbit and leaves the absolute scale and vacuum energy free. Untargeted flavor loops yield a CP-conserving local five-phase minimum, with spectra and angles supplied. The massless Goldstone bubble has a regulator-free limit at massive external momentum; invariant tadpole and UV matching remain open. A constructed native edge Fierz–Pauli action on a supplied four-dimensional base carries two uniform massless polarizations and 79 massive spin-two modes, without a nonlinear completion. The declared identity-gluing history has $\chi = 0$, $b_1 = 82$ and therefore no smooth closed Riemannian Ein-

stein metric: Einstein plus $\chi = 0$ would force a flat finite torus cover, whose b_1 is at most four. This does not exclude walls, matter, Lorentzian histories or other gluings. A constructed local smooth neck requires radial null-energy violation and is not a global membrane saddle. COMPUTATION (Passes11313–11319; exact obstructions and conditional EFT constructions)

Extending the real two-active-flavour Yukawa family gives eight exact support patterns, including an unequal noncommuting branch, but all retain a positive scalar AF barrier. A supplied full24-field family-invariant EFT has numerical local minima with spontaneous relative CP and $2 + 1$ singular-value splitting; commuting Gram matrices still preclude a CKM prediction. Finite invariant hard-sector spectral counterfunctions give conditional invariant hard tadpole matching; nonlinear covariant quotient maps, the rank11 soft logarithm and complete resummation remain open. Actual native-graph shift elimination in an added nonlinear pair-metric action gives a rank78 lapse Hessian; no symmetry-invariant spanning tree exists. A distinct, chosen $S^2 \times T^2$ history admits an exact positive-tension Maxwell-supported wall satisfying the bulk Einstein equations, both wall junctions and magnetic Dirac quantisation. An explicit native cochain selects one primitive handle; this is not the full81-handle history. Its coupled fluctuation spectrum, state selection and observed vacuum energy remain open. COMPUTATION (Passes11320–11324; scoped EFT witnesses and exact constraint/-geometry tests)

Complex two-active Yukawa fixed rays phase-lock to the prior real families; paired two/four/six/eight-flavour rays retain positive scalar AF barriers. A supplied bounded CP-even angular interaction yields a local noncommuting 24-field flavour minimum with sixteen positive normal Hessian eigenvalues and Jarlskog 0.09622. Its nearly maximal mixing and strong engineered EFT coupling are not an observed CKM prediction. An explicit numerical local 22-coordinate D-flat quotient now supplies the metric, connection and covariant scalar/Weyl/vector mass maps missing in Pass11322. Reference spectra, changed-frame covariance and quadratic remainders against the old linear jets are checked; full hard 1PI Ward resummation remains open. The published determinant multivielbein construction has a restricted equal-boost constraint proof. Applied to eighty vielbeins it gives 397 modes on that branch, but a single vertex couples 3160 pairs rather than the native 160 edges. On the actual Maxwell-wall saddle, both torus-shape metric polarizations have positive Euclidean Rayleigh forms and two exact constant zero modes. The remaining coupled fluctuation block, a generic native nonlinear gravity completion and observed physical scales remain open. COMPUTATION (Passes11325–11329; scoped local witnesses and algebraic tests)

An exact complex-isotropic four-flavour Yukawa ray evades the old scalar lower-bound tests, but an actual scalar background with fourth trace $1/5$ gives a new positive Riccati minimum $372423967259/182902583664$ and excludes simultaneous seven-quartic AF fixed points on this branch. Gaussian positive-mass adjoint mediation instead forces commuting flavour Grams. Compensated family vector fields on the local quotient fix the eight hard Ward columns; the normal hard1PI block remains uncomputed. On a collinear native pair-vielbein slice, relative boosts are fixed independently of lapses and the reduced action is lapse-linear despite cycles; generic transverse constraints remain open. The coupled Maxwell/metric torus-vector block of the original wall has four exact zero modes, hence six with the previous shapes. A supplied two-axion winding extension admits a backreacted quantised wall with $q^2 - q = hb_{\mathrm{wall}}$ and lifts both shapes: its smallest shape eigenvalue lies in $[8/625, 1/50]$. The remaining coupled sectors, nonlinear integrability, matter origin and physical scale or vacuum-energy selection remain open. COMPUTATION (Passes11335–11341; scoped exact and

numerical constructions)

Both stored four-active complex Yukawa witnesses that escaped the older scalar portal tests fail the stronger physical-background bound. A bounded, positive-mass two-adjoint mediator sector selects noncommuting two-family Grams at fixed spectra, but gives no three-family CP or RG completion. The eight quotient Ward columns leave 105 independent entries of the normal hard Hessian unfixed; the native graph's three transverse boost directions have local shift rank 237, without establishing nonlinear Dirac closure. On the backreacted axion-winding wall, $\chi = 1/b - 4/5$ and $V_\phi = -F/(qcb^2)$ solve the coupled vector zero equations exactly, even though winding lifts the two torus-shape zeros. The same exact family has a positive, quantised $\rho = 0$ bare-cosmological-parameter point at $h = 16/45$, $q = 4/3$, with supplied matter strength and wall tension. Neither construction fixes an observed mass, absolute scale or vacuum energy. COMPUTATION (Passes11342–11349; exact restricted identities and finite controls)

The prior three-qutrit-ray Bargmann loop admits an explicit algebraic lift to two full-rank positive Yukawa Grams: their CP-odd commutator cubic factors as $-6\det(G)\,\mathrm{Im}\,\mathrm{Tr}(P_1P_2P_3)$ times three real weight minors. Supplied positive weights make it nonzero, but neither the weights nor an observed CKM phase follow from the substrate. An invariant supplied heavy scalar produces one actual hard one-loop normal Hessian direction while leaving all eight quotient Ward columns unchanged. On the native metric-cycle model, an exact equal-lapse null explains the earlier rank-77 control and modular arithmetic certifies generic rank 78; this is not a generic multivielbein Dirac count. The quantised winding wall stays positive throughout $1 \le q \le 4/3$, with its unique zero-bare-ρ point at $q = 4/3$; its Ricci scalar there remains $32/(45b^2) > 0$ from supplied axion stress. A restricted coupled-vector Ritz test retains the exact zero, but full wall stability and absolute physical scale remain open. COMPUTATION (Passes11361–11368; exact algebraic witnesses and restricted numerical control; `w33_pass11361_11368_physical_frontier.py`)

The computational interface needs an additional classical datum: a Clifford frame for the selected ray. The 72 rays split into eight nine-ray Pauli orbits, two per MUB axis. Uniformly forgetting the displacement within even one such orbit gives $I/3$, by the symbolic Pauli-twirl identity. Retaining a representative C permits $C^\dagger$ followed by SUM with a fresh $|0\rangle$ to prepare $(I \otimes T)|\Phi_3\rangle$. All $72 \times 9 = 648$ ray/Bell branches replay the existing injection identity $C_{ab}K_{ab} = T/3$. If the frame is uniformly forgotten, the same protocol instead gives the entanglement-breaking channel $\rho \mapsto \mathrm{diag}\,\rho$. Thus orbit selection and frame retention are separate resources; a state-preparation Hamiltonian and its noise model remain open. COMPUTATION (`w33_20261001_tmagic_orbit_factory_contract.py`)

The twelve stabilizer mirrors form the four qutrit mutually unbiased bases. Complex-linear G_{26} reflections induce A_4 on those four bases; complex conjugation supplies an odd transposition and completes the image to S_4. This identifies the line-stabiliser S_4 flavour tetrahedron objectwise: four flavon axes are four MUBs and the twelve oriented tetrahedral chords are the ordered MUB pairs. It does not fix Yukawa amplitudes; in particular a vacuum-tilt parameter does not determine θ_{13} or δ by itself. `w33_pass11262_g26_mub_s4_yukawa_bridge.py`

There is also a metric-independent spectral statement. The nonzero spectrum of N^3 is the same in all three grades, so the 234 nonzero modes form 78 exact $\mathbb{Z}_3$ triplets. Every positive-power root-of-unity graded trace cancels; the residual graded index is $8 + 3\omega + 3\omega^2 = 5$ and

lies wholly in the kernel. This is a character identity, not a boson–fermion supertrace or a vacuum-energy calculation. `w33_pass11263_full_graded_spectrum.py`

5.7 Clock gradings of E_8

A clock gives more than a $\mathbb{Z}_3$-grading: once an origin and an orientation are chosen it gives an *integer* grading. On the committed Chevalley basis:

Theorem 5.5 (Hesse clocks grade E_8) COMPUTATION

(i) A single Hesse line (a "tick") induces the *contact grading*

$$E_8 = \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2, \qquad 1 + 56 + 134 + 56 + 1,$$

with $\mathfrak{g}_0 = E_7 \oplus \mathbb{C}$.

(ii) An oriented striation (three parallel ticks) induces the $|3|$-*grading*

$$2 + 27 + 54 + 82 + 54 + 27 + 2,$$

with Levi $\mathfrak{g}_0 = E_6 \oplus A_1 \oplus \mathbb{C}$, $\mathfrak{g}_{\pm 1} = \mathbf{27} \otimes \mathbf{2}$, $\mathfrak{g}_{\pm 2} = \mathbf{27}$, $\mathfrak{g}_{\pm 3} = \mathbf{2}$.

(iii) The repository's frozen $\mathbb{Z}_3$-grading $86 + 81 + 81$ is exactly this integer grading reduced modulo three; all 8 347 nonzero committed structure constants obey the integer degree law.

`w33_20260925_hesse_clock_e8_gradings.py`, `w33_20260925_e8_parabolic_cubic_clock_lift.py`

This parabolic spectrum was found independently by Kraft, Regeta and Zimmermann in a study of small varieties with group actions [28]. What is repository-specific is the identification with the Hesse clocks and the committed structure constants.

5.8 The cubic ladder: three steps to the top

In the $|3|$-grading, positive degrees form a nilpotent algebra $\mathfrak{g}_+ = \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_3$ of dimension $54 + 27 + 2 = 83$. It is generated by $\mathfrak{g}_1$, and the way it fills up is governed by the E_6 cubic.

Theorem 5.6 (the bracket ladder) COMPUTATION

On the committed Chevalley bracket:

(i) $[\mathfrak{g}_1, \mathfrak{g}_1] = \mathfrak{g}_2$: 270 nonzero pairs hit all 27 grade-two roots, each ten times;

(ii) $[\mathfrak{g}_1, \mathfrak{g}_2] = \mathfrak{g}_3$: 54 nonzero pairs hit both top roots, each 27 times;

(iii) the nonzero nested triple brackets exhibit exactly the 45 triads of the cubic, each 12 times, and for every signed triad d_{ijk}, $[[e_{(i,0)}, e_{(j,1)}], e_{(k,a)}] = -d_{ijk}\, T_a$ for $a \in \{0, 1\}$.

So the positive part grows $54 \to 81 \to 83$, and the committed cubic is exactly the rule by which three first-order moves reach the two-dimensional top. `w33_20260925_cubic_clock_bracket_ladder.py`

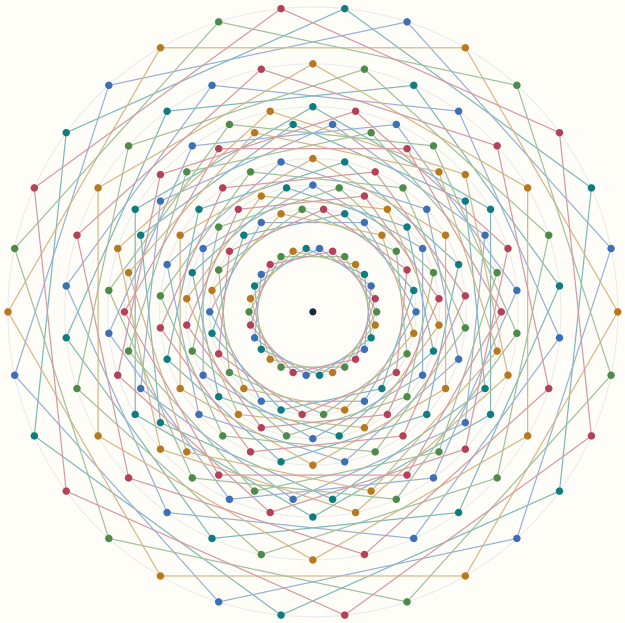


Figure 14. The 240 roots of E_8 projected to the Coxeter plane of a Coxeter element c (order 30), which acts on the plane as a rotation by 12°. The roots lie on eight rings of thirty. The orbits of c^5 — rotation by 60° — are regular hexagons, five per ring: these 40 hexagons are the 40 fibres of Theorem 5.3, one for each point of $W(3,3)$. Each hexagon is $\{\pm v, \pm\omega v, \pm\omega^2 v\}$, since $c^{15} = -1$ and c^{10} has order three. All coordinates computed (`make_figures.py`).

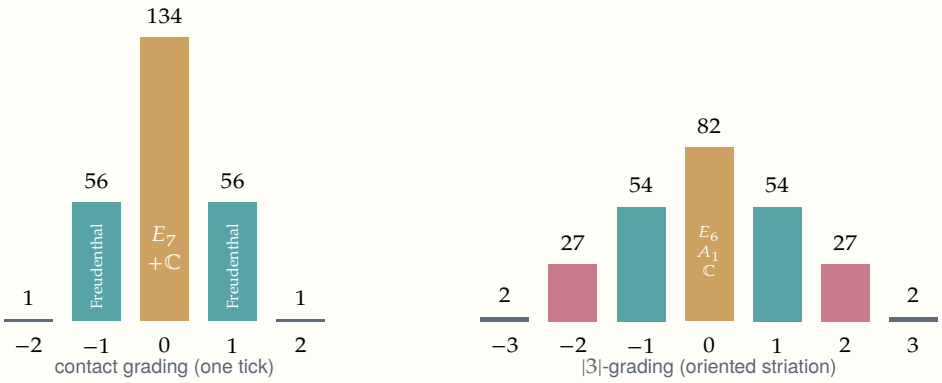


Figure 15. Two ways a clock slices E_8. Heights are dimensions. On the left, the Freudenthal "56" of Section 6 appears at degree ± 1; on the right, the 27 of E_6 appears doubled at degree ± 1 and single at degree ± 2.

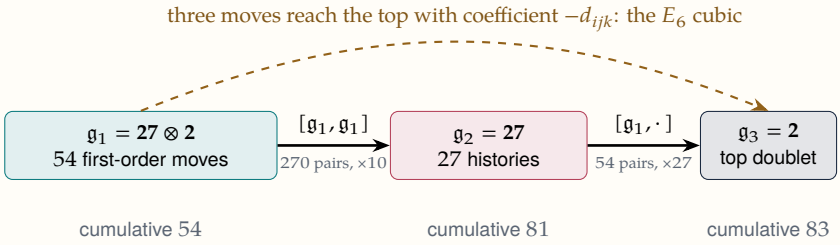


Figure 16. The cubic ladder. First-order moves combine in pairs to histories, and histories combine with one more move to reach the two-dimensional top. The only way three moves reach the top is through a triad of the cubic.

What it means physically: cubic time

Read degree as "order in time". A first-order move is a state update; two moves compose into a second-order *history*; and three reach the top doublet, whose coefficient is the cubic volume N spanned by the three moves. In this reading the E_6 cubic — in GUT language the Yukawa coupling — is the rule by which three independent updates produce a net displacement of the clock. The algebra is exact; the reading is a bridge, and it lacks what would make it physics: a Hamiltonian, a state and an energy scale. BRIDGE

One qutrit's nine histories build E_8 as $80 + 84 + 84$. The four Hesse clocks are four orthogonal A_2's whose glue is the tetracode; a clock tick grades E_8 as $1 + 56 + 134 + 56 + 1$, an oriented clock as $2 + 27 + 54 + 82 + 54 + 27 + 2$; and E_8's roots fibre six-to-one over the forty points.

6 The Freudenthal 56 and a 57-dimensional light cone

The contact grading of Theorem 5.5 has a 56-dimensional layer $\mathfrak{g}_{-1}$. In the 1950s Hans Freudenthal showed [15] that the 56-dimensional representation of E_7 carries a remarkable structure — a symplectic form and a quartic form — now called a *Freudenthal triple system*. It has since turned up in the physics of black holes in supergravity and in the entanglement of qubits. This section computes it directly from the Hesse clock, on the committed E_8 basis, and finds a light cone one dimension larger.

In plain words

Think of the 56 as the space of "charges" of a black hole in a theory with many electric and magnetic fields: 28 electric and 28 magnetic. The symplectic form pairs each electric charge with its magnetic partner. The quartic form is a single number built from all 56 charges; in string theory its square root gives the black hole's entropy. What we show is that the forty-point geometry, through its clock, produces exactly this structure — including the right signs, with nothing adjusted by hand.

6.1 Two clocks in one table

Let θ be the highest root of E_8 and α_8 the simple root with $\theta = \omega_8$. The tick line's contact grading is the α_8-coefficient a_8 of a root; the striation's $|3|$-grading is its α_7-coefficient a_7. Counting roots by both at once gives Figure 17.

Theorem 6.1 (the 56 does not extend the 54) COMPUTATION

The contact layer $a_8 = -1$ sits at clock degrees $0, -1, -2, -3$ with sizes $1, 27, 27, 1$; the clock layer $a_7 = -1$ sits at contact degrees $0, -1$ with 27 each. The two share exactly one 27. The second 27 of the Freudenthal system is the clock's *second-order* layer, and its

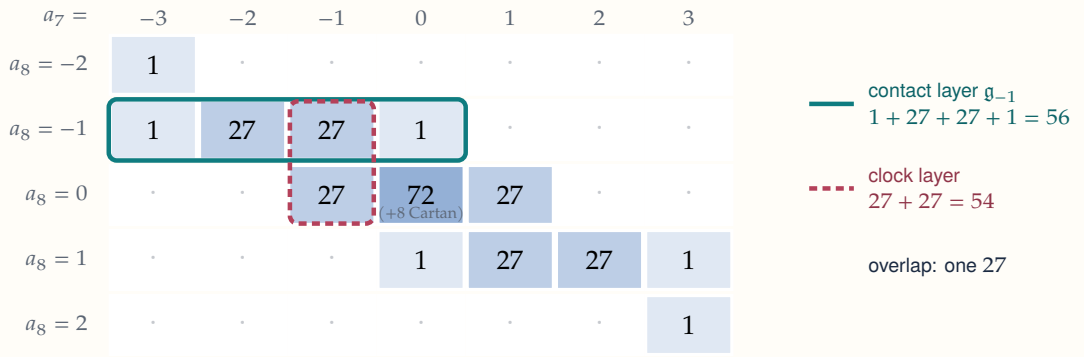


Figure 17. The 240 roots of E_8 counted by contact degree a_8 (rows) and clock degree a_7 (columns). The Freudenthal 56 (solid outline) and the clock's first-order 54 (dashed outline) share exactly one 27.

two scalar "poles" are $\alpha = e_{-\alpha_8}$ (clock degree 0) and $\beta = e_{-(\theta-\alpha_8)}$ (clock degree −3, a member of the clock's top doublet). `w33_pass10949_freudenthal_quasiconformal_clock_cone.py`

A count that misled

A proposal circulating in the programme read the 56 as "the clock's 54 plus the 2 top directions". The dimensions agree ($54+2=56$), but the objects do not: the 56 contains only half of the 54. REFUTED (as an identification)

6.2 The symplectic form and the Heisenberg algebra

Brackets of two elements of the 56 land in the one-dimensional $\mathfrak{g}_{-2} = \mathbb{C}e_{-\theta}$: $[x_i, x_j] = \Omega_{ij}\, e_{-\theta}$. On the root basis, Ω is a signed perfect matching: $\alpha \leftrightarrow \beta$, and $X_k \leftrightarrow Y_k$ where X_k and Y_k carry the *same* 27-label. Its determinant is ± 1, so it is unimodular over $\mathbb{Z}$. Since $e_{-\theta}$ is central in $\mathfrak{g}_{-1} \oplus \mathfrak{g}_{-2}$, the two together form a 57-dimensional *Heisenberg algebra* — the same algebra that, in quantum mechanics, is generated by positions, momenta and $i\hbar$. COMPUTATION

6.3 The quartic, in Freudenthal's normal form

Define $Q(x) = \frac{1}{6}$ coefficient of $e_{-\theta}$ in $\mathrm{ad}_x^4 e_\theta$ for $x \in \mathfrak{g}_{-1}$.

Theorem 6.2 (the tick line is a Freudenthal triple system) COMPUTATION

Q is an integral polynomial with 1036 monomials (primitive coefficients $\{1:28,\ \pm 2:378,\ \pm 4:630\}$), exactly invariant under all 126 root generators of the Levi E_7. Writing $x = (\alpha, X, Y, \beta)$ with X, Y in the two 27's,

$$\boxed{Q = (\alpha\beta + T(X,Y))^2 - 4\beta\, N(X) - 4\alpha\, N(Y) + 4\, T(X^{\#}, Y^{\#})}$$

where N is the repository's committed 45-triad E_6 cubic, $X_k^{\#} = \partial N / \partial X_k$, and $T(X,Y) = \sum_k t_k X_k Y_k$ with $t_k = \Omega(X_k, Y_k)\,\Omega(\alpha, \beta)$. No sign has been adjusted by hand. Moreover

$(X^{\#})^{\#} = N(X)\,X$ and $X \cdot X^{\#} = 3N(X)$, so N is the norm of a cubic Jordan algebra (a split Albert algebra) over $\mathbb{Q}$ [29]. `w33_pass10949_freudenthal_quasiconformal_clock_cone.py`

The theorem makes one structural point vivid: the cubic "history volume" $N(X)$ of Section 5 enters the quartic *only* multiplied by a clock pole β.

Black holes and qubits

In $\mathcal{N} = 8$ supergravity the 56 electric and magnetic charges of a black hole transform under $E_{7(7)}$, and the Bekenstein–Hawking entropy is $S = \pi\sqrt{|Q|}$ with Q Cartan's quartic invariant [26]; in the "magic" $\mathcal{N} = 2$ supergravity built on the octonionic Albert algebra the same formula holds with $E_{7(-25)}$. Duff and Ferrara showed that the same quartic measures the tripartite entanglement of seven qubits [11], and Borsten, Duff and collaborators built a dictionary between black holes and qubit entanglement on it [5]. Theorem 6.2 places that quartic, with its signs, inside the forty-point geometry's clock. The two real forms named here — $E_{7(7)}$ and $E_{7(-25)}$ — are exactly the two that appear, for two different reasons, in Section 7. BRIDGE

6.4 Jordan frames are the forty-five triads

A *frame* of a cubic Jordan algebra is a decomposition of the identity into three orthogonal idempotents. On the committed basis, in each of four 27-layers, the 45 triads of the cubic are exactly the 45 strongly orthogonal triples of roots: the coordinate frames. (This corrects an older reading of "27 minimal idempotents": 27 counts the coordinate rank-one directions; the frames are the 45 triads.) For every coordinate idempotent, the Peirce decomposition is $27 = 1 + 16 + 10$. COMPUTATION

6.5 The 57-dimensional light cone

Now use the 57-dimensional Heisenberg algebra as a space of points $p = (X, \tau)$, with $X \in \mathfrak{g}_{-1}$ and τ the central coordinate, and move the top vector e_θ around: $v_p = \exp(\mathrm{ad}(X + \tau e_{-\theta}))\, e_\theta$.

Theorem 6.3 (quasiconformal light cone) COMPUTATION

On the committed basis, symbolically:

(i) the $e_{-\theta}$-component of v_p is $\mathcal{N}(X, \tau) = \frac{1}{4}Q(X) - \tau^2$;

(ii) the points compose by the Heisenberg law $(X_1, \tau_1)(X_2, \tau_2) = (X_1 + X_2,\ \tau_1 + \tau_2 + \frac{1}{2}\Omega(X_1, X_2))$;

(iii) the *separation* $D(p, p') = \mathcal{N}(X - X',\ \tau - \tau' - \frac{1}{2}\Omega(X', X))$ equals $B(v_p, v_{p'})/B(e_{-\theta}, e_\theta)$ for the Killing form B;

(iv) the Weyl reflection w exchanging e_θ and $-e_{-\theta}$ acts as an *inversion* ι, and

$$D(\iota p, \iota p') = \frac{D(p,p')}{\mathcal{N}(p)\,\mathcal{N}(p')}.$$

This is the Günaydin–Koepsell–Nicolai quasiconformal realisation of E_8 [19], now computed objectwise on the repository's basis. `w33_pass10949_freudenthal_quasiconformal_clock_cone.py`

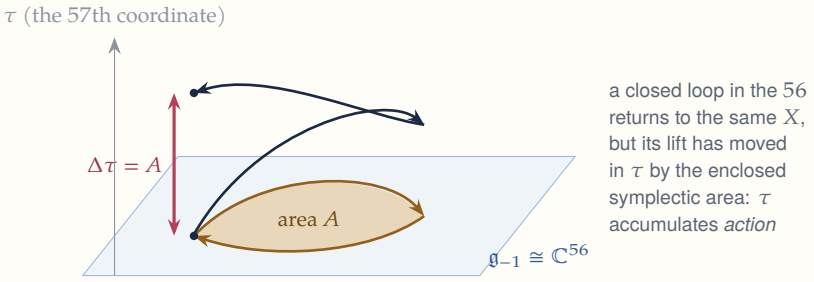


Figure 18. Why the 57th coordinate matters. In a Heisenberg group the central coordinate of a lifted path grows by the symplectic area its projection encloses. The quasiconformal separation D depends on $\tau - \tau'$ only through this corrected difference $\tau - \tau' - \frac{1}{2}\Omega(X', X)$.

What it means physically: a light cone, and its inversions

Ordinary Minkowski space has a *conformal* symmetry group, $SO(4,2)$, larger than the Lorentz group; its most striking element is the inversion $x \mapsto x/x^2$, under which the interval between two events transforms as

$$(x' - y')^2 = \frac{(x-y)^2}{x^2\,y^2}.$$

Theorem 6.3(iv) is exactly this law, with the Minkowski interval replaced by the quartic separation D on a 57-dimensional space. So E_8 acts on the forty-point clock's Heisenberg algebra as a group of "conformal" symmetries of a quartic light cone. The central coordinate τ plays the role of accumulated action: the same pattern met in Section 4, where the finite present forgot a common mode that the lifted history remembered. BRIDGE (the reading) COMPUTATION (the formulas)

A single tick of a Hesse clock turns E_8 into $1+56+134+56+1$. The 56 is Freudenthal's triple system, whose quartic — computed from the committed brackets with no sign adjusted — is the black-hole and seven-qubit invariant; and with one more coordinate it becomes a 57-dimensional light cone on which E_8 acts by conformal-type inversions.

Where this stops

The formulas are exact, but they are the formulas of the *complexified* (or split) structure. Whether the light cone is Lorentzian — one time direction, the rest space — depends on a choice of real form, which the finite field cannot see. That is the subject of Section 7.

Nothing here supplies a black hole, a charge lattice of a physical theory, or dynamics. OPEN

7 Real forms: where time's signature comes from

Every computation so far could have been done over the complex numbers — or over $\mathbb{F}_3$. But physics needs more: it needs to know which directions are *time-like* and which *space-like*. Mathematically, that is the choice of a *real form*.

In plain words

A complex Lie algebra is like a map drawn without saying which way is north. A real form chooses, for every direction, whether it behaves like a rotation (you come back to where you started — "compact", like space) or like a boost (you never come back — "non-compact", like the relation between observers moving at different speeds). The same complex E_8 has three real forms, and they differ precisely in how many directions are rotation-like.

7.1 Three real forms of E_8

A real form is fixed by a *Cartan involution*, which splits the algebra into a compact part $\mathfrak{k}$ (rotations) and a non-compact part $\mathfrak{p}$ (boosts). The difference $\dim \mathfrak{p} - \dim \mathfrak{k}$ is the *signature*.

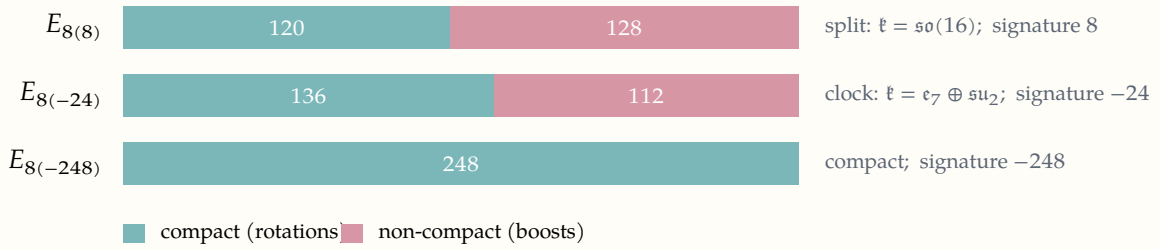


Figure 19. The three real forms of E_8, drawn as the split $248 = \dim \mathfrak{k} + \dim \mathfrak{p}$. The committed "split" form of the repository is $E_{8(8)}$; the Hesse clocks select $E_{8(-24)}$.

7.2 A sign law, and the height twist

On the committed Chevalley basis the Killing form obeys an exact law,

$$K(e_\alpha, e_{-\alpha}) = -60\,(-1)^{\mathrm{ht}\,\alpha},$$

where $\mathrm{ht}\,\alpha$ is the height of the root (the sum of its simple-root coefficients). From it one reads off the *compact* conjugation $\sigma_c(e_\alpha) = (-1)^{\mathrm{ht}\,\alpha}\,\overline{e_{-\alpha}}$, verified to be an automorphism on all 30 628 basis pairs. The repository's certified *split* involution Θ turns out to be σ_c composed with the *height parity* $(-1)^{\mathrm{ht}} = \exp(i\pi\rho^\vee)$, where $\rho^\vee$ is the Weyl co-vector. COMPUTATION (`w33_pass10949_freudenthal_quasiconformal_clock_cone.py`; prior certificate `w33_e8_split_real_form_involution.py`)

7.3 The three tick lines are a Klein four-group

The three roots θ, $-(\theta-\alpha_8)$ and $-\alpha_8$ that define the tick line and its companions sum to zero, have pairwise products -1, and are orthogonal to E_6: they form the *external* A_2 of $E_6 \times A_2$. Each defines a parity: $(-1)^{a_8}$, $(-1)^{a_7}$, $(-1)^{a_7+a_8}$.

Theorem 7.1 (tick lines give $E_{8(-24)}$) COMPUTATION

Composed with σ_c, each of the three tick-line parities defines the real form $E_{8(-24)}$. The three commute and form a Klein four-group whose simultaneous eigenspaces decompose

$$E_8 \;=\; \underbrace{80}_{\mathfrak{e}_6 \oplus \mathfrak{t}_2} \;+ 56 \;+\; 56 \;+\; 56,$$

one Freudenthal 56 per tick line. `w33_pass10949_freudenthal_quasiconformal_clock_cone.py`

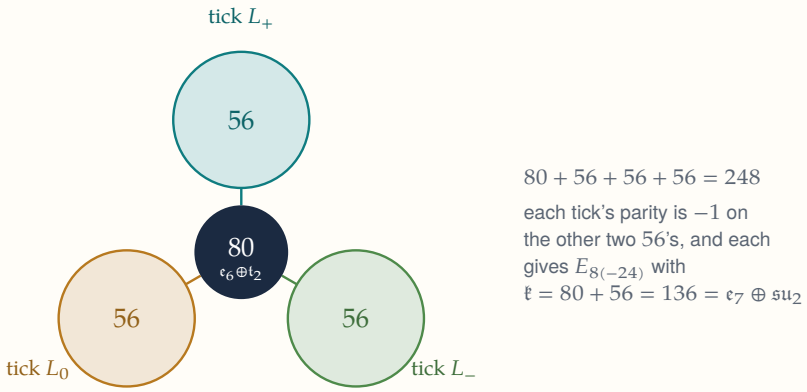


Figure 20. The three tick lines of a Hesse clock, and the decomposition of E_8 under the Klein four-group of their parities.

7.4 What each twist does to E_7 and E_6

twist of σ_c	E_8	**contact Levi E_7**	E_6
tick L_+: $(-1)^{a_8}$	$E_{8(-24)}$	compact	compact
tick L_0: $(-1)^{a_7}$	$E_{8(-24)}$	$\mathbf{E_{7(-25)}}$	compact
tick L_-: $(-1)^{a_7+a_8}$	$E_{8(-24)}$	$E_{7(-25)}$	compact
height $(-1)^{\mathrm{ht}}$ (committed split)	$E_{8(8)}$	$\mathbf{E_{7(7)}}$	$E_{6(2)}$
height $\cdot L_+$ / height $\cdot L_0$	$E_{8(8)}$	$E_{7(7)}$ / $E_{7(-5)}$	$E_{6(2)}$
height $\cdot L_-$	$E_{8(-24)}$	$E_{7(-5)}$	$E_{6(2)}$

Two supergravities in one table

$E_{7(7)}$ is the hidden symmetry ("U-duality") of maximal $\mathcal{N} = 8$ supergravity in four dimensions, and $E_{8(8)}$ that of maximal supergravity in three. $E_{7(-25)}$ is the duality group of the "magic" $\mathcal{N} = 2$ supergravity built on the octonions. The committed split

form of the repository lands on the first; the Hesse clock lands on the second. The black-hole quartic of Section 6 is the entropy formula in both.
More precisely, the clock's chain $E_{6(-26)} \subset E_{7(-25)} \subset E_{8(-24)}$ (all three computed here, the first in Section 8) is exactly the chain of duality groups of the octonionic magic $\mathcal{N} = 2$ supergravity of Günaydin, Sierra and Townsend in five, four and three space-time dimensions [21]: $E_{6(-26)}$ acts on the 27 charges in five dimensions, $E_{7(-25)}$ on the 56 in four, and $E_{8(-24)}$ is the symmetry of the three-dimensional theory obtained by one more reduction. The split chain plays the same role for maximal supergravity. BRIDGE

7.5 The Lorentzian slice lives only in the clock real form

The contact Levi $E_{7(-25)}$ has a Hermitian symmetric structure: on the 27-dimensional space $\mathfrak{p}^+ = \{a_8 = 0,\ a_7 = +1\}$ the triple product

$$\{x, y, z\} = -[[x, \sigma_c\, y], z]$$

makes $\mathfrak{p}^+$ a Hermitian Jordan triple system. A *tripotent* is an element with $\{e, e, e\} = 2e$ (the normalisation used here).

Theorem 7.2 (signature is a real-form datum) COMPUTATION

(i) Under the clock conjugation all 27 root vectors of $\mathfrak{p}^+$ are tripotents; under the committed split conjugation only 12 of 27 are.

(ii) For the tripotent e of a triad frame, $A = \{x : \{e, x, e\} = 2x\}$ with $x \circ y = \frac{1}{2}\{x, e, y\}$ is a 27-dimensional commutative Jordan algebra with *positive-definite* trace form, inertia $(27, 0, 0)$: a Euclidean Jordan algebra of rank three — the Albert algebra $H_3(\mathbb{O})$ [13].

(iii) For a primitive idempotent of the frame, the Peirce spaces have dimensions $1 + 16 + 10$, and the rank-two determinant on the 10-dimensional Peirce-0 space has inertia $(\mathbf{1}, \mathbf{9})$ — Lorentzian.

(iv) In the committed split form the same construction gives inertia $(5, 5)$. Over $\mathbb{F}_3$ the two quadrics are both of type $O^+(10, 3)$ and cannot be told apart.

`w33_pass10949_freudenthal_quasiconformal_clock_cone.py`

What it means physically: one time, not five

Tegmark observed that only space-times with exactly one time direction support predictable physics: with several, the equations of motion are ultrahyperbolic and initial data do not determine the future [43]. Theorem 7.2 says the finite geometry does not decide between one time and five — over $\mathbb{F}_3$ they are the same — but that the real form picked out by its own clock is the one-time, Lorentzian choice. The committed split form, chosen earlier for computational convenience, is the other one. BRIDGE

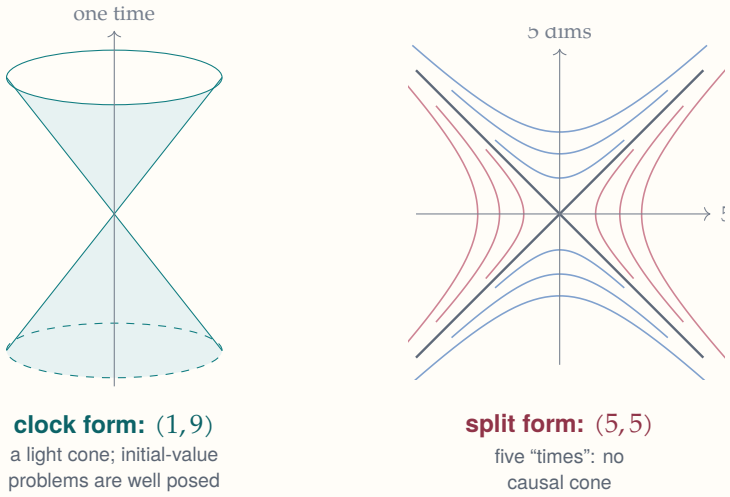


Figure 21. The same Peirce-0 quadric in two real forms of E_8. Only the real form selected by the Hesse clock is Lorentzian.

Signature is not visible over a finite field. It is a choice of real form, and the Hesse clock's tick lines make that choice: they select $E_{8(-24)} \supset E_{7(-25)}$, and with it a Euclidean Albert algebra whose Peirce slice is a $1 + 9$ light cone.

8 The Albert algebra: ten dimensions, a spinor, and matter parity

Section 7 produced, from the clock, a 27-dimensional Euclidean Jordan algebra. By the Jordan–von Neumann–Wigner classification there is exactly one such algebra that is "exceptional": the *Albert algebra* $H_3(\mathbb{O})$ of 3×3 Hermitian matrices over the octonions. Since the 1930s it has been suspected of having something to do with quantum mechanics and particle physics; this section computes its symmetries on the committed basis and finds the Lorentz group of ten-dimensional space-time, a chiral spinor, and the repository's own matter parity.

In plain words

The *octonions* are the largest of the four number systems in which you can divide: the real numbers, the complex numbers, the quaternions, and the octonions, of dimensions $1, 2, 4, 8$. Octonions do not even obey $(ab)c = a(bc)$, yet 3×3 "Hermitian" matrices of them — real numbers on the diagonal, octonions above it, their conjugates below — form a consistent algebra of dimension $3 + 3 \times 8 = 27$. Its symmetries are the exceptional groups F_4 and E_6 [3]. Manogue, Dray and Wilson have recently described the Standard Model through octonionic E_8 structures of this kind [35].

8.1 Anatomy of a 3×3 octonionic matrix

Choose the idempotent $c = E_{11}$ (a 1 in the top-left corner). Every element of the algebra splits by how it behaves under multiplication by c — its *Peirce decomposition*, with eigenvalues $1, \frac{1}{2}, 0$:

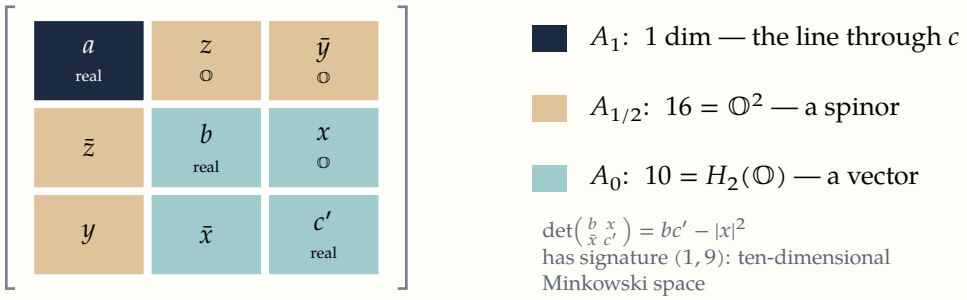


Figure 22. The Peirce decomposition $27 = 1 + 16 + 10$ of the Albert algebra with respect to a primitive idempotent. The lower-right 2×2 block is a copy of ten-dimensional Minkowski space; the two octonions in the first row and column form a sixteen-component spinor.

That the 2×2 Hermitian octonionic matrices form ten-dimensional Minkowski space, with the determinant as the Minkowski interval, is a classical observation (Sudbery; Kugo and Townsend; Baez and Huerta), and it is the algebraic reason the superstring lives in ten dimensions [4, 30]. The new content is that the forty-point clock reaches it, on the committed basis.

8.2 The symmetries, computed

Theorem 8.1 (the clock Albert algebra) COMPUTATION

For the 27-dimensional clock algebra A of Theorem 7.2:

(i) the derivations $\mathrm{Der}(A) = \mathrm{span}\,[L_x, L_y]$ form a 52-dimensional Lie algebra with negative-definite trace form $(0, 52)$: the compact $F_{4(-52)}$;

(ii) the reduced structure algebra $\mathrm{str}_0(A) = \mathrm{Der}(A) \oplus L(A_{\mathrm{tr}=0})$ is 78-dimensional, obeys $[D, L_x] = L_{Dx}$ for all 52×27 pairs, and has inertia $(26, 52)$: $E_{6(-26)}$;

(iii) the derivations fixing c form a compact 36-dimensional $\mathfrak{spin}(9)$; adding the nine "boosts" L_y ($y \in A_0$, orthogonal to $e - c$) gives a 45-dimensional algebra that preserves A_0 and its determinant of inertia $(1, 9)$, and acts faithfully: it is $\mathfrak{so}(1, 9)$;

(iv) on $A_{1/2}$ the commutant of $\mathfrak{so}(1, 9)$ is one-dimensional: the 16 is absolutely irreducible — the Majorana–Weyl spinor of $\mathfrak{so}(1, 9)$.

`w33_pass10950_clock_albert_lorentz_spinor.py`

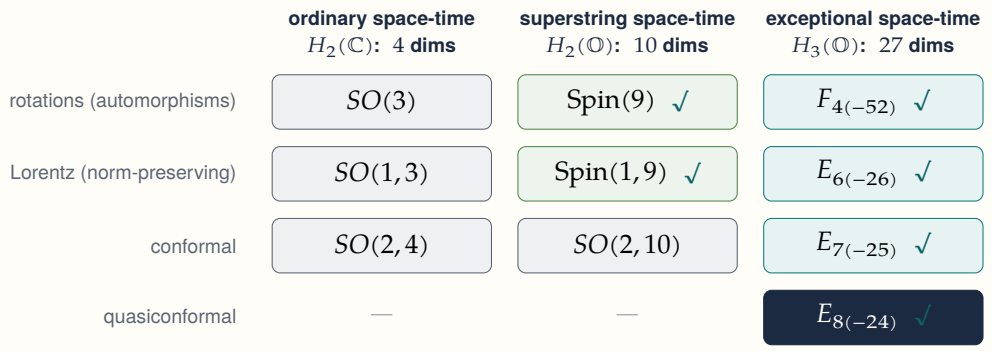


Figure 23. The ladder of space-times. A Jordan algebra of Hermitian matrices is a space-time; its determinant is the interval. For 2×2 complex matrices it is ordinary Minkowski space, for 2×2 octonionic matrices the ten-dimensional space of the superstring, and for 3×3 octonionic matrices the 27-dimensional "exceptional space-time" of Günaydin [22]. Each has rotations, a Lorentz group, a conformal group and (for the last) a quasiconformal group. The Hesse clock's real form of E_8 produces the whole right-hand column, and the middle column inside it.

What it means physically: the clock picks the division algebra

The last row of Freudenthal's "magic square" of Lie algebras can be built from the octonions in two ways: from the division algebra $\mathbb{O}$, giving the chain $F_{4(-52)} \subset E_{6(-26)} \subset E_{7(-25)} \subset E_{8(-24)}$ of automorphism, Lorentz, conformal and quasiconformal groups of $H_3(\mathbb{O})$, or from the split octonions, giving the split forms. The Hesse clock selects the first: the *division-algebra* octonions, the ones with a positive norm, the ones that make $H_2(\mathbb{O})$ a genuine Minkowski space. The repository's earlier "split" choice lands on the other side. COMPUTATION (the four real forms) BRIDGE (the reading)

8.3 Down to four dimensions

Choose a further splitting of the 10 into $4+6$: time $u = e - c$, one frame direction, and a complex line inside the octonions supplied by the complex structure of $\mathfrak{p}^+$ itself. The subalgebra of $\mathfrak{so}(1,9)$ preserving the 4 and the 6 is 21-dimensional, $\mathfrak{so}(1,3) \oplus \mathfrak{so}(6) \cong \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{su}(4)$, and its commutant on the 16 is two-dimensional and contains a complex structure. Hence

$$\mathbf{16} \;=\; (\mathbf{2},\mathbf{4}) \;\oplus\; \overline{(\mathbf{2},\mathbf{4})}.$$

COMPUTATION (`w33_pass10950_clock_albert_lorentz_spinor.py`)

Two readings of the same branching

A ten-dimensional chiral spinor, seen by a four-dimensional observer, becomes four left-handed Weyl spinors transforming as the **4** of $SU(4)$. If $SU(4)$ is read as the rotation group of six hidden dimensions, this is the fermion content of maximally supersymmetric Yang–Mills theory. If it is read as Pati–Salam colour, with lepton number as the fourth colour, it is one generation of quarks and leptons. The branching is exact; which reading nature uses is not decided by it. BRIDGE

8.4 Matter parity is a rotation by 2π

The repository carries an independent certificate for a "matter charge" Q_ψ on the 27: it splits $27 = 16_1 + 10_{-2} + 1_4$, and on the 45 triads the charges fall into the pattern $40 \times (-2,1,1) + 5 \times (-2,-2,4)$. Matter parity is $(-1)^{Q_\psi}$: odd on the 16, even on the 10 and 1. In the Minimal Supersymmetric Standard Model a parity of exactly this kind is what forbids the proton from decaying through dimension-four operators.

Theorem 8.2 (matter parity is the Peirce symmetry) COMPUTATION

For every one of the 27 coordinate idempotents c, the Peirce charge

$$Q = 6L_c - 2 \qquad \text{(eigenvalue 4 on } A_1,\ 1 \text{ on } A_{1/2},\ -2 \text{ on } A_0\text{)}$$

reproduces the certified spectrum of Q_ψ, is conserved on all 45 triads, and gives exactly the pattern $40 \times (-2,1,1) + 5 \times (-2,-2,4)$. Its parity $(-1)^Q$ is the Peirce symmetry U_{2c-e}, an automorphism of A that is $+1$ on $A_1 \oplus A_0$ and -1 on the spinor $A_{1/2}$. For a single-plane rotation generator $D \in \mathfrak{spin}(9)$, $\exp(2\pi D) = U_{2c-e}$ (numerical agreement 2.7×10^{-15}). `w33_pass10950_clock_albert_lorentz_spinor.py`

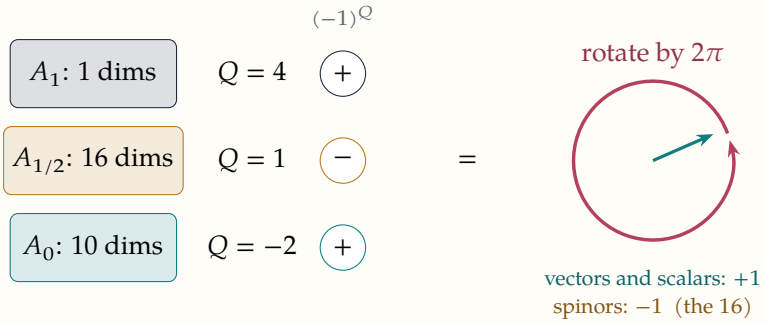


Figure 24. Matter parity in the clock Albert algebra: the sign that a full turn gives a spinor.

What it means physically

A rotation by 2π returns every vector to itself, but multiplies every spinor by -1: the mathematical root of the difference between fermions and bosons. In $SO(10)$ grand unification it is known that matter parity can be realised by exactly such a central element of Spin(10), acting as -1 on the matter **16** and $+1$ on the Higgs **10** (cf. [36]). Theorem 8.2 is the objectwise version of that fact inside the forty-point clock: the repository's matter parity *is* the 2π rotation of the Spin(9) fixing a primitive idempotent. The same $27 = 1 + 10 + 16$ that a GUT reads as singlet, Higgs and matter is, in the clock real form, scalar, Lorentz vector and Weyl spinor. BRIDGE (reading) COMPUTATION (identity)

8.5 The clock's fourth tick is the same sign

Section 4 met a clock whose fourth tick is a sign flip. The other research track has since shown that this is not an analogy but one spin structure, seen at both ends of the ladder. We

cite their results (certificates named) rather than re-derive them.

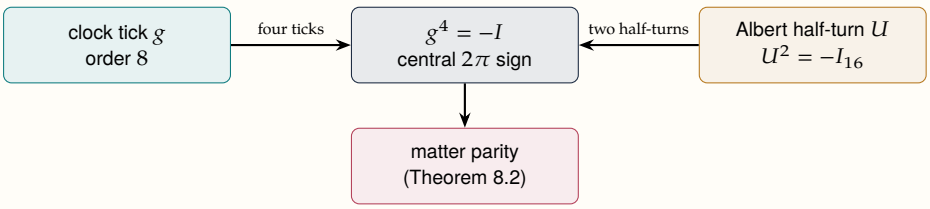
Theorem 8.3 (clock, spin and matter parity; Passes 10951–10959) COMPUTATION

(i) The negacyclic clock tick lifts uniquely to $g = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right) \in GL(2,3)$ of order eight, with $g^4 = -I$; under an explicit isomorphism $SL(2,3) \cong 2T \subset \mathrm{Spin}(3)$ the element $-I$ is the quaternion -1, the central 2π spin sign. `w33_pass10951_clock_pin_spin_central_sign_bridge.py`

(ii) As a signed permutation of the four clock coordinates, determinant-odd ticks exchange the two D_4 half-spin sheets and determinant-even ticks preserve them. `w33_pass10955_d4_halfspin_clock_bridge.py`

(iii) On the Albert Peirce 16, the $\mathrm{Spin}(8)$ fixing a spatial axis splits it as $8_s + 8_c$; an exact rational $\mathrm{Spin}(9)$ half-turn U exchanges them, with $U^2 = -I_{16}$ — the matter-parity sign of Theorem 8.2 — and $U^4 = I_{16}$. `w33_pass10956_albert_spin8_half-spin_normalizer.py`, `w33_pass10958_exact_albert_mu4_missing_mode_inter-twiner.py`

(iv) One Albert 16 cannot carry the whole order-eight clock as a $GL(2,3)$ representation; the minimal carrier is $16 \oplus 16$. `w33_pass10959_albert_clock_extension_obstruction.py`



Where this stops

The $\mathfrak{so}(1,9)$ here is the stabiliser of an idempotent inside $E_{6(-26)}$, an *internal* symmetry of the algebra; it is not derived to be the Lorentz group of physical space-time, and the 2π rotation is a rotation in that internal $\mathrm{Spin}(9)$. The identification of Q_ψ with $6L_c - 2$ holds up to the $W(E_6)$-transitive choice of coordinate idempotent. Neither chirality selection, masses nor gauge dynamics follow.

The exceptional ladder closes: from one qutrit, through $W(3,3)$, E_8, a Hesse clock and its real form, to the Albert algebra $H_3(\mathbb{O})$ — whose automorphisms are F_4, whose structure group is $E_{6(-26)}$, and inside which sit ten-dimensional Minkowski space, its Lorentz algebra $\mathfrak{so}(1,9)$, a Majorana–Weyl spinor, and, as a rotation by 2π, the matter parity that protects the proton.

9 The Standard Model on forty points

Sections 2–8 established exact structure. This section asks the question the programme exists to answer: how much of the actual physics of our universe does that structure carry? We give the answer at three levels — what the finite geometry itself contains, what a string compactification built on it produces, and what is still missing — and we end with a scorecard.

The string route, completed. *Nine families of heterotic orbifolds carry the forty-point twist* ($\mathbb{Z}_3$, $\mathbb{Z}_3 \times \mathbb{Z}_3$, $\mathbb{Z}_6$-I, $\mathbb{Z}_6$-II, $\mathbb{Z}_{12}$-I, $\mathbb{Z}_2 \times \mathbb{Z}_6$-I, $\mathbb{Z}_2 \times \mathbb{Z}_6$-II, $\mathbb{Z}_3 \times \mathbb{Z}_6$, $\mathbb{Z}_6 \times \mathbb{Z}_6$). *They produce* 560 *inequivalent three-generation Standard Models. On the selection rules the string's own worldsheet establishes, not one has a supersymmetric vacuum with a stable proton and a clean spectrum, and every escape we could find — other discrete symmetries, rotations of the hidden dimensions, heavy superpartners, flavour suppression, hidden-sector condensates — fails for a stated, computed reason. In eight families the mechanism is the same: the one field that can cancel the anomalous* $U(1)$*'s Fayet–Iliopoulos term has the quantum numbers of a right-handed sneutrino, and its vacuum value switches on the couplings that make the proton decay. In the ninth,* $\mathbb{Z}_3 \times \mathbb{Z}_3$*, the parity survives and the vacua are supersymmetric — but every one of them carries six massless particles of electric charge* $1/3$*, forced by a counting argument no hidden dynamics removes. The forty points host the Standard Model; this route does not make it our universe.*

EXACT	Quantum measurement structure (qutrit Paulis, contexts, Clifford group)
EXACT	Exceptional algebras E_6, E_7, E_8, F_4 and the Albert algebra
EXACT	A finite light cone; clocks; one Lorentzian real form $\mathfrak{so}(1,9)$
HOSTED	Gauge group $SU(3) \times SU(2) \times U(1)$
HOSTED	Three generations
EXACT	One chiral generation as a spinor (16 of Spin$(1,9)$)
NO-GO	Which chirality: exchanged only by antiunitary (CPT-type) symmetries
EXCLUDED	Matter parity in a supersymmetric vacuum (seven orbifold families)
CLOSED	A realistic vacuum of the heterotic orbifold route
OPEN	Masses, mixing angles, coupling constants
OPEN	Gravity, a dynamical space-time, the cosmological constant
NO-GO	Lepton mixing from the finite symmetry alone (residual symmetries)
HOSTED	An entanglement area law for networks of the perfect tick

Figure 25. Scorecard. EXACT: a theorem or exact computation in this paper. HOSTED: realised in an explicit construction built on the geometry, not forced by it. EXCLUDED: exhaustively ruled out in a stated scope (Section 9.5). NO-GO: proved impossible from inside. OPEN: not addressed by any exact result.

9.1 What the finite geometry itself contains

The anatomy of one symmetry. Pick a point p_0 of $W(3,3)$ and the *transvection* R centred on it: the symplectic map $x \mapsto x + \langle x, v\rangle v$, of order three. It fixes the 13 points of the plane $p_0^{\perp}$ — p_0 and its 12 neighbours — and permutes the other 27 points in nine orbits of three.

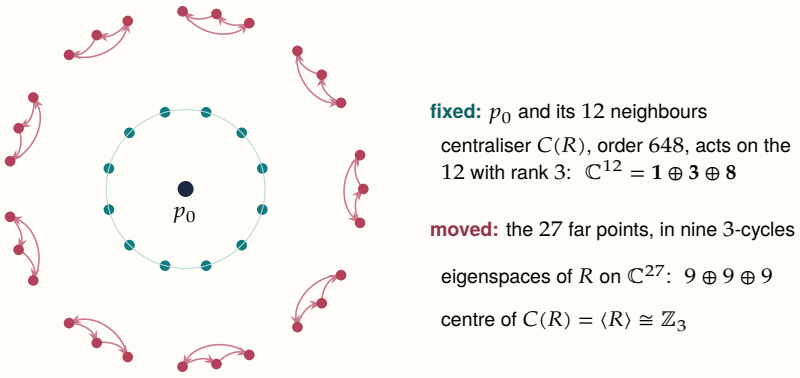


Figure 26. One transvection of $W(3,3)$. Its centraliser acts on the twelve fixed neighbours with the dimensions of the Standard Model's gauge algebra, $1 + 3 + 8$, and it splits the twenty-seven moved points into three equal eigenspaces.

Proposition 9.1 (the transvection spine) COMPUTATION

The centraliser $C(R)$ of a transvection has order 648 (it is the point stabiliser $3^{1+2}{:}2A_4$) and centre $\langle R \rangle \cong \mathbb{Z}_3$. Its permutation module on the 12 neighbours of p_0 decomposes as $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{8}$. On the 27 far points, R has eigenspaces $9 \oplus 9 \oplus 9$. `bt886_standard_model_spine.py`

How far this goes

The dimensions $1, 3, 8$ are those of the Lie algebras $\mathfrak{u}(1)$, $\mathfrak{su}(2)$, $\mathfrak{su}(3)$, and the corpus has sometimes read Proposition 9.1 as "the Standard Model gauge group follows with zero free parameters". That is an over-read. The $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{8}$ here are representations of a *finite* group of order 648; they are not Lie algebras, they do not come with gauge fields, and nothing fixes their couplings. The honest statement is that the local symmetry of one point of the geometry has the *shape* of the Standard Model's gauge algebra. BRIDGE

Three from the Steinberg module. The first homology of $W(3,3)$ (its 40 points, 240 edges and 160 triangles in lines) is 81-dimensional and irreducible under $\mathrm{PSp}(4,3)$: it is the *Steinberg module*, the most rigid representation a group of Lie type has, and the logical space of the repository's $[[240, 81, 3]]_3$ quantum code. COMPUTATION (`bt861_code_register_is_steinberg.py`)

Theorem 9.2 (every order-three symmetry splits matter $27 + 27 + 27$) THEOREM

Let $g \in \mathrm{PSp}(4,3)$ have order three. Then g acts on the 81-dimensional Steinberg module with eigenvalues $1, \omega, \omega^2$, each with multiplicity exactly 27.

Proof. The Steinberg character vanishes on every element whose order is divisible by the defining characteristic $p = 3$ (a theorem of Steinberg; see [24]). So $\chi(g) = \chi(g^2) = 0$, and the multiplicity of each eigenvalue ω^j is $\frac{1}{3}(\chi(1) + \omega^{-j}\chi(g) + \omega^{-2j}\chi(g^2)) = 81/3 = 27$. □

Three is forced, but not selected

Theorem 9.2 holds for *every* element of order three, so it produces a threefold splitting without choosing one. And the string-theory track (below) found that in the explicit compactifications the number of families is the number of complex planes of the internal torus, not a property of $W(3,3)$. We therefore record the Steinberg splitting as a structural host for three generations, not a derivation of them. BRIDGE

The qutrit register is $SU(9)$, not E_6. One might hope that the E_6 of grand unification acts directly on qutrit registers.

Theorem 9.3 (no Pauli E_6) THEOREM COMPUTATION

Conjugation by the three-qutrit Pauli group splits $\mathfrak{gl}(27)$ into 729 distinct one-dimensional character spaces, so every conjugation-stable subalgebra is spanned by Pauli operators. A simple Pauli-spanned Lie algebra has dimension $3^a - 3^b$, i.e. one of $8, 24, 72, 80, 216, 240, 648, 720, 728$. Since $\dim \mathfrak{e}_6 = 78$ is not among them, E_6 cannot act on a qutrit register through Pauli-stable generators. `Holotrade 1cde1f1`

The natural gauge algebra of the qutrit register is instead $\mathfrak{sl}_9$ — the 80 of Section 5 — which is exactly what appears in the string construction.

9.2 The string that carries the geometry

The most developed attempt to turn this structure into physics uses the $E_8 \times E_8$ heterotic string compactified on an orbifold [10]: six extra dimensions curled up as three two-dimensional tori, divided by a finite rotation group. Its advantage is that everything is computable exactly, with a public tool, the ORBIFOLDER [37].

Theorem 9.4 ($W(3,3)$ lives in the twisted sector of E_8) COMPUTATION

For the $\mathbb{Z}_3$ twist of the E_8 lattice vertex algebra, the commutator form of the twisted-sector Pauli group equals twice the lattice form on every rung, and the resulting commutation geometry is $W(3,3)$. The twist is anomalous on a single E_8 (ground-state energy $\rho = 4/9$), consistent exactly when $3 \mid k$ on E_8^k; the E_8^3 orbifold is the Niemeier lattice vertex algebra $V_{A_8^3}$ with algebra $\mathfrak{sl}_9^3$. `Holotrade 68df33f`

Order three fails, for a reason. In the $T^6/\mathbb{Z}_3$ orbifold built on this twist, Wilson lines produce three-family $SU(5)$ and $SO(10)$ models; but an order-three Wilson line keeps at most one of Q, u^c, e^c in each **10**, so no breaking completes a Standard-Model family. The general law: for a $\mathbb{Z}_N$ orbifold, u^c and e^c survive together only when N is even. This explains, from

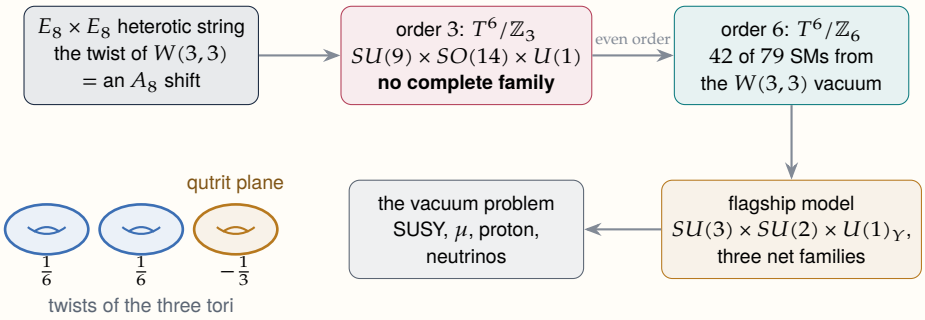


Figure 27. The heterotic route. At order three no Wilson-line breaking completes a Standard-Model family (a theorem); at even order the $W(3,3)$ vacuum is the most productive source of Standard Models; the open problem is a realistic vacuum. The third torus, the unique order-three plane, is where the Higgs lives.

first principles, why heterotic model builders moved to even-order orbifolds. An exhaustive census of 127 980 classes confirms it, and 64 000 fresh Wilson-line draws on the unique modular-invariant $W(3,3)$ twist of $T^6/\mathbb{Z}_3$ find no Standard Model (P10979). The law is a statement about a *cyclic* group: two independent order-three twists, $T^6/\mathbb{Z}_3 \times \mathbb{Z}_3$, evade it, and there the $W(3,3)$ twist gives 13 Standard Models — consistent with the known $\mathbb{Z}_3 \times \mathbb{Z}_3$ landscape of model builders [38]. THEOREM COMPUTATION (Holotrade `ed1eaca`, `a6e1c69`, `a779053`)

Order six succeeds. Among the 58 $\mathbb{Z}_6$-I and 61 $\mathbb{Z}_6$-II shifts, the $W(3,3)$ vacuum $SU(9) \times SO(14) \times U(1)$ is the most productive class: of 79 inequivalent Standard Models found in it, 42 come from the $\mathbb{Z}_6$-I $W(3,3)$ vacuum. The flagship has gauge group $SU(3)_C \times SU(2)_L \times U(1)_Y$ with three net families and no anomalies, and at one fixed point the local gauge group is $SU(9)$ with all four simple roots of the Standard Model's $SU(5)$ inside it: the Standard Model's unification group descends from the twist's own $\mathfrak{sl}_9$. COMPUTATION (Holotrade `5b3f3ad`)

Measured regularities of the $W(3,3)$ Standard Models COMPUTATION

- **The breaking is an entangling gate.** The order-three Wilson line's $\mathfrak{sl}_9$ holonomy is of two-qutrit controlled-Z type $(5,2,2)$ in 45 of 45 $\mathbb{Z}_6$-I models, against a 33% background (chance $\approx 2.6 \times 10^{-22}$). (`d772153`)
- **Two commuting holonomies cut out $SU(3) \times SU(2)$**: in all 25 models with parity-type θ^3, against a 30% background ($\approx 8.6 \times 10^{-14}$). (`94df376`)
- **A heavy top is allowed**: 57 of 60 models admit a renormalisable top Yukawa coupling. (`93b34e1`)
- **The Higgs sits alone in the qutrit plane**: in all 87 $\mathbb{Z}_6$-I models the weak doublets come only from the third torus (the unique order-three plane) and colour triplets only from the other two; this one fact gives both the top coupling and **doublet–triplet splitting**, solved in 55 of 87 models by a missing-partner mechanism. (`3caf15e`, `a6f1cae`)

What it means physically

The Standard Model's most puzzling structural feature — that the Higgs doublet is light while its would-be colour-triplet partner, which would make protons decay, is absent — is here a consequence of geometry: the Higgs lives on the one internal plane whose twist has order three, the "qutrit plane", and that plane carries no colour triplet. And the symmetry breaking that produces the Standard Model is, in qutrit language, an entangling gate. These are measured regularities of an explicit string construction, with controls; they are not theorems about all constructions.

9.3 Where the string route stands

A model is not a universe until it has a stable vacuum with the right low-energy properties. Here the honest record is mixed, and it has been corrected more than once.

Requirement	Current exact status	Badge
Supersymmetric vacuum	In the doublet–triplet-solving vacuum, F- and D-flatness conflict: one $U(1)$ direction gives every mass singlet a positive charge. Gordan's alternative ("D-flat or empty superpotential, never both") verified on all 215 models.	REFUTED (that vacuum)
D-flat directions	Exist in 23 of 87 ($\mathbb{Z}_6$-I) and 105 of 128 ($\mathbb{Z}_6$-II); 83 of the 211 anomalous models have none.	COMPUTATION
Proton stability	No supersymmetric singlet vacuum preserves a matter parity built from the gauge torus and the space group: 0 of 215 (Section 9.5). Baryon triality is impossible in all 215. Heavy superpartners would need $m_t \leq 171$ GeV.	COMPUTATION (excluded)
Parity from rotations	Appeared in one model under the old per-plane rule; with the established rules (Bizet et al.) it closes: 0 of 128.	COMPUTATION (excluded)
The μ term	Arises at order 3 or 4 in every model; none is μ-free through order five. Must come from a vacuum symmetry. In the $\mathbb{Z}_6$-I class no continuous vacuum symmetry can forbid it (0/23, exact); in $\mathbb{Z}_6$-II an unbroken $U(1)'$ does so in 54 models, never with all exotics massive (Section 9.4).	OPEN
Neutrino masses	In the splitting-solving models, one to two orders of magnitude too light.	OPEN

Figure 28 gathers the whole search in one picture. Each band is an exact filter; the number is how many models or vacua pass it, and the tag names the computation that decided it.

9.4 The Higgs mass term and vacuum symmetries

A vacuum symmetry that forbids $\mu = H_u H_d$ must survive the singlet vacuum values that cancel the Fayet–Iliopoulos term. For its continuous part this is exactly decidable. A $U(1)$ direction t survives a vacuum S iff $t \cdot q_s = 0$ for every condensing singlet. A coupling of charge Q is then forbidden to all orders iff Q is not in the rational span of the q_s. Minimal vacua, the extreme rays of the D-flat cone, leave the most symmetry. We examined all of

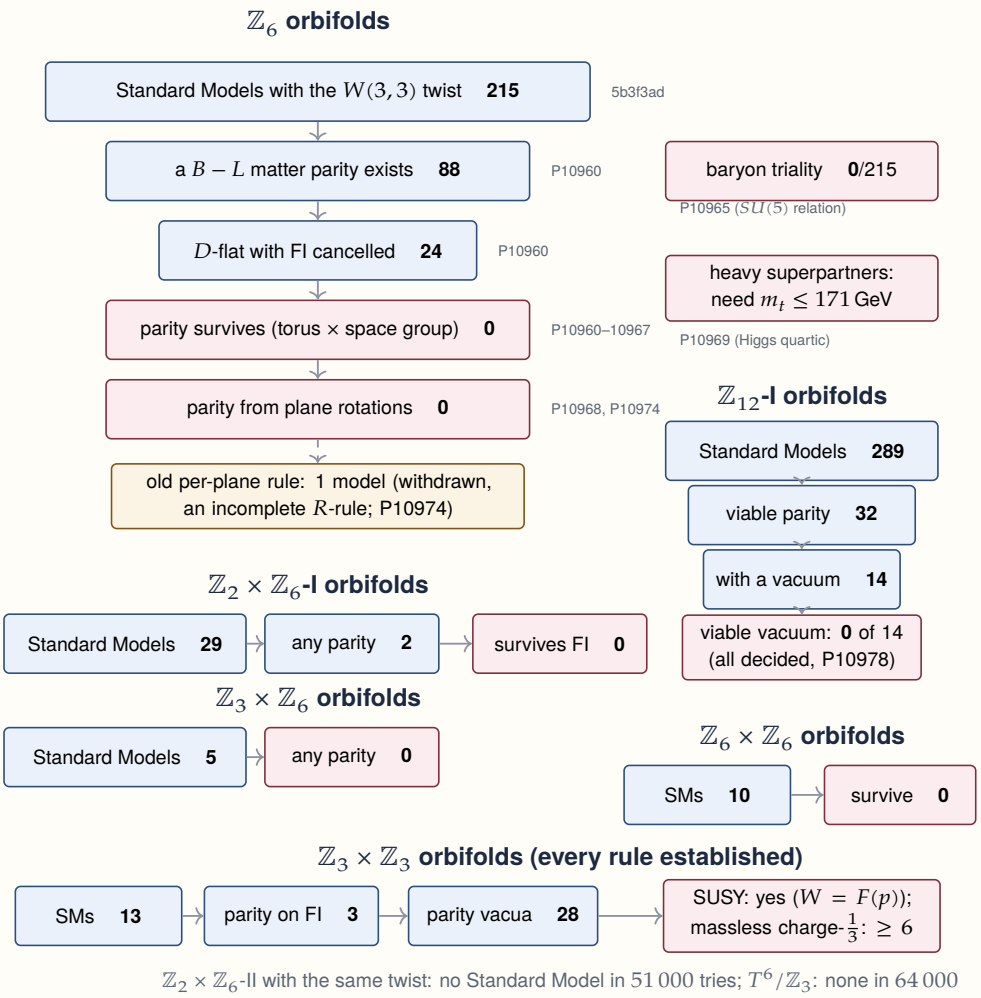


Figure 28. The vacuum gauntlet. Blue: filters passed; red: dead ends; gold: a door that looked open and was withdrawn. The last row is the one family where the parity survives: its vacua are supersymmetric, and a hidden-representation index leaves six massless charge-$\frac{1}{3}$ states in each (P10979, P11024). Every count is exact (Farkas certificates, Smith normal forms, lattice tests and integer programs), and each is regenerated by the script named in the ledger.

them, exactly, in 125 of the 128 models with D-flat directions, and exactly verified sampled vertices in the other three.

- **$\mathbb{Z}_6$-I, the class of the $W(3,3)$ vacuum:** the earlier integer-lattice audit, which lacked the plane R charges, found that all 6 695 116 vacua allowed μ. Those charges have now been restored from the Orbifolder archives and cross-validated against the frozen $\mathbb{Z}_6$-II data. With the full $U(1)$-remnant, space-group and plane-R lattice, *every one* of the 6 695 116 vacua protects μ while allowing a top Yukawa. The field-choice census reduces exactly to 1464 support/lattice states. Nevertheless 0 vacua are clean: the restored symmetry that forbids μ also leaves unwanted states light. COMPUTATION (`w33_20261001_z6i_rcharge_lattice_quotient.py`)
- **$\mathbb{Z}_6$-II:** such a $U(1)'$ exists in 429 vacua of 54 models. In all but one, the same $U(1)'$ also keeps a second Higgs pair or a vector-like exotic massless. This is the continuous counterpart of the R-symmetry that, in the one model whose FI-cancelling vacua keep a matter parity, keeps exotics massless.
- A first, rational-span test left one vacuum with every exotic massive. Integer powers of the vacuum fields (the discrete remnants of the broken $U(1)$s) already keep four vector-like doublet pairs massless there, and with the space-group and R rules quarks and Higgs pairs stay light too (Pass 11240). No clean vacuum remains.

The R-symmetry that makes such a vacuum F-flat is the one that keeps the exotics massless, as in Pass 10968. The reason is general. If an anomaly-free $U(1)'$ charges $H_u H_d$, the $SU(3)^2$–$U(1)'$ anomaly forbids every complete set of colored masses. In any perfect matching of triplets with antitriplets, the numbers of pairs using H_u and H_d are equal (both ≥ 3), so the anomaly is $-a\,(t(H_u) + t(H_d)) \neq 0$. This holds in all 812 protected cases checked (`w33_pass11242_u1prime_mu_anomaly_theorem.py`). With the full discrete and R selection rules, the same obstruction first appeared in all 16 326 sampled protected cases (`w33_pass11246_discrete_anomaly_escapes.py`). The restored $\mathbb{Z}_6$-I R data now make that statement exhaustive on this class: all 6 695 116 vacua have exactly one protected H_u case, and 0 admit a complete colored matching for any H_d. THEOREM COMPUTATION (`w33_20261001_z6i_colored_escape_exhaustive.py`)

9.5 The Fayet–Iliopoulos term forces a right-handed sneutrino

Every model of the class has an anomalous $U(1)$, whose Fayet–Iliopoulos term must be cancelled by giving vacuum values to some singlet fields (D-flatness), with $\langle n \rangle / M_s \approx 0.2$–$0.4$. Proton stability needs a *matter parity* — a $\mathbb{Z}_2$ under which quarks and leptons are odd and Higgs fields even — to survive that condensation, so the fields that condense must all be even. The earlier record found every tested D-flat direction contained an odd singlet, but it had *sampled* 60 choices of $B - L$ per model and said explicitly that this was not a proof; the raw spectra needed to finish it were in no repository. They were recovered for this paper from the original scan and frozen as an exact-rational ledger of all 215 models.

Theorem 9.6 (no matter parity survives the FI term) COMPUTATION

Over the *entire* rational solution set $B - L = x_0 + Nt$ of each model:

(i) 64 of the 88 models with a $B - L$ direction have no D-flat singlet direction at all (an exact Farkas vector certifies that the FI term cannot be cancelled);

(ii) in the other 23 $\mathbb{Z}_6$-I models, the singlets that are even for *some* choice of $B - L$ are never D-flat (one exact Farkas vector each); the only singlets that are never even carry $3(B - L) = 3$, i.e. $B - L = +1$, fixed by the Standard Model itself, and every D-flat direction uses one;

(iii) in the one remaining ($\mathbb{Z}_6$-II) model, all 304 FI-cancelling extreme rays of the D-flat cone (3334 rays, computed exactly) are obstructed by an integer (Smith normal form) parity certificate; its never-even singlets carry $B - L = \pm 1$.

Independently, all 1024 $\mathbb{Z}_2$ characters of the charge lattice that are odd on every q, u^c, d^c, e^c give no even D-flat direction. Controls: the same exact machinery finds the $23 + 105 = 128$ D-flat models and the 88 $B - L$ models of the earlier record. `w33_pass10960_matter_even_dflat_closure.py`

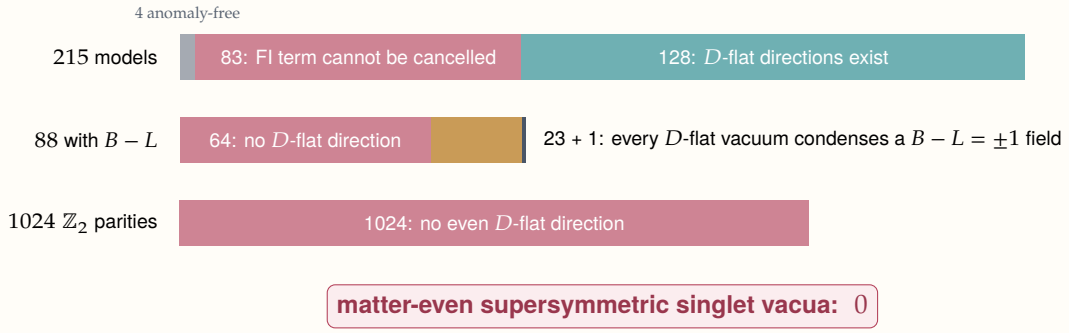


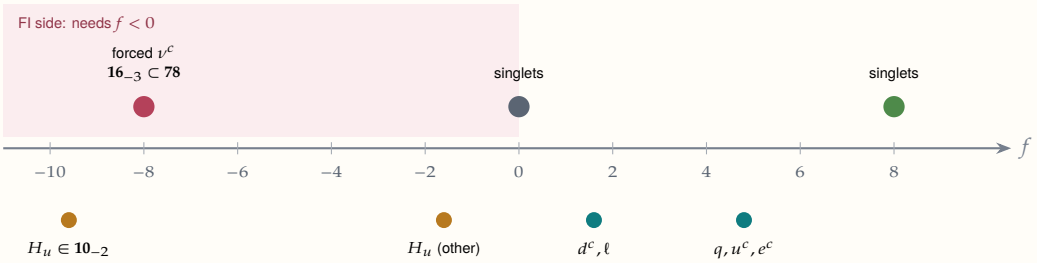
Figure 29. The exhaustive closure of Theorem 9.6. Every one of the 215 models is accounted for; no branch leaves a matter-parity-preserving supersymmetric vacuum in singlet directions.

What it means physically: the sneutrino that breaks R-parity

A singlet with $B - L = +1$ has the quantum numbers of a right-handed (s)neutrino. Theorem 9.6 says the anomalous $U(1)$ *forces* such a field to condense. Martin's criterion [36] is exactly relevant here: matter parity survives the breaking of $B - L$ only if the fields that break it have even $3(B - L)$ — which is why realistic $SO(10)$ models break $B - L$ with **126**-type fields ($B - L = \pm 2$) rather than **16**-type ones ($B - L = \pm 1$). In this class the FI term leaves no choice: $B - L$ is broken by one unit, and the baryon- and lepton-number-violating couplings that matter parity was protecting are no longer forbidden by any symmetry, at the scale of the singlet vacuum values.

Why: one E_6-shaped charge. The obstruction has a single mechanism. In each of the 23 models there is an exact certificate f — a combination of the anomalous and ordinary $U(1)$ charges — with a prescribed physical shape: $f = \frac{24}{5}$ on every q, u^c, e^c and $f = \frac{8}{5}$ on every d^c, ℓ (in units of a positive scale), and $f \geq 0$ on every singlet that could be even. On the singlets f takes only the values -8, 0 and 8, and the negative ones are exactly the forced

$B-L=+1$ fields. Because D-flatness makes the f-charge-weighted sum of condensates equal the (negative) FI contribution, one of them must condense. On the families $f = 4Q_\psi - \frac{4}{5}Q_\chi$, the $U(1)$ combination of $E_6 \supset SO(10)\times U(1)_\psi \supset SU(5)\times U(1)_\chi \times U(1)_\psi$ with $Q_\chi = 4Y - 5(B-L)$; the value -8 at $(B-L, Y) = (1, 0)$ then identifies the forced field as the right-handed-sneutrino direction of the matter-odd $\mathbf{16}_{-3}$ in the adjoint $\mathbf{78}$ of E_6. COMPUTATION (`w33_pass10964_e6_shaped_fi_mechanism.py`) BRIDGE (the E_6 reading)



singlets above the axis, charged fields below (units of μ; all 23 models)

Figure 30. The canonical certificate $f = 4Q_\psi - \frac{4}{5}Q_\chi$ on the families. Among singlets, only the right-handed-sneutrino direction of the E_6 adjoint's $\mathbf{16}_{-3}$ sits on the side that can balance the Fayet–Iliopoulos term.

The hidden sector does not rescue it. Theorem 9.6 covers singlets of every nonabelian group. Fields that are Standard-Model singlets but charged under the *hidden* gauge group can also condense, and their parity can be compensated by a hidden gauge element. Pass 10962 attacks this loophole in tiers:

- **exact, for every hidden direction:** a parity-preserving vacuum must be D-flat for the hidden maximal torus with component-wise parity (up to a hidden torus element); for 32 of the 88 models even this relaxation has an exact Farkas certificate;
- **exact over explicit invariant generators:** by the Luty–Taylor and Kempf–Ness correspondence a parity-preserving D-flat vacuum exists iff an FI-cancelling product of matter-even hidden-group invariants does; over singlets, all quadratic invariants, $SU(N)$ baryons and $SU(5)$ cubics, the other 55 models close by exact Farkas vectors, and the last by a parity obstruction;
- the four anomaly-free models (no FI term) admit no matter parity at all.

COMPUTATION (`w33_pass10962_hidden_sector_dflat_tiers.py`)

Scope

For the 56 models not closed at the first tier, completeness of the invariant generators is not proved (they contain bi-fundamental, $SU(4)$ **6** or $SU(5)$ **10** hidden fields), so for them the hidden-sector statement is "no rescue among the standard invariants" rather than a theorem. Whether higher-order selection rules suppress the regenerated couplings in a particular model is not addressed. OPEN

The other discrete guardian is excluded too. Matter parity is not the only discrete symmetry that can keep the proton alive. *Baryon triality* $B_3 = \exp(2\pi i(B - 2Y)/3)$ (Ibáñez–Ross) forbids the baryon-number violating couplings $u^c d^c d^c$ and $qqql$ while allowing the lepton-number violating ones. Could the class carry it instead? As a gauge symmetry it would have to be a $\mathbb{Z}_3$ character of the $U(1)$ charge lattice, equal to B_3 on every quark and lepton up to a hypercharge rotation. Pass 10965 shows that this linear system over GF(3) is inconsistent in all 215 models, and names the reason: in 214 of them single copies obey the exact identity

$$Q(u^c) + Q(e^c) = 2\,Q(q)$$

of $U(1)$ charge vectors — the extra $U(1)$s act on the $SU(5)$ **10** through the **10** itself. On that combination B_3 is $1 + 1 - 0 = 2$ while every hypercharge shift is $2 + 0 - 2 = 0 \bmod 3$, so nothing repairs it; the remaining model has an explicit integer relation with the same property. The zero-target control is solvable in all 215. This is the structural reason behind Holotrade's observation that the three dimension-four operators are locked together (b81ef8c). COMPUTATION (`w33_pass10965_baryon_triality_impossible.py`)

> **In plain words**
>
> Of the two textbook discrete symmetries that protect the proton, the first dies at the Fayet–Iliopoulos term and the second is ruled out by the grand-unified shape of the charges. What remains is an R-symmetry — the anomaly-free $\mathbb{Z}_4^R$ of Lee, Raby, Ratz, Ross, Schieren, Schmidt-Hoberg and Vaudrevange — which in a string orbifold lives in the internal rotation sector rather than in the gauge lattice. That is the one open route.

The vacuum switches the forbidden couplings back on. Two gaps remained. First, the searches above used the $B - L$ family and $\mathbb{Z}_2$ characters that are ± 1 on *every* field. A matter parity only has to be -1 on quarks and leptons and $+1$ on the condensate; exotic fields may carry any phase. Second, the string has discrete symmetries of its own, the *space-group selection rules*, and these are not part of the gauge lattice. Pass 10967 closes both. It takes the most general candidate: any element of the gauge torus times the space-group group (the point-group $\mathbb{Z}_6$, a $\mathbb{Z}_3$, and in $\mathbb{Z}_6$-II a further $\mathbb{Z}_2 \times \mathbb{Z}_2$ from fixed points). It then decides exactly, with a Smith normal form, which singlets such an element can leave even. The answer is unchanged: such parities exist in the same 88 models and survive the FI term in none.

The reason is a single identity. In 290 of the 302 forced singlets, single copies of the fields satisfy

$$Q(n) \;=\; -Q(u^c d^c d^c) \;=\; -Q(q\,\ell\,d^c) \;=\; -Q(\ell\,\ell\,e^c),$$

which is the $SO(10)$ quartic invariant $\mathbf{16}^4$ evaluated at ν^c. Each cubic is odd under every matter parity, so n is too. The string's own coupling rules then finish the argument (Figure 31). In each of the 23 D-flat $\mathbb{Z}_6$-I models, none of the three R-parity-violating cubics is allowed at the renormalisable level. Yet all nine forced singlets of every model appear in allowed quartics $n\,u^c d^c d^c$, $n\,q\,\ell\,d^c$ and $n\,\ell\,\ell\,e^c$. The vacuum that supersymmetry requires turns all three back on, with strength $\langle n\rangle/M_s \approx 0.2$–$0.4$. Holotrade had noted that its class closure (b81ef8c) held only if the vacuum breaks matter parity; this is that vacuum, named field by field. COMPUTATION (`w33_pass10967_fi_vacuum_regenerates_rpv.py`)

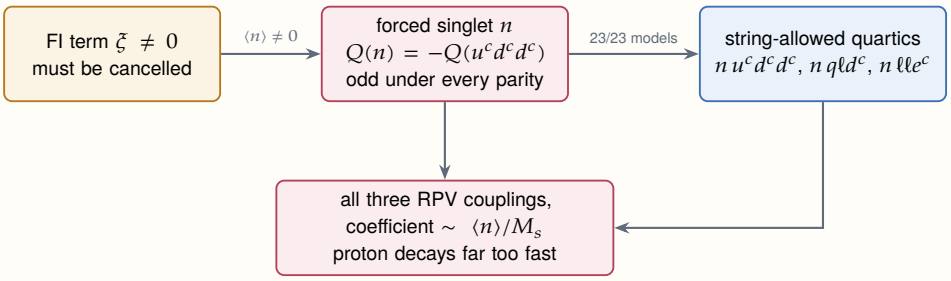


Figure 31. Why supersymmetric vacua of the class violate R-parity. The only fields that can cancel the Fayet–Iliopoulos term carry exactly the conjugate charge of the three dimension-four operators. Every string selection rule allows the resulting quartic couplings, so the vacuum turns the cubic operators back on.

Could heavy superpartners make it harmless? Another part of this programme argues that the geometry's supersymmetry is *structural*, with no light superpartners. Heavy squarks do tame R-parity violation, because proton decay through $\lambda'\lambda''$ falls as $1/\tilde{m}^4$. With $\lambda' \sim \lambda'' \sim \langle n \rangle / M_s \approx 0.2$–$0.4$, the bound $|\lambda'\lambda''| \lesssim 10^{-27}\,(\tilde{m}/100\,\mathrm{GeV})^2$ needs $\tilde{m} \gtrsim 6\times10^{14}$–$1.2\times10^{15}$ GeV. The Higgs sets a ceiling. Running the Standard-Model couplings up at two loops (checked against the published three-loop value at the Planck scale), the Higgs self-coupling turns negative at 3.2×10^{10} GeV (Figure 32). Supersymmetric matching requires it to be non-negative wherever the superpartners appear. Pushing that scale to 10^{15} GeV would need a top quark of at most 171.0 GeV, 5.4 standard deviations below the measured 172.57 ± 0.29 GeV.
COMPUTATION (`w33_pass10969_rpv_heavy_superpartners_vs_higgs_quartic.py`)

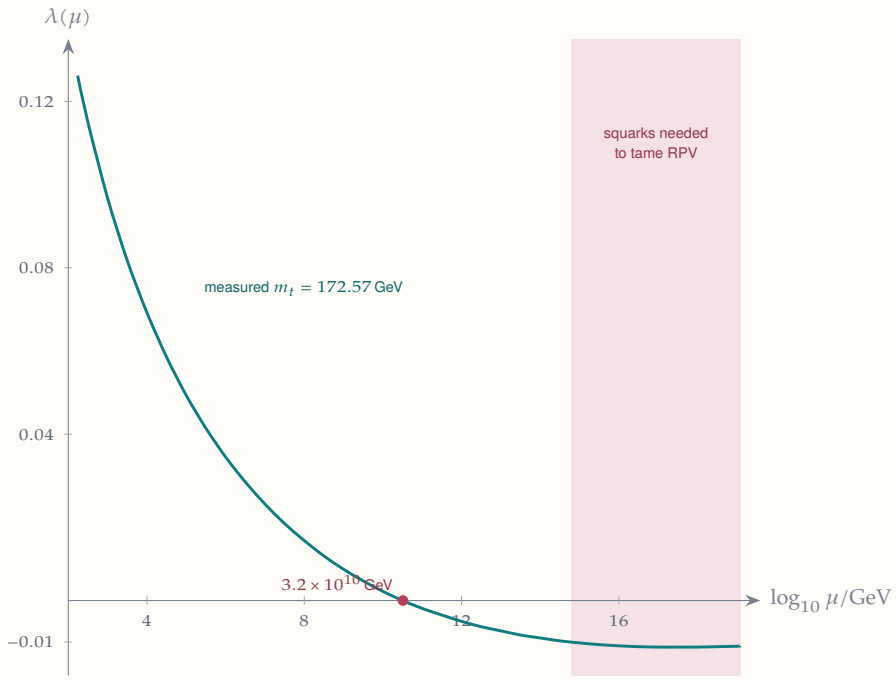


Figure 32. The Higgs self-coupling run to high energy (two loops, NNLO matching). It turns negative near 3×10^{10} GeV, far below the $\gtrsim 6\times10^{14}$ GeV that heavy squarks would need to make the regenerated R-parity violation harmless (shaded).

The last door: a parity made of rotations — and why it closes. The discrete groups tested so far live in the gauge lattice and the space-group rules. The orbifold also has *rotational* symmetries, turning each internal plane by its own angle, and these act on the supersymmetry charges (*R*-symmetries). Combinations of them that leave the superpotential alone can act as a matter parity. With the rules built into the standard orbifold code, this reopens exactly one model, $\mathbb{Z}_6\text{II}_{23}$, and the model then shows a striking two-sided trade: vacua kept flat by an unbroken rotation keep the exotic colour triplets massless at every order (202 found, 202 with massless exotics), while larger vacua that break the rotation make every exotic heavy but either lose the light Higgs pair or keep it only at the price of a superpotential that is not zero:

vacuum (old rotation rules)	flat by symmetry	exotics	a light Higgs pair
rotation unbroken (202 found)	yes	massless	protected
rotation broken, 68 directions	no	heavy	not protected
rotation partly broken, 23 directions	no	heavy	protected

Even the best of these, the certified 23-direction vacuum, would not have described our world: the Higgs pair it protects is protected by an exact symmetry that also forbids any mass for its superpartners (massless Higgsinos, long excluded), and although its top-quark coupling is of order one, its down-quark and lepton couplings show no hierarchy at all.

Then the rules themselves were checked. Bizet, Kobayashi, Mayorga Peña, Parameswaran, Schmitz and Zavala derived the *R*-charge selection rule directly from the symmetries of the worldsheet instantons that mediate string couplings, and found that the textbook rule — conservation of the rotation charge plane by plane — holds only for planes whose rotation has prime order. On a plane of order six the rule acquires an extra contribution from the phases with which twisted strings at different fixed points combine. The door of $\mathbb{Z}_6\text{II}_{23}$ had used the uncorrected rule on its order-six plane. Keeping only the rotation symmetries that are established (the order-3 and order-2 planes), and separately adding the corrected order-six charge, the model closes: all 304 Fayet–Iliopoulos cancelling directions break the parity. The whole $\mathbb{Z}_6$-II class is then excluded, 0 of 128, and three further orbifold families built on the same twist, $\mathbb{Z}_2 \times \mathbb{Z}_6$ (29 Standard Models), $\mathbb{Z}_3 \times \mathbb{Z}_6$ (5) and $\mathbb{Z}_6 \times \mathbb{Z}_6$ (10), have no surviving parity either, and $\mathbb{Z}_2 \times \mathbb{Z}_6$-II produces no Standard Model at all. For $\mathbb{Z}_{12}$-I, whose lattice is not a product of planes, the admissible rule follows from the symmetries of the E_6 lattice itself: the only rotation commuting with the twist beyond the twist's own powers is a half-turn of one plane, and with it every one of the 14 candidate vacua is decided — eleven keep exotic triplets massless, and the most promising one, which does protect a single Higgs pair, has a singlet entering its superpotential linearly at fourth order whose absorption destroys that protection. A final escape — that flavour structure might make the regenerated couplings tiny for the first generation, where the proton is sensitive — fails as well: the first-generation couplings stay at 10^{-3}–10^{-1}, far above the 10^{-10} the proton would need.

The one family in which every rule is beyond doubt is the last to be told. In $T^6/\mathbb{Z}_3 \times \mathbb{Z}_3$ every plane has prime order, so every rotation symmetry the orbifold reports is established. Here, for the first time, the parity *survives* the Fayet–Iliopoulos term: three of the 13 Standard Models keep it on cancelling directions, 36 ways. Each minimal vacuum repeats the familiar pattern — the superpotential vanishes on the vacuum and an exotic triplet stays massless, not one because of the other but both because the vacuum leaves a discrete *R*-symmetry

unbroken (the pattern is common, not a law: P11086 finds $\mathbb{Z}_{12}$-I vacua where it fails in each direction) — but singlets outside it enter linearly at third and fourth order, and absorbing them yields 28 vacua with the parity intact, the exotics massive, the Fayet–Iliopoulos term cancelled with every field switched on, and in 22 of them a Higgs pair massless to all orders.

Are they supersymmetric? On each of these vacua the superpotential turns out to be a function of just one or two composite fields p_i — products of three or four of the vacuum fields — and, to all orders, only the powers with $a_1 + a_2 \equiv 1 \pmod 3$ appear: $W = F(p)$ with $F(\omega p) = \omega F(p)$, a $\mathbb{Z}_3$ rotation symmetry acting on the composites. That is why no two terms can balance below twelfth order ($12 = 4 \times 3$), and it is also why they *can* balance there: the F-terms vanish exactly where $\nabla_p F = 0$, a system of one or two equations with the Bezout number of solutions. Every solution, for random couplings, lifts to an exact supersymmetric vacuum with every field switched on — in all 28 vacua — at field values near the string scale (0.8–1.1 M_s for couplings of order one). The door was real. What closes it is not supersymmetry but charge. A mass term needs a particle and an antiparticle with conjugate charges under *every* gauge group, the hidden ones included; counting both sides class by class, these models carry 48 more states of electric charge $\pm 1/3$ on one side than on the other, six of them neutral under every hidden group. In any vacuum that leaves the hidden sector unbroken — all 28 do — those six are exactly massless, and both hidden groups are infrared free, so no confinement binds them. Breaking the hidden $SU(4)$ is the only way to give them partners; adding any one of its 70 possible meson condensates to any of the 28 vacua never pairs them all. Massless particles of charge $1/3$ are excluded outright. Letting hidden-sector condensates into the vacuum itself is the last thing to try, and the counting says exactly which ones can help: the charge-$\frac{1}{3}$ excess disappears if and only if one particular hidden factor is broken. Such vacua exist — 54 of them keep the parity with the hidden $SU(4)$ condensed, all supersymmetric, all with exactly the Standard Model's quarks and one Higgs pair light — and in every one the excess is gone. Yet the couplings that would pair the balanced states do not exist: at least 120 fractionally charged particles stay massless in each, and most of them are protected by an exact discrete symmetry the vacuum leaves unbroken (always a 3-group, of order 3^4 or 3^6), so no correction of any kind can give them mass. A sibling model offers a near miss in which the needed condensate pairs every class in number, but its parity cannot survive two condensates of the same hidden group. **COMPUTATION** (`w33_pass10979_z3xz3_parity_vacua_f_obstruction.py`, `w33_pass11024_z3xz3_susy_vacua_fractional_charge_index.py`)

Four questions remain about that verdict, and each has an exact answer.

What protects those massless states? Not one symmetry. The unbroken group is assembled from three kinds of remnant: of the broken $U(1)$s, of the translations of the orbifold's space group, and of its discrete R-rotations. Almost half of the forbidden mass terms are forbidden twice over, so there is no single symmetry whose breaking would release them.

Would a different condensate help? No. The census was rerun over every hidden invariant, including the baryonic ones and mixtures of two kinds. It finds 63 further parity-preserving vacua in which the curing factor is condensed, some with both $SU(4)$s broken. Every one keeps at least 142 fractionally charged multiplets massless.

Could smoothing out the orbifold help? No. The fractional part of electric charge turns out to be a character of the orbifold's space group. On every one of the 946 fixed points of all 13 models, the class $3Q + t \pmod 3$ is constant (t is the colour triality), and it is a single

linear function of the fixed point's labels. The 516 fixed points where it is nonzero carry only fractionally charged states. So no neutral field lives there, and they cannot be blown up without breaking colour or electromagnetism. Even discarding every discrete selection rule and keeping only gauge invariance leaves at least fifteen fractional multiplets massless in every vacuum.

Would giving up supersymmetry help? Not here.

- The obstructions concern fermions and gauge charges, which supersymmetry breaking does not touch.
- The $\mathbb{Z}_3$ orbifolds admit no non-supersymmetric twin with the same order-three shifts. For every order-three shift of E_8, $9V^2$ is even, while attaching $(-1)^F$ to an order-three twist makes $9v^2$ odd, and modular invariance needs the two to agree. A non-supersymmetric model on the $\mathbb{Z}_3$ lattice needs a shift of order six, which makes it a different model.
- The one $W(3,3)$ family whose Standard Models do have such twins is $\mathbb{Z}_6$-I: all 87 of them, with the twist $(\frac{1}{6}, \frac{1}{6}, \frac{2}{3})$ of Font and Hernández [14]. Without superpartners they have no dimension-four or dimension-five proton-decay operators.

We computed the 87 twins after repairing five faults that stop the standard spectrum program at $N = 0$. Every twin is anomaly-free up to a single Green–Schwarz $U(1)$, keeps its parent's gauge group, and has exactly three net generations. And every one has tachyons at its θ fixed points. The reason is one level-matching fact: the tachyonic right-moving level lies exactly $\frac{1}{6}$ below the massless one, and $\frac{1}{6}$ is the smallest twisted oscillator mode. So a fixed point of the twin is tachyonic exactly when its supersymmetric parent has a massless oscillator-excited state there, and all 87 parents do. Orbifolds of the separate non-supersymmetric $SO(16)\times SO(16)$ string, which has its own software [12], are a different construction and are not covered by this statement. COMPUTATION (`w33_pass11087_unbroken_three_group_protection.py`, `w33_pass11088_baryonic_hidden_condensates.py`, `w33_pass11089_w33_twist_without_supersymmetry.py`, `w33_pass11090_resolution_cannot_free_fractional_charges.py`, `w33_pass11092_nonsusy_z6i_twins_are_tachyonic.py`)

The same level-matching fact decides the whole $\mathbb{Z}_6$-I scan, not only its Standard Models.

- For 47 of its 58 base shifts, the fixed point without a Wilson line is already tachyonic, so every model built on those shifts is tachyonic whatever its Wilson lines.
- On the other 11 shifts we drew 66 000 random Wilson lines. 9 576 twins are tachyon-free and 705 parents are Standard Models, and no model is both. If the two properties were independent, about 79 would be.

The spectrum shows why. The Standard-Model Wilson lines always leave half-charged exotic states one oscillator step above the tachyonic level, and those states become the tachyons of the twin. Every tachyon carries electric charge $\pm\frac{1}{2}$ or colour, so any condensate of them breaks electromagnetism. The twins of the other $W(3,3)$ families behave the same way (333 Standard Models of $\mathbb{Z}_{12}$-I, $\mathbb{Z}_2\times\mathbb{Z}_6$-I, $\mathbb{Z}_3\times\mathbb{Z}_6$ and $\mathbb{Z}_6\times\mathbb{Z}_6$): none is both tachyon-free and a three-generation Standard Model. COMPUTATION (`w33_pass11093_tachyon_free_filter_z6i_scan.py`, `w33_pass11094_tachyons_break_electromagnetism.py`, `w33_pass11096_twins_of_the_other_w33_families.py`)

The separate non-supersymmetric string does better. In the $SO(16)\times SO(16)$ heterotic string [12,

39] the $(-1)^F$ twist is a freely acting $\mathbb{Z}_2$ of its own, with its own gauge shift V_0.

- The $W(3,3)$ shift of $T^6/\mathbb{Z}_3$, the A_8 Kac pair, admits 793 508 such V_0, in six gauge-group classes. The Witten shift always breaks the $SU(9)$.
- On the four classes the code accepts we drew 80 000 random Wilson lines. Every model is tachyon-free, and 153 of them (104 inequivalent) pass the Standard-Model test of the non-supersymmetric orbifolder.
- We re-derived every one of them with an independent engine. Each is tachyon-free and anomaly-free, and has exactly three chiral generations, no net fractionally charged fermion, and at least 6 Higgs-doublet scalar fields.

These are non-supersymmetric, tachyon-free, three-generation string models built on the $W(3,3)$ twist. Three things remain to be shown: masses for their vector-like exotics, some of them fractionally charged; a stabilised vacuum, which requires the one-loop cosmological constant and dilaton tadpole of a non-supersymmetric string; and realistic Yukawa couplings. Standard-Model-like models of this string are known in large numbers [39]; what is new here is that the $W(3,3)$ shift is among them. Two independent codes agree on the massless spectra of every sample model the non-supersymmetric orbifolder ships. COMPUTATION (`w33_pass11095_so16xso16_a8_standard_models.py`)

How good are these models? We asked the three questions that decide a string vacuum.

Can the fractional charges be removed? Unlike in $\mathbb{Z}_3 \times \mathbb{Z}_3$, they are not protected by a symmetry. Let the Standard-Model-neutral, hidden-singlet scalars take expectation values, and impose the full set of string selection rules (gauge, the Witten $\mathbb{Z}_2$, point group, space group and H-momentum).

- Masses for *all* fractionally charged fermions are allowed in 33 of the 104 models.
- In 6 of those 33, a single scalar suffices, through cubic couplings. In every one of the 33 all such masses appear by order 7.

Is the top heavy, and alone? In 73 models, no. Their up-type Yukawas are untwisted and carry the ϵ-structure of the ten-dimensional gauge coupling. The singular values are $(1,1,0)$ for every Higgs direction: top and charm degenerate, the up quark massless. In the other 31 the quarks and Higgs doublets sit at fixed points. A light Higgs localized at one fixed point then gives a single heavy top, with charm and up suppressed by $e^{-\mathrm{area}}$. The two are degenerate at tree level, because the three fixed points of the $\mathbb{Z}_3$ torus form an equilateral triangle; this is the known two-valued structure of $\mathbb{Z}_3$ Yukawas. In 27 models the bottom behaves the same way. Twelve models pass every test so far: they are tachyon-free, have three generations, can give all their fractional charges masses (at third order), and have a heavy top and bottom from a localized Higgs.

Is the vacuum stable? We recomputed the ten-dimensional $SO(16)\times SO(16)$ one-loop integral: $\Lambda_{10} = +363\,M^{10}$, positive, as Alvarez-Gaumé, Ginsparg, Moore and Vafa found [2]. In every model the large-volume vacuum energy is one third of it times the volume, and the massless Bose–Fermi counts are never degenerate. The potential runs away, as it does for every symmetric non-supersymmetric orbifold [20].

The $W(3,3)$ models of the $SO(16)\times SO(16)$ string thus remove the proton-decay and fractional-charge obstructions of the supersymmetric routes. Twelve of them even single out the top,

but charm and up are degenerate at tree level. All of them meet the universal vacuum-energy problem of non-supersymmetric strings. The other $W(3,3)$ geometries give no tachyon-free Standard Model at all: 77 600 draws on $\mathbb{Z}_3 \times \mathbb{Z}_3$, $\mathbb{Z}_6$-I and $\mathbb{Z}_{12}$-I. COMPUTATION (`w33_pass11097_fractional_fermion_masses.py`, `w33_pass11098_yukawa_textures.py`, `w33_pass11099_vacuum_energy_so16_a8.py`, `w33_pass11100_so16_other_w33_geometries.py`, `w33_pass11101_localized_higgs_heavy_top.py`)

Can the scalar vacuum split charm from up? Extending every Yukawa entry to all orders with the same scalars that give the fractional charges their masses, the answer is no. In all 12 survivors, and for each of the 72 Higgs doublets with a tree-level coupling, the charm–up (and strange–down) block is symmetric at every order under the point reflection of the generation line of fixed points about the top's. The two lighter quarks of each type stay parametrically degenerate; a hierarchy would need a fine-tuned cancellation. COMPUTATION (`w33_pass11102_higher_order_quark_hierarchy.py`)

The reflection has a name. The symmetry group of the $\mathbb{Z}_3$ space-group rule on one torus, found by brute force over every permutation and phase of the three fixed points, has order 54: it is $\Delta(54)$, the flavour group of a Wilson-line-free $\mathbb{Z}_3$ torus [27]. Its Heisenberg subgroup $H_{27} = \langle X, Z \rangle$ (translation of the fixed points and the phase of the selection rule) has as its centre the orbifold twist itself, $ZX = \omega^1 XZ$, and the reflection is the $-I$ of $SL(2,3)$ acting on $H_{27}/Z = \mathbb{F}_3^2$. This is the qutrit Heisenberg group of the E_8 chapters, now realised on the three generations of an explicit string vacuum—with only $\{\pm I\}$ of its Clifford normaliser (index 12), because the space-group rule forbids quadratic phases. In all 104 models exactly one torus is free of Wilson lines, so the geometry realises one qutrit and never two: the two-qutrit Pauli group, whose commutation graph is $W(3,3)$, is not the flavour group of these vacua. A Higgs localised at one fixed point leaves an S_3 acting irreducibly on the other two generations, so by Schur's lemma $m_c = m_u$ exactly while it is unbroken.

Breaking it spontaneously—giving the twisted scalars at the three fixed points different expectation values—does not help the up sector: for every alignment we tried (six patterns, small and large torus area, hidden group unbroken or broken), charm and up stay at the same order in all 36 Higgs cases. The strange–down pair splits only through mixing with vector-like $\bar{d}$ states, in exactly the 8 models that have them. The heavy generation fares better. Top and bottom share a doublet only if H_u and H_d sit at the same fixed point; then $V_{tb} \approx 1$, $|V_{cb}| \sim |V_{ub}| \sim \epsilon^2$ and the τ is singled out—but $V_{cb} \sim m_c/m_t$, a factor of ~ 10 below the data, and μ and e are degenerate like charm and up. COMPUTATION (`w33_pass11103_reflection_breaking_vacua.py`, `w33_pass11104_ckm_and_leptons.py`, `w33_pass11105_delta54_flavour_join.py`)

Finally, the vacuum energy at the orbifold point. We built the full one-loop integrand: all 36 sectors of $\mathbb{Z}_2^W \times \mathbb{Z}_3$ with Wilson lines, the twisted sectors as 24 modular images of one $\Gamma_1(6)$-invariant block per fixed point, the Witten sectors as sums over the Narain lattice. It is modular invariant to 10^{-12}, reproduces the ten-dimensional integral and the large-volume limit, and its massless coefficient matches the spectrum sector by sector (which exposed that the earlier Bose–Fermi counts had omitted the $U(1)$ gauge bosons; corrected, they are never zero). The Kähler modulus of the family torus is stabilised at its self-dual point $T = \rho$: the potential in that direction is exactly $SL(2,\mathbb{Z})$-invariant and is lowest there. The Wilson-line moduli are not: the potential falls monotonically as their tori shrink, stays positive, and below a critical radius a winding state of the Witten-twisted sector becomes tachyonic—in every one of the 12 survivors, at a radius between 2 and the $SU(3)$ point. The instability of

the $O(16) \times O(16)$ string on small tori [18] is where these vacua roll. COMPUTATION (w33_pass11106_one_loop_orbifold_point.py)

Built state by state, the tachyon has the right-moving ground state of the Witten-twisted sector and a winding in the Wilson-line tori; its level at $\mathrm{Im}\, T = 1$ is $\Delta = -(3-\sqrt{3})/6$, which the one-loop integrand confirms. The critical radius is $\sqrt{3}$ or 1.51 (one model is safe on the whole imaginary axis and fails only at ρ). Every tachyonic state is a colour singlet, and in 10 of the 12 models every one carries fractional electric charge $(\pm\frac{1}{3}, \pm\frac{2}{3})$: a condensate would break electromagnetism, the disease of the non-supersymmetric twins again. Over the moduli the potential can reach, it has no minimum in the tachyon-free region: $\Lambda > 0$ at all 20 sampled points of model 2, lowest right at the tachyon boundary, and in the Einstein frame the dilaton runs to weak coupling. Among all 104 models only one stays tachyon-free down to the $SU(3)$ point, and it fails the earlier tests (massless fractional charges, top = charm, no down-type Yukawa); a fresh scan of 387 new models finds 4 more—all five with untwisted quarks, hence top = charm and no down-type Yukawa. Classifying all 491 quark sectors gives 0 of 152 twisted-quark models safe against 5 of 339 untwisted: suggestive, not significant (Fisher $p = 0.16$). COMPUTATION (w33_pass11107_winding_tachyons.py, w33_pass11108_tachyon_free_moduli_rescan.py, w33_pass11110_moduli_map.py)

The stabilised family modulus also undoes the heavy top. Worldsheet instantons make the $\mathbb{Z}_3$ Yukawa couplings theta functions of the A_2 lattice in T [31]; the one-loop minimum $T = \rho$ is a fixed point of the duality group, where $|Y_{\text{distinct}}/Y_{\text{same}}| = \frac{1}{2}$ exactly. At tree level the vacuum therefore gives $m_c = m_u = m_t/2$; the observed m_c/m_t needs $\mathrm{Im}\, T \approx 3.2$, where the potential is 2.4 times higher. BRIDGE (w33_pass11109_instanton_yukawas_at_rho.py)

Nor can the generations carry more than one qutrit. The k-qutrit Pauli group 3^{1+2k} acts non-trivially through its centre only in dimensions divisible by 3^k; the centre here is the orbifold twist, so three twisted generations force $k = 1$—the single free torus of all 104 models is a theorem, not an accident. The two-qutrit group whose commutation graph is $W(3,3)$ cannot be the family group, and it cannot be assembled from family and colour either: the twist is not a gauge-centre element on the quarks of any survivor. BRIDGE (w33_pass11111_no_243_family_group.py)

Three further attempts, and one opening. *The Higgs direction.* With a general light Higgs in the $\Delta(54)$ triplet the up-type mass determinant is a member of the Hesse pencil $x^3 + y^3 + z^3 - 3\lambda xyz$ in the Higgs direction (forced by $\Delta(27)$, whose invariant cubics are exactly that pencil—the other track's Hesse configuration is the geometry of the quark masses), and the modulus picks the member: at the stabilised point $T = \rho$ it is exactly the singular one, $\lambda = \omega^2$. Yet no alignment beats a localised Higgs: the second singular value never falls below $|Y_{\text{dist}}/Y_{\text{same}}| = \frac{1}{2}$ of the first. *Duality.* Along the zero-B axis of the Wilson-line tori the potential must turn around at small radius, but the turnaround never lands in a tachyon-free window: one model's duality extremum is its tachyonic self-dual radius, and the survivor safe furthest down rolls into a charged tachyon. *The other track's groups.* The centre of the family H_{27}—the orbifold twist—acts on the massless spectrum as a gauge element in all 104 models (as the anomalous-$U(1)$ $\mathbb{Z}_3$ itself in three), cf. the gauge origin of $\Delta(54)$ [6]; no $SU(3)$ factor, visible or hidden, can be glued to it, so the two-qutrit group is realised on no matter at all.

The opening. In the two survivors with neutral winding tachyons, the neutral ones wind in one Wilson-line torus and the charged ones in the other, and the one-loop potential pulls

the neutral torus down faster. The first instability these vacua meet is therefore Standard-Model-neutral: its condensation breaks extra $U(1)$s only, and electromagnetism survives. Where it ends is the open question. COMPUTATION (w33_pass11112_neutral_tachyon_corridor.py, w33_pass11113_rho_tachyon_correlation.py, w33_pass11114_hesse_mass_determinant.py, w33_pass11115_axis_duality_and_window.py, w33_pass11116_one_qutrit_vs_other_track.py)

Following the opening further. The condensate breaks exactly one $U(1)$ (a combination with a component along the anomalous one), leaves the Standard Model and every hidden non-abelian factor intact, and is a singlet of the family $\Delta(54)$—so it cannot help the flavour problem; no cubic term exists, so the instability is Higgs-like. The full gradient flow over both Wilson-line radii of model 57 reaches the neutral boundary from equal radii, but the margin is narrow: a 1.5% head start for the charged torus reverses it. Across all 491 models, 60 place their neutral tachyons on a Wilson-line torus of their own (38 with no charged tachyon on either torus alone), and every one of them has twisted quarks, the heavy-top class; and 19 of the new ones pass every earlier test—fractional charges removable, a single heavy top and bottom from a localised Higgs—giving 21 survivors with a neutral exit. In 8 of them only neutral tachyons appear even when both Wilson-line tori shrink together, and in 6 the whole zero-B direction stays neutral down to $\mathrm{Im}\,T = 0.6$: charged tachyons need a B-field at small radius or the $SU(3)$ point, far below the neutral onset at $\mathrm{Im}\,T \approx 1.5$–$1.7$. For these six the first instability the potential leads to is Standard-Model-neutral with a wide margin, not a knife-edge; their open problems are the class-wide ones (family degeneracy, positive vacuum energy). No closed-form lattice criterion decides the $SU(3)$-point tachyon: the gauge cells give a sharp necessary condition, but the torus offsets are needed too. COMPUTATION (w33_pass11117_neutral_condensate.py, w33_pass11118_wilson_line_flow.py, w33_pass11119_neutral_tori_across_491.py, w33_pass11120_rho_cell_criterion.py, w33_pass11121_hesse_gauge_bridge.py)

Following the six to the end of the opening. Along the zero-B diagonal their one-loop potential falls monotonically to the neutral onset with no barrier; it is a volume law, $\Lambda_\beta \approx 160\,y^2 - (5\text{–}13)$ in units of M^4, identical across the six, so the drive is the large-volume Bose–Fermi term and the approach is steep ($d\Lambda/dy \approx 600$), not slow roll. The critical radii are exact: $y_c = \sqrt{3}$ in two models and $\varphi^2/\sqrt{3}$ (φ the golden ratio) in four, where the neutral states have $P_L^2 = 2$, $p_R^2 = 1$; no pair of them makes a massless vector, so the onset is not an enhanced-symmetry point. Their tree-level four-point amplitude is a Kawai–Lewellen–Tye beta function whose low-energy limit is pure exchange, $4C\pi(1/t + 1/u) + 3C\pi s^2/(tu)$, with no contact quartic, and the tachyon mass falls as the radius shrinks ($dm^2/dy > 0$): condensate and radius run together, into the winding-tachyon regime where the cycle pinches off (Adams–Liu–McGreevy–Saltman–Silverstein), outside perturbative control. A B-field does not change the order: charged tachyons need both $B \neq 0$ and $\mathrm{Im}\,T \leq 1$, always at least 0.4 below the neutral onset, and in the four golden models $B \geq 0.4$ removes the neutral tachyon at $\mathrm{Im}\,T = 1.4$. The condensate carries no quantum numbers on the family torus, so it cannot move T^* from ρ or touch $m_c = m_u$. What it does change is the selection rules: its charge lies outside the span of the singlet vacuum, and every Yukawa target in the new direction needs exactly one condensate insertion. With it, charged-lepton, down-quark and neutrino-Dirac entries forbidden at every order become allowed (orders 2–6, 1863 entries, 1854 with a twisted field), while up-type Yukawas and the μ-term stay forbidden. These are individual entries of matrices that already have allowed ones: the

light doublets have tree-level lepton and down Yukawas (see below), so this is not a new source of the missing masses, as first read. A first encoding that treated the winding as an exactly conserved $\mathbb{Z}_3$ found nothing; that zero was an artefact, since winding modulo $(1-\theta)\Lambda$ is the fixed-point class. COMPUTATION (w33_pass11122_six_potentials.py, w33_pass11123_critical_point_amplitude.py, w33_pass11124_condensate_unlocked_couplings.py, w33_pass11125_bfield_charged_onset.py, w33_pass11126_family_modulus_under_condensation.py)

Where the unlocked entries matter, and where the opening ends. For the three light Higgs doublets (those with a tree-level down coupling) the lepton and down textures are $[0,1,1]$ with and without the condensate, and the cubic lepton entries are symmetric under the point reflection that forces $m_c = m_u$, so $m_\mu = m_e$ survives. But in a non-supersymmetric model with one light Higgs h—the doublet that gives the top—the bottom and tau come from h^*, and h^* is not a tree-level down doublet: in all six the conjugates of the three top doublets are exactly the heavier doublets. There the condensate lowers m_b/m_t from ϵ^3 to ϵ^2 in four models and brings the tau to the same order, so $m_b/m_\tau = O(1)$ with $\epsilon \approx 0.12$–0.15, i.e. $\langle T\rangle\langle S\rangle/M^2 \approx 0.01$–$0.02$; the light generations stay degenerate. Across all 21 neutral-exit survivors the condensate never unlocks an up-type Yukawa or a μ-term (down entries in 14, lepton in 17). The endpoint itself is not a nearby vacuum: shrinking the neutral torus deepens the tachyon along an exact winding law, $p_R^2 = y_1/\sqrt{3}$ (plus $1/(3\sqrt{3}\,y_2)$ in the golden models), whose diagonal root $y^2 - \sqrt{3}\,y + \frac{1}{3} = 0$ is where the golden radius $\varphi^2/\sqrt{3} = (3+\sqrt{5})/(2\sqrt{3})$ comes from; as $y \to 0$ it approaches the $\Delta = -\frac{1}{2}$ ground state of the T-dual tachyonic string (cf. Fraiman–Graña–Parra De Freitas–Sethi), a single complex species that exists only through winding, with no tachyon-free island. The B-field is a near-flat direction ($|\partial_B\Lambda| \lesssim 5\times10^{-5}|\partial_y\Lambda|$), with $B = 0$ a shallow minimum just above the onset. COMPUTATION (w33_pass11127_lepton_texture_under_condensation.py, w33_pass11128_pinched_cycle.py, w33_pass11129_offaxis_potential.py, w33_pass11130_single_higgs_bottom_tau.py, w33_pass11131_unlocked_across_21.py)

The geometry of the exit is exact. On the diagonal of the two Wilson-line moduli $T = B + iy$ every tachyon domain is a hyperbolic disk centred at a special point of $\Gamma_0(3)$: the charged one at the elliptic point $(3+i\sqrt{3})/6$ with radius $\ln(2+\sqrt{3})$, the neutral one either the horodisk $\mathrm{Im}(-1/T) > 1/\sqrt{3}$ at the cusp 0 or the disk at the Fricke point $i/\sqrt{3}$ with radius $2\ln\varphi$—which is where the golden ratio comes from. The neutral top lies at least 0.197 above the charged one for every B, so the exit is SM-neutral on the whole B circle. Its consequence for flavour is now class-wide: with one light Higgs the bottom falls to ϵ^2 in exactly the 14 of 21 models with unlocked down entries, with the tau at the same order for at least two of the three candidate Higgs, so $m_b/m_\tau = O(1)$ with $\epsilon_S\epsilon_T \approx m_b/m_t$. The light generations do not follow: every entry of the conjugate sector has the same options, so no ratio of the two VEV scales splits them (anarchy). The size of the condensate is not fixed: the supersymmetric FI formula would give $\langle T\rangle/M_P = 0.15$–$0.20$, close to what is needed, but without supersymmetry there is no FI potential for a complex scalar, so this is a coincidence of scales, not a prediction. COMPUTATION (w33_pass11132_bottom_tau_across_21.py, w33_pass11134_dual_tachyon_species.py, w33_pass11135_fi_scale_no_mechanism.py, w33_pass11136_tachyon_domains.py, w33_pass11137_neutral_exit_scorecard.py)

The end of the exit and the flavour question, closed from four sides. The duality group of the Wilson-line modulus is $\Gamma_0(3)$: $T \mapsto T/(3T+1)$ and the order-three elliptic element are

exact symmetries of the partition function, S and $\Gamma^0(3)$ are broken, and the Fricke map is only approximate. Its one elliptic point is exactly the centre of the charged tachyon disk and its cusp at 0 is tachyonic in 15 of 21 models (in the other six it is the approximate Fricke image of the large-volume region, below the golden disk), so every point the symmetry singles out is tachyonic or a runaway: the Wilson-line modulus cannot be stabilised at a self-dual point. At zero radius of the neutral torus the massless vectors form $\mathfrak{su}(16) \oplus \mathfrak{u}(1)$ in all 21 models — the non-supersymmetric $U(16)$ heterotic string, whose $2^1 = 2$ tachyons are exactly the two norm-one states found there (Fraiman, Graña, Parra De Freitas and Sethi [51]) — so the endpoint of the neutral condensation, perturbatively, is that string. Read thermally, the exit is a Hagedorn transition that exists only because of the Wilson-line holonomy: with the Wilson lines switched off no Witten-sector tachyon appears at any radius. On flavour: no pattern of singlet expectation values (108 random hierarchies) and no two-doublet mixture of Higgs fields (576 tried) splits the light generations; the degeneracy is structural. A new hypothesis — that the condensate makes the neutrino Dirac matrix anarchic, the known route to large lepton mixing — fails: the light-Higgs Dirac texture is the up-quark one at tree level ($m_D(\nu) \sim m_u$ in texture, as in $SO(10)$) and is untouched by the condensate, while the anarchic conjugate sector that would give large lepton mixing gives large quark mixing too. COMPUTATION (`w33_pass11138_singlet_vev_hierarchies.py`, `w33_pass11139_wilson_line_duality_group.py`, `w33_pass11140_zero_radius_u16.py`, `w33_pass11141_no_symmetric_stabilization.py`, `w33_pass11142_higgs_mixing_up_sector.py`, `w33_pass11145_neutrino_anarchy.py`, `w33_pass11146_thermal_reading.py`)

Two refinements. The zero-radius vector lattice is identified, not only counted: it is even, of rank 16 and determinant 4 with the hyperbolic discriminant form on $\mathbb{Z}_2^2$ — the genus of the vector lattices of the non-supersymmetric heterotic strings, which that genus exhausts [51] — and its root system is A_{15}, so it is the $U(16)$ lattice, in all 21 models. And the tachyon-free region near the cusp at 0 in the six golden-type models is, to 0.3–0.5%, the Fricke mirror of the large-volume region: the potential falls steeply into the golden disk from below, so it holds no minimum either. COMPUTATION (`w33_pass11150_cusp0_potential.py`, `w33_pass11151_u16_lattice.py`)

The apparent opening was an artefact of an incomplete selection rule; it was caught by reading the literature on the rules themselves, not by the computation. COMPUTATION (`w33_pass10974_valid_r_rules_close_the_rotation_door.py`; the conditional analysis is `w33_pass10968_r_symmetry_parity_and_massless_exotics.py`)

9.6 Handedness: host, not select

The weak force acts only on left-handed particles. Can $W(3,3)$ explain why?

Theorem 9.5 (the chirality no-go) THEOREM COMPUTATION

The two half-spin representations $S^{\pm}$ of the D_5 structure built on the geometry are exchanged by the substrate's own outer controller T, which generates $\mathrm{PGSp}(4,3) = \mathrm{PSp}(4,3){:}2 = W(E_6)$ together with $\mathrm{PSp}(4,3)$ and has $\det T = -1$. Consequently no invariant of the full automorphism group of $W(3,3)$ can distinguish S^+ from S^-. `w33_pass346_the_chirality_no_go.py`

In plain words

The forty-point geometry is perfectly capable of *carrying* a left-handed world: the chiral spinors are there (Section 8). What it cannot do is *choose* left over right, because one of its own symmetries swaps them. Any explanation of the handedness of the weak force must come from outside the geometry — from a choice of vacuum, of orientation, or of dynamics. This is the same distinction as in Section 2: the two groups of order 51 840 differ precisely by whether that swapping symmetry is present.

What it means physically: the swap is antiunitary

Which symmetries do the swapping? For the qutrit, the determinant of a symplectic label map decides: determinant $+1$ is a unitary Clifford operation, determinant -1 can only be implemented antiunitarily (Pass 5730, `w33_pass5727_5730_torsion_family_heisenberg.py`). The decisive statement was proved on the 32 orientations of a complete two-qutrit factorisation frame (Holotrade `e92a047`, `frame_chirality_is_reversed_only_by_antiunitaries.py`): they split $16+16$ exactly as the two half-spin weight sets of D_5, unitary Cliffords preserve each half, and antiunitary ones — and nothing else — exchange them. The other track has since found the same law on the clock: determinant-odd ticks exchange the two D_4 half-spin sheets (Pass 10955) and are not completely positive (Pass 10952). So the geometry's chirality is invariant under every *unitary* symmetry and exchanged only by *antiunitary* ones. That is the structure of a chiral quantum theory obeying CPT: the mirror of a chiral world is related to it by an antiunitary map, and which of the two is called "left" is a convention that no internal measurement can fix. The no-go is the finite form of this, not a defect peculiar to $W(3,3)$. BRIDGE (`w33_pass10952_clock_complete_positivity_firewall.py`, `w33_pass10955_d4_halfspin_clock_bridge.py`)

Corollary 9.7 (one coset reverses time, acts antiunitarily, and swaps chirality) THEOREM

Fix a "now" (the Bell line B). Its stabiliser in $\mathrm{PGSp}(4,3)$ has order 1296, and its elements outside $\mathrm{PSp}(4,3)$ form a single coset of size 648. That coset is exactly

(i) the set of symmetries that *reverse* the unique global orientation of the history graph (Theorem 4.4);

(ii) the set that can only be implemented *antiunitarily* on qutrits (Pass 5730);

(iii) the set that *exchange* the two chiral half-spin modules (Theorem 9.5; Holotrade `e92a047`).

Proof. Each item identifies its set with the complement of $\mathrm{PSp}(4,3)$ in the relevant group: (i) is the statement of Theorem 4.4(ii) (the 648 elements of the Bell stabiliser in $\mathrm{PSp}(4,3)$ fix the orientation cycle, the other 648 negate it); (ii) symplectic similitudes of multiplier $+1$ lift to unitary Clifford operations and those of multiplier -1 only to antiunitary ones; (iii) $\mathrm{PSp}(4,3)$ preserves S^+ and S^- (they are inequivalent) while the outer element T exchanges them. Intersecting with the stabiliser of B gives the same coset in all three cases. □

What it means physically: a finite CPT theorem

In quantum field theory, the CPT theorem says that the only way to relate a chiral world to its mirror image by a symmetry is antiunitary, and that the same operation reverses the direction of time. Corollary 9.7 is that structure, exactly, in the forty-point geometry: the operations that reverse time's orientation are the antiunitary ones, and they are precisely the ones that exchange left and right. Choosing an orientation of time does not choose a chirality — but it makes chirality a conserved label, and the world of opposite chirality is the time-reversed one. This answers, in the finite model, the question of whether the datum that fixes the arrow and the datum that fixes handedness are related: they live in the same coset. BRIDGE

The coincidence in Corollary 9.7 is forced rather than accidental. The similitude multiplier μ (defined by $M^{\mathsf{T}}JM = \mu J$) maps $\mathrm{PGSp}(4,3)$ onto $\mathbb{Z}_2$ with kernel $\mathrm{PSp}(4,3)$. Because $\mathrm{Sp}(4,3)$ is perfect — its commutators generate all 51 840 of its elements — μ is the *only* nontrivial sign character on the symmetries of the forty points. Holotrade had already made this identification (`5419c27`, `98cafff`): frame orientation parity is the index-two character of $W(E_6)$, and the Pass 346 half-spin swap is the same character. It also separated this relative chirality from two other $\mathbb{Z}_2$ labels that are *not* characters: the point/line type, which is invariant and can be selected, and the local charge conjugations of the 45 factorisations. The other track later found two further labels. Its cubic "tick" orientation is again μ. Its chamber-sheet sign is flipped by elements with $\mu = +1$, so it is defined only relative to a chosen flag. So however the geometry is probed, it has exactly one global sign to fix: which chirality goes with which direction of time. THEOREM COMPUTATION (`w33_pass11091_one_global_orientation_bit.py`)

9.7 What the finite symmetry alone gives: flavour and area

Mixing angles from residual symmetry. The principled way for a finite symmetry to fix flavour is the residual-symmetry method. If the three generations carry a three-dimensional representation and the charged-lepton and neutrino mass terms each keep a subgroup, the mixing matrix is fixed by group theory alone. We applied this to every three-dimensional irreducible representation of every subgroup of $W(E_6) = \mathrm{Aut}(W(3,3))$ and of $\mathrm{Sp}(4,3)$. That is 606 representations with 34 distinct images in $U(3)$, among them A_4, S_4, A_5, $\Delta(27)$, $\Delta(54)$ and the single-qutrit Clifford group $\Sigma(216\times 3)$. We compared every pattern with the current global fit, allowing any assignment of generations.

No pair of residual symmetries fixes all three lepton angles inside their 3σ ranges, for Majorana or Dirac neutrinos. None gives the Cabibbo angle, and none reproduces the older manuscripts' $\sin^2\theta_{12} = 4/13$, $\sin^2\theta_{23} = 7/13$, $\sin^2\theta_{13} = 2/91$. What survives fixes a single column:

- the trimaximal TM_1 column $(2/3, 1/6, 1/6)$, from the S_4 by which the stabiliser of a line of $W(3,3)$ (a measurement context) acts. It predicts $\sin^2\theta_{12} = 0.317$–$0.320$, within 1σ, and $\cos\delta \in [-0.33, 0.45]$;
- the qutrit Clifford group's column $(\frac{2}{3}\sin^2\frac{4\pi}{9}, \frac{2}{3}\sin^2\frac{\pi}{9}, \frac{2}{3}\sin^2\frac{2\pi}{9})$, which forces $\delta \approx \pi$;
- two golden-ratio columns from A_5, and TM_2 from A_4.

These last columns sit 2–3σ from the fit. The symmetry hosts the classic flavour groups and their partial patterns. It cannot fix the mixing matrix: that needs dynamics. COMPUTATION (w33_pass11225_flavour_from_symmetry.py)

TM_1 makes a sharp prediction. Fixing the first column fixes $\cos\delta$ as a function of θ_{23}: $\cos\delta = 0$ exactly at maximal θ_{23}, for every θ_{13}, and the sign of $\cos\delta$ is the octant. Over the current range, $|\delta - 90°| < 26°$ (or the mirror value near 270°). The Clifford column fits only at the upper edge, $\sin^2\theta_{23} \in [0.586, 0.596]$, with $\delta \approx \pi$. Whether dynamics selects TM_1 is a separate question. TM_1 needs a neutrino flavon fixed by exactly one involution, $\mathrm{diag}(1,-1,-1)P_{23}$, i.e. pointing along $(0,1,-1)$.

- One flavon with the most general renormalisable S_4-invariant potential: never.
- One flavon with sextic terms: sometimes $(37/150)$.
- Two flavons with every allowed cross-coupling: in 0 of 1500 random vacua, because the cross-couplings align the two vacuum directions.

This is the known vacuum-alignment problem of flavour symmetry, here for the line stabiliser's S_4. Solving it needs structure the geometry does not supply here. COMPUTATION (w33_pass11230_tm1_alignment.py)

In the geometry the required alignment is an incidence statement. The stabiliser of a line of $W(3,3)$ acts on the line's four points as the full S_4 (with kernel 3^3 times the centre). Relative to the charged-lepton $\mathbb{Z}_3$, which fixes one point X of the line, the three classes of neutrino involution give:

- TM_1 exactly when the involution swaps X with a second point Q of the same line;
- $\theta_{13} = 0$ when it fixes X;
- TM_2 for the double transpositions.

In the triplet $\mathbf{3}'$ the two flavons point along the point itself, d_X, and along the chord $d_X - d_Q$. The chord's stabiliser is exactly $\langle(XQ)\rangle$. Generic potentials miss this pair because it is fixed only by $(RS)\times(-1,-1)$, so TM_1 is a critical point only on a codimension-two set of couplings. In the sequestered limit TM_1 ties 48 : 48 with the excluded $\theta_{13} = 0$ orientation. The only orientation-sensitive quartic invariant that tells them apart is $(\varphi\cdot\chi)^2$, equal to 16 against 0, so a single sign, $\varepsilon < 0$, selects TM_1 up to $O(\varepsilon)$ corrections. COMPUTATION (w33_pass11243_tm1_line_geometry.py) An explicit sequestered model, with sextic-to-octic terms for the chord, confirms this: $\varepsilon < 0$ selects TM_1, $\varepsilon > 0$ the $\theta_{13} = 0$ orientation, and the vacua tilt by $O(\varepsilon)$ (w33_pass11248_sequestered_tm1.py). Adding equivariant Yukawa couplings fixes what this does and does not predict. At $\varepsilon = 0$ the first column is exactly TM_1 for every coupling, while θ_{23} and δ range freely ($\sin^2\theta_{23} \in [0.29, 0.71]$, $\cos\delta \in [-1,1]$ over fits to θ_{13} and the mass splittings); the neutrino mass matrix needs the unique equivariant cubic, without which an accidental reflection freezes the whole mixing matrix. The TM_1 relation $|U_{e1}|^2 = 2/3$ breaks linearly in the vacuum tilts, about 13 and 17 per radian on the neutrino and charged-lepton sides, so keeping $\sin^2\theta_{12}$ inside the 3σ window needs both vacua aligned to about 0.1°, i.e. $|\varepsilon| \lesssim 10^{-3}$ in this potential: the sign of ε selects TM_1, its size is a tuning. COMPUTATION (w33_pass11254_tm1_yukawa_breaking.py)

An area law. Perfect tensors are the building blocks of holographic codes, in which the entropy of a boundary region equals a minimal cut through the network: a discrete Ryu–Takayanagi law. A network built only from the geometry's perfect two-qutrit tick p, whose Choi state is AME$(4,3)$, satisfies it exactly on every boundary interval of a hyperbolic $\{4,5\}$ patch, and so does a $\{5,4\}$ code of AME$(6,3)$ tensors. The same network built from the non-perfect SUM gate fails on more than half the intervals. Flat grids of p up to 5×5 satisfy the law too, so the area law does not select negative curvature, and it does not fix the sign of the cosmological constant. Gravity is hosted here as an exact area law, not derived. COMPUTATION (`w33_pass11226_holographic_area_law.py`)

A dynamical space-time needs the law in time as well. A brick-wall circuit of ticks is itself a tensor network in space-time, so the entropy of any output interval is at most the least number of worldline segments that must be cut to separate it. We computed both exactly on twelve qutrits, through depth ten, for every interval of up to half the chain (570 cases per run). No tick saturates the bound everywhere, so the law in time is approximate, not exact:

- random perfect gates acting on Bell pairs come closest, 548/570, with the half-chain entropy growing at the maximal rate;
- the arrow-carrying three-qutrit tick of Section 4 comes next, 534–539 from product states;
- the bare perfect tick p is fine-tuned: it keeps $|0\cdots0\rangle$ a product state (15/570);
- the non-perfect SUM gate does worst.

The cases that fall short are not confined to the edges of the chain. COMPUTATION (`w33_pass11231_dynamical_area_law.py`)

9.8 The forty points as the level-3 structure of a modulus

The open entries of the scorecard — masses, mixing, and the datum that would choose a chirality — are what a *modulus* supplies in string compactifications: its vacuum value breaks the symmetries, and modular forms evaluated there are the couplings. The geometry already is the level-three structure of such a modulus. Let $A = \mathbb{C}^2/(\mathbb{Z}^2 + \Omega\mathbb{Z}^2)$ be an abelian surface with period matrix Ω in the Siegel half-space. Its nine level-three theta functions $f_c(z) = \theta[c/3, 0](3z, 3\Omega)$, $c \in \mathbb{F}_3^2$, are a two-qutrit register: translation by the 81 points of order three acts as the Pauli group, the Weil pairing on those points is the commutator, so $A[3]$ *is* the symplectic space of $W(3,3)$.

Theorem 9.8 (the Siegel modular group is the Clifford group) COMPUTATION

On the theta-null state $\psi(\Omega)_c = f_c(0)$, the modular map $\Omega \mapsto -\Omega^{-1}$ acts as the qutrit Hadamards $F \otimes F$ (with weight $\frac{1}{2}$), the shear $\Omega_{12} \mapsto \Omega_{12} + 1$ as CZ_3, the shift $\Omega_{11} \mapsto \Omega_{11} + 2$ as a phase gate, exchange of the two tori as SWAP, and $\Omega \mapsto -\bar{\Omega}$ as complex conjugation. Reduced mod 3 these generate $\mathrm{Sp}(4,3)$, the finite Siegel modular group $\Gamma_{2,3}$; on the forty points they act as $\mathrm{PSp}(4,3)$, and adding $\Omega \mapsto -\bar{\Omega}$ (multiplier -1) gives $\mathrm{PGSp}(4,3) = W(E_6)$. `w33_pass11869_11873_siegel_level3_two_qutrits.py`

So the one antiunitary coset of Corollary 9.7 — time reversal, chirality exchange — is the

generalised CP of the modulus. Three consequences are exact.

- **The arrow is the CP sign of the modulus.** For one qutrit and an elliptic curve E_τ, the Hesse cubic through the theta-null state is E_τ itself, $j(E_{\psi(\tau)}) = j(\tau)$; with Pass 11645, $\operatorname{sign} W = \operatorname{sign} \operatorname{Im} j(\tau)$, which vanishes exactly on the CP-conserving locus of modular flavour models.
- **The Hesse space is the space of tori.** Every two-qutrit theta-null state is a singular point of a unique Heisenberg-invariant cubic (the Coble cubic), and its coefficients lie on the Burkhardt quartic; random states are singular points of none.
- **Factorisation is factorisation.** $\psi(\Omega)$ is a product state exactly when the torus splits, $\Omega_{12} = 0$; near such a point its Schmidt spectrum is $(1, \varepsilon, \varepsilon^2)$ with $\varepsilon \propto \Omega_{12}$.

The symmetric points and the order parameters are now computed. The six isolated fixed points of $\mathrm{Sp}(4, \mathbb{Z})$ have stabilisers of orders $10, 24, 24, 32, 48, 72$, all embedded in $\mathrm{Sp}(4, 3)$, and every one is isomorphic to its CP image; at the three product points $(E_i^2, E_\rho^2, E_i \times E_\rho)$ the theta-null texture has rank one (a single heavy generation), at the three Jacobian points (including Bolza's curve) it has rank three with no hierarchy. Near a product point the hierarchy obeys the exact law $\sigma_1\sigma_3/\sigma_2^2 = r(\tau_1)r(\tau_2)/2$, and because this and σ_2/σ_1 would be the same in every fermion sector, one modulus with one theta characteristic is excluded by the measured masses (the shape invariants of up quarks, down quarks and charged leptons differ by a factor 34). The theta-nulls carry the even Weil module (scalar, weight $\frac{1}{2}$) and the odd theta gradients the odd one, with an extra holomorphic index — the two singletons of Section 4. The magic of $\psi(\Omega)$ is a modular function with no interior maximum (its supremum sits at a cusp), so it does not fix the modulus; and ratios of Burkhardt invariants of the Coble point are level-one Siegel modular functions, real exactly on CP-conserving moduli: the arrow of a two-qutrit register is the CP phase of its torus, and two tori with opposite arrows compose to a CP-even one. `w33_pass11874_11878_siegel_fixed_points_cp_magic.py`

Where does the modulus come from? From the $\mathbf{84} = \Lambda^3\mathbb{C}^9$, the $\mathbb{Z}_3$-twisted half of E_8 (Theorem 9.4). Its Vinberg Cartan subspace is the odd Weil module, and its points are exactly the odd-characteristic theta-nulls of a torus: the Cartan $\mathbb{P}^3$ is the moduli of level-three genus-two curves with a marked Weierstrass point, sixfold over the Burkhardt quartic (the degree is Bruin–Filatov's; the theta realisation is ours). In an $SU(9)$ gauge theory with one $\mathbf{84}$, the most general renormalizable potential has only two quartic invariants, $(\operatorname{tr}\rho)^2$ and $\operatorname{tr}\rho^2$ with $\rho = xx^\dagger$; for $\lambda_2 > 0$ its vacua are the Kempf–Ness set $SU(9) \cdot \mathfrak{h}$, so *the genus-two modulus is a classical flat direction*. A generic vacuum breaks $SU(9)$ to a finite group containing the two-qutrit Pauli group — $W(3,3)$ is the commutation geometry of the unbroken discrete gauge symmetry — and at the forty Witting rays, the $W(3,3)$ points, the unbroken group is $SU(3)^3$: trinification. The first holomorphic invariant on the Cartan has degree 12, so the modulus and its CP phase are lifted only by dimension-twelve operators. `w33_pass11879_11883_e8_higgs_siegel_modulus.py` Quantum corrections then choose the point. Along the flat direction $\operatorname{Tr} M^2$ and $\operatorname{Tr} M^4$ of the vector masses are constant and the heavy scalars track the vectors exactly (ratio $8/3$, the moment-map identity), so every bosonic Coleman–Weinberg term is a positive multiple of $\sum_i e_i^2 \log e_i$; its global minimum, $-6\log 3$, sits exactly at the forty Witting rays, where $SU(9) \to SU(3)^3$. The one-loop vacuum is therefore trinification at the $W(3,3)$ points, strictly stable except for the phase of the $\mathbf{84}$, a physical axion-like direction lifted only with the dimension-twelve CP phase. Charged fermions, which enter with the opposite sign, are not included. `w33_pass11884_11886_one_loop_trinification.py` Read

inside E_8, that vacuum is a single $W(3,3)$ point: the centraliser of the vev is $E_6 \oplus \mathfrak{u}(1)^2$, the 248 splits into three **27**s, three $\overline{\mathbf{27}}$s and six singlets whose $U(1)^2$ charges are the weights and roots of a family $SU(3)$, and for each of the forty rays exactly one $W(3,3)$ point has a Pauli operator fixing it, inside that point's $E_6 \times A_2$. The dictionary entry "a point of $W(3,3)$ is $E_6 \times SU(3)$" becomes the output of a vacuum computation, with the Higgs breaking the family $SU(3)$ to its Cartan. Only the $SU(3)^3$ part is gauged in the four-dimensional model; the enhancement to E_6 would need a higher-dimensional (gauge–Higgs) completion. `w33_pass11887_11889_e6_family_at_w33_point.py` Which vacuum is realised depends on the matter, and each regime picks a different distinguished object: bosonic loops a point of $W(3,3)$ (trinification); a negative soft mass in the supersymmetric theory the same point; a positive one the most symmetric pointed Jacobian, the curve $y^2 = x^5 - 1$ at its $\mathbb{Z}_5$-fixed Weierstrass point; in the gauge–Higgs (Hosotani) completion on $T^2/\mathbb{Z}_3$, bosons or Scherk–Schwarz gauginos restore $SU(9)$ while periodic adjoint fermions break to $D_4 \oplus D_4$ at a tensor factorisation of the two qutrits. Moving the vev a distance δ off a trinification ray moves the torus toward its cusp as $\operatorname{Im}\Omega \simeq (3/\pi)\ln(1/\delta)$, so the theta-texture hierarchy parameter is linear in δ. `w33_pass11890_11892_vacuum_phase_diagram.py` In the chiral heterotic realisation ($T^6/\mathbb{Z}_3$, three untwisted **84**s and twenty-seven twisted $\overline{\mathbf{9}}$s, cubic coupling the E_8 bracket) the trinification vacuum is F- and D-flat and leaves $3[(3,3,3)+1]+27[(\bar{3},1,1)+(1,\bar{3},1)+(1,1,\bar{3})]$ massless: three families, one per plane, but of the wrong **27** — every state coloured. No embedding of the weak $SU(2)$ yields three quark doublets without chiral coloured quartets, so this point carries no Standard Model; the trinification bifundamentals that would form one are exactly the states the vacuum eats or lifts. `w33_pass11893_trinification_chiral_spectrum.py` At order six the modulus survives in the family planes (a four-dimensional Kempf–Ness Cartan of the $\mathbb{Z}_6$ grading), and one loop now selects a vacuum whose unbroken group is exactly $SU(3) \times SU(2) \times U(1)$. Its colour, however, is the diagonal of a Georgi–Glashow and a Pati–Salam triplet ($\mathbf{5} = 3 + 2$, $\mathbf{4} = 3 + 1$ on $\mathbb{C}^9$), so the matter carries colour sextets and octets and no lepton doublet: the right gauge group with the wrong matter. `w33_pass11894_11895_z6_sector_selection.py` Adding the flagship's CZ_3 Wilson line restores the Standard-Model gauge group and near-complete families in the family planes, but then every flat direction is carried by Standard-Model-charged fields and breaks $SU(3) \times SU(2)$: in the visible untwisted sector the modulus and the Standard Model exclude each other, which points the modulus to a hidden sector that communicates with visible matter only through moduli-dependent (theta-function) couplings. `w33_pass11896_flagship_tension.py` The geometry itself supplies a neutral carrier. On $T^4/\mathbb{Z}_3 = (\mathbb{C}/\mathbb{Z}[\omega])^2/\mathbb{Z}_3$ the complex structure is frozen and the untwisted moduli are the Hermitian Kähler-plus-B matrix Z (off-diagonal entries included), a point of the Hermitian half-space over $\mathbb{Z}[\omega]$ with duality group $U(2,2;\mathbb{Z}[\omega])$. The nine fixed points are $(\mathbb{Z}[\omega]/\sqrt{-3})^2 = \mathbb{F}_3^2$, and the twisted instanton sums are the theta functions $\theta_\mu(Z) = \sum_{x \in \mathbb{Z}[\omega]^2 + \mu/\sqrt{-3}} e(x^* Z x)$. They obey $\theta(-Z^{-1}) = \det(Z/i)\, F\theta(Z)$, with F the two-qutrit Fourier transform; a diagonal B-shift is a phase gate, an off-diagonal one the entangling gate $\omega^{2c_1c_2}$, and $GL(2,\mathbb{Z}[\omega])$ permutes μ. The finite group generated has projective order 51840 and is isomorphic, by its action on the Pauli group, to $Sp(4,3) = \operatorname{Aut} W(3,3)$. The diagonal moduli alone give only $T' \times T'$ (order 576), so the $W(3,3)$ structure is carried by the off-diagonal Kähler moduli. With the space-group Heisenberg group of two Wilson-line-free tori this is the full two-qutrit Clifford group: the eclectic flavour group of the two tori, extending the one-torus $\Delta(54)$ and T' of Nilles, Ramos-Sánchez and Vaudrevange. Since $\theta_\mu = \theta_{-\mu}$, the couplings span the even Weil **5**. These five weight-one thetas map the moduli onto $\mathbb{P}^4$, and the real-symmetric

(Siegel) slice onto the dual of the Burkhardt quartic, as Freitag–Salvati Manni's description of $\mathcal{A}_2(3)$ requires. On $T^6/\mathbb{Z}_3$ the same construction acts on the 27 fixed points, which carry the twenty-seven $\bar{\mathbf{9}}$s, through $Sp(6,3)$. Kähler moduli are gauge singlets, so this is a Standard-Model-neutral home for the Siegel modulus. It needs two Wilson-line-free tori: the $SU(9)$ sector has them, but the three-generation spectra of Pass 11105 have Wilson lines on two of their three tori. `w33_pass11897_11898_kahler_moduli_two_qutrit_weil.py` Modular-invariant potentials have critical points at the symmetric points, and these form a $W(3,3)$ dictionary. The Burkhardt group has 13805 isolated special points on the coupling space $\mathbb{P}^4$, in eleven orbits with stabilisers $648, 576, 162, 120, 108, 48, 36, 16, 12, 9, 5$. The 40 points of stabiliser 648 are the cusps (large volume); each is fixed together with exactly one line of $W(3,3)$, a maximal commuting Pauli set, in which the couplings become diagonal. The 45 nodes of the Burkhardt quartic fix one tensor factorisation, the 160 a flag, the 240 a point. All ten non-cusp orbits are interior Kähler moduli. Every special point is CP-conserving, while a generic Kähler point violates CP by an amount that falls exponentially toward the cusp ($4 \cdot 10^{-3}, 3 \cdot 10^{-5}, 10^{-7}$ as the volume is scaled by $1, 2, 3$). Distance from the cusp therefore sets both the family hierarchy and the size of CP violation, as for the Siegel fixed points. `w33_pass11899_kahler_special_points_cp.py` Three generations sit over a point of $W(3,3)$. A three-generation $\mathbb{Z}_3$ model needs Wilson lines on two tori and leaves one free family torus. A Wilson line on a second torus labels the fixed points by its holonomy Z_2, a nontrivial Pauli operator and so a point p of $W(3,3)$. The compatible Kähler dualities form its centraliser, of projective order 648: the point stabiliser, which the corpus had already matched to the qutrit Clifford group. The families are the eigenspaces of Z_2, one qutrit each, namely $p^\perp/p$. On the Wilson-line-neutral class the dualities act only as T'. On the two charged classes they alone generate the full one-qutrit Clifford group: the $GL(2,\mathbb{Z}[\omega])$ transvection restricts to the shift X_1^a, and the off-diagonal B-shift to the clock Z_1^{2a}. The "traditional" $\Delta(54)$ of the families is thus what is left of the entangling dualities. For the renormalisable twisted masses this means the following. When left- and right-handed families share a Wilson-line class, the light pair is degenerate exactly, for all moduli, because θ is even. When the classes differ, the pair is degenerate at diagonal moduli and is split by the off-diagonal modulus at third order: each torus rotation fixes its cosets, so the first odd term is $(\bar{x}_1 x_2)^3$. Mixing is still absent at this level (the diagonal selection rule). `w33_pass11900_three_generations_over_a_w33_point.py` Applied to the twelve surviving $SO(16) \times SO(16)$ A_8 models, this closes the Kähler escape route left open for $m_c = m_u$. In all 72 Higgs choices, up and down, the renormalisable twisted triangles sit at one point of both Wilson-line tori: $a = 0$, and only the family torus separates the light entries. Their couplings are $\theta_{(\pm 1,0,0)}(Z)$ of the full nine-modulus Hermitian matrix, and the instanton sum is even under $x \mapsto -x$ on all three tori. So the two are equal for every Z (to 10^{-15}; with $a \neq 0$ they differ at $O(1)$), and $m_c = m_u$, $m_s = m_d$ hold for all Kähler moduli. The escape needs $a \neq 0$ on some Wilson-line torus. Because the Wilson-line classes are common to the three families, the top then pays that torus's instanton factor $\theta_{(0,a)}/\theta_{(0,0)}$, which is $\frac{1}{2}$ at the self-dual point (the normalisation agrees with Pass 11109). A light-pair split is bought with a top suppression, and this is the criterion for the next scan. `w33_pass11901_kahler_loophole_closed.py` The same holds across the whole class. All 104 A_8 $SO(16) \times SO(16)$ Standard Models of the scan fall into two kinds, and each has an up-sector degeneracy that no Kähler modulus lifts at the renormalisable level. In 73 the quark doublets are untwisted, with the families on the three planes. There every up coupling, for each of 219 Higgs rows, is the E_8 cubic $g\varepsilon_{ijk}$, an antisymmetric matrix with spectrum $(1,1,0)$: top equals charm and the up quark is massless. The untwisted

metric $(T + T^\dagger)^{-1}$ normalises by congruence AMA^T, which preserves antisymmetry, so this spectrum holds for the full Hermitian T (a non-antisymmetric control does move). In the other 31 the families are twisted on one torus, and every up triangle is single-pointed on both Wilson-line tori, so $m_c = m_u$ by theta evenness. Only one model has a down sector able to split. The lever for the up-quark hierarchy is therefore not the Kähler sector of this class. `w33_pass11902_a8_census_kahler_independent_degeneracy.py` The reason is local grand unification. The census was widened to the 387 inequivalent models of the 400,000-draw rescan, 491 models in all, with dumps regenerated by engines that reproduce the archived ones byte for byte. In every one of the 152 twisted-family models, u^c and e^c sit at the same point of the Wilson-line tori, and in the same twisted sector, as Q: each family is a complete local $SU(5)$ **10**. If Q and u^c share a point g, the space-group rule forces the up Higgs to $g_H = -2g \equiv g \pmod 3$. The triangle is then single-pointed, and $m_c = m_u$ holds for every Kähler modulus. In prime $\mathbb{Z}_3$ orbifolds, a localised **10** is therefore incompatible with a tree-level charm–up hierarchy. No model escapes in the up sector $(0/491)$. The $\bar{\mathbf{5}}$ is split across points, and this is why the down sector escapes in five models. Splitting charm from up through the $W(3,3)$ modulus requires splitting the **10**. `w33_pass11903_local_ten_locks_charm_up.py` The magnetized home behaves differently. On T^4 with flux 3 on each torus the zero modes are the level-3 Siegel thetas above, and the lock is lifted at first order in two ways. The off-diagonal Siegel modulus Ω_{12} lifts it when $a \neq 0$, at linear order where heterotic $\mathbb{Z}_3$ gives cubic order, because a real period lattice lacks the Eisenstein coset rigidity. A continuous Wilson line lifts it even at $a = 0$: heterotic Wilson lines are quantised Pauli displacements, while magnetized ones are continuous Weyl displacements. A theta-zero law caps how far this goes. $\theta[a,0](z,\Omega)$ vanishes at $z^* = \frac{1}{2} + (\frac{1}{2} - a)\Omega$, and there the other two level-3 thetas have equal modulus: the substitution $n \mapsto -1 - n$ exchanges their sums. With a localised Higgs, an exactly massless up quark therefore forces top = charm. This one statement covers the Hesse alignments of Pass 11114, the symmetric Kähler points of Pass 11899 and the magnetized Wilson-line zeros. The hierarchy comes instead from the near-cusp Gaussian regime. There $m_u m_c / m_t^2 = e^{-2\pi \operatorname{Im}\Omega/9}$, independently of the Wilson-line offset δ, while $m_u/m_c = e^{-4\pi \operatorname{Im}\Omega\, \delta/3}$: the torus area sets the product and the Wilson line sets the split. Up ratios of the observed order are reached at $\operatorname{Im}\Omega \approx 25$, $\delta \approx 0.06$ (two parameters, two ratios: this shows existence, not a prediction). Non-renormalisable couplings in the heterotic survivors give charm and up at equal order (Pass 11103), and their coefficients differ only through the $O(Z_{12}^3)$ Kähler breaking, so they give no parametric split. `w33_pass11904_magnetized_unlock_and_theta_zero_law.py` The theta-zero law is one case of a general statement about symmetric Wilson lines. The level-3 Heisenberg–reflection group acts on the Wilson-line argument by $z \mapsto z + 1$, $z \mapsto z + \Omega/3$ (shifting the characteristic) and $z \mapsto -z$. Its reflections $z \mapsto -z + m + k\Omega/3$ fix the points $z = m/2 + k\Omega/6$. At each of them, two of (top, charm, up) coincide exactly, because the reflection pairs $c \leftrightarrow -c - k \pmod 3$:

- for $k \equiv 0$, $m_c = m_u$;
- for $k \not\equiv 0$, the top equals one of the light quarks.

These coincidences occur only at those points, and the theta-zero law is the case $k = 3$. Invariant potentials for the Wilson line have their symmetric critical points exactly there, so no symmetric Wilson-line vacuum gives $m_t \gg m_c \gg m_u$. The hierarchy point of Pass 11904 ($\delta \approx 0.06$; the symmetric offsets are 0 and $\frac{1}{6}$) is generic. In the magnetized home the up hierarchy is therefore a property of a dynamically selected Wilson line, not of a symmetric one.

w33_pass11905_symmetric_wilson_lines_degenerate.py These statements hold in true normalisation, as an audit by direct overlap integrals shows. For normalised flux-3, 3 and 6 zero modes on T^2 with Wilson lines, the masses for a localised Higgs equal $|\theta[c/3,0](3\zeta,6\tau)|$ up to a permutation, to $6\cdot10^{-16}$. Here ζ is the relative Wilson line, so the level-3 texture is exact with $z=3\zeta$ and $\Omega=6\tau$. Consequently:

- the theta-zero law holds, with a vanishing mass and the other two equal;
- the symmetric points are degenerate;
- the Gaussian invariant $m_1m_2/m_0^2=e^{-(4\pi/3)\mathrm{Im}\,\tau}$ holds;
- the observed-order up ratios come out at $\mathrm{Im}\,\tau=4.21$, $\mathrm{Im}\,\zeta/\mathrm{Im}\,\tau=0.121$, which is the earlier point, found again independently.

The parallel track's equal-background overlap is reproduced. w33_pass11906_magnetized_normalisation_audit.py What fixes the Wilson line? The minimal potential, the two-loop vacuum energy $\pm\sum_c|Y_c|^2$ with the light Higgs, always selects a degenerate texture. Its maximum is at $\zeta=0$, where $m_c=m_u$; its minimum is at the theta zero, with spectrum $(0,1,1)$. Adding a competing quartic term, $V=-F_2+rF_4$ with $F_p=\sum_c|Y_c|^p$, changes this. The symmetric point becomes unstable at exactly $r=\frac{1}{2}$ (in units of the top coupling). For every larger r tested, the vacuum moves to a non-symmetric Wilson line with a hierarchical spectrum. The observed up ratios are the vacuum at $r^*=0.890$ with $\mathrm{Im}\,\tau=4.21$: an $O(1)$ coefficient, with no hierarchy put in by hand. The coefficients themselves are not derived. w33_pass11907_wilson_line_dynamics.py The heterotic lock is not special to the A_8 class. Three further $\mathbb{Z}_3$ shift classes of the $SO(16)^2$ string were scanned: $E_6\times SU(3)$ with E_8, $E_6\times SU(3)$ squared, and $E_7\times U(1)$ with $SO(14)\times U(1)$. This gave 109 new three-generation models. All 73 with twisted families carry a local **10**, and all 36 with untwisted families have top = charm. Across 600 models in four shift classes, none can split charm from up through a Kähler modulus at tree level. The shifts and Wilson lines of all 600 are now frozen with source hashes. w33_pass11908_split_ten_scan_beyond_a8.py The coefficient r follows from the up sector's own loops. At fixed v the Wilson-line potential is the ζ-dependent part of the one-loop top-sector potential. With an SM cutoff, or with superpartners at any mediation scale at or above the stop mass, the vacuum is the symmetric point, so $m_c=m_u$. The hierarchical phase exists only where the Wilson-line-dependent Higgs-mass term nearly cancels, in a band of width $\sim(v/\tilde{m})^2$. The up-quark hierarchy is therefore tied to the naturalness of the top-sector Higgs mass. w33_pass11909_up_sector_coleman_weinberg.py Across sectors, the minimal home makes a sharp prediction. Every sector with fluxes $(3,3,6)$ on the shared family torus has the same invariant $m_1m_2/m_3^2=e^{-(4\pi/3)\mathrm{Im}\,\tau}$. Running masses at M_Z give $1.9\cdot10^{-5}$ for the down quarks and $1.6\cdot10^{-5}$ for the leptons, which agree (this is the product form of the Georgi–Jarlskog pattern). For the up quarks they give $2.7\cdot10^{-8}$, so the minimal home is falsified by the up/down comparison: the up Yukawas need a larger effective area. With a localised Higgs the CKM is a permutation ($V_{us}=0$). A Higgs admixture of $\epsilon\approx3\cdot10^{-3}$ in a second selection class gives a Cabibbo-sized angle, but such a Higgs is not supplied by the dynamics above. w33_pass11910_cross_sector_invariant_and_ckm.py In the supersymmetric $E_8\times E_8$ $\mathbb{Z}_3$ orbifolds the question does not arise. Four shift classes were scanned (the standard embedding, $E_6\times SU(3)$ squared, $E_7\times U(1)$ with $SO(14)\times U(1)$, and the A_8 Kac shift), with 20,000 random Wilson-line draws each. They gave no Standard-Model-like vacuum by the orbifolder criterion, which recognises a known

$\mathbb{Z}_6$-II Standard Model as a positive control. The standard-embedding space saturates. The $\mathbb{Z}_3$ Standard Models of this programme therefore live in the non-supersymmetric $SO(16)^2$ class, where the lock holds in all 600. `w33_pass11911_susy_z3_scan.py` A Wilson line is not the only possible source of the hierarchy. The two-qutrit theta decomposes exactly as $\Theta[(\alpha_1,\alpha_2)](0,\Omega) = \sum_n e^{i\pi(n+\alpha_2)^2\Omega_{22}}\,\theta[\alpha_1]((n+\alpha_2)\Omega_{12},\Omega_{11})$. Take families in a Pauli class a of the second torus (the three-generation sector of Pass 11900). For large $\mathrm{Im}\,\Omega_{22}$ their texture is the one-qutrit texture with Wilson line $z_{\mathrm{eff}} = \pm\Omega_{12}/3$ when $a \neq 0$, and $z_{\mathrm{eff}} = 0$ (locked) when $a = 0$. The entangling, $W(3,3)$-carrying modulus Ω_{12} is therefore the family Wilson line. The observed up ratios follow from $\mathrm{Im}\,\Omega_{12} = 4.57$ in units where $\mathrm{Im}\,\Omega_{11} = 25.25$, with no Wilson line. Complex-structure moduli are stabilised by three-form fluxes, so the hierarchy can be fixed by flux quanta rather than by the radiative balance above. This is a route, not a constructed vacuum. `w33_pass11912_entangling_modulus_is_the_wilson_line.py` On the toroidal orientifold the route fails. Primitive $(2,1)$ type IIB fluxes on $T^4 \times T^2$ exist at rational points near the hierarchy point, with tadpoles as small as $N = 8$. But they leave the Siegel block flat. Fixing all complex-structure moduli costs $N \gtrsim 500$, and 10^4–$5\cdot10^5$ near the point, far above the $T^6/\mathbb{Z}_2$ budget. `w33_pass11913_flux_vacua_near_hierarchy_point.py` The quark-flavour problem of the two-qutrit home therefore reduces to the Higgs, for three reasons.

- With Q's families an eigenspace of one Pauli operator, the invariant $m_1 m_2/m_3^2$ is the same for every choice of right-handed $W(3,3)$ eigenspace and Higgs class (2214 configurations). A shared Q cannot give the observed up/down ratio with localised Higgs fields.
- The spread phase of the minimal Higgs potential gives near-degenerate down quarks and anarchic mixing, $|V| \approx 1/\sqrt{3}$.
- The exact magnetic-translation $\mathbb{Z}_3$ forbids the small cross-class Higgs admixture radiatively.

A realistic quark sector needs a weak explicit breaking of that symmetry in the Higgs profile. Two corrections from the parallel audit are incorporated: the earlier Cabibbo statistic was not the mass-ordered V_{us}, and the mediation-scale band depends on scheme. `w33_pass11914_higgs_is_the_flavour_lever.py` The two theta systems of this programme turn out to be the two flavour sectors of SU(5). Three generations on magnetized T^4 need a flux matrix of determinant 3. Positive binary forms of determinant 3 fall into exactly two classes, the A_2 (Eisenstein) form and x^2+3y^2: these are the Hermitian and the level-3 systems used above. With the **10** and the $\bar{\mathbf{5}}$ in these classes, the down and lepton Yukawas are both of type $\mathbf{10}\cdot\bar{\mathbf{5}}$, so their invariants agree automatically, as the data require. Direct four-dimensional overlap integrals give further consequences.

- The Gaussian slopes are quantised: $-4/3$, $-8/3$ (or $-2/3$), and -2, -1 or -4, in units of π. The observed up/down log-ratio 1.605 is therefore not attainable with a scalar period matrix; the entangling modulus is required.
- The down Higgs lattice has determinant coprime to 3. It breaks the class selection geometrically, so the CKM is nonzero without any Higgs admixture.
- For generic Wilson lines, however, the mixing is too small.

`w33_pass11915_su5_matrix_fluxes_on_t4.py`

What this does and does not give

The datum the chirality no-go asks for from outside now has a name, the vacuum value of Ω, which breaks generalised CP unless it lies on the real locus; and a three-generation hierarchy has a geometric source. Neither is a prediction yet: no moduli stabilisation, no fit, no string model is claimed, and the natural home is a magnetized or intersecting-brane T^4 rather than the heterotic orbifolds of this section. The mathematics (Mumford, Coble, Burkhardt, Freitag–Salvati Manni) is classical; the identification with the qutrit objects is the new step. BRIDGE

9.9 What we do not count as evidence

The older manuscripts contain many numerical coincidences: $1/\alpha \approx 137$ written as a combination of cyclotomic values of $q = 3$, spectral indices of the cosmic microwave background, spanning-tree counts read as gravitational entropies. Some are striking. None is derived from a dynamical principle with its uncertainties, and a finite geometry with many natural integers can match many targets. We therefore do not use any of them as evidence in this paper. (The value $\sin^2\theta_W = 3/8$ at unification, often quoted alongside them, is the standard prediction of any $SU(5)$ or $SO(10)$ grand unification and is not specific to $W(3,3)$.) BRIDGE at most.

The geometry carries the Standard Model's shapes — its gauge algebra's dimensions, three equal matter sectors, a chiral spinor, a matter parity — and a string compactification built on its twist produces genuine three-family Standard Models whose Higgs sits in the "qutrit plane". It cannot choose the handedness of the weak force — only an antiunitary, CPT-type symmetry exchanges the two — and, exhaustively, none of its supersymmetric singlet vacua preserves a matter parity: the anomalous $U(1)$ forces a $B - L = \pm 1$ condensate.

10 The machine: a computer read off the geometry

The same forty points that carry E_8 and a light cone also describe, with no design choices, a quantum computer: an instruction set, a memory, a network and a supply of the non-classical resource that makes quantum computation powerful. The Holonet papers develop this at length; here we give the parts that are exact, and mark carefully what has and has not been built.

10.1 The carrier: two qutrits in one photon

A single photon has several independent degrees of freedom. Two of them — which of three *time bins* it arrives in, and which of three *frequency bins* it occupies — give two qutrits, and together the $\mathbb{C}^9$ on which the two-qutrit Paulis of Section 2 act. Its path and polarisation give a second register, $\mathbb{C}^4$, whose 40 Witting rays realise $W(3,3)$ as *states* (Theorem 5.3 and

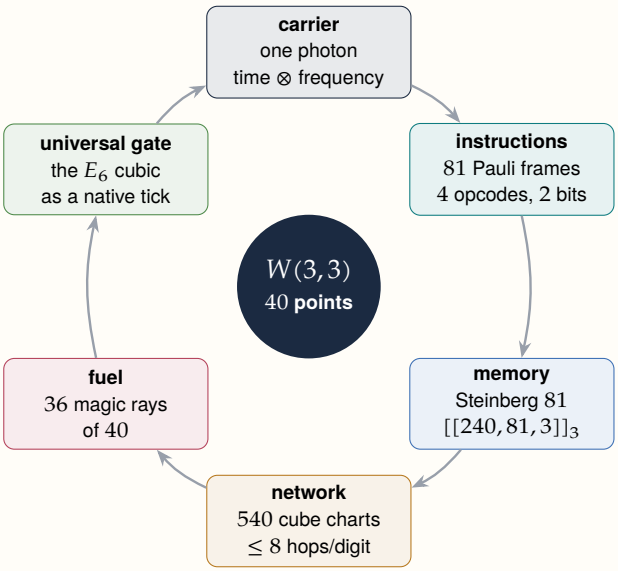


Figure 33. The Holonet stack. Every layer is a structure of $W(3,3)$ read as engineering: the Pauli frames are the 81 points of $\mathbb{F}_3^4$, the memory is the Steinberg module, the network is the geometry's atlas of cube charts, the fuel is its set of magic states, and the universal gate is the E_6 cubic of Section 5.

the box after it).

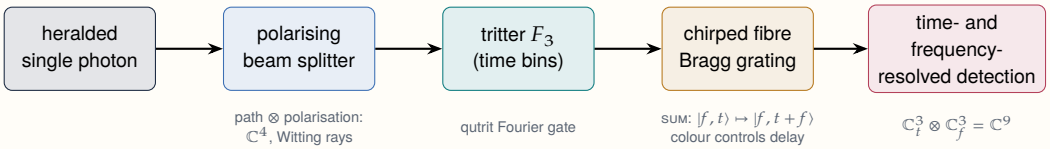


Figure 34. One photon, two qutrits. The controlled-add (SUM) gate between the frequency and time qutrits has been demonstrated with a single passive element, a chirped fibre Bragg grating, at basis fidelity 0.92 ± 0.01 [25].

10.2 The instruction set

The machine's classical control state is a *Pauli frame*: a label $x \in \mathbb{F}_3^4$ saying which Pauli correction is pending. There are 81 frames. Instructions are symmetries of the geometry: the two Fourier gates, the two phase gates, the two controlled-add gates, and the Pauli translations.

Theorem 10.1 (the complete instruction group) COMPUTATION

The six Clifford opcodes generate exactly $\mathrm{Sp}(4,3)$ (order 51 840); adjoining one translation generates the affine group

$$\mathrm{ASp}(4,3) = \mathbb{F}_3^4 \rtimes \mathrm{Sp}(4,3), \qquad |\mathrm{ASp}(4,3)| = 81 \times 51\,840 = 4\,199\,040.$$

No one or two linear opcodes suffice; exactly six triples do, so *four* instructions (three Clifford, one translation) in a *two-bit* opcode field generate everything, with worst-case program length 19 for the best triple. The minimal arithmetic core synthesises to 43 logic cells at 208.86 MHz on an iCE40 HX8K FPGA. `holonet_machine_blueprint_`

`body.tex`, Passes 2778, 2789, 2882

In plain words

Four instructions are enough to perform every one of more than four million possible frame transformations, and no program needs more than nineteen steps. The instruction set was not designed: it is the list of symmetries of the forty-point shape, and the theorem says which symmetries you can leave out.

Optimal perfect-gate compiler (Pass 11193) COMPUTATION

Let $H = (\mathrm{SL}(2,3)\times\mathrm{SL}(2,3)){:}C_2$ be the local/swap subgroup and p one fixed order-four tetracode perfect gate. Exact word replay on all 51 840 elements gives

$$\mathrm{Sp}(4,3) = H \sqcup HpH \sqcup Hph_*pH,$$

with optimal p-depths $0,1,2$ and respective counts 1 152, 13 824, 36 864. Thus every two-qutrit Clifford uses at most two copies of one fixed perfect interaction; the standard qutrit SUM has a frozen minimal word Lph_*pR. The Hecke identity $A^2 = 12I+3A+3(J-I-A)$ proves the depth bound. Relative to the E_6 cubic split $45 = 5+40$, its uniformly locally dressed walk has quotient $\left(\begin{smallmatrix}4 & 8\\ 1 & 11\end{smallmatrix}\right)$, stationary weights $(1/9, 8/9)$ and transient eigenvalue $1/4$. This evolves cubic-term labels, not Yukawa amplitudes, masses or mixing angles. `w33_pass11193_optimal_perfect_gate_compiler.py`

10.3 The fuel: matter is magic

Clifford operations alone can be simulated efficiently on an ordinary computer; quantum advantage needs a non-Clifford resource, usually supplied as *magic states*; contextuality is what supplies them [23].

Theorem 10.2 (magic is everything except one line) COMPUTATION

Of the 40 Witting rays, exactly the 4 computational-basis rays are stabiliser states; the other 36 are magic. The 36 split by stabiliser fidelity into grades $8 + 24 + 4$. A classical $\{0,1\}$ assignment can satisfy the "exactly one" rule on at most 36 of the 40 contexts, so the contextual fraction is exactly $1/10$; the independence number of the collinearity graph is 7, attained by heptads of seven rays with all pairwise overlaps $1/3$. `bt822_magic_census_matter_is_magic.py`, `bt818_ovoid_nogo_theta_gap.py`

The cubic gate. Clifford gates are "degree two": they act on frame labels by linear symplectic maps and on phases by quadratic forms. The classical continuous-variable theorem of Lloyd and Braunstein [34] says that quadratic operations plus a single cubic one are universal. In Section 5 the cubic appeared as the E_6 form N, the rule by which three first-order moves reach the top of the ladder. Pass 10947 compiles that signed cubic into a native gate on a seven-qutrit virtual machine: ten operations in seven parallel layers, both ancillas returned to zero. COMPUTATION (`w33_pass10947_five_front_execution.py`) The same object

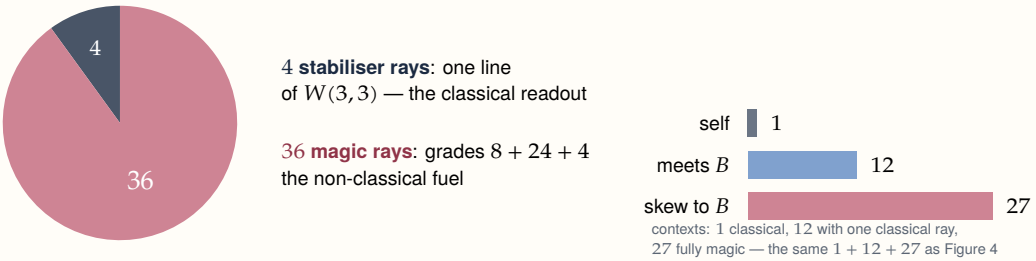


Figure 35. The machine's quantum resource is the geometry itself: remove one line and every remaining point is a magic state.

is, in the language of Section 5, the Yukawa coupling of E_6 unification, and in the language of this section, the machine's universal non-Clifford gate. BRIDGE

10.4 Memory and error correction

Theorem 10.3 (the memory is the Steinberg module) COMPUTATION

The CSS code $[[240, 81, 3]]_3$ on the edges of $W(3,3)$, with distances $(d_X, d_Z) = (3, 4)$, stores its 81 logical qutrits in $H_1(W(3,3))$, which is the irreducible Steinberg module of $\mathrm{PSp}(4,3)$. By Schur's lemma any symmetry-covariant logical error acts as a scalar. Separately, a ternary Golay $[[11, 1, 5]]_3$ lane with two verified syndrome rounds has 552 fault locations in a declared model, 2 576 205 malignant fault triples, and a conservative threshold crossing at $p \approx 5.32 \times 10^{-4}$. `bt861_code_register_is_steinberg.py`, `w33_pass10947_five_front_execution.py`

10.5 The network

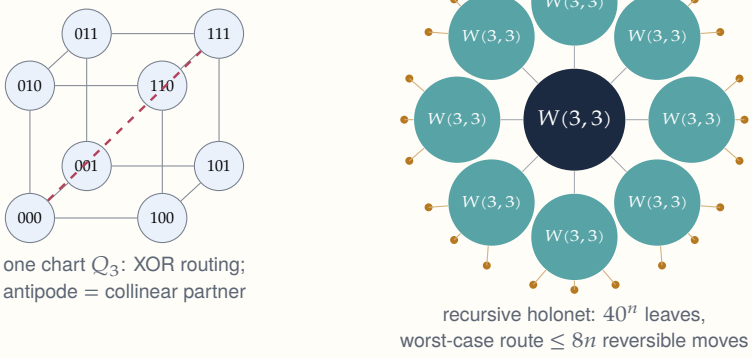


Figure 36. Left: a local routing chart. Each of the 540 pairs of disjoint lines of $W(3,3)$ carries a cube of eight points with XOR addressing. Right: substituting a copy of $W(3,3)$ at every point gives a network with 40^n leaves and logarithmic routing.

$W(3,3)$ contains exactly 540 pairs of disjoint lines; each pair's eight points carry the cube graph Q_3, in which the antipode of a vertex is its unique collinear partner. The charts are

glued along the 1620 apartments of the Tits building of $\mathrm{Sp}(4,3)$. A route needs at most 3 in-chart moves plus 5 chart hops, so each digit of a recursive address costs at most 8 reversible moves. COMPUTATION (`bt773_involution_cube_theorem.py`, `bt777_hypercube_chart_web.py`, `bt827_holonet_fractal_architecture.py`)

What is measured, what is proved, what is not built

Measured in a laboratory (prior art): the single-photon time–frequency SUM gate, fidelity 0.92 ± 0.01 [25]. **Proved exactly:** the group theory, the instruction semantics, reachability, code parameters, routing bounds, and FPGA synthesis figures. **Not built:** an integrated device; a magic-state injection circuit with a threshold; a physical demonstration of the network. The machine is a specification whose every number can be checked, not a device.

The forty points are also a computer: 81 frames, four instructions generating 4 199 040 operations, a memory that is the Steinberg module, a network of 540 cube charts, and a quantum fuel that is every point off one line. The E_6 cubic that is the Yukawa coupling in physics is the universal gate in the machine.

11 Epilogue: what a theory of everything still needs

We began with a qutrit acting on itself and followed one chain of exact identifications: forty points; a finite present with a light cone; four clocks; E_8 from nine histories; the Freudenthal quartic and a 57-dimensional light cone; a real form in which time has one dimension; the Albert algebra, the Lorentz algebra $\mathfrak{so}(1,9)$, a chiral spinor, and matter parity as a 2π rotation. On top of that chain a heterotic string construction produces three-family Standard Models, and the same geometry specifies a quantum computer.

exact

- $W(3,3)$ as qutrit commutation geometry
- Bell line = unitary channels
- 27-event light cone, $t^2-x^2-y^2$
- tetracode = four clocks
- $E_8 = 80+84+84$; $240 = 40 \times 6$
- clock gradings; cubic ladder
- Freudenthal quartic; 57-cone
- clock real form $\Rightarrow (1,9)$
- F_4, $E_{6(-26)}$, $\mathfrak{so}(1,9)$, 16
- matter parity $= 2\pi$ rotation
- chirality no-go (antiunitary swap)
- string route closed: no parity-keeping vacuum in 7 families
- ISA, memory, magic census

bridge / hosted

- "null" = shared measurement
- lifted history = proper time
- forgetting 2 trits = arrow
- cubic ladder = cubic time
- quartic = black-hole entropy
- $1+3+8$ = gauge shape
- $27+27+27$ = generations
- $W(3,3)$ twist $\Rightarrow$ heterotic SMs
- Higgs in the qutrit plane
- SM breaking = CZ gate

open

- a dynamical principle
- a continuum limit; gravity
- which chirality, and why
- a stable universe beyond heterotic orbifolds
- masses, mixings, couplings
- a physical 54 chamber gate
- magic-state injection
- a built device

Figure 37. The state of the programme, in three columns.

11.1 What stands, and what closed

Two different kinds of result are in this paper, and they deserve to be read differently.

The mathematics stands. Everything in Sections 2–8 is exact and independent of any physical reading: the forty points as the commutation geometry of a qutrit acting on itself, the finite light cone, the four clocks and the tetracode, E_8 from nine histories and its 6:1 fibration over the forty points, the Freudenthal quartic and its 57-dimensional cone, the real form in which time has one dimension, and the Albert algebra with its $\mathfrak{so}(1,9)$ and chiral **16**. None of it depends on a string construction, and none of it was touched by the corrections below.

The string route closed. Section 9 asked whether heterotic orbifolds carrying the forty-point twist give our universe. The answer, reached by exact computation over nine orbifold families and checked against the selection rules the worldsheet establishes, is no in all nine: in eight, their supersymmetric vacua cannot keep the proton stable; in the ninth, $\mathbb{Z}_3 \times \mathbb{Z}_3$, the parity survives and the vacua are supersymmetric, but a counting argument over hidden representations leaves six massless particles of charge $1/3$ in every one. That is a finished negative result — it tells the programme where the forty points do *not* lead, and it isolates what a successful route must do differently: provide a vacuum in which the Fayet–Iliopoulos term is cancelled by something other than a sneutrino-like field, with a hidden sector whose representations pair up.

11.2 Seven open problems, stated sharply

1. **A realistic vacuum.** Every parity built from the gauge torus, the space group and the established rotation symmetries is excluded in $\mathbb{Z}_6$-I, $\mathbb{Z}_6$-II, $\mathbb{Z}_{12}$-I (where the E_6 lattice itself fixes the admissible rule), $\mathbb{Z}_2 \times \mathbb{Z}_6$, $\mathbb{Z}_3 \times \mathbb{Z}_6$ and $\mathbb{Z}_6 \times \mathbb{Z}_6$. The one family where it survives is $\mathbb{Z}_3 \times \mathbb{Z}_3$: its 28 parity vacua are supersymmetric ($W = F(p)$, $\mathbb{Z}_3$-graded) but carry six massless charge-$\frac{1}{3}$ states; its 54 parity vacua with the hidden $SU(4)$ condensed cure that count but leave at least 120 fractional charges massless, protected by an unbroken discrete 3-group. Baryonic and mixed condensates do not free them, and neither does any resolution of the orbifold or any breaking of supersymmetry. The 87 non-supersymmetric twins of the $\mathbb{Z}_6$-I Standard Models are consistent three-generation models, but every one is tachyonic. What remains:

    - for the $SO(16)\times SO(16)$ models with the $W(3,3)$ shift, a source of the light-generation hierarchies (the $\Delta(54)$ flavour symmetry keeps $m_c = m_u$ and $m_\mu = m_e$ at the same order under every alignment of the scalar vacuum, and the stabilised family modulus $T = \rho$ gives $m_c = m_u = m_t/2$) and a vacuum at all (no minimum in the tachyon-free region: the Wilson-line moduli roll into a winding tachyon and the dilaton runs to weak coupling—but 21 survivors have a neutral exit and in six of them every nearby instability is Standard-Model-neutral; there the potential drives them steeply to it, the condensate never touches the up sector but, for a single light Higgs, lowers m_b/m_t to ϵ^2 with the tau at the same order in 14 of 21 models; the exit is SM-neutral on the whole B circle (exact hyperbolic disks at the $\Gamma_0(3)$ points, the duality group itself), its endpoint is the non-supersymmetric $U(16)$ string, not a nearby vacuum, every symmetric point of the modulus is tachyonic or a runaway, and the light generations stay degenerate under every vacuum and Higgs choice tried);

- twists and lattices not yet scanned;
- a mechanism other than a parity for proton stability.

2. **A dynamical principle.** Every result in this paper is kinematics. What action, Hamiltonian or spectral principle on $W(3,3)$ has the structures found here as its symmetries, and what does it predict? In particular: does such a principle select the clock real form $E_{8(-24)}$ over the split form — does the geometry *choose* one time?

3. **A continuum limit.** The finite light cones (in $2+1$, $3+1$, $1+9$ and the quartic 57) are exact algebraic objects. Is there a limit, a tower over $W(3,q)$ or a lattice construction, in which one of them becomes a smooth Lorentzian manifold with dynamical curvature?

4. **Chirality from outside.** Corollary 9.7 shows that reversing time's orientation, acting antiunitarily and exchanging chirality are one coset, so fixing an arrow makes chirality conserved but does not choose it. The chamber law now supplies an exact candidate order parameter: the two nonsplit cocycle orientations $s = \pm 1$ are exchanged by qutrit complex conjugation, $\Psi(x \star_s y) = \Psi(x) \star_{-s} \Psi(y)$, and their class-three central tick reverses sign. `w33_20261001_u81_antiunitary_orientation.py` The remaining problem is dynamical rather than combinatorial: what selects one sign, and does that selected finite orientation couple to the unique global chirality character of $\mathrm{PGSp}(4,3)$ (Holotrade 98cafff, P11091) in a physical vacuum?

5. **A physical 54 chamber gate.** The literal H_{27}-equivariant 54×54 intertwiner is impossible (rank ceiling 36), but the symmetry-changing replacement is now explicit. Four single-foliation 27-triangle factors transport through both nonzero $K_{81} \to U_{81}$ cocycle orientations and remain rank 54, with $S_2 \oplus L$ projection ranks $27 + 27$ at both split primes. Thus the algebraic class-three transducer exists. `w33_20261001_u81_54d_cocycle_transducer.py` What remains is physical: derive a local Hamiltonian or optical circuit whose finite evolution realises one of these transported weighted factors, with an error model and measurable calibration protocol.

6. **Numbers.** The up-type Yukawa couplings of the $\mathbb{Z}_6$-I class have a $2+1$ block structure qualitatively like the observed quark mixing. Can any mass ratio or mixing angle be computed, with an error, from a vacuum of the class?

7. **The machine.** Specify a magic-state injection circuit for the 36 Witting rays with a decoder and a threshold, and build the two-qutrit photonic core.

11.3 Corrections we kept on the page

A research record is trustworthy only if it keeps its errors. These were believed, written down, and then shown false; each correction made the structure *more* natural, not less.

Was believed	Is true
there is a natural bijection 240 edges ↔ 240 roots	no equivariant bijection; a 6:1 fibration of roots over *points* (§5.4)
the Freudenthal 56 is the clock's 54 plus 2	the 56 contains only one 27 of the 54 (§6.1)
the Albert algebra has "27 minimal idempotents" as frames	the frames are the 45 triads (§6.4)
three time bins hold a past and a future qutrit	$\mathbb{C}^3$ cannot hold two qutrits; use time ⊗ frequency (§3.4)
the 9 fixed modes of the spectral clock are the qutrit's 9 histories	same count, different module (§4.4)
the independence number of $W(3,3)$ is 10 (an ovoid)	it is 7; $W(3,3)$ has no ovoid (§10.3)
at most 34 of 40 contexts are classically satisfiable	exactly 36 (§10.3)
the $\mathbb{Z}_6$-I class is excluded unconditionally	excluded only if matter parity is broken (§9.3)
every D-flat support contains a matter-odd singlet (from 60 samples)	exhaustively true, and the odd singlet always has $B - L = \pm 1$ (§9.4)
the three families come from $W(3,3)$	in the orbifolds they come from the three tori (§9.1)
directions each compatible with a parity form a parity-even vacuum	they must be even under *one* element; the 84-direction vacuum was withdrawn, a 68-direction one certified (§9.4)
the Higgs quartic is positive at unification (+1/40)	measured running gives −0.011 at 10^{16} GeV (§9.4)
a parity from plane rotations reopens one model	it used an R-rule valid only on prime planes; with the established rules it closes (§9.4)

The forty points do not yet give a theory of everything. They give something rarer: a single finite object in which quantum measurement, a light cone, clocks, E_8, the octonionic Albert algebra, ten-dimensional Lorentz symmetry, a chiral spinor, matter parity and a quantum computer are exact aspects of one structure — with every claim checkable, and the remaining distance to physics mapped rather than hidden.

A The ledger: every claim, its status, its witness

Every script named below lives in `analysis/` of the repository `wilcompute/W33-Theory` and writes its certificate to `data/`; most have a regression test in `tests/`. Commit hashes marked "Holotrade" refer to the second research track and are recorded row by row in the ledger of the $W(3,3)$ paper.

Claim	Badge	Witness
The object and one qutrit (§§2–3)		
$W(3,3)$: 40 points, 40 lines, SRG$(40,12,2,4)$, spectrum $12, 2^{24}, (-4)^{15}$	THEOREM	Props. 2.1–2.2 (proofs in text)
two-sided Weyl maps span $\mathfrak{sl}_9$; commutation geometry is $W(3,3)$	THEOREM COMPUTATION	`w33_20260924_temporal_su9_e8_compiler.py`
Bell line = unitary-channel diagonal; swap is anti-symplectic	THEOREM COMPUTATION	`w33_20260924_liouville_swap_cp_outer_bridge.py`
transpose induces the outer class (up to labelling gauge)	COMPUTATION	`w33_20260924_intrinsic_transpose_outer.py`
memory witness: Bell fidelity $1 - 8p/9$, threshold $p = 3/4$	COMPUTATION	`w33_20260924_temporal_quantum_memory_witness.py`
modular flow of a qutrit state has A_2 frequency pattern	COMPUTATION	`w33_20260924_modular_a2_event_clock.py`
space and time are two $GQ(3,3)$ on the same 40 points sharing the 16 product lines; partial transpose = bijection 216 entangled states $\leftrightarrow$ 216 Clifford gates; common symmetries 2304	THEOREM COMPUTATION	`w33_pass11143_space_time_two_quadrangles.py`
CGLMP: $W(3,3)$ resources give exactly 2 in space and time; onset $2 + \frac{\pi^2}{6}t^2$; one qutrit in time $3.1628 > 2.9149$	COMPUTATION	`w33_pass11144_spookiness_budget.py`
thermofield double = Choi state of Euclidean half-evolution; negativity $((\sum \sqrt{p_i})^2 - 1)/2$	COMPUTATION	`w33_pass11146_thermal_reading.py`
polygamy in time: pairwise temporal negativities sum to 3; space ≤ 1.139, $N_{AB} + N_{AC} \leq 4/\sqrt{15}$	COMPUTATION	`w33_pass11148_polygamy_in_time.py`
temporal CGLMP beats spatial for every $d \geq 3$ $(3.163, 3.283, 3.377$ vs $1 + \sqrt{11/3}, 2.973, 3.016)$	COMPUTATION	`w33_pass11147_temporal_cglmp_by_dimension.py`
Pauli-only CGLMP optimum $2\sqrt{2}\cos(\frac{1}{3}\arccos\frac{1}{2\sqrt{2}})$ (root of $x^3 - 6x - 2$): magic in the state suffices	COMPUTATION	`w33_pass11149_magic_for_spookiness.py`
perfect space–time gates: $\det S_{AA} + \det S_{BA} = 1$; $13824 = 24^3$ perfect (AME(4,3)), one local orbit; tetracode gate is the time-reversing representative	THEOREM COMPUTATION	`w33_pass11156_perfect_spacetime_gates.py`
Pauli-only Bell operator: characteristic polynomial $(x^3 - 6x - 2)(x^3 - 3x + 1)^2$ exactly	COMPUTATION	`w33_pass11152_pauli_bell_charpoly.py`

Claim	Badge	Witness
in time, Pauli measurements give exactly 2 for every start: magic must sit in the measurements	COMPUTATION	w33_pass11153_temporal_magic_budget.py
$4/\sqrt{15}$ optimum: spectra $(7,4,4)/15$, $(11,2,2)/15$; $N_{AB} = N_{AC} = 2/\sqrt{15}$, $N_{BC} = 1/15$	COMPUTATION	w33_pass11154_polygamy_optimum_structure.py
temporal CGLMP $d = 6, 7$: 3.455, 3.521 (corrected in Pass 11171; 3.453, 3.517 were local optima; spatial 3.050, 3.078); deficit $\approx 3.4/d$	COMPUTATION	w33_pass11155_temporal_cglmp_trend.py
ninth roots explained: $B^3 - 3B + 1 = (B+1)(I+U+U^2)$, $U = (XZ^2) \otimes (XZ^2)$	COMPUTATION	w33_pass11157_ninth_roots_explained.py
general law $\sum_i \det S_{ij} = 1$; perfect three-qutrit gates exist ($\approx 3.1\%$ of $\mathrm{Sp}(6,3)$), columns $(1,1,-1)$	THEOREM COMPUTATION	w33_pass11158_perfect_three_qutrit_gates.py
$4/\sqrt{15}$ derived: $N(x) = \sqrt{(1+15x)(1-x)}/4$, max at $x = 7/15$; explicit optimal state	COMPUTATION	w33_pass11159_explicit_polygamy_state.py
temporal CGLMP $d = 8, 9, 10$: 3.576, 3.621, 3.658 (spatial 3.101, 3.122, 3.140); deficit $\approx 3.4/d$ (Pass 11171; the $d^{-0.75}$ fit is superseded)	COMPUTATION	w33_pass11160_temporal_cglmp_high_d.py
$I_3 = 2 - \operatorname{rank} S_{AA} - \operatorname{rank} S_{BA} \le 0$ for every two-qutrit Clifford gate; perfect = maximal scramblers; tick + forgetting the partner = Theorem 4.7	COMPUTATION	w33_pass11161_scrambling_and_arrow.py
$4/\sqrt{15}$ a certified local maximum: partial-transpose gap 2/15, Hessian 32 negative, 20 flat = local-unitary orbit; 96/96 restarts reach it	COMPUTATION	w33_pass11162_polygamy_local_maximum.py
exactly $286\,654\,464 = 2^{17}3^7$ perfect three-qutrit gates, 128/4095 of $\mathrm{Sp}(6,3)$ (control: 13824 for two)	COMPUTATION	w33_pass11163_perfect_three_qutrit_count.py
no $\mathbb{F}_3$ tetracode analogue for three qutrits; over $\mathbb{F}_9$, 12288 perfect unitaries in $U(3,\mathbb{F}_9)$, e.g. $K = P^2 + i\mathbf{1}$ (Grassl–Gulliver's code, not new)	COMPUTATION	w33_pass11164_three_qutrit_tetracode.py
perfect three-qutrit tick secret-shares each past (0 to every proper subset of futures, 2 to all three); cf. [54]	COMPUTATION	w33_pass11165_three_qutrit_arrow.py
spectral clock tick = Clifford propagator $e^{-iq(p)}$; translation-invariant clocks never perfect (off-diagonal rank ≤ 1)	COMPUTATION	w33_pass11166_clock_tick_clifford.py

Claim	Badge	Witness
$4/\sqrt{15}$ exact on its symmetry class $(U \otimes \bar{U} \otimes \bar{U})$; rigorous global bound $25/18$; 3000/3000 dual see-saws reach $4/\sqrt{15}$	THEOREM COMPUTATION	`w33_pass11167_polygamy_global.py`
clock + potential kick + Lorentz group: order 576, no perfect tick; kick–tick–kick perfect for 108/729 potentials, all with π = light-cone reflection	THEOREM COMPUTATION	`w33_pass11168_clock_dual_pair.py`
perfect three-qutrit gates: six local orbits labelled by $\pi \in S_3$ (stabiliser 4), one up to relabelling	COMPUTATION	`w33_pass11169_perfect_gate_local_orbits.py`
perfect qutrit ticks for $1, 2, 3, 5$ qutrits, never 4 (AME$(8, 3)$ excluded [56]); explicit five-qutrit circulant over $\mathbb{F}_9$	COMPUTATION	`w33_pass11170_perfect_gate_ladder.py`
temporal CGLMP never reaches 4 at finite d (proof); $3.715, 3.787, 3.831, 3.853$ at $d = 12, 16, 20, 24$; deficit $\approx 3.4/d$; $d = 6, 7$ corrected to $3.455, 3.521$	THEOREM COMPUTATION	`w33_pass11171_temporal_cglmp_large_d.py`
polygamy frontier: family = global for every weight; corner $(1, 0)$ optimal up to $\lambda_c = 0.9214$; bulge beyond $N_{AB} + N_{AC} = 1$ at most 0.0328	COMPUTATION	`w33_pass11172_polygamy_frontier.py`
temporal CGLMP: $I_d = 4 - \frac{2}{d-1}L$ (ramp); explicit strategy $L = 3 + 2(\sqrt{d} - 2)/(\sqrt{d}(\sqrt{d} - 1))$, so $\sup_d I_d = 4$	THEOREM COMPUTATION	`w33_pass11173_temporal_cglmp_to_four.py`
perfect interacting tick's pattern = Sym^2 of the det -1 swap of $GL(2, 3)$; 36/81 Hesse-clock potentials perfect, one oblique clock suffices	THEOREM COMPUTATION	`w33_pass11174_lightcone_reflection_time_reversal.py`
Sp$(10, 3)$ perfect gates: five admissible orientation patterns; none in 7×10^6 samples; 2 642 411 520 $\mathbb{F}_9$-linear ones realise two patterns	COMPUTATION	`w33_pass11175_five_qutrit_patterns.py`
temporal entanglement through a Clifford tick is all-or-nothing; perfect tick: amnesia and echo (tick 9); Bell $3.061 > 2$ with zero temporal negativity	THEOREM COMPUTATION	`w33_pass11176_temporal_echo.py`
perfect two-qutrit gate $\Leftrightarrow$ collinear factorisations in GQ$(4, 2)$ (tritangent planes sharing a line); rank-3 action, subdegrees $1, 12, 32$	THEOREM COMPUTATION	`w33_pass11177_perfect_gates_tritangent.py`

Claim	Badge	Witness
three-qutrit factorisations: rank 20 under $\mathrm{Sp}(6,3)$; signature separates 15 of 20 relations; perfect = one orbital (3456)	COMPUTATION	`w33_pass11178_three_qutrit_orbitals.py`
four-qutrit near miss: best stabilizer registers fail exactly 2 of 35 balanced cuts (rank 2), 3-uniform, paired structure; minimum found every time	COMPUTATION	`w33_pass11179_four_qutrit_near_miss.py`
quantum mereology (two qutrits): intrinsically entangling (local in no split) $\Leftrightarrow$ order 5 or 9: 19/45; unique split 19/48; some ticks local in 6 splits, perfect in 27	THEOREM COMPUTATION	`w33_pass11180_mereology.py`
split stabilisers are $\{2,3\}$-groups: every tick of period with a prime ≥ 5 is intrinsically entangling; three qutrits 7922/12285, unique split 7/48 (GAP, all classes)	THEOREM COMPUTATION	`w33_pass11181_intrinsic_entanglement.py`
the clock is local in 108 of 110565 splits (none light-cone; 4 are the orthogonal frames of q); the perfect interacting tick in none	COMPUTATION	`w33_pass11182_paper_ticks_mereology.py`
intrinsic arrow $A = \min_F \sum_q \mathrm{rank}\, S_F[\text{others}, q]$: two qutrits 0 or 2 trits (227/360 have 2); three qutrits $\{0,2,3\}$, never 1; clock 0, perfect tick 2, $\mathbb{F}_9$ gate 3	THEOREM COMPUTATION	`w33_pass11183_intrinsic_arrow.py`
three-qutrit relations: 20 orbits recomputed independently; image/kernel incidences separate all but two pairs, which are exactly time-reversal pairs (gate vs inverse)	COMPUTATION	`w33_pass11184_merged_orbitals.py`
E_6 sectors: $45 = 5 + 40$ ($1 \cdot 10 \cdot 10$, $16 \cdot 16 \cdot 10$); partial ticks never keep a singlet-sector split; $W(D_5)$ (order 1920) acts locally or perfectly	COMPUTATION	`w33_pass11185_e6_yukawa_sectors.py`
five qutrits beyond $\mathbb{F}_9$: 71 AME$(10,3)$ graph states (tabu), 17 892 perfect gates, same two patterns; determinant law of every order $\sum_{\lvert I\rvert=k} \det S[I,J] = 1$, $\det S[I,J] = \det S[I^c, J^c]$	THEOREM COMPUTATION	`w33_pass11186_five_qutrit_beyond_f9.py`
$4/\sqrt{15}$ is the maximum on the torus-symmetric class: certified branch and bound terminates (265725 boxes)	THEOREM COMPUTATION	`w33_pass11187_torus_class_bnb.py`

Claim	Badge	Witness
exact intrinsic arrow on all 74 classes of $\mathrm{PSp}(6,3)$: $A \in \{0,2,3\}$ with fractions 55241/884520, 581/1080, 8836/22113; orders divisible by 7 or 13 have $A = 3$	THEOREM COMPUTATION	`w33_pass11188_exact_arrow_by_class.py`
oriented holonomy (N vs $-N$ in $\mathrm{SL}(2,3)$) plus a symplectic twist $Q = \pm 1$ classify all 20 three-qutrit split relations with time direction; $\tau = (x,z) \mapsto (x,-z)$ swaps exactly the two unseparated pairs	COMPUTATION	`w33_pass11189_oriented_holonomy.py`
AME$(10,3)$ stabilizer states: 6-sets split 3|3 or 6|0; local graphs are prism$+K_1$ (Klein quadric + CP-SAT); uniform 4-sets $= S(3,4,10)$; unique sign structure, Aut $= \mathrm{PGL}(2,9)$; five-qutrit patterns only C_{10} or cross+perm (11186 conjecture proved)	THEOREM COMPUTATION	`w33_pass11190_ame10_sign_structure.py`
polygamy prior art: family and global conjecture are Allen–Meyer (2017); normalised maxima $1.0448, 1.0328, \ldots, 1.0159$ for $D = 2..7$: the bulge is not a qutrit effect	COMPUTATION	`w33_pass11191_polygamy_across_dimension.py`
time reversal (anti-symplectic half of $W(E_6)$): arrow $\in \{0,2,4\}$; $A = 4$ exactly for the 36 reflections; 19/45 local in no split	COMPUTATION	`w33_pass11192_time_reversal_mereology.py`
AME$(10,3)$ local geometry: every 3-set carries a unique regular spread of $\mathrm{PG}(3,3)$; its reguli meet the Steiner blocks in 2 of 5 (not the same plane)	COMPUTATION	`w33_pass11199_ame10_local_spreads.py`
sign method: AME$(6,3)$ pentagons, unique, $\mathrm{PGL}(2,5)$; AME$(8,3)$ signs consistent (30 structures, $S(3,4,8)$, $\mathrm{AGL}(3,2)$) — non-existence is not a sign obstruction	COMPUTATION	`w33_pass11200_sign_method_ame6_ame8.py`
third-moment local invariants of stabilizer Choi states are Gauss sums; with labelled registers the twist is $I = Q\, i\sqrt{3}/243$ on the 256 pair ($\pm i\sqrt{3}/729$ on the 6912 pair), flipping under time reversal; fully symmetrised it cancels	COMPUTATION	`w33_pass11201_twist_chiral_invariant.py`
E_6 reflections: moved frames = a double-six ($36 \leftrightarrow 36$); 15 fixed splits = planes of fixed lines $= \mathrm{GQ}(2,2)$; SWAP there	COMPUTATION	`w33_pass11202_reflection_double_six.py`

Claim	Badge	Witness
$4/\sqrt{15}$ on the torus class with directed rounding: 9 219 253 boxes, all closures outward-rounded	THEOREM COMPUTATION	`w33_pass11203_certified_bnb.py`
arrow-free fraction 1, 133/360, 55241/884520, 0.0058 (sampled), 0/3000 (sampled) for $n = 1..5$	COMPUTATION	`w33_pass11204_second_law_counting.py`
directed split relations have non-real spectra (Eisenstein frequencies), reverses conjugate; configuration non-commutative	COMPUTATION	`w33_pass11205_one_way_clock.py`
no Clifford T-violation (Wonenburger); only-anti-unitary reversal for 77/162 and 84013/157464 (orders divisible by 3)	COMPUTATION	`w33_pass11206_t_violation.py`
cubic layer breaks T: fraction without substrate time reversal $0, 1/12, \ldots, 0.92$ for 0..8 cubic gates; minimal violators $= \zeta_9^{x^3} \cdot X^{a\neq 0}$(diag Clifford) (18/216)	COMPUTATION	`w33_pass11213_cubic_t_violation.py`
AME sign structures as cross-ratio geometry: harmonic pairing on PG(1, 5); Baer sublines + chiral cross-ratio rule on PG(1, 9) (all 71)	COMPUTATION	`w33_pass11214_projective_line_rule.py`
$E_n[c]$ exact for all n (Fulman cycle index; Steinberg-checked, matches $n \le 5$ class sums); $\lim E_n[c] = 0.72553535079905834944\ldots$	THEOREM COMPUTATION	`w33_pass11215_protected_qutrit_limit.py`
clock circulation: current spectra $\pm 24\sqrt{3}i, \pm 36\sqrt{3}i$ (256) and $\pm 72\sqrt{3}i, \pm 324\sqrt{3}i$ (6912); oriented 3-cycles 7168 per split (256)	COMPUTATION	`w33_pass11216_clock_circulation.py`
two qutrits: $(T \otimes T)$SUM T-violating; one cubic gate + SUM reversible (exact, 51840 × 81 anti-unitary Cliffords; the general single-sector claim is corrected by Pass 11238)	COMPUTATION	`w33_pass11227_two_qutrit_cp_mixing.py`
T-violation quantum: best time-reversal fidelity $(1 + 2\cos 2\pi/9)/3$ for every minimal violator (and the mildest with ≤ 2 cubic gates); cubic Pauli invariant identically real	COMPUTATION	`w33_pass11228_t_witness.py`
independent brute force (no closed form, no class list): qubit $E_2[c] = 29/40$, $E_3[c] = 5981/8960$ over all of Sp(4, 2), Sp(6, 2); $A = n - c$ against all splits	THEOREM COMPUTATION	`w33_pass11234_qubit_bruteforce_check.py`

Claim	Badge	Witness
random two-qutrit Clifford+cubic words: violating 5, 39, 56, 59 of 60 for 1–4 cubic gates; mildest violation $(1 + 2\cos 2\pi/9)/3$ up to two gates, 0.939 from three	COMPUTATION	`w33_pass11235_two_qutrit_t_spectrum.py`
cubic gate as residual flavour symmetry: no complete PMNS or Cabibbo pattern (Clifford, $\langle CTC^\dagger \rangle$, 148 one-gate magic eigenbases)	COMPUTATION	`w33_pass11236_cubic_flavour.py`
fidelity levels: $F^2_{\min}$ root of $81x^3 - 81x^2 + 18x - 1$; 0.8100955 in $\mathbb{Q}(\cos 2\pi/9)$; level 0.712386 $= F^2_{\min}$ to 2×10^{-16} (120-digit findpoly + PSLQ for the first two)	THEOREM COMPUTATION	`w33_pass11237_fidelity_levels.py`
short words (exhaustive): control-only cubic never violates (13/13); target-only from 3 gates (12/66); both from 2 (20/70)	THEOREM COMPUTATION	`w33_pass11238_one_sector_law.py`
three qutrits: product reversals give $F \geq 0.8440$ for $[(T \otimes T)\mathrm{SUM}] \otimes I$ and $\otimes T$; generic search reaches 0.46 (lower bounds only)	COMPUTATION	`w33_pass11239_three_qutrit_t.py`
$F_T((1 \otimes T)\mathrm{SUM}(1 \otimes T^2)) = F^2_{\min}$ exactly: residual phases $= \{-1, 0, 1\} + \{-1, 0, 1\}$, trace $(1 + \zeta + \zeta^{-1})^2$; Galois gap 6×10^{-13}	THEOREM COMPUTATION	`w33_pass11251_exact_reversal.py`
reversibility decided exactly: $c(Lp) = \mu\, \omega^{\langle b, Lp \rangle} c(p)$, L anti-symplectic; agrees with brute force on 204/204 two-qutrit words; three qutrits with 1..4 cubic gates violate in 10/100, 94/100, 100/100, 100/100	THEOREM COMPUTATION	`w33_pass11252_exact_reversibility.py`
every Clifford+T reversal fidelity has $F^2 \in \mathbb{Q}(\cos 2\pi/9)$ (any n); 148 levels exact, 147 cubic, 1 rational ($F = 2/3$); $F \notin \mathbb{Q}(\cos 2\pi/9)$ for 5	THEOREM COMPUTATION	`w33_pass11253_cubic_field_levels.py`
intrinsic arrow for every n: $A(S) = n - c(S)$, c = orthogonal invariant planes $= \sum_{\pm 1}(m_1/2 + m_2) + m_1^{\mathbb{F}_9}$; exact on all 20/74/278 classes ($n \leq 4$), plane formula, constructions for every indecomposable	THEOREM COMPUTATION	`w33_pass11207_arrow_theorem.py`
arrow law class by class for qubits ($n \leq 5$), 5-, 7-dits and all 940 classes of $\mathrm{PSp}(10, 3)$; $E[c] = 0.7255\ldots$; arrow-free fraction exact: 0.00503 ($n = 4$), 0.00020 ($n = 5$)	COMPUTATION	`w33_pass11208_arrow_universality.py`

Claim	Badge	Witness
Kashiwara–Maslov chirality of split relations (∓ 54, ± 18, else 0); stabilizer Bargmann phase $= (-i)^{\tau}$	COMPUTATION	`w33_pass11209_maslov_chirality.py`
Kramers time reversals ($\theta^2 = -1$) exist only for even n, are local only by exchange; for two qutrits exactly the 36 reflections; $\theta^2 = +1$ is always a local conjugation	THEOREM COMPUTATION	`w33_pass11210_kramers_time_reversal.py`
$S(3,4,10)$ theorem replicated with an independent CNF and two SAT solvers (CaDiCaL, Glucose)	COMPUTATION	`w33_pass11211_ame10_sat_replication.py`
one chirality: Maslov count $= 54Q$ on all 512 members of the 256 pair; block-type-sorted Choi phase is label-free, chiral exactly on the four time-directed relations, separates all 20	COMPUTATION	`w33_pass11212_one_chirality.py`
qubit arrow law: $A = n - c$ in characteristic 2 via a bi-Lagrangian criterion and $\mathbb{F}_2[s]$-lattices for $V(2k)$, $W(k)$; closed form $c = m_1/2 + \chi_2 m_2 + m_1(x^2 + x + 1)$ exact on 320 classes	THEOREM COMPUTATION	`w33_pass11217_qubit_arrow_law.py`
Gaussian arrow law: $A = n - c$ over $\mathbb{R}$ (minimal inter-mode quadrature channels); $c =$ elliptic + hyperbolic normal modes; $A = 2\times$ loxodromic quartets; switches on at opposite-signature Krein collisions	THEOREM COMPUTATION	`w33_pass11219_gaussian_arrow.py`
qubit mean protection exact through $n = 6$: $E_6[c] = 0.665161$, limit ≈ 0.66517 ($\mathrm{Sp}(12,2)$, 477 classes)	COMPUTATION	`w33_pass11220_qubit_mean_protection.py`
Gaussian arrow as operator-entanglement rate: Choi mode entropy $= 2\,\mathrm{rank} \cdot r + O(1)$; $A = \frac{1}{2}$ min total rate	COMPUTATION	`w33_pass11221_gaussian_choi_entropy.py`
Maslov factor 54 on the 256 pair: after quotienting the ineffective $\{\pm I\}$ the relation stabiliser is $U_{81}{:}C_2$; the Sylow-3 orbit defect is $\chi = \pm(81 - 27) = \pm 54$, so $\lvert\chi\rvert/27 = 3 - 1 = 2$	THEOREM COMPUTATION	`w33_20261001_maslov_borel_factor54.py`
arrows of named ticks: 17/48 of perfect two-qutrit gates arrow-free (incl. p); $A = 2$ local gates all swap; three-qutrit VKV $A = 2$, $\mathbb{F}_9$ gate $A = 3$	COMPUTATION	`w33_pass11223_named_tick_arrows.py`

Claim	Badge	Witness
best time-reversal fidelity is a Weyl-coefficient Fourier maximum; for the hard $(T^{\otimes 3})\mathrm{SUM}_{23}\mathrm{SUM}_{12}$ case the exact Weyl support is $x_1 = 0$, the three magnitudes are the z_3 fibres, and an anti-symplectic witness reaches $F^2_{\min}$; every support-nonpreserving map is bounded by $F^2_{\min}$, and the full 314 928 magnitude-preserving census has the same maximum (global layer-scrambling parabolic still open)	THEOREM COMPUTATION	w33_20261001_three_qutrit_fidelity_flag.py
one qutrit Clifford–cubic words violating reversal: $18/216 = 1/12$, $18\,954/46\,656 = 13/32$, $5\,524\,362/10\,077\,696 = 421/768$ for one, two, three cubic gates	COMPUTATION	w33_pass11266_one_magic_gate.py
fixed perfect $\mathbb{F}_9$-circulant ticks at $n = 2, 3$: one magic leg never violates; all but one always violates (exhaustive for those representatives; all-perfect-tick theorem open)	COMPUTATION	w33_pass11268_perfect_magic_ticks.py
shared level: $\langle +\|T\|+\rangle = \mathrm{tr}\,T/3$, so T-magic stabilizer fidelity = reversal level = $\|\mathrm{tr}\,T/3\|^2$; $F_T \geq 2F^2_{\mathrm{Cl}} - 1$ (proof), $F_T \geq F^2_{\mathrm{Cl}}$ on 1503/1503 words, $F_T \geq F_{\mathrm{Cl}}$ false	THEOREM COMPUTATION	w33_pass11308_shared_level_and_clifford_bound.py
one gate, two qutrits: 6480 five-frame-flagged classes checked at all81 frames; affine complements there; $P = \frac{223}{2430}$ conditional on the unflagged-class affine law	COMPUTATION	w33_pass11309_one_gate_law.py
$\langle X, S, T\rangle$/scalars = SmallGroup(81, 9) (18 of order 9) $\ncong U_{81} = (81, 7) = C_3 \wr C_3$: cocycle orientation $\neq$ magic orientation	THEOREM COMPUTATION	w33_pass11310_magic_orientation_cocycle.py
sampled edge law: one magic leg breaks 430/6000 non-$\mathbb{F}_9$ perfect ticks; three magic legs break all sampled $\mathbb{F}_9$-linear perfect ticks (6000/6000), about91% of others	COMPUTATION	w33_pass11311_perfect_edge_law.py
one qutrit, $k = 4$ cubic gates (exact, 2 985 984 coset-reduced words): $P = \frac{1425}{2048}$	COMPUTATION	w33_pass11312_depth_law.py
one gate, two qutrits, unconditional: 1944-element centralizer of $T \otimes 1$, 4110 orbits, 1000/1000 direct checks; 385 344 violators, $P = \frac{223}{2430}$; affine law on all 51 840 classes	THEOREM COMPUTATION	w33_pass11330_orbit_census.py

Claim	Badge	Witness
twisted fixed point $T_1 E T_1^{-1} \propto C^{-1} E C^T$: E symplectically in Stab(z_1) (648); independent census agrees on every class; all-frame classes (1296) = symplectically unsolvable	THEOREM COMPUTATION	`w33_pass11331_twisted_fixed_point.py`
long magic words: P(reversible) $\to 0$ (Kawada–Itô, portmanteau); $T^9 = 1$ gives reversible words at every depth $9m$; no rate	THEOREM	`w33_pass11332_reversibility_limit.py`
three qutrits, one gate: flagged bad fraction 0.111 ± 0.009 over 1100 sampled classes (seven-frame screen)	COMPUTATION	`w33_pass11333_three_qutrit_one_gate.py`
$\mathbb{F}_9$-linear perfect ticks, exhaustive: three magic legs 165888/165888, one magic leg 3456/41472 (reduced via $T^3 = Z$)	THEOREM COMPUTATION	`w33_pass11334_f9_three_leg_law.py`
linear decider: $Ms^{-k}Q(JMJ) = Q$ plus an affine frame condition; reversible frames = union of affine pieces; 51 838/51 840 classes + the two trivial $M = \pm 1$; 300/300 Weyl checks	THEOREM COMPUTATION	`w33_pass11350_linear_decider.py`
non-collinear cell: bad only with exactly 3 symplectic solutions (a transvection coset, 2/9), half of them; M bad $\Leftrightarrow -M$ good; $\frac{1}{9} = \frac{2}{9} \cdot \frac{1}{2}$	THEOREM COMPUTATION	`w33_pass11351_one_ninth.py`
three qutrits (31994 classes, linear decider; 660/660 Weyl checks): violating fraction 0.1333 ± 0.0019, not 1/8 (4.4σ); fixed/reversed rules persist; affine law fails (2 classes)	COMPUTATION	`w33_pass11352_three_qutrit_geometry.py`
level statistics: reversible magic ticks COE ($\langle r \rangle = 0.536$), violating CUE (0.600), $n = 5$; $\Theta^2 = +1$ forced by odd dimension	COMPUTATION	`w33_pass11353_level_statistics.py`
Clifford-averaged T on $PU(3)$ tensor spaces to dimension 2187: contraction exactly 3/8 on the first Clifford-anisotropic sector; reversibility decays ≈ 0.72 per gate	COMPUTATION	`w33_pass11354_decay_rate.py`
substrate Jarlskog: reversible $\Leftrightarrow U^T$ Clifford-conjugate to U; $J_{2t} = \text{avg}\lvert \operatorname{tr} U^\dagger C U C^\dagger \rvert^{2t} - \text{avg}\lvert \operatorname{tr} U^\dagger C U^T C^\dagger \rvert^{2t}$; qutrit $J_2 = J_4 = 0$; J_6 complete on all one-qutrit words to 3 gates (per word) and 4 (2 077 650/2 985 984); one gate 9/8	THEOREM COMPUTATION	`w33_pass11355_substrate_jarlskog.py`

Claim	Badge	Witness
J_6 on two qutrits: 240 ticks, 0 mismatches; $(T\otimes T)$SUM: 0.0030; $(1\otimes T)$SUM$(1\otimes T^2)$: 363/160	COMPUTATION	w33_pass11356_jarlskog_two_qutrits.py
witness degree by dimension: first non-vanishing J_{2t} has degree 10 ($d=2$), 6 ($d=3$), 4 ($d=5$; complete at 6); the 2-design property forces only $J_2=0$	COMPUTATION	w33_pass11357_jarlskog_degree_by_dimension.py
three-qutrit mechanism: no (S) solution ⇒ violating, exactly one ⇒ reversible (2021/2021); half-rule and -1 exchange loosen	COMPUTATION	w33_pass11358_three_qutrit_mechanism.py
higher-degree contraction (matrix-free, to dim 59 049): 3/8 at $(4,4),(6,3)$; 11/24 at $(5,5)$; 1/2 at $(9,0)$	COMPUTATION	w33_pass11359_higher_degree_rates.py
Jarlskog spectrum: all J_6 values of 1–2-gate words in $\mathbb{Q}(\cos 2\pi/9)$ (PSLQ, 10^{-50}), denominators $\mid 216$; rational values 9/8, 15/8	THEOREM COMPUTATION	w33_pass11360_jarlskog_arithmetic.py
J_6 incomplete on $PU(3)$: 4-dim family of non-reversible zeros ($J_8>0$; Gauss–Newton to 10^{-14}, rank $dF=4$); J_8 search clean; $d=5$ control 400/400; complete on 1.4M words to depth 20; hierarchy complete at $t^*\in[4,216]$	THEOREM COMPUTATION	w33_pass11369_j6_completeness.py
witness degree = twisted Frobenius–Schur count: $10,6,4,4$ for $d=2,3,5,7$ ($d=7$ predicted); qubit spin $5=E+2T_1+T_2$; 7 odd qutrit invariants	THEOREM COMPUTATION	w33_pass11370_twisted_indicator_degree_law.py
arrow = Altland–Zirnbauer class (A, AI, AIII, D, BDI; $T^2=C^2=+1$); one-gate violators AIII; no D on words; $J_{\mathrm{CKM}}(U^T)=J_{\mathrm{CKM}}(U)$; one gate: violators ⇒ $J_{\mathrm{CKM}}=0$	THEOREM COMPUTATION	w33_pass11371_flavour_vs_arrow.py
$\mathbb{E}\lvert\operatorname{tr}U_k\rvert^6=6+(3/8)^k$, $\mathbb{E}\lvert\operatorname{tr}U_k\rvert^8=23+16(3/8)^k$; t-design rates $0,0,\frac{3}{8},\frac{3}{8},\frac{11}{24},\frac{7}{12},\frac{1}{8}+\sqrt{7/32},\rho_{(8,8)}$; $\rho_{(18,0)}=0.781>0.72$	THEOREM COMPUTATION	w33_pass11372_irrep_contraction_rates.py
three-qutrit one gate exact: violating fraction $\frac{240857}{2388204}$, bad classes $\frac{437}{3276}$ (2308 stabiliser orbits, GAP)	COMPUTATION	w33_pass11373_three_qutrit_exact_fraction.py
J_8 survives: no spurious zero on pseudo-reflections (0/36), none from 307 J_6-spurious starts, $\min J_8\lvert_{\mathrm{dist}\ge 0.3}=7\times10^{-7}>0$; near 1 every witness is $O(\mathrm{dist}^{12})$	COMPUTATION	w33_pass11418_j8_completeness.py

Claim	Badge	Witness
$J_6(1+\lambda P) = \lvert\lambda\rvert^{12}\Delta_6(\psi)$; first T-odd Clifford invariant of a ray has bidegree $(6,6)$ (unique); two 4-dim spurious components	THEOREM COMPUTATION	`w33_pass11419_spurious_family.py`
one law behind the magic-axis rules: $M^2 z_1 = z_1 \Rightarrow$ frames with $\omega(a, z_1 + Mz_1) \neq 0$ violate ($n = 2$ all, $n = 3$ all orbits, $n = 4$ sample); lemmas L1–L4	THEOREM COMPUTATION	`w33_pass11420_magic_axis_rules.py`
four qutrits (sparse decider): four rules hold on all decided targeted classes; bad-class fraction 0.118 ± 0.006	COMPUTATION	`w33_pass11421_four_qutrits.py`
depth 5 exact: reversible 11003/49152; denominators $2^{3k-1}3^{[k\,\mathrm{odd}]}$; sampled drift withdrawn in Pass 11459 (threshold artefact)	COMPUTATION	`w33_pass11422_depth5_exact.py`
magic-axis law: $W(z_1 + Mz_1)$ commutes with $V_M T_1$, grades violating frames (proved); fixed axis at $n = 4$: all 549 classes of $\mathrm{Stab}(z_1)$ reached by the decider (97.6%), 0 failures	THEOREM COMPUTATION	`w33_pass11433_magic_axis_law.py`
$h_6 = \frac{1}{432}\sum_g \mathrm{sgn}(g) p_{ga}^3 p_{gb}^2 p_{gc}$ (three distinct MUBs), $\Delta_6 = 349920 h_6^2$: a chirality of the MUB geometry	COMPUTATION	`w33_pass11434_h6_explicit.py`
$J_8 = \lvert\lambda\rvert^{14}[24\Delta_7 + \lvert\lambda\rvert^2(\Delta_8 - 8\Delta_7)]$ on J_6's pseudo-reflection component, $\Delta_7 > 0$, $\Delta_8/\Delta_7 \geq 3.76$: $J_8 > 0$; correction of the earlier displayed expansion	THEOREM COMPUTATION	`w33_pass11435_j8_positivity.py`
$\lvert K\rvert = 27$ Cliffords pass through T: 8 effective choices per gate; exact $P_6 = 20327/131072$, $P_7 = 353927/3145728$; denominators $2^{3k-1}3^{[k\,\mathrm{odd}]}$ for $k \leq 7$	THEOREM COMPUTATION	`w33_pass11436_depth_reduction.py`
four-qutrit cells: same-line cell $77/342 \to 0.14$, others $\to \frac{1}{8}$; $P_4 = 0.128 \pm 0.005$	COMPUTATION	`w33_pass11437_four_qutrit_cells.py`
magic-axis law at $n = 4$: fixed axis 97.6% decided +0.6% proved, 1.8% open; same line: 166 decided, 0 failures	COMPUTATION	`w33_pass11456_n4_closure.py`
THEOREM: $\mathrm{Im}(M-1)$ nondegenerate $\Rightarrow$ magic-axis law (Weyl-coefficient moduli 0.844, 0.449, 0.293); sound on 376 classes; magnitudes reach 82%	THEOREM COMPUTATION	`w33_pass11457_magic_axis_theorem.py`

Claim	Badge	Witness
generic blind spot of J_6: rank 4 at 6 refined points; no continuous hidden symmetry; achiral eigenvector at some points only	COMPUTATION	`w33_pass11458_generic_component.py`
deep decay, tight cut $3 - 10^{-7}$ (10^7 samples per depth to 30): constant local rates 0.7209, 0.7231, 0.7278 ($\rho_{(8,8)} = 0.7228$); loose-cut drift withdrawn	COMPUTATION	`w33_pass11459_deep_decay.py`
five qutrits: one-gate bad-class fraction 0.138 ± 0.004 (3.3σ above 1/8); fixed-axis law on 24/24 decided classes	COMPUTATION	`w33_pass11460_five_qutrits.py`
THEOREMS: magic-axis law proved uniformly on all 1296 two-qutrit classes (352 moduli, 704 level hyperplane, 48 + 168 phases, 24 transvection); $n = 4$ open 0.24%	THEOREM COMPUTATION	`w33_pass11486_blind_classes_phases.py`
level-hyperplane extension ($z_1 \in \mathrm{Im}(M-1)$, $q \neq 0$): sound 752/752; $n = 4$ undecided classes 70 more proved	THEOREM COMPUTATION	`w33_pass11487_magic_axis_extension.py`
exact reversible fraction at eight cubic gates $P_8 = 665275/8388608$	COMPUTATION	`w33_pass11488_exact_depth8.py`
six qutrits: one-gate bad-class fraction 0.123 ± 0.011 (1000 classes)	COMPUTATION	`w33_pass11489_six_qutrits.py`
cut audit: deciders agree on 119 998 deep words; J_6 cut holds with < 1 decade margin	COMPUTATION	`w33_pass11490_cut_audit.py`
time-odd Hilbert–Molien series of a qutrit state $t^6(1+t)/\prod_{d\in\{1,2,3,4,6\}}(1-t^d)$, proved; generators $(3,2,1)$, $(4,2,1)$ MUB chiralities	THEOREM COMPUTATION	`w33_pass11491_time_odd_molien.py`
first time-odd state invariant: degree 9, 6, 6, 5, 5 ($d = 2, 3, 5, 7, 11$); two qutrits: unique in degree 6, restricts to $\Delta_6/108$ (1/120 + 1/1080, derived)	THEOREM COMPUTATION	`w33_pass11492_time_odd_by_dimension.py`
THEOREM: magic-axis law for every n ($Av = -v$ for $A = QJ$; $QJv = -v$ on all 4684 solution spaces, $n \leq 4$)	THEOREM COMPUTATION	`w33_pass11498_magic_axis_law_all_n.py`
THEOREM (frame level): good rules $Mz_1 = -z_1$, $M^2 z_1 = -z_1$; covers all 2592 classes at $n = 2$, $\approx 96\%$ at $n = 3$	THEOREM COMPUTATION	`w33_pass11499_good_rules.py`
exact $P_9 = 11519231/201326592$; cubed-phase test exact on all words to depth 9	COMPUTATION	`w33_pass11500_exact_depth9.py`

Claim	Badge	Witness
six qutrits: bad-class fraction 0.128 ± 0.007 (2500 classes)	COMPUTATION	`w33_pass11501_six_qutrits_extended.py`
$\Delta^{(2)} = \Delta_6/108$ proved; odd module generated in degrees 6–10, even ring in 1, 3–9 (to degree 13)	THEOREM COMPUTATION	`w33_pass11502_time_odd_module.py`
rule-cell shares $2, 1, 3/(3^{2n}-1)$, exact at $n = 2, 3$	THEOREM COMPUTATION	`w33_pass11503_rule_cell_shares.py`
cubed-phase test on $PU(3)$: 4 unitaries on non-affine relations, 0 counterexamples	COMPUTATION	`w33_pass11504_cubed_phase_test.py`
THEOREM: good rules for every n when the unipotent part at 1 and the z_1-block have dim ≤ 4 (all of $n = 2, 3$)	THEOREM COMPUTATION	`w33_pass11511_good_rules_all_n.py`
cubed-phase test NOT exact on $PU(3)$: explicit non-reversible unitary passes it (overlap 2.985); exact on words	COMPUTATION	`w33_pass11512_cubed_phase_exhaustive.py`
exact $P_{10} = 21873707/536870912$ (32 445 792 distinct depth-8 operators)	COMPUTATION	`w33_pass11513_exact_depth10.py`
six qutrits: 0.131 ± 0.006 (4000 classes)	COMPUTATION	`w33_pass11514_six_qutrits_4000.py`
time-odd generators $= (k, 2, 1)$ MUB chiralities, $k = 3..7$; free to degree 12, one syzygy in degree 13	THEOREM COMPUTATION	`w33_pass11515_odd_generators_and_syzygy.py`
UNION LAW: reversible frames $= \bigcup_Q ((1-(QJ)^{-1})K_0)^{\perp}$; necessity proved all n; exact on all $n = 2$ classes and 99.97% of $n = 3$	THEOREM COMPUTATION	`w33_pass11537_union_law.py`
one-gate W-law: $P_2 = 1/8$, $P_3 \ge 97/728$ (exact 437/3276); $P_4 \ge 0.1337 \pm 0.0003$, $P_5 \ge 0.1342 \pm 0.0003$, $P_6 \ge 0.1347 \pm 0.0005$; $(1-9^{1-n})/8$ refuted	THEOREM COMPUTATION	`w33_pass11533_one_gate_law.py`
cubed test exact on words: non-affine relations only at T-count ≤ 1 (depth ≤ 7)	COMPUTATION	`w33_pass11531_cubed_test_words_vs_pu3.py`
invariants proved to degree 20: stabiliser moments $M_1, M_3..M_9$ and five chiralities generate	THEOREM COMPUTATION	`w33_pass11534_time_arrow_module_exact.py`
good rules: dim-6 indecomposables (and J_8) witnessed	COMPUTATION	`w33_pass11532_good_rules_dim6.py`
decay: no linear recurrence of order ≤ 4 in $P_1..P_{10}$; rates still rising	COMPUTATION	`w33_pass11535_decay_exact_data.py`
Hesse CP $W = -108^3 \prod_{a<b}(\Pi_a - \Pi_b)$ (MUB triple products); $\rho = M_3 - 1 \in [0, 1/3]$; chambers = MUB orderings	THEOREM COMPUTATION	`w33_pass11641_hesse_cp_is_mub_vandermonde.py`

Claim	Badge	Witness
label-blind time: even invariants spectral to degree 20; arrows of degree 6–8 need outcome labels	THEOREM COMPUTATION	w33_pass11642_label_blind_time.py
Hamming null cube = eight transvection ticks; Weil phase = trace phase; arrow = handedness	COMPUTATION	w33_pass11643_hamming_cube_is_transvection_clock.py
union law: $k = 0$ off isotropic locus; splitting when $3 \nmid$ ord; proved off eigen-cells ($n = 2$), 99.67% ($n = 3$)	THEOREM COMPUTATION	w33_pass11644_union_law_splitting.py
time reversal = $j \mapsto \bar{j}$ of the Hesse cubic through ψ; sign W = sign Im j	THEOREM COMPUTATION	w33_pass11645_time_reversal_is_conjugation_of_j.py
union law THEOREM: union over semisimple reversers (semisimple-part lemma; A^3 in eigen-cells); all $n = 2$, 99.97% $n = 3$	THEOREM COMPUTATION	w33_pass11647_union_law_closed.py
label-sensitive arrows = arrows on the equal-spectra locus; series $t^6(1 + t - t^3)/((1 - t)(1 - t^2)(1 - t^3))$	COMPUTATION	w33_pass11648_label_blind_sigma.py
max $\lvert h_6 \rvert = \sqrt{3}/62208$ at $(0, \cos\frac{\pi}{8}, \sin\frac{\pi}{8} e^{5\pi i/6})$; orbit 72 + 72; $W = 0$ there	THEOREM COMPUTATION	w33_pass11649_maximal_arrow_state.py
no reversal ⇔ affine system unsolvable; 1/40, 1/28, ≈ 0.037 ($n = 4, 5$)	THEOREM COMPUTATION	w33_pass11650_no_reversal_fraction.py
n-qutrit Hesse space = dual even Weil; G_6 ($n = 1$), G_{33} ($n = 2$); 40 stabiliser bases → SRG(40, 12, 2, 4) of $Q(4, 3)$ (dual of $W(3, 3)$)	THEOREM COMPUTATION	w33_pass11651_n_qutrit_hesse_space.py
shear-twisted reversers: one Y with $YMY^{-1} = s^k M$ per class (Lemma C); same-line case closed at $n \leq 3$	THEOREM COMPUTATION	w33_pass11655_shear_twisted_reversers.py
label-sensitive arrows generated by g_6, g_7 with one degree-9 relation; series derived	COMPUTATION	w33_pass11656_sigma_module_two_arrows.py
two-qutrit Hesse $\mathbb{P}^4$: ticks = Burkhardt j-planes, factorisations = nodes, stabiliser bases = points; full $W(3, 3)$ incidence	THEOREM COMPUTATION	w33_pass11657_hesse_space_geometrises_w33.py
Hesse CP sign first at bidegree $(6, 6)$ for $n = 1, 2$; reflection tower stops at $n = 2$	THEOREM COMPUTATION	w33_pass11658_hesse_cp_tower.py
Burkhardt nodes = determinants (concurrences); frame sum rule $\sum \lvert \det_F \rvert^2 = (M_3 - 2)/6$	THEOREM COMPUTATION	w33_pass11659_burkhardt_nodes_are_concurrences.py
Pauli-invariant cubic tensors ≅ conjugate Hilbert space; Weil parity = transposition sign (all n)	THEOREM COMPUTATION	w33_pass11680_heisenberg_cubic_tensors.py

Claim	Badge	Witness
$\mathfrak{e}_8 = \mathfrak{sl}_9 \oplus \Lambda^3 \oplus \Lambda^{3*}$ of two qutrits: Pauli-singlet Cartan; roots = Witting; orthogonality = commutation	THEOREM COMPUTATION	w33_pass11681_e8_from_two_qutrits.py
Pauli centralisers: point E_6A_2, line A_2^4, factorisation $D_4 \times D_4$; triality = $\mathbb{Z}_3$ grading	THEOREM COMPUTATION	w33_pass11687_11691_two_qutrit_e8_dictionary.py
T gate = E_8 |3|-grading ($E_6A_1T_1$); chirality character $\varepsilon\tau$ ($C \times T$); antilinear Witting intertwiner 40/40	THEOREM COMPUTATION	w33_pass11687_11691_two_qutrit_e8_dictionary.py
tick centralisers: SM-shaped $A_2 + A_1$ only with magic ($\approx 0.2\%$; Clifford 0/45000)	COMPUTATION	w33_pass11692_tick_symmetry_in_e8.py
$\sum_{v\neq 0}(\mathrm{Im}\langle D_v\rangle)^2 = \frac{d}{2}(1 - \langle\Pi\rangle^2)$ (all n); quartic $\leq 27/8$, equality exactly on 320 line stabiliser states (flag × chirality)	THEOREM COMPUTATION	w33_pass11697_11698_chirality_vacuum.py
GSW Coble covariant of the E_8 Cartan = $-2\times$ Maschke quartic; equivariant quartics odd→even: dimension 1	THEOREM COMPUTATION	w33_pass11699_coble_covariant_is_maschke_map.py
no gate realises $8_s \leftrightarrow 8_c$; triality on $D_4 \times D_4$ only the central $\mathbb{Z}_3$ lift	COMPUTATION	w33_pass11700_triality_is_central.py
Kac: SM-shaped centraliser needs order ≥ 16; third-level clocks (order ≤ 9) give only GUT shapes; two commuting clocks give SM shape (5 037 660 pairs); $E_6A_2 \to D_5A_1 \to$ SM	THEOREM COMPUTATION	w33_pass11701_11702_clock_symmetry_breaking.py
SM-shaped commuting-clock breaking from $Z \otimes I$ = trinification on one line (1944/1944; colour, weak $\subset$ distinct $SU(3)_q$); compatible with the chirality vacua	COMPUTATION	w33_pass11703_vacuum_and_clock_share_a_line.py
line-trinification SM excludes every pure-state vacuum (colour $\subset \Lambda^3$, vacuum centraliser $\mathfrak{su}(8) \oplus \mathfrak{u}(1) \subset \mathfrak{sl}(9)$); clock-Hamiltonian ground state does not align	THEOREM COMPUTATION	w33_pass11706_11709_vacuum_clock_hypercharge.py
operator-type SM (83 592 clock pairs): Georgi–Glashow $SU(5)$ on five levels, vacuum on the other four ($9 = 5 + 4$); unique vacuum-annihilating hypercharge; standard spectrum	COMPUTATION	w33_pass11706_11709_vacuum_clock_hypercharge.py
single SM tick $R^2 \otimes R^{-1}$, $R = \zeta_{27}^{x^3}$ (648 of 6561, order 27); quartic bound $\frac{9}{8}3^{n-1}$, vacua = Lagrangian stabiliser states (all n)	THEOREM COMPUTATION	w33_pass11710_11711_single_tick_and_n_qutrit_vacua.py
family $SU(3)$ on three levels has centraliser E_6; each family $27 = 15 + \bar{6} + \bar{6}$ across the three grades	COMPUTATION	w33_pass11713_family_e6_grade_dictionary.py

Claim	Badge	Witness
$T^6/\mathbb{Z}_3$ W(3,3) vacuum $= 3(84) + 27(\bar{9})$ (orbifolder); 31 of 33 $\mathbb{Z}_6$-I SMs: levels $\{a\}\|\{b\}\|\{c,d\}$, 2+1 family levels, $y_t = g$ the nine-level determinant	COMPUTATION	`w33_pass11714_11716_heterotic_two_qutrit_levels.py`
Q_{em} is level-diagonal: integral occupation ⇒ standard charges; every fractional charge is a fractional-occupation twisted state (33 models); twisted families are single-level	THEOREM COMPUTATION	`w33_pass11802_11807_levels_charges_textures.py`
fractional class conserved: exotics unbalanced in all 33, so decoupling needs twisted condensates; cubic up Yukawa $g\varepsilon_{\alpha\beta}$, $m_t = m_c$ at tree level; $\sin^2\theta_W = 3/8$; trivector cubic = nine-level determinant	THEOREM COMPUTATION	`w33_pass11802_11807_levels_charges_textures.py`
all-order Yukawa textures: tree-level $Y_{\text{up}} = g\varepsilon$ ($m_t = m_c$) in all 33; exactly $g\varepsilon$ to all orders in 10; down/lepton zero to all orders (hidden-neutral) in 22, else degree ≥ 12 (corrected R)	COMPUTATION	`w33_pass11810_11813_yukawa_textures.py`
fractional-charge classes $= \langle[V]\rangle \times \langle[W]\rangle = \mathbb{Z}_2 \times \mathbb{Z}_3$ (47/47): one bit and one qutrit; Lean: level-charge integrality, $\sin^2\theta_W = 3/8$, $1680, 3 \cdot 9 = 27$	THEOREM COMPUTATION	`w33_pass11810_11813_yukawa_textures.py`
$SU(9)$ flavour unification: W(3,3) vacuum $= 3 \times (84 + 9 \times \bar{9})$; alternating cubic ⇒ copy-families give $(\|h\|, \|h\|, 0)$; $\mathbb{Z}_9$ centre forbids $84 \cdot \bar{9}^2$ (bottom needs $84 \cdot \bar{9}^3$)	THEOREM COMPUTATION	`w33_pass11818_11821_su9_flavour_and_corrected_textures.py`
corrected R rule (Cabo Bizet et al.): $\mathbb{Z}_6$-I lock exact in 10 models, hidden $SU(5)$ $\Lambda/M_s \approx 4 \times 10^{-9}$, class excluded; $\mathbb{Z}_6$-II $SU(9)$ models hierarchical with a single cubic top	COMPUTATION	`w33_pass11818_11821_su9_flavour_and_corrected_textures.py`
$\mathbb{Z}_6$-II ranking: no model fits; single-cubic-top models give the mass hierarchy, invert V_{us}/V_{cb} and allow renormalisable QLd^c; Pauli invariants 9:0, 80:0, 84:4; $\mathbb{Z}_3^2$ line flavour ≤ 1 entry per row	COMPUTATION	`w33_pass11826_11830_z6ii_candidate_and_qutrit_flavour.py`

Claim	Badge	Witness
Clifford quadric $J_aJ_b + J_bJ_a = 2\beta$: 121 = 40 contexts +45 splits +36 Kramers reversals; orthogonal = anticommuting explains 11210 (15, swap), 11177 (perfect gates), 27 frames, Round 23/24 sectors (Cartesian square of $J(4,2)$)	THEOREM COMPUTATION	w33_pass11831_11833_finite_ads4.py
Kramers stabiliser $SL(2,9)\cdot 2$ (finite Lorentz group); celestial sphere = 10 invariant contexts = $P^1(\mathbb{F}_9)$ with $S_6 \cong P\Sigma L(2,9)$ action; split 1152, context 1296	THEOREM COMPUTATION	w33_pass11831_11833_finite_ads4.py
tangent Minkowski space: Lorentz orbits 1 + 20 + 30 + 30; spinor map 4 : 1 onto the light cone; light-cone graph Brouwer–Haemers SRG(81, 20, 1, 6); reversal pairs compose to Fourier (270) or unipotent ticks (360)	THEOREM COMPUTATION	w33_pass11831_11833_finite_ads4.py
exact linear Weil rep: Rac 5 ⊕ Di 4; Rac⊗Rac = 15 ⊕ 10 multiplicity-free, Sym^2Rac = the unique bulk∩boundary mode; Rac⊗$\overline{\text{Rac}}$+Di⊗$\overline{\text{Di}}$ = 1 + ℂ[40]	THEOREM COMPUTATION	w33_pass11834_11837_singletons_bulk_poincare_e8.py
bulk ℂ[36] = 1 + 15 + 20 (SRG(36, 15, 6, 6)); contexts see rank 16 (the 15), splits rank 21 (the 20): reconstruction needs both	THEOREM COMPUTATION	w33_pass11834_11837_singletons_bulk_poincare_e8.py
finite Poincaré $M \rtimes SL(2,9)$: invariant operators separate mass shells; massless little group $3^2 \rtimes \mathbb{Z}_4$ (4 helicities +8 two-dim); massive $SL(2,3)$	THEOREM COMPUTATION	w33_pass11834_11837_singletons_bulk_poincare_e8.py
E_8 under the Weil-lifted Clifford group: $\mathfrak{sl}_9$ = ℂ[80], Λ^3 has 7 constituents; centraliser of finite Lorentz = $\mathfrak{u}(1)_P$ only; Lean Clifford quadric	THEOREM COMPUTATION	w33_pass11834_11837_singletons_bulk_poincare_e8.py
crosscap holography: $T_P^2 = P$; B_p symmetric on Rac, antisymmetric on Di, exactly equivariant; Sym^2Rac → ℂ[36] injective onto the bulk 15; two-point 5/+1/−1	THEOREM COMPUTATION	w33_pass11839_11842_crosscap_fields_couplings_spin.py
free fields: massless propagator distinct exactly on the light cone (−7/81); four irreducible inequivalent helicity modules on the 80 spinors (finite spinor-helicity)	THEOREM COMPUTATION	w33_pass11839_11842_crosscap_fields_couplings_spin.py

Claim	Badge	Witness
cubic couplings: Rac·Rac·15 = 1; bulk 20 has no Rac·Rac vertex, couples via $(\text{Rac}\otimes\overline{\text{Di}})^2$; massive Di = $2\cdot 2_0$ or 2_1+2_2: no three doublets	THEOREM COMPUTATION	w33_pass11839_11842_crosscap_fields_couplings_spin.py
E_8 commutant of the Tits extension (Lorentz × torus 2-elements) = $\mathfrak{su}(3)\oplus\mathfrak{su}(2)$ (dim 11, rank 3); Weil route gives $\mathfrak{u}(1)$; corrected in 11845	THEOREM COMPUTATION	w33_pass11843_e8_tits_lorentz_commutant.py
faithful A_6 Lorentz lift (order 360): E_8 commutant $\mathfrak{su}(5)$; torus signs break to $\mathfrak{su}(3)\oplus\mathfrak{su}(2)$ along exactly the Georgi–Glashow hypercharge (full ad-spectrum); $A_6\times\mathbb{Z}_7(Y)$ commutant = $\mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1)_Y$	THEOREM COMPUTATION	w33_pass11844_11846_e8_poincare_lifts_matter.py
matter: (Q,u^c,e^c) in Lorentz 5, (d^c,L) in Lorentz 10; translations only as $\mathbb{Z}_3\cdot M$ (no 4-dim submodule); finite Poincaré commutant $\mathfrak{su}(3)\oplus\mathfrak{su}(2)$	THEOREM COMPUTATION	w33_pass11844_11846_e8_poincare_lifts_matter.py
bulk path integral: $\langle O_x O_y\rangle = 216\,\Pi_{15}$; tree contact diagram = 186624× boundary cubic; hidden 20 in loops only; Lean $G^2 = 12G$ and crosscap symmetry	THEOREM COMPUTATION	w33_pass11847_finite_bulk_path_integral.py
Lorentz A_6 × central $\mathbb{Z}_3$ translation: E_8 commutant = $\mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1)_Y$ exactly (dim 12, centre 1)	THEOREM COMPUTATION	w33_pass11849_11851_spinor_lorentz_central_z3_generations.py
spinorial $SL(2,9)\subset Sp(4)$ via quaternionic Weil Di (reproduces parallel Rounds 34/35): commutant $\mathfrak{so}(11)$; $E_8 = (55,1)+(11,5)+(32,\text{Di})+(1,10)$, -1 only on the 128; one $SO(10)$ family per spinor; 27-index is colour	THEOREM COMPUTATION	w33_pass11849_11851_spinor_lorentz_central_z3_generations.py
boundary four-point: $\dim(\text{Sym}^4 15)^G = 3$, spanned by contact, 15- and 20-exchange; hidden 20 detected (residual $1/\sqrt{10}$); Lean GG hypercharge spectrum	THEOREM COMPUTATION	w33_pass11852_bulk_four_point.py
family no-go: $c_{E_8}(\mathfrak{su}(5)\oplus\mathfrak{su}(3)_{\text{fam}}) = \mathfrak{su}(2)\oplus\mathfrak{u}(1)$, so no $A_6/SL(2,9)$ commutes with GUT and family	THEOREM COMPUTATION	w33_pass11854_11857_family_nogo_chirality_firewall_combined.py
chirality firewall: Lorentz-commuting projections act on Di by ± 1; quaternionic 32 ⇒ vector-like (200 trials); no E_8 automorphism is chiral (corrected in 11859)	THEOREM COMPUTATION	w33_pass11854_11857_family_nogo_chirality_firewall_combined.py

Claim	Badge	Witness
bootstrap: s-channels $1, 15, 20, 24, 60$; 60 only 15-exchange, 24 no contact: boundary data fix all three couplings; $SL(2,9) \times \mathbb{Z}_3$ commutant $\mathfrak{su}(3) \oplus \mathfrak{so}(5) \oplus \mathfrak{u}(1)$; Lean A_5 mod 3	THEOREM COMPUTATION	w33_pass11856_boundary_bootstrap_channels.py
parity (Lorentz-normalising) and GUT-phase×parity orbifolds vector-like: E_8 adjoint is real; three central-charge sectors $1+20+30+30$ / $6+15+60$ twice; E_8 multiplicity-free under $\mathfrak{so}(11) \times SL(2,9)$	THEOREM COMPUTATION	w33_pass11859_11862_parity_orbifold_sectors_breaking.py
central translation $t_c = \exp(8\pi i Y) =$ colour triality ($q \equiv 12Y \bmod 3$); spinorial centre is $B-L$ (1/4 vs 3/4), rank 5 survives Lorentz-invariant breaking	THEOREM COMPUTATION	w33_pass11859_11862_parity_orbifold_sectors_breaking.py
reflection positivity: integer channel data; $g_3^2 \le 1/48$, $g'^2 \le 1/8 + 6g_3^2$, $g_4 \le 1/12$; Lean: $g^2 = 1$ in $SL(2,\mathbb{C}) \Rightarrow g = \pm 1$, no injective $A_6 \to SL(2,\mathbb{C})$	THEOREM COMPUTATION	w33_pass11861_bootstrap_positivity_bounds.py
$SL(2,9)$ ambivalent (all irreps self-conjugate, no Weyl spinors); Di has zero helicity on the light cone (continuous-spin type) $\Rightarrow$ no chiral Di-matter	THEOREM COMPUTATION	w33_pass11864_11866_weyl_chirality_bl_quark_sectors.py
$B-L$: all charged SM singlets are spinors; unique Lorentz-invariant (antisymmetric) $\nu^c\nu^c$ condensate; Levi stabiliser $= \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)_Y$ (seesaw)	THEOREM COMPUTATION	w33_pass11864_11866_weyl_chirality_bl_quark_sectors.py
charged-sector little groups A_5, S_4, S_3 (rotation type); one loop: bubble, box positive, positivity restored (tree bounds = perturbative limit); Lean: no injective $A_6 \to GL(2,\mathbb{C})$	THEOREM COMPUTATION	w33_pass11867_one_loop_positivity.py
AME$(10,3)$ = Glynn's $[10,5,6]_9$ arc (all 71 census states, Schur square 10); no Hermitian self-dual GRS; $\lvert\mathrm{Aut}_{LC}\rvert = 2880$, PGL$(2,9)$; one orbit, 79 888 260 016 373 760 perfect gates	THEOREM COMPUTATION	w33_pass11224_ame10_f9_structure.py
residual-symmetry flavour from all 606 three-dimensional irreps of all subgroups of $W(E_6)$, Sp$(4,3)$: no complete pattern, no Cabibbo; five surviving columns (TM_1 from the line stabiliser, $\sin^2\theta_{12} \approx 0.318$; Clifford $\frac{2}{3}\sin^2\frac{k\pi}{9}$; golden; TM_2)	COMPUTATION	w33_pass11225_flavour_from_symmetry.py

Claim	Badge	Witness
discrete Ryu–Takayanagi S = min-cut exact on all intervals of $\{5,4\}$, $\{4,4\}$ and perfect-tick $\{4,5\}$ networks and flat $\{4,4\}$ grids of p up to 5×5 (curvature not selected); non-perfect control fails	COMPUTATION	`w33_pass11226_holographic_area_law.py`
TM_1 sum rule: $\cos\delta=0$ at maximal θ_{23}, sign = octant ($\delta\in65°$–$108°$ or mirror); Clifford column only at $\sin^2\theta_{23}\geq0.586$; no renormalisable S_4 flavon potential selects TM_1 (0/1500 two-flavon vacua; sextic one-flavon 37/150)	COMPUTATION	`w33_pass11230_tm1_alignment.py`
μ from a vacuum $U(1)$: all FI-cancelling extreme rays of 128 D-flat models; $\mathbb{Z}_6$-I 0/23; $\mathbb{Z}_6$-II 54 models (429 vacua) protect μ with top allowed; exotics always stay light (rational-span "clean" vacuum withdrawn in 11240: integer remnants and R rules keep v, q, d, Higgs pairs light)	COMPUTATION	`w33_pass11232_mu_vacuum_symmetry.py`
char-2 cycle index: unitary (x^2+x+1, $Q=2$) contribution 0.27261186563783750 …; unipotent factor $P(u)=\prod_i(1-u2^{-1-2i})^{-1}$; $F_1..F_6$ exact; reproduces all E_n, $n\leq6$; E_n not monotone ($E_7<E_6$), so 11220's extrapolation 0.665166 was high; bracket $[0.6626,0.6777]$	COMPUTATION	`w33_pass11233_qubit_limit_cycle_index.py`
TM_1 on a line: stabiliser of a line acts as S_4 on its 4 points (order 1296, kernel 54); TM_1 iff the neutrino involution swaps the charged-lepton point X with a second point; flavons d_X and chord d_X-d_Q; criticality codimension 2; sequestered tie 48:48 broken by the sign of $(\varphi\cdot\chi)^2$ (16 vs 0)	COMPUTATION	`w33_pass11243_tm1_line_geometry.py`
clean $U(1)'$ μ-vacuum of 11232 withdrawn: integer lattice keeps 4 $v\bar{v}$ pairs light; with space group and R, also $q\bar{q}$, $d\bar{d}$, Higgs pairs; μ stays forbidden, vacuum F-flat by lattice	COMPUTATION	`w33_pass11240_clean_vacuum_discrete_rules.py`
μ-protecting anomaly-free $U(1)'\Rightarrow$ some colored state massless (pair-counting proof); $SU(3)^2$–$U(1)$ anomaly zero on all non-anomalous $U(1)$s of 215 models; 812 protected cases, 0 violations	THEOREM COMPUTATION	`w33_pass11242_u1prime_mu_anomaly_theorem.py`

Claim	Badge	Witness
char-2 unipotent identities (Jordan weights = odd-q formula at $q = 2$, 92/92; $P(\chi_2{=}0) = 2^{-m_2}$, 53/53) ⇒ qubit limit 0.66516065 (conditional); MC F_7 consistent; alternating/symmetric form ratio 2^{-k} proved	COMPUTATION	`w33_pass11244_hesselink_qubit_limit.py`
both char-2 identities are theorems (Lusztig pieces = Jordan strata, characteristic-free point counts; uniform size-two form) ⇒ qubit limit 0.66516065 unconditional; brute force on all 979200 elements of $\mathrm{Sp}(4,4)$ agrees	THEOREM COMPUTATION	`w33_pass11250_qubit_identities_proved.py`
sequestered S_4 model: $\varepsilon(\varphi\cdot\chi)^2$, $\varepsilon<0$ selects TM_1, $\varepsilon>0$ $\theta_{13}=0$; misalignment $O(\varepsilon)$	COMPUTATION	`w33_pass11248_sequestered_tm1.py`
TM_1 Yukawa layer: exact at $\varepsilon=0$, θ_{23},δ free; $\lvert U_{e1}\rvert^2$ breaks at ≈ 13 (ν) and 17 (e) per radian of vacuum tilt ⇒ alignment $\lesssim 0.1^\circ$, $\lvert\varepsilon\rvert \lesssim 10^{-3}$; unique equivariant cubic needed	COMPUTATION	`w33_pass11254_tm1_yukawa_breaking.py`
TM_1 selector: $\kappa(\varphi\cdot\chi)$ selects a TM_1 orientation for either sign on the symmetric vacuum manifold; in a scoped triplet-messenger model boson loops generate $\varepsilon<0$ of order 10^{-4}–10^{-3}	COMPUTATION	`w33_pass11267_tm1_selector.py`
AME(10, 3) with a twisted ten-cycle symmetry (twist order 1, 2, 4): 1600 invariant codes, 48 AME, all $\mathbb{F}_9$-linear	COMPUTATION	`w33_pass11249_ame10_cyclic_search.py`
Z6-I plane R-charges were absent from the earlier frozen snapshot; the old Z6-I μ verdict was therefore scoped to $U(1)$s, remnants and point group	COMPUTATION REFUTED	`w33_pass11247_z6i_r_symmetry_data.py`
restored Z6-I plane R charges (87 models, 30,980 fields): all 6 695 116/6 695 116 D-flat vacua protect μ with a top Yukawa; exact quotient to 1464 support/lattice states; 0 clean	THEOREM COMPUTATION	`w33_20261001_z6i_rcharge_lattice_quotient.py`
independent AME(10, 3) census: 138 new states, all $\mathbb{F}_9$-linear (Glynn)	COMPUTATION	`w33_pass11229_ame10_f9_census.py`

Claim	Badge	Witness
discrete/R colored-mass obstruction: 16 326 sampled protected cases had 0 escapes; restored Z6-I upgrade is exhaustive, 6 695 116 protected-H_u cases and 0 complete colored matchings	THEOREM COMPUTATION	`w33_20261001_z6i_colored_escape_exhaustive.py`
dynamical area law: brick-wall circuits of ticks ($N = 12$, depth 10, 570 intervals) never saturate the space-time min-cut everywhere; best 548/570 (random perfect gates on Bell pairs), VKV 534–539; deficits not only at edges	COMPUTATION	`w33_pass11231_dynamical_area_law.py`
point/line firewall (27/108/36 coincidences, non-isomorphic)	COMPUTATION	`w33_20260924_history_pointline_cospectral_firewall.py`
Time (§4)		
$40 = 1 + 12 + 27$; big cell of $Q(4,3)$	COMPUTATION	`w33_20260924_history_bigcell_q43_compactification.py`
finite light cone $t^2 - x^2 - y^2$; $1 + 8 + 6 + 12$	THEOREM COMPUTATION	same
$\mathrm{Sym}_2(\mathbb{F}_3) = \mathbb{F}_3^4/\langle 1111\rangle$; $D_0 + D_\infty + D_+ + D_- = 3I$	THEOREM COMPUTATION	`w33_20260924_positive_null_counter_cover.py`
null graph spectra; $U^3 = I$ at $t^\star = 2\pi/9$	COMPUTATION	`w33_20260924_null_history_spectral_clock.py`
fixed 9 is not the qutrit module	REFUTED	`w33_20260924_spectral_clock_fixed9_group_probe.py`
36 triangles; unique orientation cycle; $82 = 1 + 81$, $81 = 35 + 46$	COMPUTATION	`w33_20260924_history_invariant_cycle_orientation.py`
filled temporal H_1 injects into $H_1(W(3,3))$	COMPUTATION	`w33_20260924_history_to_global_h1_intertwiner.py`
universal cover: 8-regular tree, $\pi_1 = F_{82}$	THEOREM COMPUTATION	`w33_20260924_temporal_universal_cover.py`
Weil phase census $1^{13} i^4 (-1)^6 (-i)^4$; $(FG)^3 = iF^2$	COMPUTATION	`w33_20260924_temporal_maslov_mu12.py`
common S_4 of null directions, Hesse clocks, A_2 factors	COMPUTATION	`w33_20260924_null_hesse_4a2_s4_intertwiner.py`
tetracode = clock evaluation code	COMPUTATION	`w33_pass10946_clock_code_cone_objectwise.py`
negacyclic hyperbolic pair, $N^4 = -I$ (other track)	COMPUTATION	`w33_pass10948_clock_tetracode_negacyclic_hyperbolic_bridge.py`

Claim	Badge	Witness
Page–Wootters diagonal weld; 24 vs 248	COMPUTATION	`w33_20260924_page_wootters_diagonal_weld.py`
Hermitian 3 + 1: 21 + 30 + 30; null directions = spread	COMPUTATION	`w33_20260924_hermitian_3p1_spread_bridge.py`
no compatible complex structure J	REFUTED	same
forgetting two trits gives the depolarising channel	THEOREM	Thm. 4.7 (proof in text); `bt823_the_closure.py`
E_8 (§5)		
$E_8 = \mathfrak{sl}_9 + \Lambda^3 + \Lambda^6$ as packet operations	COMPUTATION	`w33_20260924_temporal_a8_e8_process_bracket.py`
Hesse $4A_2$; $240 = 72 + 24 + 144$	THEOREM COMPUTATION	`w33_20260924_temporal_hesse_4a2_a8_bridge.py`
no equivariant edge–root bijection	COMPUTATION	`w33_pass1020_e8_transitive_51840.g`
$240 = 40 \times 6$ fibration onto points	COMPUTATION	`w33_pass1021_e8_fibration_over_forty.g`
Hesse clock gradings 1+56+134+56+1 and |3|	COMPUTATION	`w33_20260925_hesse_clock_e8_gradings.py`
$\mathbb{Z}_3$ grading is the mod-3 shadow	COMPUTATION	`w33_20260925_e8_parabolic_cubic_clock_lift.py`
cubic ladder $54 \to 81 \to 83$; 45 triads ×12	COMPUTATION	`w33_20260925_cubic_clock_bracket_ladder.py`
explicit signed-3 ⊗ 27 regular semisimple Cartan plane: cubic-Jacobian rank 78, commuting dimension 3, $\operatorname{rank}(18 \operatorname{ad} v) = \operatorname{rank}(18 \operatorname{ad} v)^3 = 234$	COMPUTATION	`w33_pass11260_exact_semisimple_cartan.py`
exact Cartan-to-qutrit G_{26} map: restricted Pfaffian $2^{17} C_3 N_9 S_9^3$; stabilizer/SIC mirror multiplicities 1/3 versus Jacobian 2/1	COMPUTATION	`w33_pass11261_g26_cartan_coordinate_map.py`
four stabilizer MUBs: complex-linear G_{26} image A_4; complex conjugation completes it to line-stabiliser S_4; tetrahedral chords = oriented MUB pairs	COMPUTATION	`w33_pass11262_g26_mub_s4_yukawa_bridge.py`
234 nonzero adjoint modes form 78 exact $\mathbb{Z}_3$ spectral triplets; positive-power character traces cancel, kernel index $8 + 3\omega + 3\omega^2 = 5$; no $\mathbb{Z}_2$ supertrace cancellation	COMPUTATION	`w33_pass11263_full_graded_spectrum.py`
G_{26} Jacobian $= 19440 \prod_{9\,\mathrm{SIC}} \ell \prod_{12\,\mathrm{MUB}} m^2$ symbolically in $\mathbb{Q}[\omega]$; reflection group order 1296, projective order 216, all elements qutrit Clifford	COMPUTATION	`w33_pass11269_g26_qutrit_dictionary.py`

Claim	Badge	Witness
E_8 Pfaffian and G_{26} Jacobian have mirror valuations $(3,1)$ and $(1,2)$; determinant 5 isolates the two strata: $P^2/J \propto A^5$, $J^3/P \propto B^5$ (divisor-lattice index only, no physical $\mathbb{Z}_5$ claim)	COMPUTATION	w33_20261001_g26_e8_mirror_separation.py
intrinsic Cartan grade-pair form $G = \left(\begin{smallmatrix} 590 & 20 & 0 \\ 20 & 980 & 0 \\ 0 & 0 & 300 \end{smallmatrix}\right) > 0$; equal 21-mirror projective repulsion has a strict local maximum at $[0{:}0{:}1] \mapsto [1, \zeta_9, \zeta_9^{-1}]$, whose cubic interface is CP-conserving	THEOREM COMPUTATION	w33_20261001_cartan_kinetic_wall_vacuum.py
balanced 21-mirror logarithmic barrier: the same qutrit ray is exactly $\lvert T_3\rangle = (1, \zeta_9, \zeta_9^{-1})/\sqrt{3}$, with exact projective Hessian spectrum $30^2, 54^2$; its projective Clifford orbit has 72 rays, stabiliser C_3, and selects one MUB/tetrahedral axis with purity pattern $(1/3, 5/9, 5/9, 5/9)$; conjugates stay in one orbit, so scalar G_{26} invariants cannot choose CP/time orientation	THEOREM COMPUTATION	w33_20261001_balanced_g26_tmagic_vacuum.py
four sparse rank-54 single-foliation transducers survive both nonzero $K_{81} \to U_{81}$ cocycle orientations, retaining 27 triangles and $S_2 \oplus L$ ranks $27 + 27$ at split primes $103, 109$	THEOREM COMPUTATION	w33_20261001_u81_54d_cocycle_transducer.py
qutrit conjugation $\tau(a,b,c) = (-a,b,-c)$ leaves the chamber cocycle invariant; $\Psi(h,d) = (\tau h, -d)$ exchanges U_{81} orientations $s \leftrightarrow -s$ and reverses the class-three central tick	THEOREM COMPUTATION	w33_20261001_u81_antiunitary_orientation.py
Freudenthal, real forms, Albert (§§6–8)		
56 shares one 27 with the 54; $54 + 2 = 56$ refuted	COMPUTATION REFUTED	w33_pass10949_freudenthal_quasiconformal_clock_cone.py
Freudenthal normal form of the quartic, no sign repair	COMPUTATION	same
57-cone, Heisenberg law, GKN inversion	COMPUTATION	same
$K(e_\alpha, e_{-\alpha}) = -60(-1)^{\mathrm{ht}\,\alpha}$; split = height twist	COMPUTATION	same
tick lines: $E_{8(-24)}$, Klein four-group, $80 + 3 \times 56$	COMPUTATION	same
Euclidean Albert algebra, Peirce $1 + 16 + 10$, $(1,9)$ vs $(5,5)$	COMPUTATION	same

Claim	Badge	Witness
Der $= F_{4(-52)}$, $\mathrm{str}_0 = E_{6(-26)}$	COMPUTATION	`w33_pass10950_clock_albert_lorentz_spinor.py`
$\mathfrak{so}(1,9)$, Majorana–Weyl 16; $16 = (2,4) + \overline{(2,4)}$	COMPUTATION	same
$Q_\psi = 6L_c - 2$; matter parity $= \exp(2\pi D)$	COMPUTATION	same
clock $g^4 = -I$ is the central 2π spin sign (other track)	COMPUTATION	`w33_pass10951_clock_pin_spin_central_sign_bridge.py`
Albert $16 = 8_s + 8_c$; half-turn $U^2 = -I_{16}$; minimal full-clock carrier $16 \oplus 16$ (other track)	COMPUTATION	`w33_pass10956_albert_spin8_halfspin_normalizer.py`, `w33_pass10959_albert_clock_extension_obstruction.py`
Physics (§9)		
transvection spine: 648, $1 \oplus 3 \oplus 8$, $9 \oplus 9 \oplus 9$	COMPUTATION BRIDGE	`bt886_standard_model_spine.py`
Steinberg: every order-3 element splits $27 + 27 + 27$	THEOREM	Thm. 9.2 (proof in text)
no Pauli-spanned E_6 on qutrit registers	THEOREM COMPUTATION	Holotrade `1cde1f1`
$W(3,3)$ = Pauli geometry of $\mathbb{Z}_3$-twisted V_{E_8}	COMPUTATION	Holotrade `68df33f`
order-3 orbifold completes no SM family	THEOREM COMPUTATION	Holotrade `ed1eaca`, `a6e1c69`, `a779053`
$\mathbb{Z}_6$: 42 of 79 SMs from the $W(3,3)$ vacuum	COMPUTATION	Holotrade `5b3f3ad`
CZ-type breaking 45/45; $SU(3) \times SU(2)$ cut 25/25	COMPUTATION	Holotrade `d772153`, `94df376`
Higgs alone in the qutrit plane; DTS 55/87	COMPUTATION	Holotrade `3caf15e`, `a6f1cae`
F/D conflict; Gordan alternative 215/215	COMPUTATION	Holotrade `babfd48`
matter parity in all 87; class closure conditional	COMPUTATION	ledger commit `78ec0a0fc`
every sampled D-flat support has a matter-odd singlet	COMPUTATION OPEN	ledger commit `a3dbdf2d7`
chirality no-go	THEOREM COMPUTATION	`w33_pass346_the_chirality_no_go.py`
chirality swap is exactly the antiunitary (determinant-odd) coset	COMPUTATION BRIDGE	Holotrade `e92a047`; `w33_pass10952_clock_complete_positivity_firewall.py`, `w33_pass10955_d4_halfspin_clock_bridge.py`

Claim	Badge	Witness
no matter-even SUSY singlet vacuum: 0/88 over full $B-L$, 0/1024 $\mathbb{Z}_2$ parities; forced $B-L=\pm1$ condensate	COMPUTATION	`w33_pass10960_matter_even_dflat_closure.py`
hidden sector: 32 exact (all directions), 55 over invariant generators, 1 by 62,309 rays; anomaly-free models have no parity	COMPUTATION	`w33_pass10962_hidden_sector_dflat_tiers.py`, `w33_pass10962_tierc_rays.py`
time reversal = antiunitary = chirality swap (one coset)	THEOREM	Corollary 9.7 (proof in text)
FI acts through $f=4Q_\psi-\frac{4}{5}Q_\chi$; singlet values $\{-8,0,8\}$; forced field $=\nu^c$ of $\mathbf{16}_{-3}\subset\mathbf{78}$	COMPUTATION BRIDGE	`w33_pass10964_e6_shaped_fi_mechanism.py`
Baryon triality impossible in 215/215: $Q(u^c)+Q(e^c)=2Q(q)$ $(SU(5))$	COMPUTATION	`w33_pass10965_baryon_triality_impossible.py`
Most general matter parity (torus × space group) survives FI in 0/215; forced $Q(n)=-Q(u^cd^cd^c)$ in 290/302; quartic RPV regenerated 23/23	COMPUTATION	`w33_pass10967_fi_vacuum_regenerates_rpv.py`
(Withdrawn by P10974) Plane-rotation R-parity under the old per-plane rule: 3/128; $\mathbb{Z}_6\mathrm{II}_{23}$ 38 vacua, exotics massless	COMPUTATION	`w33_pass10968_r_symmetry_parity_and_massless_exotics.py`
68-direction vacuum: one parity element, exotics full rank by order 5, Higgs rank 6	COMPUTATION	`w33_pass10968_leaf1_vacuum_certificate.json`
23-direction vacuum: exotics full rank, Higgs rank 5/6 at all orders, $W\|_S\neq0$	COMPUTATION OPEN	`w33_pass10968_approxR_vacuum_certificate.json`
$\mathbb{Z}_{12}$-I: 289 SMs, 14 parity vacua, exotics massive in none; $\mathbb{Z}_2\times\mathbb{Z}_6$-I: 29 SMs, 0 survive	COMPUTATION	`w33_pass10968_z12_left_chiral_ledger.json.gz`
Harmless RPV needs $\tilde{m}\gtrsim6\times10^{14}$ GeV; $\lambda<0$ above 3.2×10^{10} GeV, so $m_t\leq171.0$ GeV (5.4σ low)	COMPUTATION	`w33_pass10969_rpv_heavy_superpartners_vs_higgs_quartic.py`
Established R-rules (prime planes + γ-corrected G2 plane): $\mathbb{Z}_6\mathrm{II}_{23}$ closes, $\mathbb{Z}_6$-II 0/128; $\mathbb{Z}_3\times\mathbb{Z}_6$ 0/5	COMPUTATION	`w33_pass10974_valid_r_rules_close_the_rotation_door.py`
$C_{\mathrm{Aut}(E_6)}(c)=\langle c\rangle\times\langle-1\rangle$: single admissible $\mathbb{Z}_2^R$; all 14 $\mathbb{Z}_{12}$-I parity vacua decided, 0 viable	COMPUTATION	`w33_pass10978_z12_e6_admissible_r_rule.py`
$T^6/\mathbb{Z}_3$: 0 SMs in 64 000; $\mathbb{Z}_3\times\mathbb{Z}_3$ (all rules established): 13 SMs, parity on FI rays in 3, 28 parity vacua, W exponent vectors independent through degree 11 in all 28	COMPUTATION	`w33_pass10979_z3xz3_parity_vacua_f_obstruction.py`

Claim	Badge	Witness
$\mathbb{Z}_3 \times \mathbb{Z}_3$ parity vacua: $W\|_S = F(p)$, $F(\omega p) = \omega F(p)$ to all orders (28/28); SUSY roots exist (28/28, $\langle\phi\rangle \approx M_s$); hidden-refined index: 48 chiral charge-$\frac{1}{3}$ states, 6 hidden singlets, massless in every hidden-unbroken vacuum; hidden groups IR-free; 54 SU(4)-condensed parity vacua cure the index but keep ≥ 120 fractional states massless (unbroken discrete 3-group of order 3^4 or 3^6); exact-lattice certificate of the 10978–10980 integer programs (0 differences)	COMPUTATION	`w33_pass11024_z3xz3_susy_vacua_fractional_charge_index.py`
"Flat-or-massive" is an unbroken R-symmetry, not a law: where $W\|_S \equiv 0$ and the exotic triplet is massless, both are forbidden by the same unbroken symmetry (46/48); counterexamples both ways in $\mathbb{Z}_{12}$-I (2 and 11)	COMPUTATION	`w33_pass11086_flat_or_massive_is_an_unbroken_r_symmetry.py`
The unbroken 3-group protecting the $\mathbb{Z}_3 \times \mathbb{Z}_3$ fractional charges is assembled from broken-$U(1)$ remnants, space-group translations and $\mathbb{Z}_3^R$ (point group and hidden centres contribute nothing); 47% of forbidden mass terms redundantly protected; the 28 singlet vacua also leave a massless $U(1)'$ coupled to SM matter	COMPUTATION	`w33_pass11087_unbroken_three_group_protection.py`
Baryonic and mixed hidden condensates (all 775/802/277 single-factor invariants, exact subset-sum parity engine): 63 more parity vacua with the curing factor condensed, every one keeping ≥ 142 fractional multiplets massless	COMPUTATION	`w33_pass11088_baryonic_hidden_condensates.py`
No non-SUSY twin of $T^6/\mathbb{Z}_3$ or $\mathbb{Z}_3 \times \mathbb{Z}_3$ with an order-3 shift ($9V^2$ even, $9v'^2$ odd); $\mathbb{Z}_6$-II has no non-SUSY twin; all 87 $\mathbb{Z}_6$-I W(3,3) SMs have twins with the same shifts ($N = 0$)	THEOREM COMPUTATION	`w33_pass11089_w33_twist_without_supersymmetry.py`

Claim	Badge	Witness
$3Q + t$ mod 3 is constant on all 946 fixed points of the 13 $\mathbb{Z}_3 \times \mathbb{Z}_3$ SMs and is a linear character of the space group; 516 fractional fixed points host no neutral field; gauge-only rules still leave ≥ 15 fractional multiplets massless	COMPUTATION	`w33_pass11090_resolution_cannot_free_fractional_charges.py`
$\mathrm{Sp}(4,3)$ is perfect, so the multiplier is the unique nontrivial character of $\mathrm{PGSp}(4,3)$ (identification with half-spins: Holotrade); the other track's cubic tick is this character, its chamber-sheet sign is flag-relative	THEOREM COMPUTATION	Holotrade `98cafff`; `w33_pass11091_one_global_orientation_bit.py`
The 87 non-SUSY $\mathbb{Z}_6$-I twins (five orbifolder $N = 0$ faults repaired): anomaly-free up to one GS $U(1)$, parent gauge group, 3 net generations, 87/87; all tachyonic (2–20 states in the θ sector); tachyonic fixed point $\Leftrightarrow$ parent has a massless oscillator-excited state there (261/261)	COMPUTATION	`w33_pass11092_nonsusy_z6i_twins_are_tachyonic.py`
Tachyon-freedom and the SM exclude each other in the $\mathbb{Z}_6$-I $W(3,3)$ scan: 47/58 base shifts tachyonic for any Wilson lines; 66 000 draws on the rest give 9 576 tachyon-free twins and 705 SM parents, overlap 0 (78.7 expected); SM exotics v, w, x become the tachyons	COMPUTATION	`w33_pass11093_tachyon_free_filter_z6i_scan.py`
Every θ-sector tachyon of the 87 $\mathbb{Z}_6$-I twins has charge $\pm\frac{1}{2}$ or colour, none neutral (two engines; mass-level orbifolder validated 87/87 and against the non-SUSY orbifolder)	COMPUTATION	`w33_pass11094_tachyons_break_electromagnetism.py`
Non-SUSY orbifolder cross-check 19/19 identical; $SO(16) \times SO(16)$ with the A_8 shift: 793 508 Witten shifts in 6 classes; 153 tachyon-free three-generation models (104 inequivalent), all re-verified	COMPUTATION	`w33_pass11095_so16xso16_a8_standard_models.py`
Twins of $\mathbb{Z}_{12}$-I (289/289), $\mathbb{Z}_3 \times \mathbb{Z}_6$ (5/5), $\mathbb{Z}_6 \times \mathbb{Z}_6$ (10/10) all tachyonic; $\mathbb{Z}_2 \times \mathbb{Z}_6$-I: 8 tachyon-free, none with SM chirality; corrects P11089's mixed condition	COMPUTATION	`w33_pass11096_twins_of_the_other_w33_families.py`

Claim	Badge	Witness
$SO(16) \times SO(16)$ A_8 models: fractional fermions not symmetry-protected – all massive in 33/104 (gauge, Witten $\mathbb{Z}_2$, PG, SG, H-momentum), 6/33 at cubic order, all by order 7	COMPUTATION	`w33_pass11097_fractional_fermion_masses.py`
Tree-level Yukawas, generic Higgs: up-type singular values $(1, 1, 0)$ (ϵ-structure) in 73/104, full rank in 31; down-type zero in 77	COMPUTATION	`w33_pass11098_yukawa_textures.py`
$\Lambda_{10}(SO(16)^2) = +363\,M^{10}$ (massless term $-2112 = n_B - n_F$); A_8 models: $\Lambda_4 \to (V/3)\Lambda_{10} > 0$, $n_B - n_F \in [-480, -48]$ (corrected in Pass 11106)	COMPUTATION	`w33_pass11099_vacuum_energy_so16_a8.py`
$SO(16) \times SO(16)$ with $W(3,3)$ shifts on $\mathbb{Z}_3 \times \mathbb{Z}_3$, $\mathbb{Z}_6$-I, $\mathbb{Z}_{12}$-I: 77 600 draws, no tachyon-free SM (only $T^6/\mathbb{Z}_3$ works)	COMPUTATION	`w33_pass11100_so16_other_w33_geometries.py`
Localized light Higgs: single heavy top $(0, w, w)$ in all 31 twisted-up models, single heavy bottom in 27; $m_c = m_u$ at tree level (equilateral fixed points); 12 models pass every test (fractional masses at $T^* = 3$)	COMPUTATION	`w33_pass11101_localized_higgs_heavy_top.py`
Higher orders do not split charm from up: in all 12 survivors the non-top 2×2 block is symmetric under the point reflection of the generation line at every order (72/72 Higgs cases); $m_c/m_u = O(1)$ parametrically	COMPUTATION	`w33_pass11102_higher_order_quark_hierarchy.py`
No alignment of the scalar VEVs (fixed-point dependent, 6 patterns, 2 area regimes, hidden group unbroken/broken) splits charm from up (0/36 cases); strange–down split only via vector-like $\bar{d}$ (the 8 models with them)	COMPUTATION	`w33_pass11103_reflection_breaking_vacua.py`
Top and bottom in one doublet iff H_u, H_d at the same fixed point (36/108); then $V_{tb} \approx 1$, $V_{cb} \sim V_{ub} \sim \epsilon^2 \sim m_c/m_t$; single heavy τ, $m_\mu = m_e$ at the same order	COMPUTATION	`w33_pass11104_ckm_and_leptons.py`

Claim	Badge	Witness
Symmetry group of the $\mathbb{Z}_3$ space-group rule $= \Delta(54) = H_{27} \rtimes \langle -I \rangle$ (order 54; centre of H_{27} = point-group twist; index 12 in Clifford-648); one Wilson-line-free torus in 104/104: two-qutrit 3^{1+4} ($W(3,3)$ graph) realised in 0/104	BRIDGE	`w33_pass11105_delta54_flavour_join.py`
Full one-loop integrand (36 sectors, Wilson lines) modular invariant, massless term $= -252$ sector by sector; family modulus minimal at $T = \rho$; Wilson-line moduli roll to small radius, winding tachyon in 12/12; $\Lambda > 0$ throughout	COMPUTATION	`w33_pass11106_one_loop_orbifold_point.py`
Winding tachyons built explicitly: $\Delta(\mathrm{Im}\, T = 1) = -(3 - \sqrt{3})/6$; critical radius $\sqrt{3}$ (4 models) or 1.51 (7), model 77 only at ρ; colour singlets, fractionally charged in 10/12 (neutral directions in 53, 57); corrects Pass 11106's onset	COMPUTATION	`w33_pass11107_winding_tachyons.py`
Tachyon-free over the Wilson-line moduli: 104/104 at $2i$, 55 at i, 1 at ρ (model 18, which fails fractional charges, top = charm, no down Yukawa); rescan: 4/387 new models free at ρ, all with untwisted quarks (correlation not significant, Pass 11113)	COMPUTATION	`w33_pass11108_tachyon_free_moduli_rescan.py`
$\mathbb{Z}_3$ Yukawas = A_2 theta functions of T (duality closes only for the double-triangle action); $\lvert Y_{\mathrm{dist}}/Y_{\mathrm{same}}\rvert = \frac{1}{2}$ at ρ, $(\sqrt{3}-1)/2$ at i; m_c/m_t needs $\mathrm{Im}\, T \approx 3.2$	BRIDGE	`w33_pass11109_instanton_yukawas_at_rho.py`
Model 2 potential at 20 tachyon-free moduli points: $\Lambda > 0$ everywhere, family minimum at ρ (2.4× lower than at $3.2i$), Wilson-line minimum on the tachyon boundary; dilaton runaway	COMPUTATION	`w33_pass11110_moduli_map.py`
3^{1+2k} has faithful degree 3^k: three twisted generations force $k = 1$; family⊗colour gluing impossible (twist $\notin$ SM gauge centre on quarks, 12/12)	BRIDGE	`w33_pass11111_no_243_family_group.py`

Claim	Badge	Witness
Models 53, 57: neutral tachyons wind in one Wilson-line torus, charged in the other (degenerate for any common B); the one-loop potential shrinks the neutral torus first ($\Delta\Lambda = -1.3, -2.3; -0.9, -1.7$): SM-neutral first instability	COMPUTATION	`w33_pass11112_neutral_tachyon_corridor.py`
ρ-safety vs quark sector over 491 models: 0/152 twisted vs 5/339 untwisted, Fisher $p = 0.16$ (not significant; corrects Pass 11108's reading)	COMPUTATION	`w33_pass11113_rho_tachyon_correlation.py`
$\det M(h)$ is a Hesse-pencil cubic in the Higgs direction; at $T = \rho, \lambda = \omega^2$ (singular member); $\min_h \sigma_2/\sigma_1 = \lvert Y_{\text{dist}}/Y_{\text{same}}\rvert = \frac{1}{2}$ (localised Higgs optimal)	BRIDGE	`w33_pass11114_hesse_mass_determinant.py`
$B = 0$ axis: approximate Fricke duality in model 10 (extremum at tachyonic $1/\sqrt{3}$); model 77 tachyon-free to 0.87, Λ monotone to its onset	COMPUTATION	`w33_pass11115_axis_duality_and_window.py`
Family-H_{27} centre (orbifold twist) is a gauge element in 104/104 ($U(1)$s alone 97; FI $\mathbb{Z}_3$ exactly in 3); no $SU(3)$ gluing in any model: two-qutrit group on no matter	COMPUTATION	`w33_pass11116_one_qutrit_vs_other_track.py`
Neutral condensate (53, 57): breaks exactly one $U(1)$ (component along the anomalous one), SM and hidden non-abelian intact, $\Delta(54)$ singlet, no cubic term	COMPUTATION	`w33_pass11117_neutral_condensate.py`
Model 57 gradient flow on a 6×6 grid: neutral boundary from equal radii; separatrix at a 1–2% head start for the charged torus; no interior minimum	COMPUTATION	`w33_pass11118_wilson_line_flow.py`

Claim	Badge	Witness
Neutral tachyons on their own Wilson-line torus in 60/491 models (38 neutral-only), all with twisted quarks; gauntlet passes: A8SM_20260928_2233_TF, A8SM_20260928_533_TF, A8SM_20260941_7061_TF, A8SM_20260941_7243_TF, A8SM_20260942_36621_TF, A8SM_20260942_46043_TF, A8SM_20260951_24165_TF, A8SM_20260971_14027_TF, A8SM_20260971_18172_TF, A8SM_20260971_19247_TF, A8SM_20260971_3818_TF, A8SM_20260971_40521_TF, A8SM_20260971_5285_TF, A8SM_20260971_5904_TF, A8SM_20260972_17224_TF, A8SM_20260972_19688_TF, A8SM_20260972_2080_TF, A8SM_20260972_31386_TF, A8SM_20260972_5273_TF, A8SM_20260982_1645_TF, A8SM_20260982_47048_TF	COMPUTATION	`w33_pass11119_neutral_tori_across_491.py`
ρ-safety: necessary sparse-cell condition (2 norm-1, 10 norm-$\frac{5}{3}$ cells), not sufficient (15 tachyonic share it); no norm invariant decides	COMPUTATION	`w33_pass11120_rho_cell_criterion.py`
Bridge to the other track: Hesse parameter $\lambda(i) = 1 + \sqrt{3}$ ($j = 1728$), $\lambda(\rho) = \omega^2$ (singular); object table (H27 realised, Clifford-648 partial, 243 not)	BRIDGE	`w33_pass11121_hesse_gauge_bridge.py`
Six neutral-exit survivors on the $B = 0$ diagonal: monotone to the onset, volume law $\Lambda_\beta \approx 160y^2 - (5\text{–}13)$, $d\Lambda/dy \approx 600$	COMPUTATION	`w33_pass11122_six_potentials.py`
Critical radii $\sqrt{3}$ (2) and $\varphi^2/\sqrt{3}$ (4); no massless vector; KLT amplitude with no contact quartic; radion slope $1/(2\sqrt{3})$, 0.247	BRIDGE	`w33_pass11123_critical_point_amplitude.py`
Condensate unlocks forbidden lepton (6/6), down (4/6), ν Dirac (6/6) entries at orders 2–6, one insertion each; up and μ untouched; separate-winding zero was an artefact; "missing Yukawas" reading corrected in 11127	COMPUTATION	`w33_pass11124_condensate_unlocked_couplings.py`

Claim	Badge	Witness
B-field map: charged tachyons need $B \neq 0$ and $\mathrm{Im}\, T \leq 1$; $B \geq 0.4$ lifts the neutral tachyon at 1.4 in exactly the four golden models	COMPUTATION	`w33_pass11125_bfield_charged_onset.py`
Condensate has zero family-torus quantum numbers: $T^* = \rho$ and $m_c = m_u$ unchanged	BRIDGE	`w33_pass11126_family_modulus_under_condensation.py`
Light-doublet lepton/down textures $[0,1,1]$ unchanged by the condensate; reflection symmetry keeps $m_\mu = m_e$ (18/18); corrects the 11124 reading	COMPUTATION	`w33_pass11127_lepton_texture_under_condensation.py`
Pinched cycle: exact winding law $p_R^2 = y_1/\sqrt{3}\,(+1/(3\sqrt{3}y_2))$; golden radius from $y^2 - \sqrt{3}y + \frac{1}{3} = 0$; $\Delta \to -\frac{1}{2}$, no tachyon-free island	BRIDGE	`w33_pass11128_pinched_cycle.py`
Off-axis potential: $\Lambda(B) - \Lambda(0) = A(1 - \cos 2\pi B)/2$, $\lvert A\rvert \leq 5 \times 10^{-5}$ of the radial force; $B = 0$ a minimum near onset	COMPUTATION	`w33_pass11129_offaxis_potential.py`
Top Higgs conjugates are the heavier doublets; single-Higgs m_b/m_t: $\epsilon^3 \to \epsilon^2$ (4/6), tau same order; light generations degenerate	COMPUTATION	`w33_pass11130_single_higgs_bottom_tau.py`
Across 21 neutral-exit survivors: up and μ never unlocked; down 14, lepton 17, ν 20; reproduces 11124	COMPUTATION	`w33_pass11131_unlocked_across_21.py`
Single light Higgs across 21: bottom $\epsilon^3 \to \epsilon^2$ in exactly the 14 down-unlocked models, tau at the same order for ≥ 2 of 3 Higgs	COMPUTATION	`w33_pass11132_bottom_tau_across_21.py`
Two VEV scales ($r = \log \epsilon_T / \log \epsilon_S$): 0 splittings; conjugate sector anarchic	COMPUTATION	`w33_pass11132_bottom_tau_across_21.py`
Zero-radius tachyon: one complex species ($N = \pm 1 \bmod 3$, none at $N = 0$), $I_1^2 \in \{\frac{2}{9}, \frac{4}{9}, \frac{5}{9}, \frac{7}{9}\}$; not a vector multiplet	BRIDGE	`w33_pass11134_dual_tachyon_species.py`
FI scale $\langle T\rangle/M_P = 0.15$–$0.20$ (SUSY formula) near the needed ϵ, but no FI mechanism without SUSY: coincidence, not prediction	COMPUTATION	`w33_pass11135_fi_scale_no_mechanism.py`
Tachyon domains = hyperbolic disks at $\Gamma_0(3)$ points: charged at $(3 + i\sqrt{3})/6$, radius $\ln(2 + \sqrt{3})$; neutral horodisk at cusp 0 or Fricke point $i/\sqrt{3}$, radius $2 \ln \varphi$; neutral first on the whole B circle	BRIDGE	`w33_pass11136_tachyon_domains.py`
Scorecard of the neutral-exit vacua (works / fails / open)	COMPUTATION	`w33_pass11137_neutral_exit_scorecard.py`

Claim	Badge	Witness
per-singlet VEV hierarchies (108 draws): light generations stay degenerate	COMPUTATION	`w33_pass11138_singlet_vev_hierarchies.py`
Wilson-line duality group $\Gamma_0(3)$; S, $\Gamma^0(3)$ broken; Fricke approximate	COMPUTATION	`w33_pass11139_wilson_line_duality_group.py`
zero-radius gauge algebra $\mathfrak{su}(16) \oplus \mathfrak{u}(1)$ in 21/21: the $U(16)$ string, 2 tachyons	COMPUTATION	`w33_pass11140_zero_radius_u16.py`
elliptic point = charged-disk centre ($\Delta = -\frac{1}{6}$); cusp 0 tachyonic in 15/21, Fricke image of large volume in 6: no symmetric stabilisation	COMPUTATION	`w33_pass11141_no_symmetric_stabilization.py`
no two-doublet light Higgs splits charm from up (0/576)	COMPUTATION	`w33_pass11142_higgs_mixing_up_sector.py`
neutrino anarchy refuted: Dirac texture up-like $[0,1,1]$ at tree level, unchanged by the condensate; conjugate-sector anarchy mixes quarks too	COMPUTATION	`w33_pass11145_neutrino_anarchy.py`
no Wilson-line holonomy ⇒ no Witten-sector tachyon at any radius (21/21)	COMPUTATION	`w33_pass11146_thermal_reading.py`
zero-radius vector lattice: even, rank 16, det 4, hyperbolic $\mathbb{Z}_2^2$ form, roots A_{15} = the $U(16)$ lattice (21/21)	COMPUTATION	`w33_pass11151_u16_lattice.py`
near cusp 0: potential = Fricke mirror of large volume to 0.3–0.5%, runs into the disk	COMPUTATION	`w33_pass11150_cusp0_potential.py`
First-generation regenerated RPV in the 23 D-flat $\mathbb{Z}_6$-I vacua $\geq 2 \times 10^{-3}$ (upper bounds); none below 10^{-10}	COMPUTATION	`w33_pass10980_rpv_flavour_upper_bound.py`
Regular polynomial G26 potential: exact global 72-ray T orbit; 8×9 reduced intersection; location independent of positive angular coefficients, model masses still free	COMPUTATION	`w33_20261001_g26_global_polynomial_vacuum.py`
Balanced 21-mirror barrier: exact T-magic local minimum, Hessian $30, 30, 54, 54$; known 72-ray orbit and one-MUB signature; no physical preparation mechanism	COMPUTATION	`w33_20261001_balanced_g26_tmagic_vacuum.py`
Framed magic interface: 8 Pauli orbits of 9; 648 injection branches give $T/3$; uniformly forgotten frame yields dephasing, not a magic resource	COMPUTATION	`w33_20261001_tmagic_orbit_factory_contract.py`

Claim	Badge	Witness
Sparse normalized Cartan plane: exact integer moments; universal SIC/MUB mass law by signed-permutation blocks; global invariant-circuit rank81 EFT completion	COMPUTATION	`w33_20261001_exact_cartan_mirror_completion.py`
Declared gauge/scalar/anomaly-paired-fermion model: exact inverse-Gram mass identity, full one-loop T saddle or angular cancellation; no selected quantum vacuum	COMPUTATION	`w33_20261001_complete_gauge_scalar_fermion_loop.py`
Global 81-field invariants: signed-cubic I_6, Aronhold/I_6^2 combination I_{12}; exact generator invariance and restrictions; analytic covectors	COMPUTATION	`w33_20261001_global_e6_cartan_covariants.py`
Global degree18 Hessian/Aronhold circuit; exact signed81-field normalization; real phase lift stays in one gauge orbit and does not select CP	COMPUTATION	`w33_20261001_degree18_phase_completion.py`
Three completed light modes are exact unbroken-gauge singlets; eight surviving generators cannot contain the Standard Model; paired matter has zero chiral index	COMPUTATION	`w33_20261001_chiral_mass_assignment_audit.py`
Completed EFT: full162-real-scalar determinant, negative finite-step curvature and Goldstone infrared boundary; no smooth stability theorem	COMPUTATION	`w33_20261001_completed_eft_one_loop.py`
Directed continuum heat response at $t = 1/5$: exact rational endpoints, width 3.4×10^{-18}; supplied metric variation retains conformal/CC boundaries	COMPUTATION	`w33_20261001_metric_dynamics_and_heat_bound.py`
Actual paired inventory: $b_{E_6} = 17$, $b_{SU(3)} = -59/2$; mirror coupling match is not RG invariant	COMPUTATION	`w33_20261001_uv_running_scale_audit.py`
Separate chiral $(27,3) + (1,\overline{10})$ anomaly repair; generic SIC-wall E_6 enhancement; conditional degree18 condensate interface	COMPUTATION	`w33_20261001_chiral_decuplet_and_condensate_bridge.py`

Claim	Badge	Witness
Conditional canonical Cartan condensate: exact six-direction positive Hessian, angular squared-mass ratio3; full moduli/physical assignment remain open	COMPUTATION	w33_20261001_condensate_cartan_potential.py
Eleven-complex-modulus condensate control:14 positive scalars and8 global Goldstones; two radial Kähler deformations tested; canonical spectrum is published prior art	COMPUTATION	w33_pass11270_full_condensate_moduli.py
Symmetric family-sextet Yukawa and rank-ten spectator mass tensor; changed Higgs inventory and UV coefficients are explicit	COMPUTATION	w33_pass11271_chiral_symmetric_yukawa_completion.py
Canonical signed27 Higgs orbit: exact12-dimensional SM stabilizer, hypercharges and invariant rank-ten exotic mass block; imposed selecting potential	COMPUTATION	w33_pass11271_canonical_sm_higgs_bridge.py
Actual paired-EFT soft infrared coefficient; separate11-modulus hard tadpole and Ward-resummed global Goldstones; nonzero poles remain open	COMPUTATION	w33_pass11272_resummed_condensate_radial_control.py
Uniform Wilson curvature domination proves fixed-time response limit; directed $K = 64$ tail $< 1.083 \times 10^{-25}$ at$t = 1/5$; supplied geometry	COMPUTATION	w33_pass11273_wilson_uniform_tail.py
Condensate $\mathrm{STr}\, M^2 = 0$ but $\mathrm{STr}\, M^4 = 852930m^4/2401 > 0$; external sequestering shift test, no derived CC	COMPUTATION	w33_pass11274_scale_and_sequestering.py
Polynomial SM Higgs EFT: exact120 normal and66 gauge directions; local orbit, degree8 EFT/degree4 large scalar lift	COMPUTATION	w33_pass11275_polynomial_sm_higgs.py
Sextet composite: rank-ten spectator mass, family $b = -79/3$ still non-asymptotically free	COMPUTATION	w33_pass11276_sextet_composite_spectator.py
Radial-only scalar EFT momentum bubble: squared pole shift -0.144%; full modulus poles excluded	COMPUTATION	w33_pass11277_radial_momentum_pole.py
Literal symplectic ring lift; positive invariant real metric obstruction, no spacetime selection	COMPUTATION	w33_pass11278_symplectic_ring_geometry.py

Claim	Badge	Witness
W33 clique times history circle: injective four-boundary, no intrinsic topological four-flux	COMPUTATION	w33_pass11279_history_flux_obstruction.py
Factor Higgs lift:996 real fields,905 positive normals, added auxiliary gauge factor; E6 and auxiliary SU5 AF	COMPUTATION	w33_pass11280_factor_higgs_mediator.py
Family sextet orbit selector and mixed neutral-Higgs bilinear; hierarchy targets imposed, CKM open	COMPUTATION	w33_pass11281_family_sextet_selection.py
All internal modulus radial cuts: complex pole and Goldstone width; full external matrix open	COMPUTATION	w33_pass11282_all_modulus_radial_self_energy.py
Added compatible-metric scales span ten local Lorentz variations; continuum base is input	COMPUTATION	w33_pass11283_symplectic_metric_field.py
Added closed handlebody history: one top flux, conditional linear sequestering; observed vacuum energy open	COMPUTATION	w33_pass11284_closed_history_flux.py
SO10 frame projection:1740 local positive normals; composite Yukawa UV completion open	COMPUTATION	w33_pass11285_semisimple_factor_higgs.py
CP-even decorrelation selects $J^2 = 1/108$; toy mixing, four extra flat phases	COMPUTATION	w33_pass11286_misaligned_family_vacua.py
Full22-field nonanalytic quotient matrix; independent D-flat cubic; physical poles open	COMPUTATION	w33_pass11287_quotient_self_energy_matrix.py
Imposed EH pullback: nine first-class constraints, two graviton polarizations	COMPUTATION	w33_pass11288_metric_constraint_audit.py
Prior Levi critical group applied to closed gluing; flux magnitude remains input	COMPUTATION	w33_pass11289_cycle_gram_gluing_flux.py
Family gauging moves Goldstone decay to coupling-dependent vector thresholds	COMPUTATION	w33_pass11290_gauged_family_decay.py
Fixed composite inventory: unique jointly AF rank-five E6/SO10 window	COMPUTATION	w33_pass11291_rank_five_AF_window.py
Declared Abelian cycle CS model: boundary obstruction; doubling supplies Lagrangian subgroup	COMPUTATION	w33_pass11292_cycle_discriminant_boundary.py
Economical two-frame/54 alignment:1254 normals; scalar/fermion mediators retain E6/SO10 AF	COMPUTATION	w33_pass11293_yukawa_uv_budget.py

Claim	Badge	Witness
Hierarchical CP phase locks: rank16 at four vacua; coefficients imposed	COMPUTATION	w33_pass11294_hierarchical_cp_phase_vacuum.py
Local Ward-matched leading complex poles; analytic UV/tadpole matching open	COMPUTATION	w33_pass11295_ward_matched_modulus_poles.py
Covariant quotient matter preserves fiber and continuum ADM constraints	COMPUTATION	w33_pass11296_covariant_matter_constraints.py
Membrane relaxation model; intensive ladder differs from old uniform flux branch	COMPUTATION	w33_pass11297_membrane_flux_dynamics.py
Single-channel Yukawa running; rank-one mediator obstruction; full two-channel quartics open	COMPUTATION	w33_pass11298_coupled_running_closure.py
Canonical E6 two-Higgs spectrum depends only on summed Grams; phase locks not generated	COMPUTATION	w33_pass11299_e6_flavor_spectral_obstruction.py
Invariant vector local-jet counterterm and conditional radial cut; full UV matching open	COMPUTATION	w33_pass11300_invariant_vector_matching.py
Native Levi site brackets generate edge currents; enlarged gravity closure open	COMPUTATION	w33_pass11301_graph_adm_regulator.py
Separate membrane form: two-cap junction and global flux laboratory, added geometry/inputs	COMPUTATION	w33_pass11302_sequestered_membrane_junction.py
One-scalar rank-two source; portal-independent54 UV barrier; isolated added-fermion AF escape	COMPUTATION	w33_pass11303_scalar_source_and_uv_barrier.py
Vectorlike E6-singlet probes detect four Takagi-phase flats; vacuum selection open	COMPUTATION	w33_pass11304_singlet_phase_probes.py
Matrix vector cuts and rank11 scalar Goldstone IR boundary; total UV matching open	COMPUTATION	w33_pass11305_vector_matrix_and_total_jet_audit.py
Native graph clocks: exact first-class relational oscillator constraints, not gravity	COMPUTATION	w33_pass11306_relational_graph_clock_constraints.py
Quantized membrane flux: one reduced radial negative direction; full bounce index open	COMPUTATION	w33_pass11307_quantized_membrane_radial_mode.py
Exact54+45 quartic running and two specified-Yukawa portal AF barriers	COMPUTATION	w33_pass11313_symmetric_adjoint_portal_closure.py
Untargeted loop potential: CP-conserving local five-phase minimum, fixed spectra	COMPUTATION	w33_pass11314_untargeted_phase_vacuum.py

Claim	Badge	Witness
Finite-momentum massless Goldstone limit; invariant tadpole and UV poles open	COMPUTATION	w33_pass11315_regulator_free_goldstone_poles.py
Constructed native-edge linear spin-two action:2 massless plus79 massive modes	COMPUTATION	w33_pass11316_native_edge_fierz_pauli.py
Declared81-handle closed history admits no smooth Riemannian Einstein metric	COMPUTATION	w33_pass11317_history_einstein_obstruction.py
Isolated54 radiative radial scale connected to AF ray; absolute scale input	COMPUTATION	w33_pass11318_radiative_scale_on_af_trajectory.py
Constructed smooth local neck requires radial null-energy violation	COMPUTATION	w33_pass11319_smooth_neck_stress.py
All real two-active Yukawa fixed rays retain scalar AF barriers	COMPUTATION	w33_pass11320_general_real_two_flavor_yukawas.py
Full-field local2+1 flavour splitting and relative CP; CKM unresolved	COMPUTATION	w33_pass11321_full_flavor_bifurcation.py
Finite invariant hard matching; rank11 soft IR and resummation open	COMPUTATION	w33_pass11322_hard_soft_invariant_matching.py
Native nonlinear metric cycle lapse witness; no invariant tree repair	COMPUTATION	w33_pass11323_native_cycle_lapse_hessian.py
Exact quantized positive-tension Maxwell wall on chosen one-handle history	COMPUTATION	w33_pass11324_quantized_warped_history_wall.py
Complex two-active phase locking and larger paired scalar barriers	COMPUTATION	w33_pass11325_complex_yukawa_phase_audit.py
Engineered noncommuting flavour vacuum; observed CKM not predicted	COMPUTATION	w33_pass11326_noncommuting_flavor_operator.py
Numerical nonlinear local quotient metric and covariant mass maps	COMPUTATION	w33_pass11327_nonlinear_quotient_mass_maps.py
Restricted determinant gravity count and dense/native graph obstruction	COMPUTATION	w33_pass11328_determinant_multivielbein_scope.py
Coupled Euclidean wall shape block positive with two zero modes	COMPUTATION	w33_pass11329_wall_shape_fluctuations.py
Finite larger-complex Yukawa matrix search; no completeness claim	COMPUTATION	w33_pass11335_larger_complex_flavor_search.py
Gaussian adjoint mediation forces commuting flavour Grams	COMPUTATION	w33_pass11336_gaussian_mediator_alignment.py
Actual quotient family fields and hard Ward mass columns	COMPUTATION	w33_pass11337_quotient_hard_ward_columns.py
Native pair-vielbein collinear slice has linear reduced lapses	COMPUTATION	w33_pass11338_native_boost_constraint_slice.py

Claim	Badge	Witness
Coupled Maxwell/metric wall block nonnegative; four zero modes	COMPUTATION	`w33_pass11339_coupled_maxwell_metric_modes.py`
Exact complex-isotropic Yukawa ray and stronger scalar AF barrier	COMPUTATION	`w33_pass11340_isotropic_flavor_portal.py`
Backreacted quantised winding wall lifts both torus shapes	COMPUTATION	`w33_pass11341_winding_lifted_wall.py`
Machine (§10)		
$\mathrm{ASp}(4,3)$, 4 199 040; four opcodes; length 19	COMPUTATION	blueprint, Passes 2778/2789/2882
Optimal fixed-perfect-gate compiler: $\mathrm{Sp}(4,3) = H \sqcup HpH \sqcup Hph_*pH$; depths 0, 1, 2 with counts 1 152/13 824/36 864; E_6 sector walk quotient $\left(\begin{smallmatrix} 4 & 8 \\ 1 & 11 \end{smallmatrix}\right)$, transient 1/4	COMPUTATION	`w33_pass11193_optimal_perfect_gate_compiler.py`
4 stabiliser + 36 magic rays; KS budget 36/40; $\alpha = 7$	COMPUTATION	`bt822_magic_census_matter_is_magic.py`, `bt818_ovoid_nogo_theta_gap.py`
Steinberg memory $[[240, 81, 3]]_3$	COMPUTATION	`bt861_code_register_is_steinberg.py`
Golay lane: 552 locations, threshold 5.32×10^{-4}	COMPUTATION	`w33_pass10947_five_front_execution.py`
cube-chart atlas; ≤ 8 moves per digit	COMPUTATION	`bt777_hypercube_chart_web.py`, `bt827_holonet_fractal_architecture.py`
single-photon SUM gate, 0.92 ± 0.01	measured	[25]
11342: both stored complex flavour escape witnesses fail stronger scalar bound	COMPUTATION	`w33_pass11342_11349_eight_frontier_probes.py`
11343: bounded non-Gaussian mediators select two-family misalignment	COMPUTATION	`w33_pass11342_11349_eight_frontier_probes.py`
11344: Ward columns leave 105 normal Hessian entries free	COMPUTATION	`w33_pass11342_11349_eight_frontier_probes.py`
11345: native transverse shift–boost linearisation has rank 237	COMPUTATION	`w33_pass11342_11349_eight_frontier_probes.py`
11346: winding wall retains exact mixed vector zero profiles	COMPUTATION	`w33_pass11342_11349_eight_frontier_probes.py`
11347: two-Gram CP-odd trace first occurs at degree six	COMPUTATION	`w33_pass11342_11349_eight_frontier_probes.py`
11348: dimensionless data alone leave absolute scale free	COMPUTATION	`w33_pass11342_11349_eight_frontier_probes.py`
11349: quantised positive winding cap with zero bare ρ	COMPUTATION	`w33_pass11342_11349_eight_frontier_probes.py`
11361: three-ray Bargmann phase lifts to a full-rank Gram CP cubic	COMPUTATION	`w33_pass11361_11368_physical_frontier.py`

Claim	Badge	Witness
11362: supplied heavy scalar realizes one Ward-free hard normal loop entry	COMPUTATION	w33_pass11361_11368_physical_frontier.py
11363: equal-lapse rank-77 metric null is the vertex potential	COMPUTATION	w33_pass11361_11368_physical_frontier.py
11364: native pair-metric cycle rank 78 has an exact modular witness	COMPUTATION	w33_pass11361_11368_physical_frontier.py
11365: zero-bare-ρ wall retains an exact mixed vector zero	COMPUTATION	w33_pass11361_11368_physical_frontier.py
11366: quantised wall family positive; zero bare ρ still curved	COMPUTATION	w33_pass11361_11368_physical_frontier.py
11367: transverse boosts require flat native-octagon holonomy	COMPUTATION	w33_pass11361_11368_physical_frontier.py
11368: legacy physical labels scoped; $40 \cdot 3 + 4 - 2 = 122$	COMPUTATION	w33_pass11361_11368_physical_frontier.py
11869: Siegel generators act as $F \otimes F$, CZ_3, phase, SWAP; level-3 quotient $\mathrm{Sp}(4,3)$; with CP $W(E_6)$	COMPUTATION	w33_pass11869_11873_siegel_level3_two_qutrits.py
11870: Hesse cubic through theta-null qutrit is E_τ; arrow = sign $\mathrm{Im}\, j(\tau)$	COMPUTATION	w33_pass11869_11873_siegel_level3_two_qutrits.py
11871: theta-null states are Coble singular points on the Burkhardt quartic	COMPUTATION	w33_pass11869_11873_siegel_level3_two_qutrits.py
11872: product $\Leftrightarrow \Omega_{12} = 0$; Schmidt $(1, \varepsilon, \varepsilon^2)$	COMPUTATION	w33_pass11869_11873_siegel_level3_two_qutrits.py
11874: six isolated fixed points, stabilisers $10, 24, 24, 32, 48, 72$ in $\mathrm{Sp}(4,3)$; all CP-conserving; products rank one	COMPUTATION	w33_pass11874_11878_siegel_fixed_points_cp_magic.py
11875: theta-nulls = even Weil 5, odd gradients = odd Weil 4 (weight $\det^{1/2} \otimes$std)	COMPUTATION	w33_pass11874_11878_siegel_fixed_points_cp_magic.py
11876: $\sigma_1\sigma_3/\sigma_2^2 = r(\tau_1)r(\tau_2)/2$; one modulus for all sectors excluded	COMPUTATION	w33_pass11874_11878_siegel_fixed_points_cp_magic.py
11877: magic is modular, supremum at a cusp: no stabilisation	COMPUTATION	w33_pass11874_11878_siegel_fixed_points_cp_magic.py
11878: Burkhardt invariant ratios are level-one modular CP order parameters; opposite arrows cancel	COMPUTATION	w33_pass11874_11878_siegel_fixed_points_cp_magic.py
11879: E_8 Cartan of the **84** = odd-characteristic theta-nulls; genus-2 moduli with a Weierstrass point	COMPUTATION	w33_pass11879_11883_e8_higgs_siegel_modulus.py

Claim	Badge	Witness
11880: generic **84** vev leaves the two-qutrit Pauli group; $SU(3)^3$ at the 40 Witting rays	COMPUTATION	`w33_pass11879_11883_e8_higgs_siegel_modulus.py`
11881: two quartic invariants; Kempf–Ness flat moduli = Siegel modulus (dim $88 = 80 + 8$)	COMPUTATION	`w33_pass11879_11883_e8_higgs_siegel_modulus.py`
11882: lowest holomorphic Cartan invariant has degree 12: lifted only at dimension 12	COMPUTATION	`w33_pass11879_11883_e8_higgs_siegel_modulus.py`
11884: bosonic one loop on the flat modulus $\propto \sum e^2 \log e$; minimum $-6 \log 3$ at the 40 Witting rays: $SU(3)^3$	COMPUTATION	`w33_pass11884_11886_one_loop_trinification.py`
11885: trinification vacuum strictly stable except the 84 phase (physical, lifted at dimension 12)	COMPUTATION	`w33_pass11884_11886_one_loop_trinification.py`
11887: in E_8 the one-loop vacuum's centraliser is $E_6 \oplus u(1)^2$ (grades $24, 28, 28$)	COMPUTATION	`w33_pass11887_11889_e6_family_at_w33_point.py`
11888: $248 = 80 + 3 \cdot 27 + 3 \cdot \overline{27} + 6$ with family-triplet $U(1)^2$ charges	COMPUTATION	`w33_pass11887_11889_e6_family_at_w33_point.py`
11889: each Witting ray $\leftrightarrow$ a unique $W(3,3)$ point ($40 \leftrightarrow 40$), inside its $E_6 \times A_2$	COMPUTATION	`w33_pass11887_11889_e6_family_at_w33_point.py`
11890: Hosotani on $T^2/\mathbb{Z}_3$: SU(9) restored (bosons, SS); periodic fermions $\to D_4 \oplus D_4$ at a factorisation	COMPUTATION	`w33_pass11890_11892_vacuum_phase_diagram.py`
11891: $\mathrm{Im}\Omega \simeq (3/\pi) \ln(1/\delta)$: hierarchy parameter linear in the Higgs displacement	COMPUTATION	`w33_pass11890_11892_vacuum_phase_diagram.py`
11892: SUSY flat; soft $m^2 < 0$ trinification, $m^2 > 0$ the C_{10} Weierstrass point; fermions $\to$ cusp	COMPUTATION	`w33_pass11890_11892_vacuum_phase_diagram.py`
11893: heterotic $T^6/\mathbb{Z}_3$ sector at the trinification point: $3[(3,3,3) + 1] + 27 \cdot \bar{3}$'s; no SM embedding	COMPUTATION	`w33_pass11893_trinification_chiral_spectrum.py`
11894: $\mathbb{Z}_6$ sector: 4-dim Kempf–Ness Cartan in the family planes (flat modulus survives)	COMPUTATION	`w33_pass11894_11895_z6_sector_selection.py`
11895: one loop selects $SU(3) \times SU(2) \times U(1)$ (diagonal colour); octets, sextets, no lepton doublet	COMPUTATION	`w33_pass11894_11895_z6_sector_selection.py`
11896: flagship (CZ_3 Wilson line): flat directions SM-charged, unbroken 2–4 of 16: modulus and SM exclude each other	COMPUTATION	`w33_pass11896_flagship_tension.py`

Claim	Badge	Witness
11897: Kähler moduli of $T^4/\mathbb{Z}_3$ act on the 9 twisted fixed points by the two-qutrit Weil rep: projective order 51840 $\cong Sp(4,3)$; diagonal moduli only 576	COMPUTATION	w33_pass11897_11898_kahler_moduli_two_qutrit_weil.py
11898: even thetas fill $\mathbb{P}^4$; Siegel slice → dual Burkhardt (deg 18); $T^6/\mathbb{Z}_3$: 27 fixed points, $Sp(6,3)$	COMPUTATION	w33_pass11897_11898_kahler_moduli_two_qutrit_weil.py
11899: symmetric Kähler vacua: 13805 special points, 11 orbits; 40 cusps = lines of $W(3,3)$, 45 nodes = factorisations, 160 = flags; all CP-conserving, generic CP violation exponentially small at large volume	COMPUTATION	w33_pass11899_kahler_special_points_cp.py
11900: second Wilson line = a point of $W(3,3)$ (centraliser 648); families = its eigenspace; on charged classes the Kähler dualities generate the full one-qutrit Clifford ($\Delta(54)$ included); light-pair splitting exact zero (same class) or $O(Z_{12}^3)$	COMPUTATION	w33_pass11900_three_generations_over_a_w33_point.py
11901: all 12 survivors: every quark triangle single-pointed on both Wilson-line tori (72/72), so theta evenness keeps $m_c = m_u$, $m_s = m_d$ for all nine Kähler moduli; escape needs $a \neq 0$, which costs the top ($\frac{1}{2}$ at ρ)	COMPUTATION	w33_pass11901_kahler_loophole_closed.py
11902: all 104 A_8 $SO(16)^2$ SMs: 73 untwisted-family (antisymmetric ε Yukawa, top = charm, $m_u = 0$, preserved by Kähler congruence) + 31 twisted-family ($m_c = m_u$ by theta evenness); no Kähler modulus lifts the up degeneracy	COMPUTATION	w33_pass11902_a8_census_kahler_independent_degeneracy.py
11903: 491 models: 339 untwisted (top = charm) + 152 twisted, all with a local $SU(5)$ $\mathbf{10} = Q + u^c + e^c$ at one Wilson-line point, which forces $g_H = -2g = g$ and $m_c = m_u$ for all Kähler moduli; up escape 0/491, down 5/491	COMPUTATION	w33_pass11903_local_ten_locks_charm_up.py
11904: magnetized T^4 (flux 3×3): charm–up unlocked at first order by Ω_{12} ($a \neq 0$) and by a continuous Wilson line (any a); theta-zero law: massless up forces top = charm; Gaussian invariant $m_u m_c / m_t^2 = e^{-2\pi \operatorname{Im}\Omega/9}$; non-renormalisable×Kähler gives no parametric split	COMPUTATION	w33_pass11904_magnetized_unlock_and_theta_zero_law.py

Claim	Badge	Witness
11905: at every fixed point $z = m/2 + k\Omega/6$ of the level-3 Heisenberg–reflection group two of (top, charm, up) coincide ($k \equiv 0$: $m_c = m_u$; else top = light); the up hierarchy needs a non-symmetric Wilson-line vacuum	COMPUTATION	`w33_pass11905_symmetric_wilson_lines_degenerate.py`
11906: direct flux-$(3,3,6)$ overlap integrals: normalised masses $= \lvert\theta[c/3,0](3\zeta, 6\tau)\rvert$ exactly $(6 \cdot 10^{-16})$; theta-zero law, symmetric-point degeneracy, Gaussian invariant and the 11904 hierarchy point hold in true normalisation; TOE46 overlap reproduced	COMPUTATION	`w33_pass11906_magnetized_normalisation_audit.py`
11907: Wilson-line dynamics: $\pm\sum\lvert Y\rvert^2$ always selects a degenerate texture (max at $\zeta = 0$, min at the theta zero); $-F_2 + rF_4$ destabilises the symmetric point at $r = \frac{1}{2}$ and gives non-symmetric hierarchical vacua, observed up ratios at $r^* = 0.890$	COMPUTATION	`w33_pass11907_wilson_line_dynamics.py`
11908: three further $\mathbb{Z}_3$ shift classes, 109 new models: all 73 twisted carry a local **10**, 36 untwisted top = charm; up escape 0/600 across four classes; shifts and Wilson lines of all 600 frozen with SHA-256	COMPUTATION	`w33_pass11908_split_ten_scan_beyond_a8.py`
11909: up-sector Coleman–Weinberg for the Wilson line: SM cutoff and SUSY with $Q \geq \tilde{m}$ give $m_c = m_u$; hierarchy only in a band $Q/\tilde{m} - e^{-1/2} \sim (v/\tilde{m})^2$ – the up hierarchy is tied to top-sector naturalness	COMPUTATION	`w33_pass11909_up_sector_coleman_weinberg.py`
11910: minimal magnetized home: universal invariant $m_1 m_2/m_3^2 = e^{-(4\pi/3)\mathrm{Im}\,\tau}$; data: down $\approx$ leptons (17%), up $\sim 700\times$ smaller (falsified); localised Higgs gives $V_{us} = 0$; Cabibbo needs a non-localised Higgs $(\epsilon \approx 3 \cdot 10^{-3})$	COMPUTATION	`w33_pass11910_cross_sector_invariant_and_ckm.py`
11911: supersymmetric $E_8 \times E_8$ $\mathbb{Z}_3$ (standard embedding, $E_6 SU_3^2$, $E_7 \times SO_{14}$, A_8; $4 \times 20{,}000$ draws): 0 SM-like (standard embedding saturated at 717 spectra; positive control recognised) – the $\mathbb{Z}_3$ Standard Models live in the $SO(16)^2$ class	COMPUTATION	`w33_pass11911_susy_z3_scan.py`

Claim	Badge	Witness
11912: exact decomposition $\Theta[(\alpha_1,\alpha_2)] = \sum_n e^{i\pi(n+\alpha_2)^2\Omega_{22}}\theta[\alpha_1]((n+\alpha_2)\Omega_{12},\Omega_{11})$: families in a charged Pauli class see the Wilson line $z_{\text{eff}} = \pm\Omega_{12}/3$; the up hierarchy from Ω_{12} alone (flux-stabilisable)	COMPUTATION	`w33_pass11912_entangling_modulus_is_the_wilson_line.py`
11913: type IIB primitive $(2,1)$ fluxes on $T^4 \times T^2$ near the hierarchy point: tadpoles $N = 8$–68 exist but leave the Siegel block flat; fixing all moduli needs $N \gtrsim 500$ (10^4–$5 \cdot 10^5$ near the point) – not stabilisable within the $T^6/\mathbb{Z}_2$ bound	COMPUTATION	`w33_pass11913_flux_vacua_near_hierarchy_point.py`
11914: shared-Q universality (one invariant over 2214 $W(3,3)$-eigenspace configurations); spread H_d is anarchic ($\lvert V\rvert \approx 1/\sqrt{3}$); magnetic-translation $\mathbb{Z}_3$ forbids radiative Higgs admixture; corrections to 11909/11910 from TOE49	COMPUTATION	`w33_pass11914_higgs_is_the_flavour_lever.py`
11915: SU(5)-structured det-3 fluxes on T^4 (the two classes A_2 and x^2+3y^2): down = lepton automatic; Gaussian slopes quantised ($-4/3, -8/3, -2, -1, -4$ in units of π), observed up/down 1.605 not attainable for scalar Ω; coprime Higgs lattice breaks the class selection (geometric CKM), but generic Wilson lines give too little mixing	COMPUTATION	`w33_pass11915_su5_matrix_fluxes_on_t4.py`

B Glossary

A_2 The root system of $SU(3)$: six vectors forming a regular hexagon.

Albert algebra The 27-dimensional algebra $H_3(\mathbb{O})$ of 3×3 Hermitian octonionic matrices; the only exceptional Euclidean Jordan algebra.

Bell line The line of $W(3,3)$ formed by the unitary channels $A \mapsto P_u A P_u^\dagger$; the "now" of Section 4.

Bridge A true mathematical statement together with a proposed physical reading.

Cartan involution An involution of a real Lie algebra splitting it into rotations ($\mathfrak{k}$) and boosts ($\mathfrak{p}$); it fixes the real form.

Clifford group Quantum operations mapping Pauli operators to Pauli operators; for two

qutrits it acts on labels as $\mathrm{Sp}(4,3)$.

Context A maximal set of simultaneously measurable Pauli operators; a line of $W(3,3)$.

Contextuality The impossibility of assigning pre-existing outcomes to all measurements consistently across contexts; a resource for quantum computation.

CSS code A quantum error-correcting code built from two classical codes; $[[n,k,d]]_q$ means n physical qudits, k logical, distance d.

E_8 The largest exceptional simple Lie algebra: dimension 248, 240 roots.

$\mathbb{F}_3$ The field $\{0,1,2\}$ with arithmetic modulo 3.

Freudenthal triple system A 56-dimensional space with a symplectic form and an E_7-invariant quartic; the charge space of certain supergravity black holes.

Generalized quadrangle A point–line geometry with no triangles in which a point off a line is collinear with exactly one of its points.

Grading A splitting of an algebra into pieces labelled by integers (or integers mod n) that add under the bracket.

Hesse clock One of the four parallel classes of lines of $AG(2,3)$, read as three ticks.

Heisenberg algebra The algebra of positions, momenta and a central element; here $56+1=57$ dimensions.

Heterotic string A closed string theory with gauge group $E_8 \times E_8$ (or $SO(32)$); on orbifolds it yields Standard-Model-like theories.

Lagrangian A maximal subspace on which the symplectic form vanishes; a line of $W(3,3)$.

Magic state A state that is not a stabiliser state; injecting it makes Clifford circuits universal.

Majorana–Weyl spinor A real, chiral spinor; in ten dimensions it has 16 components.

Matter parity A $\mathbb{Z}_2$ symmetry, -1 on matter fermions and $+1$ on Higgs fields, that forbids fast proton decay in supersymmetric theories.

Octonions $\mathbb{O}$ The eight-dimensional, non-associative normed division algebra.

Orbifold A space divided by a finite group; used to curl up extra dimensions.

Pauli operator For a qutrit, $X^a Z^b$; for two qutrits, a tensor product of two.

Peirce decomposition The splitting of a Jordan algebra by the eigenvalues $1, \frac{1}{2}, 0$ of multiplication by an idempotent.

Qutrit A quantum system with three perfectly distinguishable states.

Real form A real Lie algebra whose complexification is a given complex one; E_8 has three.

Root An eigen-direction of the maximal commuting part of a Lie algebra.

Signature For a quadratic form, the numbers of positive and negative directions; for a real Lie algebra, $\dim \mathfrak{p} - \dim \mathfrak{k}$.

Spread A set of pairwise disjoint lines covering every point; $W(3,3)$ has 36.

Steinberg module A distinguished irreducible representation of a finite group of Lie type; for $\mathrm{PSp}(4,3)$ it is 81-dimensional and is $H_1(W(3,3))$.

Stabiliser state A joint eigenstate of a maximal commuting set of Paulis.

Strongly regular graph A regular graph in which adjacent pairs have λ, and non-adjacent pairs μ, common neighbours.

Symplectic form An alternating, nondegenerate bilinear form; here the commutation phase of Pauli operators.

Tetracode The ternary $[4,2,3]$ code; here the evaluation code of the four clocks.

Transvection The symplectic map $x \mapsto x + \langle x, v\rangle v$.

Tripotent An element e of a Jordan triple system with $\{e,e,e\} = 2e$ (normalisation of this paper).

Witting polytope A regular complex polytope in $\mathbb{C}^4$ with 240 vertices, the E_8 roots, and 40 diameters.

References

[1] J. F. Adams, *Lectures on Exceptional Lie Groups*, University of Chicago Press, 1996.

[2] L. Alvarez-Gaumé, P. Ginsparg, G. Moore and C. Vafa, An $O(16) \times O(16)$ heterotic string, *Phys. Lett. B* **171** (1986) 155–162.

[3] J. C. Baez, The octonions, *Bull. Amer. Math. Soc.* **39** (2002) 145–205.

[4] J. C. Baez and J. Huerta, Division algebras and supersymmetry I, in *Superstrings, Geometry, Topology, and C*-algebras*, Proc. Symp. Pure Math. **81** (2010) 65–80, arXiv:0909.0551.

[5] L. Borsten, D. Dahanayake, M. J. Duff, H. Ebrahim and W. Rubens, Black holes, qubits and octonions, *Phys. Rep.* **471** (2009) 113–219, arXiv:0809.4685.

[6] F. Beye, T. Kobayashi and S. Kuwakino, Gauge origin of discrete flavor symmetries in heterotic orbifolds, *Phys. Lett. B* **736** (2014) 433–437, arXiv:1401.2310.

[7] A. E. Brouwer and H. Van Maldeghem, *Strongly Regular Graphs*, Cambridge University Press, 2022.

[8] A. Connes and C. Rovelli, Von Neumann algebra automorphisms and time–thermodynamics relation in generally covariant quantum theories, *Class. Quantum Grav.* **11** (1994) 2899–2918.

[9] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer, 1999.

[10] L. Dixon, J. A. Harvey, C. Vafa and E. Witten, Strings on orbifolds, *Nucl. Phys.* **B261** (1985) 678–686.

[11] M. J. Duff and S. Ferrara, E_7 and the tripartite entanglement of seven qubits, *Phys. Rev. D* **76** (2007) 025018, arXiv:quant-ph/0609227.

[12] E. Escalante-Notario, R. Pérez-Martínez, S. Ramos-Sánchez and P. K. S. Vaudrevange, The non-SUSY orbifolder: a tool to build promising non-supersymmetric string models, arXiv:2504.20137 (2025).

[13] J. Faraut and A. Korányi, *Analysis on Symmetric Cones*, Oxford University Press, 1994.

[14] A. Font and A. Hernández, Non-supersymmetric orbifolds, *Nucl. Phys. B* **634** (2002) 51–70, arXiv:hep-th/0202057.

[15] H. Freudenthal, Beziehungen der E_7 und E_8 zur Oktavenebene I, *Indag. Math.* **16** (1954) 218–230.

[16] S. Garibaldi, E_8, the most exceptional group, *Bull. Amer. Math. Soc.* **53** (2016) 643–671.

[17] K. S. Gibbons, M. J. Hoffman and W. K. Wootters, Discrete phase space based on finite fields, *Phys. Rev. A* **70** (2004) 062101.

[18] P. Ginsparg and C. Vafa, Toroidal compactification of non-supersymmetric heterotic strings, *Nucl. Phys. B* **289** (1987) 414–480.

[19] M. Günaydin, K. Koepsell and H. Nicolai, Conformal and quasiconformal realizations of exceptional Lie groups, *Commun. Math. Phys.* **221** (2001) 57–76, arXiv:hep-th/0008063.

[20] S. Groot Nibbelink, O. Loukas, A. Mütter, E. Parr and P. K. S. Vaudrevange, Tension between a vanishing cosmological constant and non-supersymmetric heterotic orbifolds, *Fortsch. Phys.* **68** (2020) 2000044, arXiv:1710.09237.

[21] M. Günaydin, G. Sierra and P. K. Townsend, Exceptional supergravity theories and the magic square, *Phys. Lett.* **B133** (1983) 72–76.

[22] M. Günaydin, Generalized conformal and superconformal group actions and Jordan algebras, *Mod. Phys. Lett. A* **8** (1993) 1407–1416, arXiv:hep-th/9301050.

[23] M. Howard, J. Wallman, V. Veitch and J. Emerson, Contextuality supplies the 'magic' for quantum computation, *Nature* **510** (2014) 351–355.

[24] J. E. Humphreys, The Steinberg representation, *Bull. Amer. Math. Soc.* **16** (1987) 247–263.

[25] P. Imany, J. A. Jaramillo-Villegas, M. S. Alshaykh, J. M. Lukens, O. D. Odele, A. J. Moore, D. E. Leaird, M. Qi and A. M. Weiner, High-dimensional optical quantum logic in large operational spaces, *npj Quantum Inf.* **5** (2019) 59.

[26] R. Kallosh and B. Kol, E_7 symmetric area of the black hole horizon, *Phys. Rev. D* **53** (1996) R5344, arXiv:hep-th/9602014.

[27] T. Kobayashi, H. P. Nilles, F. Plöger, S. Raby and M. Ratz, Stringy origin of non-Abelian discrete flavor symmetries, *Nucl. Phys. B* **768** (2007) 135–156, arXiv:hep-ph/0611020.

[28] H. Kraft, A. Regeta and S. Zimmermann, Small G-varieties, arXiv:2009.05559.

[29] S. Krutelevich, Jordan algebras, exceptional groups and Bhargava composition, *J. Algebra* **314** (2007) 924–977.

[30] T. Kugo and P. Townsend, Supersymmetry and the division algebras, *Nucl. Phys.* **B221** (1983) 357–380.

[31] J. Lauer, J. Mas and H. P. Nilles, Twisted sector representations of discrete background symmetries for two-dimensional orbifolds, *Nucl. Phys. B* **351** (1991) 353–424.

[32] R. Landauer, Irreversibility and heat generation in the computing process, *IBM J. Res. Dev.* **5** (1961) 183–191.

[33] O. Lebedev, H. P. Nilles, S. Raby, S. Ramos-Sánchez, M. Ratz, P. K. S. Vaudrevange and A. Wingerter, The heterotic road to the MSSM with R parity, *Phys. Rev. D* **77** (2008) 046013, arXiv:0708.2691.

[34] S. Lloyd and S. L. Braunstein, Quantum computation over continuous variables, *Phys. Rev. Lett.* **82** (1999) 1784–1787.

[35] C. A. Manogue, T. Dray and R. A. Wilson, Octions: an E_8 description of the Standard Model, *J. Math. Phys.* **63** (2022) 081703, arXiv:2204.05310.

[36] S. P. Martin, Some simple criteria for gauged R-parity, *Phys. Rev. D* **46** (1992) R2769, arXiv:hep-ph/9207218.

[37] H. P. Nilles, S. Ramos-Sánchez, P. K. S. Vaudrevange and A. Wingerter, The Orbifolder: a tool to study the low-energy effective theory of heterotic orbifolds, *Comput. Phys. Commun.* **183** (2012) 1363–1380, arXiv:1110.5229.

[38] Y. Olguín-Trejo, R. Pérez-Martínez and S. Ramos-Sánchez, Charting the flavor landscape of MSSM-like Abelian heterotic orbifolds, *Phys. Rev. D* **98** (2018) 106020, arXiv:1808.06622.

[39] R. Pérez-Martínez, S. Ramos-Sánchez and P. K. S. Vaudrevange, Landscape of promising non-supersymmetric string models, *Phys. Rev. D* **104** (2021) 046026, arXiv:2105.03460.

[40] D. N. Page and W. K. Wootters, Evolution without evolution: dynamics described by stationary observables, *Phys. Rev. D* **27** (1983) 2885–2892.

[41] S. E. Payne and J. A. Thas, *Finite Generalized Quadrangles*, 2nd ed., European Mathematical Society, 2009.

[42] T. A. Springer, Regular elements of finite reflection groups, *Invent. Math.* **25** (1974) 159–198.

[43] M. Tegmark, On the dimensionality of spacetime, *Class. Quantum Grav.* **14** (1997) L69–L75, arXiv:gr-qc/9702052.

[44] Watanabe, Belfiore, de Carvalho and Vieira Filho, a cyclotomic construction of E_8 from the negacyclic tetracode, *Int. J. Appl. Math.* **31**(1) (2018), doi:10.12732/ijam.v31i1.6 (as cited in Pass 10948).

[45] J. Fullwood and J. Li, Partial transpose as a space-time swap, *Phys. Rev. A* (2025), arXiv:2508.12256.

[46] S. Marcovitch and B. Reznik, Correlation preserving map between bipartite states and temporal evolutions, arXiv:1107.2186.

[47] D. Collins, N. Gisin, N. Linden, S. Massar and S. Popescu, Bell inequalities for arbitrarily high dimensional systems, *Phys. Rev. Lett.* **88** (2002) 040404.

[48] A. Acín, T. Durt, N. Gisin and J. I. Latorre, Quantum nonlocality in two three-level systems, *Phys. Rev. A* **65** (2002) 052325.

[49] T. Fritz, Quantum correlations in the temporal Clauser–Horne–Shimony–Holt scenario, *New J. Phys.* **12** (2010) 083055.

[50] D. Gross, Hudson's theorem for finite-dimensional quantum systems, *J. Math. Phys.* **47** (2006) 122107.

[51] B. Fraiman, M. Graña, H. Parra De Freitas and S. Sethi, Non-supersymmetric heterotic strings on a circle, arXiv:2307.13745.

[52] I. Tan, Transversal gates of the $((3,3,2))$ qutrit code and local symmetries of the absolutely maximally entangled state of four qutrits, arXiv:2601.19677.

[53] B. R. Pahari, Classification and exact local masking in finite-field Clifford dual-unitary circuits, arXiv:2607.00210.

[54] W. Helwig, W. Cui, J. I. Latorre, A. Riera and H.-K. Lo, Absolute maximal entanglement and quantum secret sharing, *Phys. Rev. A* **86** (2012) 052335.

[55] S. Rana, Negative eigenvalues of partial transposition of arbitrary bipartite states, *Phys. Rev. A* **87** (2013) 054301.

[56] F. Huber, C. Eltschka, J. Siewert and O. Gühne, Bounds on absolutely maximally entangled states from shadow inequalities, and the quantum MacWilliams identity, *J. Phys. A* **51** (2018) 175301, arXiv:1708.06298.

[57] S. M. Carroll and A. Singh, Quantum mereology: factorizing Hilbert space into subsystems with quasiclassical dynamics, *Phys. Rev. A* **103** (2021) 022213.

[58] M. Grassl and T. A. Gulliver, On circulant self-dual codes over small fields, *Des. Codes Cryptogr.* **52** (2009) 57–81.

[59] M. Grassl and M. Rötteler, Quantum MDS codes over small fields, *Proc. IEEE ISIT* (2015), arXiv:1502.05267.

[60] G. W. Allen and D. A. Meyer, Polynomial monogamy relations for entanglement negativity, *Phys. Rev. Lett.* **118** (2017) 080402, arXiv:1502.04807.

[61] H. He and G. Vidal, Disentangling theorem and monogamy for entanglement negativity, *Phys. Rev. A* **91** (2015) 012339.

[62] P. Zanardi, E. Dallas, F. Andreadakis and S. Lloyd, Operational quantum mereology and minimal scrambling, *Quantum* **8** (2024) 1406, arXiv:2212.14340.

[63] D. M. Appleby, SIC-POVMs and the extended Clifford group, *J. Math. Phys.* **46** (2005) 052107, arXiv:quant-ph/0412001.

[64] M. Planat, Pauli graphs when the Hilbert space dimension contains a square: why the Dedekind psi function?, *J. Phys. A* **44** (2011) 045301, arXiv:1009.3858.

[65] M. J. Wonenburger, Transformations which are products of two involutions, *J. Math. Mech.* **16** (1966) 327–338.

[66] J. A. Thas, Symplectic spreads in $\mathrm{PG}(3,q)$, inversive planes and projective planes, *Discrete Math.* **174** (1997) 329–336.

[67] O. Taussky and H. Zassenhaus, On the similarity transformation between a matrix and its transpose, *Pacific J. Math.* **9** (1959) 893–896.

[68] G. Lusztig, Unipotent elements in small characteristic, *Transform. Groups* **10** (2005); arXiv:math/0503739.

[69] T. Xue, On unipotent and nilpotent pieces for classical groups, arXiv:0912.3820.

[70] G. E. Wall, On the conjugacy classes in the unitary, symplectic and orthogonal groups, *J. Austral. Math. Soc.* **3** (1963) 1–62.

[71] D. G. Glynn, The non-classical 10-arc of $\mathrm{PG}(4,9)$, *Discrete Math.* **59** (1986) 43–51.

[72] W. H. Hesselink, Nilpotency in classical groups over a field of characteristic 2, *Math. Z.* **173** (1980) 165–181.

[73] J. Milnor, On isometries of inner product spaces, *Invent. Math.* **8** (1969) 83–97.

[74] G. Lion and M. Vergne, *The Weil Representation, Maslov Index and Theta Series*, Progress in Mathematics **6**, Birkhäuser (1980).